

No-clustering bootstrap screen $\text{Sharpe} > 0$ (IS 2012–2022)

Directed factor-mimicking portfolio catalogue

Abstract

This catalogue constructs one FMP for each configured field or composite and compares all factors on one factor- and control-complete monthly window. Factors are indexed by reported raw Sharpe, highest first. Direction is frozen upstream; this report performs no return-based sign selection.

Methodology

Signals are observed at month-end, earn the next calendar month's simple return, and refresh monthly. Every factor must independently pass breadth, history, coverage, and non-degeneracy gates; none is silently dropped. One factor- and control-complete window fixes the reported Sharpe order and feeds all reported analyses. Displayed signal signs define the pre-direction composite; the reported Direction then orients the whole FMP. Risk Loadings include HAC t-statistics; Return Contributions are loading times annualized control mean, plus alpha and realized total. Rank diagnostics use 0/1/3/6/12/36-month horizons; available-history statistics are diagnostic only. Return Uncertainty uses paired circular stationary-bootstrap draws of the common monthly panel. It measures sampling and path uncertainty, not out-of-sample generalisation.

Construction Configuration

Setting	Effective value
Factors	100
Universe	shared security screen; refreshed monthly
Formation filters	price $\geq$ None; 21d dollar volume $\geq$ none
Market-cap screen	top 90.0% of names monthly
Signal preparation	within-month winsorize, normalize, then demean
Composite	mean available normalized components; factor-specific coverage floor
Construction	one blanket profile
Rebalance / return	month-end / next calendar month simple return
Costs	effective configured costs; default 0 bps
Minimum sample	20 stocks/month; 24 common months; 90% factor coverage
Comparison window	2012-02-01 to 2022-12-01; factor- and control-complete
Winsor settings	1%-99%
Controls	MKT, SMB, VAL, MOM, BAB, QMJ
Return uncertainty	paired stationary bootstrap; 2000 draws; expected block 12 months; seed 0
Catalogue note	Each of the 100 admitted signals is one standalone factor. No clustering, PCA, varimax, or post-screen factor pruning is applied. Admission requires at least 70% of fixed-orientation 2012–2022 stationary-bootstrap Sharpes to exceed 0.

Returns PCA

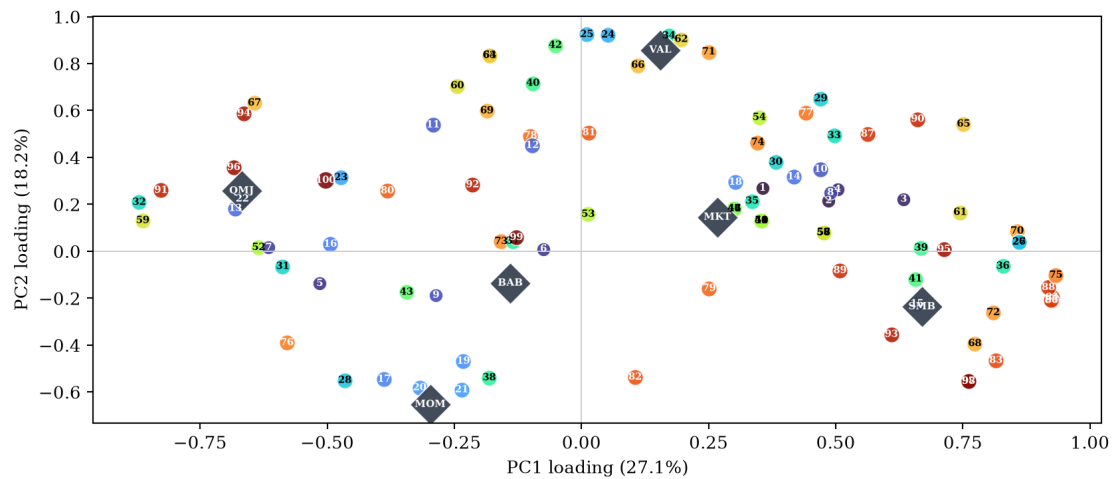


Figure: catalogue factors are numbered by reported-Sharpe index; attribution controls are named; $N=131$ common months.

Index

Idx	Factor	Dir.	N
1	ret_zscore_63d	+1	131
2	resid_ret_252v_1260c_SPY	+1	131
3	stoch_k_14d	+1	131
4	bb_pctb_20d	+1	131
5	eps_growth_yoy	+1	131
6	capex_to_revenue	+1	131
7	net_income_growth_yoy	+1	131
8	rsi_14d	+1	131
9	dpo	+1	131
10	price_zscore_63d	+1	131
11	debt_to_equity	+1	131
12	debt_to_assets	+1	131
13	gp_to_equity	+1	131
14	momentum_0_21	+1	131
15	accruals_total	+1	131
16	fcf_growth_yoy_ttm	+1	131
17	momentum_21_252_vol_63d	+1	131
18	accruals_noa_proxy	+1	131
19	momentum_21_126	+1	131
20	momentum_21_252	+1	131
21	resid_momentum_21_252	+1	131
22	share_turnover_21d	+1	131
23	asset_turnover	+1	131
24	ps_ttm	+1	131
25	sales_to_price	+1	131
26	log_market_cap	+1	131
27	market_cap	+1	131
28	revenue_growth_yoy	+1	131
29	altman_z	+1	131
30	ma_dist_63d	+1	131
31	gpoa	+1	131
32	fcf_to_equity	+1	131
33	equity_growth_yoy	+1	131
34	assets_to_mc	+1	131
35	accruals_wc_proxy	+1	131
36	ev	+1	131
37	dso	+1	131
38	is_negative_invested_capital	+1	131
39	interest_to_debt	+1	131
40	ncav_to_mc	+1	131
41	accruals	+1	131
42	evs_ttm	+1	131
43	surprisePercentage	+1	131
44	bab_252v_1260c_FRENCH_CRSP	+1	131
45	bab_shrunk_252v_1260c_FRENCH_CRSP	+1	131
46	beta_252v_1260c_FRENCH_CRSP	+1	131
47	beta_shrunk_252v_1260c_FRENCH_CRSP	+1	131
48	bab_252v_1260c_VTI	+1	131
49	bab_shrunk_252v_1260c_VTI	+1	131
50	beta_252v_1260c_VTI	+1	131
51	beta_shrunk_252v_1260c_VTI	+1	131
52	ebit_growth_yoy_ttm	+1	131
53	debt_issuance_yoy	+1	131
54	asset_growth_yoy	+1	131
55	bab_252v_1260c_SPY	+1	131
56	bab_shrunk_252v_1260c_SPY	+1	131
57	beta_252v_1260c_SPY	+1	131
58	beta_shrunk_252v_1260c_SPY	+1	131
59	fcf_margin	+1	131
60	cash_ratio	+1	131
61	dollar_volume_21d	+1	131
62	ev_to_assets	+1	131
63	gp_ev	+1	131
64	gross_profit_to_ev	+1	131
65	value_proxy_1260d	+1	131
66	pe_ttm	+1	131
67	fcf_to_market_cap	+1	131
68	net_operating_assets_proxy	+1	131
69	current_ratio	+1	131
70	amihud_illiq_21d	+1	131
71	pb	+1	131
72	roe_volatility	+1	131
73	dio	+1	131
74	gross_profit_margin_ttm	+1	131
75	net_income_margin_ttm	+1	131
76	roic_nopat	+1	131
77	growth_gmar_5y	+1	131
78	capital_intensity	+1	131
79	is_positive_revenue	+1	131
80	skew_63d	+1	131
81	accruals_percent	+1	131
82	rd_to_opex	+1	131
83	net_income_to_market_cap	+1	131
84	corr_1260d_FRENCH_CRSP	+1	131
85	corr_1260d_VTI	+1	131
86	corr_1260d_SPY	+1	131
87	growth_gpoa_5y	+1	131
88	ebit_margin_ttm	+1	131
89	working_capital_proxy	+1	131
90	momentum_21_756	+1	131
91	lottery_max_5_21d	+1	131
92	growth_acc_5y	+1	131
93	cash_earnings_ttm	+1	131
94	equity_issuance_yoy	+1	131
95	surprise	+1	131
96	sga_to_revenue	+1	131
97	earnings_yield	+1	131
98	ebit_to_ev	+1	131
99	net_income_growth_yoy_sign_flip	+1	131
100	ret_volume_corr_63d	+1	131

Returns

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ret_zscore_63d	8.24%	5.21%	1.58	4.51	-4.40%	0.179	7.32%	5.21%
resid_ret_252v_1260c_SPY	8.40%	5.88%	1.43	3.53	-4.59%	0.197	8.37%	5.88%
stoch_k_14d	8.96%	6.90%	1.30	4.22	-8.91%	0.294	10.04%	6.90%
bb_pctb_20d	8.13%	6.49%	1.25	4.20	-7.53%	0.189	9.00%	6.49%
eps_growth_yoy	5.49%	5.14%	1.07	3.04	-11.71%	0.241	2.72%	5.14%
capex_to_revenue	4.42%	4.53%	0.98	3.38	-5.61%	0.263	2.40%	4.53%
net_income_growth_yoy	4.55%	4.67%	0.97	2.88	-11.44%	0.309	1.63%	4.67%
rsi_14d	5.72%	6.20%	0.92	3.26	-12.74%	0.220	7.76%	6.20%
dpo	4.05%	4.43%	0.92	3.03	-8.94%	0.158	2.02%	4.43%
price_zscore_63d	5.57%	6.65%	0.84	2.83	-7.14%	0.340	8.57%	6.65%
debt_to_equity	4.36%	5.28%	0.83	2.62	-8.76%	0.445	2.77%	5.28%
debt_to_assets	4.05%	4.97%	0.81	2.68	-7.38%	0.252	3.12%	4.97%
gp_to_equity	4.40%	5.41%	0.81	2.51	-8.67%	0.497	0.44%	5.41%
momentum_0_21	5.26%	6.47%	0.81	3.05	-9.68%	0.309	7.40%	6.47%
accruals_total	3.92%	5.23%	0.75	2.62	-7.17%	0.394	5.70%	5.23%
fcf_growth_yoy_ttm	2.52%	3.62%	0.70	2.14	-9.35%	0.178	0.83%	3.62%
momentum_21_252_vol_63d	5.52%	7.95%	0.69	2.81	-10.16%	0.679	0.90%	7.95%
accruals_noa_proxy	3.38%	4.99%	0.68	2.19	-7.68%	0.241	4.32%	4.99%
momentum_21_126	5.05%	7.47%	0.68	2.90	-8.70%	0.485	0.76%	7.47%
momentum_21_252	5.60%	8.38%	0.67	2.87	-10.23%	0.678	1.09%	8.38%
resid_momentum_21_252	5.46%	8.35%	0.65	2.90	-11.86%	0.643	1.39%	8.35%
share_turnover_21d	5.54%	8.54%	0.65	2.05	-31.05%	0.463	4.89%	8.54%
asset_turnover	4.05%	6.26%	0.65	2.58	-7.85%	0.487	-0.82%	6.26%
ps_ttm	4.97%	8.25%	0.60	1.74	-20.72%	0.718	2.88%	8.25%
sales_to_price	5.01%	8.35%	0.60	1.74	-22.11%	0.714	2.60%	8.35%
log_market_cap	5.61%	10.24%	0.55	1.56	-15.67%	0.750	7.39%	10.24%
market_cap	5.61%	10.24%	0.55	1.56	-15.67%	0.750	7.39%	10.24%
revenue_growth_yoy	2.98%	5.51%	0.54	1.61	-16.61%	0.322	1.40%	5.51%
altman_z	3.92%	7.31%	0.54	1.70	-14.66%	0.568	6.11%	7.31%
ma_dist_63d	4.00%	7.57%	0.53	1.94	-11.14%	0.438	7.66%	7.57%
gpoa	3.07%	5.90%	0.52	1.58	-13.17%	0.511	-0.50%	5.90%
fcf_to_equity	3.74%	7.35%	0.51	1.49	-26.56%	0.629	-0.16%	7.35%
equity_growth_yoy	2.63%	5.20%	0.51	1.58	-10.74%	0.305	4.02%	5.20%
assets_to_mc	3.97%	8.26%	0.48	1.41	-22.36%	0.754	4.03%	8.26%
accruals_wc_proxy	1.79%	3.73%	0.48	1.62	-7.06%	0.168	2.65%	3.73%
ev	4.54%	9.61%	0.47	1.35	-14.69%	0.744	5.83%	9.61%
dso	2.17%	4.63%	0.47	1.54	-9.24%	0.072	0.54%	4.63%
is_negative_invested_capital	3.20%	7.00%	0.46	1.60	-16.52%	0.412	0.62%	7.00%
interest_to_debt	2.17%	4.79%	0.45	1.46	-9.71%	0.440	3.85%	4.79%
ncav_to_mc	3.05%	6.80%	0.45	1.37	-15.17%	0.588	2.16%	6.80%
accruals	2.27%	5.08%	0.45	1.53	-7.00%	0.315	4.61%	5.08%
evs_ttm	3.38%	7.64%	0.44	1.33	-20.57%	0.678	1.08%	7.64%
surprisePercentage	1.67%	3.84%	0.43	1.47	-8.69%	0.241	-0.81%	3.84%
bab_252v_1260c_FRENCH_CRSP	2.71%	6.28%	0.43	1.46	-11.84%	0.132	4.49%	6.28%
bab_shrunk_252v_1260c_FRENCH_CRSP	2.71%	6.28%	0.43	1.46	-11.84%	0.132	4.49%	6.28%
beta_252v_1260c_FRENCH_CRSP	2.71%	6.28%	0.43	1.46	-11.84%	0.132	4.49%	6.28%
beta_shrunk_252v_1260c_FRENCH_CRSP	2.71%	6.28%	0.43	1.46	-11.84%	0.132	4.49%	6.28%
bab_252v_1260c_VTI	2.50%	6.23%	0.40	1.37	-11.66%	0.118	4.52%	6.23%
bab_shrunk_252v_1260c_VTI	2.50%	6.23%	0.40	1.37	-11.66%	0.118	4.52%	6.23%
beta_252v_1260c_VTI	2.50%	6.23%	0.40	1.37	-11.66%	0.118	4.52%	6.23%
beta_shrunk_252v_1260c_VTI	2.50%	6.23%	0.40	1.37	-11.66%	0.118	4.52%	6.23%
ebit_growth_yoy_ttm	1.97%	4.91%	0.40	1.33	-13.25%	0.306	-1.38%	4.91%
debt_issuance_yoy	1.44%	3.61%	0.40	1.52	-7.71%	0.107	0.67%	3.61%
asset_growth_yoy	2.39%	6.03%	0.40	1.14	-15.85%	0.372	3.53%	6.03%
bab_252v_1260c_SPY	2.49%	6.44%	0.39	1.30	-11.82%	0.136	4.87%	6.44%
bab_shrunk_252v_1260c_SPY	2.49%	6.44%	0.39	1.30	-11.82%	0.136	4.87%	6.44%
beta_252v_1260c_SPY	2.49%	6.44%	0.39	1.30	-11.82%	0.136	4.87%	6.44%
beta_shrunk_252v_1260c_SPY	2.49%	6.44%	0.39	1.30	-11.82%	0.136	4.87%	6.44%
fcf_margin	2.48%	6.66%	0.37	0.99	-29.76%	0.525	0.50%	6.66%
cash_ratio	1.80%	5.44%	0.33	0.99	-21.24%	0.516	-1.03%	5.44%
dollar_volume_21d	2.56%	7.92%	0.32	1.00	-15.71%	0.663	7.92%	7.92%
ev_to_assets	2.56%	7.98%	0.32	0.93	-25.52%	0.731	3.27%	7.98%
gp_ev	2.30%	7.20%	0.32	0.95	-21.28%	0.637	0.08%	7.20%
gross_profit_to_ev	2.30%	7.20%	0.32	0.95	-21.28%	0.637	0.08%	7.20%
value_proxy_1260d	3.08%	9.66%	0.32	0.90	-17.75%	0.572	6.87%	9.66%
pe_ttm	2.33%	7.36%	0.32	1.03	-22.05%	0.555	2.56%	7.36%
fcf_to_market_cap	2.22%	7.01%	0.32	0.87	-30.79%	0.620	-0.21%	7.01%
net_operating_assets_proxy	3.10%	9.86%	0.31	0.91	-27.32%	0.738	4.46%	9.86%
current_ratio	1.77%	5.72%	0.31	0.93	-20.15%	0.478	0.39%	5.72%
amihud_illiq_21d	2.60%	8.72%	0.30	0.87	-15.89%	0.710	5.28%	8.72%
pb	2.11%	7.23%	0.29	0.91	-21.26%	0.747	3.47%	7.23%
roe_volatility	2.07%	7.51%	0.28	0.77	-18.98%	0.662	4.23%	7.51%
dio	1.11%	4.06%	0.27	1.08	-11.78%	0.056	0.86%	4.06%
gross_profit_margin_ttm	1.19%	4.58%	0.26	0.71	-19.19%	0.360	0.75%	4.58%
net_income_margin_ttm	2.37%	9.44%	0.25	0.72	-26.34%	0.641	5.75%	9.44%
roic_nopat	1.20%	4.77%	0.25	0.78	-9.59%	0.556	0.08%	4.77%
growth_gmar_5y	1.33%	5.48%	0.24	0.66	-18.65%	0.490	1.75%	5.48%
capital_intensity	1.05%	4.34%	0.24	0.71	-10.68%	0.243	0.76%	4.34%
is_positive_revenue	1.48%	6.16%	0.24	0.87	-14.86%	0.188	4.19%	6.16%
skew_63d	1.07%	4.53%	0.24	0.66	-16.90%	0.243	0.81%	4.53%
accruals_percent	0.89%	3.81%	0.23	0.66	-12.95%	0.289	1.33%	3.81%
rd_to_opex	1.43%	6.16%	0.23	0.73	-14.31%	0.361	3.39%	6.16%
net_income_to_market_cap	1.97%	8.91%	0.22	0.63	-27.74%	0.585	5.51%	8.91%
corr_1260d_FRENCH_CRSP	2.75%	12.49%	0.22	0.62	-29.11%	0.624	7.95%	12.49%
corr_1260d_VTI	2.71%	12.55%	0.22	0.60	-29.78%	0.626	7.90%	12.55%
corr_1260d_SPY	2.71%	12.62%	0.21	0.60	-30.26%	0.635	7.80%	12.62%
growth_gpoa_5y	1.18%	5.54%	0.21	0.58	-18.14%	0.463	2.40%	5.54%
ebit_margin_ttm	1.93%	9.23%	0.21	0.59	-28.86%	0.637	4.86%	9.23%
working_capital_proxy	0.85%	4.09%	0.21	0.77	-7.27%	0.245	2.50%	4.09%
momentum_21_756	1.74%	8.74%	0.20	0.57	-22.25%	0.533	5.96%	8.74%
lottery_max_5_21d	1.75%	9.74%	0.18	0.54	-37.05%	0.584	-1.61%	9.74%
growth_acc_5y	0.76%	4.28%	0.18	0.65	-12.10%	0.145	-0.92%	4.28%
cash_earnings_ttm	1.18%	6.92%	0.17	0.49	-18.41%	0.703	1.09%	6.92%
equity_issuance_yoy	1.09%	6.47%	0.17	0.44	-28.44%	0.735	-0.87%	6.47%
surprise	0.74%	4.70%	0.16	0.49	-17.47%	0.311	3.65%	4.70%
sga_to_revenue	0.86%	5.65%	0.15	0.41	-24.94%	0.389	-0.02%	5.65%
earnings_yield	1.37%	9.17%	0.15	0.44	-31.02%	0.607	4.89%	9.17%
ebit_to_ev	1.37%	9.17%	0.15	0.44	-31.02%	0.607	4.89%	9.17%
net_income_growth_yoy_sign_flip	0.66%	4.92%	0.13	0.45	-12.74%	0.015	1.37%	4.92%
ret_volume_corr_63d	-0.07%	5.31%	-0.01	-0.04	-21.17%	0.353	0.15%	5.31%

Return Uncertainty

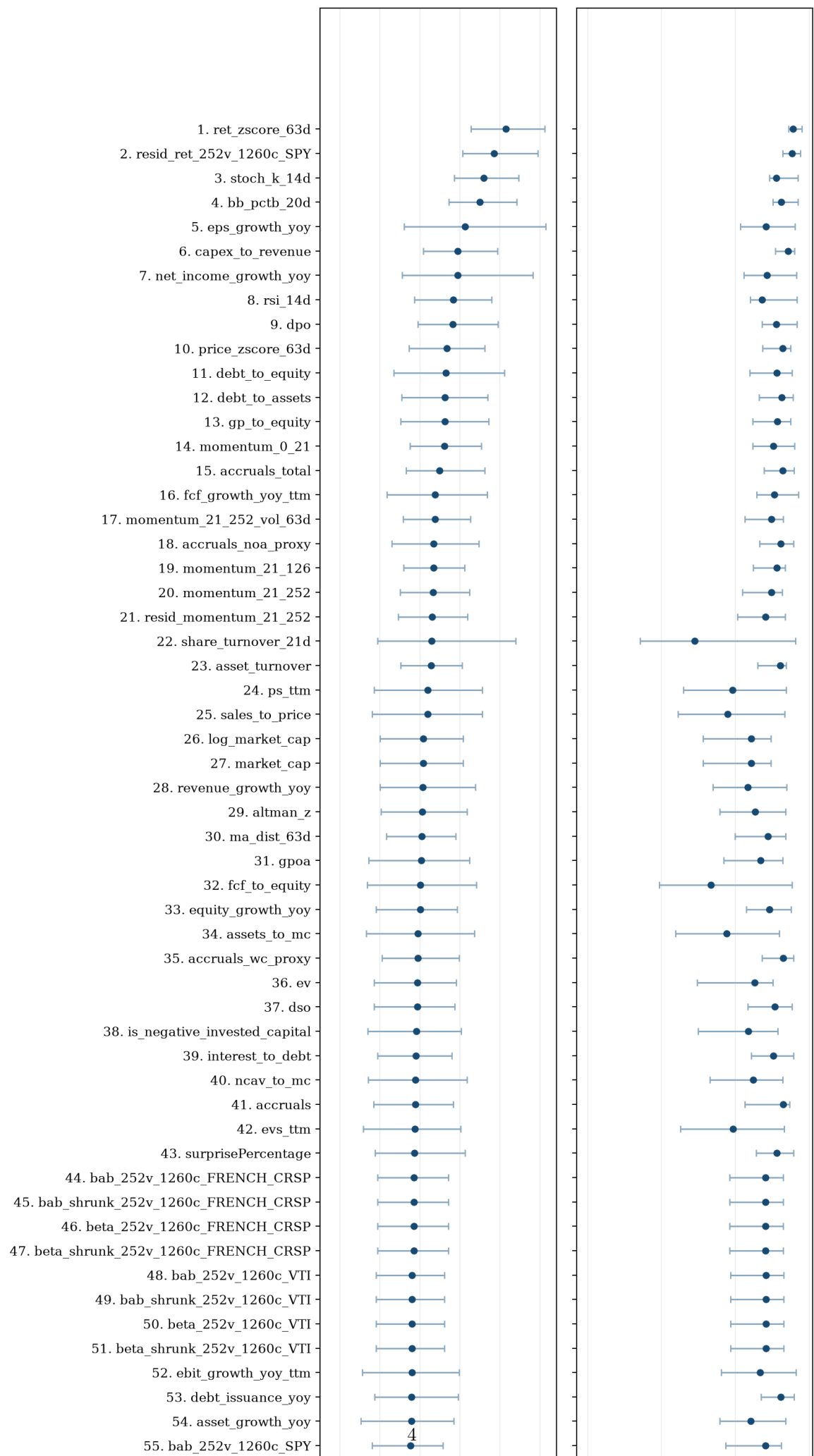


Figure: point estimates and paired stationary-bootstrap 95% intervals.

Idx	Factor	Ann. ret. [95%]	Sharpe [95%]	Pr(S<=0)	Skew [95%]	Max DD [95%]	ES 5% [95%]
1	ret_zscore_63d	8.2% [5.0%, 11.8%]	1.58 [1.14, 2.06]	0.0%	2.29 [0.50, 3.05]	-4.4% [-5.5%, -1.9%]	-1.5% [-1.9%, -1.1%]
2	resid_ret_252v_1260c_SP500	3.4% [4.5%, 13.2%]	1.43 [1.04, 1.97]	0.0%	2.96 [-0.03, 3.51]	-4.6% [-7.1%, -2.3%]	-1.8% [-2.1%, -1.3%]
3	stoch_k_14d	9.0% [5.4%, 13.2%]	1.30 [0.93, 1.74]	0.0%	1.35 [0.03, 1.77]	-8.9% [-10.8%, -3.0%]	-2.4% [-2.9%, -1.8%]
4	bb_pctb_20d	8.1% [4.8%, 12.0%]	1.25 [0.87, 1.72]	0.0%	1.26 [0.14, 1.81]	-7.5% [-9.8%, -3.1%]	-2.3% [-2.9%, -1.7%]
5	eps_growth_yoy	5.5% [1.8%, 9.2%]	1.07 [0.30, 2.07]	0.5%	-1.05 [-1.65, -0.06]	-11.7% [-18.6%, -3.8%]	-3.6% [-4.8%, -1.9%]
6	capex_to_revenue	4.4% [2.5%, 6.4%]	0.98 [0.55, 1.47]	0.0%	0.52 [-0.23, 1.18]	-5.6% [-9.2%, -3.9%]	-2.0% [-2.4%, -1.6%]
7	net_income_growth_yoy	4.5% [1.5%, 7.7%]	0.97 [0.28, 1.92]	0.4%	-1.44 [-2.45, 0.50]	-11.4% [-17.7%, -3.5%]	-2.0% [-2.4%, -1.5%]
8	rsi_14d	5.7% [2.5%, 9.1%]	0.92 [0.43, 1.40]	0.0%	0.77 [0.10, 1.25]	-12.7% [-15.9%, -3.2%]	-2.7% [-3.5%, -2.0%]
9	dpo	4.1% [2.2%, 5.9%]	0.92 [0.48, 1.48]	0.0%	-0.37 [-1.08, 0.80]	-8.9% [-12.7%, -3.2%]	-2.4% [-3.7%, -1.4%]
10	price_zscore_63d	5.6% [2.2%, 9.8%]	0.84 [0.36, 1.31]	0.0%	0.92 [0.32, 1.19]	-7.1% [-12.6%, -5.1%]	-2.4% [-2.7%, -2.1%]
11	debt_to_equity	4.4% [0.9%, 7.9%]	0.83 [0.17, 1.56]	0.7%	-0.12 [-0.84, 0.40]	-8.8% [-16.1%, -4.6%]	-3.0% [-3.9%, -2.0%]
12	debt_to_assets	4.1% [1.3%, 6.7%]	0.81 [0.27, 1.35]	0.2%	-0.50 [-1.61, 0.46]	-7.4% [-13.6%, -4.4%]	-2.8% [-4.1%, -1.8%]
13	gp_to_equity	4.4% [1.3%, 7.3%]	0.81 [0.26, 1.37]	0.2%	0.22 [-0.68, 0.70]	-8.7% [-15.3%, -5.0%]	-2.9% [-3.5%, -2.3%]
14	momentum_0_21	5.3% [2.1%, 8.5%]	0.81 [0.38, 1.27]	0.0%	0.99 [0.58, 1.33]	-9.7% [-15.3%, -3.9%]	-2.6% [-3.2%, -2.0%]
15	accruals_total	3.9% [1.6%, 6.3%]	0.75 [0.33, 1.31]	0.1%	1.41 [-0.79, 2.42]	-7.2% [-12.2%, -4.0%]	-2.5% [-3.3%, -1.7%]
16	fcf_growth_yoy_ttm	2.5% [0.3%, 4.6%]	0.70 [0.09, 1.34]	0.9%	-0.70 [-1.23, 0.29]	-9.3% [-14.2%, -2.8%]	-2.3% [-3.5%, -1.3%]
17	momentum_21_252_vol_63d	5.5% [2.4%, 8.6%]	0.69 [0.30, 1.13]	0.0%	-0.13 [-0.46, 0.21]	-10.2% [-17.4%, -7.0%]	-4.8% [-5.5%, -3.3%]
18	accruals_noa_proxy	3.4% [0.7%, 6.2%]	0.68 [0.15, 1.24]	0.5%	0.99 [-0.33, 1.88]	-7.7% [-13.5%, -4.2%]	-2.6% [-3.1%, -1.7%]
19	momentum_21_126	5.1% [2.2%, 7.8%]	0.68 [0.30, 1.06]	0.0%	-0.15 [-0.61, 0.48]	-8.7% [-15.2%, -6.5%]	-4.3% [-5.3%, -3.0%]
20	momentum_21_252	5.6% [2.3%, 8.9%]	0.67 [0.26, 1.13]	0.0%	-0.15 [-0.54, 0.21]	-10.2% [-18.1%, -7.3%]	-5.1% [-5.9%, -3.6%]
21	resid_momentum_21_252	5.5% [1.9%, 8.9%]	0.65 [0.23, 1.10]	0.1%	-0.04 [-0.52, 0.36]	-11.9% [-19.4%, -6.5%]	-5.0% [-5.8%, -3.7%]
22	share_turnover_21d	5.5% [-0.3%, 9.7%]	0.65 [-0.03, 1.70]	2.9%	-3.33 [-4.26, 0.68]	-31.1% [-45.8%, -3.7%]	-5.6% [-10.4%, -1.8%]
23	asset_turnover	4.1% [1.6%, 6.5%]	0.65 [0.26, 1.03]	0.1%	-0.03 [-0.31, 0.21]	-7.9% [-13.9%, -6.2%]	-3.4% [-3.9%, -2.6%]
24	ps_ttm	5.0% [-0.6%, 10.6%]	0.60 [-0.07, 1.28]	3.9%	0.26 [-0.42, 0.83]	-20.7% [-34.0%, -6.2%]	-4.5% [-5.7%, -2.8%]
25	sales_to_price	5.0% [-0.7%, 10.7%]	0.60 [-0.09, 1.28]	3.9%	0.25 [-0.35, 0.73]	-22.1% [-35.5%, -6.6%]	-4.7% [-5.8%, -2.8%]
26	log_market_cap	5.6% [0.0%, 12.5%]	0.55 [0.00, 1.04]	2.5%	1.02 [-0.07, 1.42]	-15.7% [-28.8%, -10.3%]	-4.4% [-5.2%, -3.5%]
27	market_cap	5.6% [0.0%, 12.5%]	0.55 [0.00, 1.04]	2.5%	1.02 [-0.07, 1.42]	-15.7% [-28.8%, -10.3%]	-4.4% [-5.2%, -3.5%]
28	revenue_growth_yoy	3.0% [0.0%, 5.5%]	0.54 [0.00, 1.19]	2.5%	-0.68 [-1.09, -0.02]	-16.6% [-26.0%, -6.0%]	-3.6% [-4.7%, -2.1%]
29	altman_z	3.9% [0.1%, 8.2%]	0.54 [0.02, 1.10]	2.1%	-0.08 [-1.72, 1.20]	-14.7% [-24.2%, -6.3%]	-4.2% [-6.2%, -2.4%]
30	ma_dist_63d	4.0% [0.6%, 8.0%]	0.53 [0.08, 0.95]	1.1%	0.92 [0.36, 1.17]	-11.1% [-20.1%, -6.4%]	-2.6% [-2.7%, -2.5%]
31	gpoa	3.1% [-0.8%, 6.6%]	0.52 [-0.14, 1.13]	5.7%	0.13 [-0.25, 0.49]	-13.2% [-23.1%, -7.2%]	-3.3% [-3.9%, -2.6%]
32	fcf_to_equity	3.7% [-1.2%, 8.0%]	0.51 [-0.16, 1.21]	6.5%	-0.40 [-1.09, 0.47]	-26.6% [-40.6%, -4.6%]	-4.6% [-6.1%, -2.7%]
33	equity_growth_yoy	2.6% [-0.2%, 5.7%]	0.51 [-0.05, 0.97]	3.6%	0.92 [0.08, 1.39]	-10.7% [-17.0%, -4.8%]	-2.5% [-2.9%, -1.8%]
34	assets_to_mc	4.0% [-1.3%, 9.6%]	0.48 [-0.17, 1.19]	7.0%	0.03 [-0.83, 0.84]	-22.4% [-36.1%, -8.1%]	-5.0% [-6.5%, -2.9%]
35	accruals_wc_proxy	1.8% [0.1%, 3.5%]	0.48 [0.03, 0.99]	2.2%	0.54 [-0.78, 1.48]	-7.1% [-12.7%, -4.1%]	-2.1% [-2.7%, -1.3%]
36	ev	4.5% [-0.5%, 10.7%]	0.47 [-0.07, 0.96]	4.2%	1.07 [0.01, 1.40]	-14.7% [-30.3%, -9.8%]	-3.9% [-4.3%, -3.4%]
37	dso	2.2% [-0.3%, 4.8%]	0.47 [-0.07, 0.94]	4.5%	0.33 [-0.12, 0.75]	-9.2% [-16.7%, -4.6%]	-2.5% [-3.1%, -1.8%]
38	is_negative_invested_capital	3.2% [-1.1%, 6.7%]	0.46 [-0.15, 1.02]	6.5%	-0.33 [-0.60, -0.05]	-16.5% [-30.1%, -8.5%]	-4.3% [-4.9%, -3.3%]
39	interest_to_debt	2.2% [-0.1%, 4.9%]	0.45 [-0.03, 0.90]	3.1%	0.80 [-0.21, 1.32]	-9.7% [-15.7%, -4.2%]	-2.4% [-2.8%, -1.8%]
40	ncav_to_mc	3.1% [-0.9%, 7.4%]	0.45 [-0.14, 1.09]	7.3%	-0.62 [-1.81, 0.49]	-15.2% [-26.9%, -7.2%]	-4.6% [-6.0%, -2.7%]
41	accruals	2.3% [-0.3%, 4.9%]	0.45 [-0.08, 0.92]	4.5%	1.36 [-0.40, 2.18]	-7.0% [-17.4%, -5.3%]	-2.4% [-3.1%, -1.8%]
42	evs_ttm	3.4% [-1.5%, 8.1%]	0.44 [-0.20, 1.01]	8.0%	0.60 [0.17, 0.90]	-20.6% [-34.9%, -6.7%]	-4.0% [-4.3%, -2.5%]
43	surprisePercentage	1.7% [-0.2%, 3.6%]	0.43 [-0.06, 1.07]	4.3%	-0.86 [-1.50, 0.02]	-8.7% [-14.3%, -4.2%]	-2.6% [-3.8%, -1.7%]
44	bab_252v_1260c_FRENCH_CRS	5.1% [4.0%, 5.7%]	0.43 [-0.03, 0.86]	3.5%	1.41 [-0.08, 2.54]	-11.8% [-21.6%, -7.0%]	-3.0% [-3.6%, -2.3%]
45	bab_shrunk_252v_1260c_FRENCH_CRS	5.1% [4.0%, 5.7%]	0.43 [-0.03, 0.86]	3.5%	1.41 [-0.08, 2.54]	-11.8% [-21.6%, -7.0%]	-3.0% [-3.6%, -2.3%]
46	beta_252v_1260c_FRENCH_CRS	5.1% [4.0%, 5.7%]	0.43 [-0.03, 0.86]	3.5%	1.41 [-0.08, 2.54]	-11.8% [-21.6%, -7.0%]	-3.0% [-3.6%, -2.3%]
47	beta_shrunk_252v_1260c_FRENCH_CRS	5.1% [4.0%, 5.7%]	0.43 [-0.03, 0.86]	3.5%	1.41 [-0.08, 2.54]	-11.8% [-21.6%, -7.0%]	-3.0% [-3.6%, -2.3%]
48	bab_252v_1260c_VTI	2.5% [-0.2%, 5.4%]	0.40 [-0.05, 0.81]	4.5%	1.50 [-0.02, 2.63]	-11.7% [-21.3%, -6.9%]	-3.0% [-3.5%, -2.2%]
49	bab_shrunk_252v_1260c_VTI	2.5% [-0.2%, 5.4%]	0.40 [-0.05, 0.81]	4.5%	1.50 [-0.02, 2.63]	-11.7% [-21.3%, -6.9%]	-3.0% [-3.5%, -2.2%]
50	beta_252v_1260c_VTI	2.5% [-0.2%, 5.4%]	0.40 [-0.05, 0.81]	4.5%	1.50 [-0.02, 2.63]	-11.7% [-21.3%, -6.9%]	-3.0% [-3.5%, -2.2%]
51	beta_shrunk_252v_1260c_VTI	2.5% [-0.2%, 5.4%]	0.40 [-0.05, 0.81]	4.5%	1.50 [-0.02, 2.63]	-11.7% [-21.3%, -6.9%]	-3.0% [-3.5%, -2.2%]
52	ebit_growth_yoy_ttm	2.0% [-1.1%, 4.9%]	0.40 [-0.21, 1.00]	11.1%	-0.59 [-1.31, 0.33]	-13.2% [-23.8%, -3.5%]	-3.2% [-4.3%, -2.1%]
53	debt_issuance_yoy	1.4% [-0.2%, 3.3%]	0.40 [-0.06, 0.98]	4.6%	0.24 [-0.52, 0.84]	-7.7% [-13.0%, -4.1%]	-2.2% [-2.8%, -1.2%]
54	asset_growth_yoy	2.4% [-1.1%, 6.2%]	0.40 [-0.23, 0.93]	9.6%	1.03 [-0.19, 1.65]	-15.8% [-24.2%, -6.3%]	-3.1% [-3.9%, -1.9%]
55	bab_252v_1260c_SPY	2.5% [-0.4%, 5.7%]	0.39 [-0.10, 0.79]	5.0%	1.71 [0.13, 2.76]	-11.8% [-22.6%, -7.5%]	-3.1% [-3.5%, -2.3%]
56	bab_shrunk_252v_1260c_SPY	2.5% [-0.4%, 5.7%]	0.39 [-0.10, 0.79]	5.0%	1.71 [0.13, 2.76]	-11.8% [-22.6%, -7.5%]	-3.1% [-3.5%, -2.3%]
57	beta_252v_1260c_SPY	2.5% [-0.4%, 5.7%]	0.39 [-0.10, 0.79]	5.0%	1.71 [0.13, 2.76]	-11.8% [-22.6%, -7.5%]	-3.1% [-3.5%, -2.3%]
58	beta_shrunk_252v_1260c_SPY	2.5% [-0.4%, 5.7%]	0.39 [-0.10, 0.79]	5.0%	1.71 [0.13, 2.76]	-11.8% [-22.6%, -7.5%]	-3.1% [-3.5%, -2.3%]
59	fcf_margin	2.5% [-2.6%, 6.5%]	0.37 [-0.32, 1.22]	14.4%	-1.60 [-2.50, 0.53]	-29.8% [-45.6%, -4.6%]	-4.8% [-7.5%, -2.1%]

Idx	Factor	Ann. ret. [95%]	Sharpe [95%]	Pr(S<=0)	Skew [95%]	Max DD [95%]	ES 5% [95%]
60	cash_ratio	1.8% [-2.2%, 5.9%]	0.33 [-0.47, 1.04]	19.9%	0.29 [-0.62, 0.81]	-21.2% [-31.0%, -15.7%]	-3.1% [-3.9%, -2.0%]
61	dollar_volume_21d	2.6% [-1.7%, 7.2%]	0.32 [-0.23, 0.86]	10.9%	0.39 [-0.11, 0.67]	-25.5% [-40.5%, -10.3%]	-3.9% [-4.6%, -3.2%]
62	ev_to_assets	2.6% [-2.7%, 7.9%]	0.32 [-0.34, 0.98]	16.4%	0.39 [-0.29, 0.88]	-21.3% [-36.4%, -8.2%]	-4.7% [-5.4%, -3.0%]
63	gp_ev	2.3% [-2.1%, 6.8%]	0.32 [-0.31, 0.90]	15.2%	0.70 [-0.02, 0.99]	-21.3% [-36.4%, -8.1%]	-3.8% [-4.1%, -2.8%]
64	gross_profit_to_ev	2.3% [-2.1%, 6.8%]	0.32 [-0.31, 0.90]	15.2%	0.70 [-0.02, 0.99]	-21.3% [-36.4%, -8.1%]	-3.8% [-4.1%, -2.8%]
65	value_proxy_1260d	3.1% [-2.4%, 9.4%]	0.32 [-0.31, 0.81]	14.4%	1.12 [-0.12, 1.67]	-17.7% [-38.2%, -11.6%]	-4.8% [-5.8%, -3.4%]
66	pe_ttm	2.3% [-2.0%, 6.7%]	0.32 [-0.28, 0.93]	14.9%	0.48 [-0.13, 0.73]	-22.1% [-33.4%, -7.2%]	-3.9% [-4.3%, -2.8%]
67	fcf_to_market_cap	2.2% [-3.8%, 7.6%]	0.32 [-0.53, 1.08]	21.9%	0.46 [-0.30, 0.76]	-30.8% [-47.2%, -6.4%]	-3.6% [-4.0%, -2.7%]
68	net_operating_assets_proxy	1% [-2.6%, 9.5%]	0.31 [-0.28, 0.89]	15.7%	0.72 [-0.19, 1.22]	-27.3% [-44.4%, -10.0%]	-5.4% [-6.0%, -3.3%]
69	current_ratio	1.8% [-2.1%, 5.7%]	0.31 [-0.40, 0.97]	19.8%	0.07 [-0.66, 0.58]	-20.1% [-29.8%, -6.4%]	-3.6% [-4.2%, -2.2%]
70	amihud_illiq_21d	2.6% [-2.3%, 8.1%]	0.30 [-0.29, 0.84]	15.4%	0.57 [-0.15, 0.89]	-15.9% [-36.3%, -11.5%]	-4.2% [-5.0%, -3.5%]
71	pb	2.1% [-2.2%, 6.6%]	0.29 [-0.32, 0.92]	16.4%	0.26 [-0.17, 0.72]	-21.3% [-34.9%, -7.8%]	-4.3% [-4.9%, -2.7%]
72	roe_volatility	2.1% [-2.6%, 8.0%]	0.28 [-0.44, 0.87]	21.3%	1.25 [-0.64, 1.85]	-19.0% [-36.8%, -9.3%]	-4.2% [-4.8%, -2.9%]
73	dio	1.1% [-1.0%, 3.0%]	0.27 [-0.19, 0.98]	14.3%	-0.02 [-0.93, 0.59]	-11.8% [-20.4%, -3.6%]	-2.6% [-3.7%, -1.3%]
74	gross_profit_margin_ttm	1.2% [-2.2%, 4.5%]	0.26 [-0.47, 1.10]	22.9%	0.32 [-0.00, 0.63]	-19.2% [-31.4%, -5.5%]	-2.4% [-2.7%, -1.6%]
75	net_income_margin_ttm	2.4% [-2.8%, 8.8%]	0.25 [-0.37, 0.79]	19.3%	1.31 [-0.57, 1.91]	-26.3% [-43.2%, -11.6%]	-5.0% [-6.0%, -3.0%]
76	roic_nopat	1.2% [-1.2%, 3.4%]	0.25 [-0.26, 0.73]	16.8%	0.23 [-0.80, 1.20]	-9.6% [-22.3%, -6.7%]	-2.8% [-3.8%, -1.9%]
77	growth_gmar_5y	1.3% [-2.1%, 4.6%]	0.24 [-0.40, 0.83]	19.6%	0.39 [-0.26, 0.84]	-18.7% [-31.1%, -6.9%]	-3.0% [-3.6%, -2.1%]
78	capital_intensity	1.1% [-1.6%, 3.8%]	0.24 [-0.41, 0.80]	24.6%	-0.15 [-0.52, 0.22]	-10.7% [-22.6%, -7.2%]	-2.7% [-3.3%, -2.0%]
79	is_positive_revenue	1.5% [-1.4%, 4.4%]	0.24 [-0.23, 0.76]	15.0%	0.15 [-0.41, 0.75]	-14.9% [-26.5%, -6.6%]	-3.5% [-4.5%, -2.2%]
80	skew_63d	1.1% [-1.6%, 3.7%]	0.24 [-0.34, 0.93]	21.1%	-0.53 [-0.99, 0.60]	-16.9% [-29.9%, -4.2%]	-3.1% [-4.2%, -1.5%]
81	accruals_percent	0.9% [-1.7%, 3.3%]	0.23 [-0.46, 0.87]	22.2%	-0.20 [-0.72, 0.38]	-12.9% [-25.5%, -5.3%]	-2.4% [-3.0%, -1.4%]
82	rd_to_opex	1.4% [-1.7%, 4.9%]	0.23 [-0.31, 0.80]	21.1%	0.46 [-0.14, 0.86]	-14.3% [-27.3%, -7.7%]	-3.1% [-3.7%, -2.5%]
83	net_income_to_market_cap	2% [-3.8%, 8.6%]	0.22 [-0.45, 0.87]	26.1%	0.86 [-0.77, 1.77]	-27.7% [-47.4%, -8.8%]	-5.3% [-6.0%, -3.0%]
84	corr_1260d_FRENCH_CREST	5% [-4.9%, 12.7%]	0.22 [-0.52, 0.81]	28.0%	1.72 [-0.42, 2.41]	-29.1% [-54.8%, -15.1%]	-6.2% [-7.8%, -4.0%]
85	corr_1260d_VTI	2.7% [-4.9%, 12.7%]	0.22 [-0.53, 0.81]	28.3%	1.70 [-0.44, 2.41]	-29.8% [-55.6%, -15.1%]	-6.3% [-7.9%, -4.0%]
86	corr_1260d_SPY	2.7% [-4.9%, 12.8%]	0.21 [-0.52, 0.81]	28.1%	1.68 [-0.42, 2.37]	-30.3% [-56.2%, -15.1%]	-6.3% [-7.9%, -4.0%]
87	growth_gpoa_5y	1.2% [-2.1%, 4.4%]	0.21 [-0.41, 0.79]	21.5%	0.71 [-0.39, 1.29]	-18.1% [-30.6%, -7.8%]	-2.9% [-3.3%, -2.1%]
88	ebit_margin_ttm	1.9% [-3.2%, 8.2%]	0.21 [-0.44, 0.77]	23.4%	1.28 [-0.41, 1.86]	-28.9% [-46.6%, -11.1%]	-4.8% [-5.8%, -3.1%]
89	working_capital_proxy	0.8% [-1.1%, 2.8%]	0.21 [-0.30, 0.64]	21.4%	0.41 [-0.30, 0.84]	-7.3% [-18.1%, -5.3%]	-2.4% [-2.7%, -1.6%]
90	momentum_21_756	1.7% [-3.1%, 6.6%]	0.20 [-0.40, 0.68]	22.9%	0.87 [0.15, 1.22]	-22.2% [-41.8%, -13.3%]	-4.1% [-4.5%, -3.5%]
91	lottery_max_5_21d	1.8% [-5.1%, 7.1%]	0.18 [-0.40, 1.04]	28.5%	-2.15 [-2.97, 0.58]	-37.0% [-60.1%, -6.8%]	-7.1% [-12.0%, -3.0%]
92	growth_acc_5y	0.8% [-1.4%, 3.4%]	0.18 [-0.39, 0.68]	29.0%	0.80 [-0.33, 1.73]	-12.1% [-19.7%, -5.6%]	-2.4% [-2.9%, -1.6%]
93	cash_earnings_ttm	1.2% [-2.8%, 6.0%]	0.17 [-0.42, 0.80]	30.4%	0.74 [0.17, 1.20]	-18.4% [-35.9%, -9.1%]	-3.5% [-4.0%, -2.4%]
94	equity_issuance_yoy	1.1% [-4.1%, 6.1%]	0.17 [-0.61, 0.93]	32.2%	0.05 [-0.52, 0.75]	-28.4% [-47.8%, -9.1%]	-4.0% [-4.8%, -2.2%]
95	surprise	0.7% [-2.0%, 3.7%]	0.16 [-0.49, 0.68]	28.5%	1.52 [-0.07, 2.14]	-17.5% [-28.1%, -5.3%]	-2.3% [-2.7%, -1.4%]
96	sga_to_revenue	0.9% [-2.9%, 4.4%]	0.15 [-0.47, 0.88]	32.0%	-1.06 [-1.76, 0.63]	-24.9% [-38.2%, -5.2%]	-4.0% [-6.4%, -1.7%]
97	earnings_yield	1.4% [-4.7%, 7.9%]	0.15 [-0.50, 0.80]	30.9%	0.54 [-0.76, 1.34]	-31.0% [-51.4%, -10.4%]	-5.7% [-6.6%, -3.4%]
98	ebit_to_ev	1.4% [-4.7%, 7.9%]	0.15 [-0.50, 0.80]	30.9%	0.54 [-0.76, 1.34]	-31.0% [-51.4%, -10.4%]	-5.7% [-6.6%, -3.4%]
99	net_income_growth_yoy	9% [-1.6%, 3.2%]	0.13 [-0.31, 0.70]	29.6%	-0.58 [-1.07, 0.32]	-12.7% [-24.6%, -6.1%]	-3.3% [-4.4%, -1.9%]
100	ret_volume_corr_63d	-0.1% [-3.1%, 2.8%]	-0.01 [-0.48, 0.64]	51.7%	-1.14 [-1.50, 0.12]	-21.2% [-38.5%, -6.6%]	-4.0% [-5.5%, -1.9%]

Risk Loadings

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ret_zscore_63d	0.03 (1.07)	0.21 (3.15)	0.07 (1.60)	-0.05 (-1.04)	0.12 (1.06)	0.07 (0.81)	6.91% (4.89)	0.188
resid_ret_252v_1260c_SPY	0.04 (1.03)	0.19 (2.26)	0.07 (1.39)	-0.07 (-1.44)	0.14 (1.08)	-0.05 (-0.52)	7.70% (4.21)	0.213
stoch_k_14d	0.12 (2.44)	0.25 (3.97)	0.04 (0.64)	-0.11 (-1.53)	0.03 (0.33)	-0.09 (-1.30)	8.49% (4.89)	0.335
bb_pctb_20d	0.13 (2.67)	0.15 (2.65)	0.03 (0.37)	-0.11 (-1.56)	0.01 (0.09)	-0.02 (-0.36)	7.23% (4.11)	0.250
eps_growth_yoy	0.04 (1.27)	-0.05 (-0.89)	0.11 (2.36)	0.23 (4.68)	0.08 (1.57)	0.07 (0.87)	2.88% (1.71)	0.243
capex_to_revenue	0.01 (0.50)	0.27 (4.01)	-0.03 (-0.90)	0.00 (0.04)	0.06 (1.33)	0.25 (4.95)	2.47% (1.96)	0.264
net_income_growth_yoy	0.03 (0.85)	-0.06 (-1.55)	0.10 (3.01)	0.19 (4.82)	0.07 (1.51)	0.15 (2.05)	1.96% (1.36)	0.313
rsi_14d	0.11 (2.64)	0.13 (2.41)	0.01 (0.20)	-0.11 (-1.48)	-0.10 (-1.42)	-0.05 (-1.05)	6.15% (3.38)	0.264
dpo	0.03 (0.87)	0.10 (1.54)	-0.07 (-2.40)	0.03 (1.09)	0.04 (0.62)	0.18 (2.27)	2.08% (1.38)	0.158
price_zscore_63d	0.09 (2.27)	0.02 (0.27)	-0.03 (-0.71)	-0.26 (-5.60)	-0.10 (-1.54)	-0.11 (-1.87)	7.15% (4.23)	0.368
debt_to_equity	0.09 (2.86)	-0.33 (-6.93)	0.14 (3.49)	-0.01 (-0.33)	0.14 (2.33)	-0.06 (-1.11)	2.47% (1.95)	0.450
debt_to_assets	0.06 (1.76)	-0.15 (-2.62)	0.16 (3.62)	0.02 (0.43)	0.15 (2.20)	-0.10 (-1.54)	2.73% (2.18)	0.261
gp_to_equity	0.05 (1.82)	-0.18 (-4.54)	-0.08 (-2.75)	-0.04 (-1.52)	0.14 (3.48)	0.32 (6.17)	0.51% (0.37)	0.498
momentum_0_21	0.15 (2.74)	0.03 (0.53)	-0.00 (-0.05)	-0.19 (-3.52)	-0.05 (-0.90)	-0.09 (-1.47)	5.38% (3.61)	0.365
accruals_total	-0.10 (-3.26)	0.19 (3.70)	0.02 (0.40)	0.01 (0.25)	0.12 (1.72)	-0.27 (-4.25)	5.81% (4.03)	0.395
fcf_growth_yoy_ttm	-0.02 (-0.59)	-0.05 (-1.23)	0.02 (0.93)	0.05 (1.36)	0.09 (1.92)	0.08 (1.50)	1.31% (1.26)	0.192
momentum_21_252_vol_63d	0.03 (0.98)	-0.12 (-2.33)	-0.05 (-1.20)	0.49 (9.65)	0.17 (3.37)	-0.05 (-0.60)	1.12% (0.84)	0.680
accruals_noa_proxy	-0.11 (-3.11)	0.07 (1.16)	0.20 (3.56)	0.06 (1.39)	0.05 (0.41)	-0.06 (-0.80)	4.94% (3.57)	0.265
momentum_21_126	0.04 (1.16)	0.10 (1.22)	0.04 (0.61)	0.46 (6.82)	0.11 (2.43)	0.06 (0.81)	1.03% (0.61)	0.486
momentum_21_252	0.02 (0.77)	-0.06 (-1.11)	-0.04 (-0.70)	0.53 (9.02)	0.19 (3.21)	-0.09 (-1.65)	1.31% (1.03)	0.679
resid_momentum_21_252	0.02 (0.45)	-0.04 (-0.66)	-0.04 (-0.64)	0.51 (8.68)	0.19 (2.72)	-0.13 (-2.46)	1.61% (1.23)	0.644
share_turnover_21d	-0.01 (-0.23)	-0.31 (-2.63)	-0.00 (-0.05)	-0.03 (-0.54)	-0.27 (-1.48)	0.44 (3.61)	5.47% (2.97)	0.469
asset_turnover	0.08 (2.45)	0.17 (3.03)	0.03 (0.75)	-0.01 (-0.20)	0.08 (1.35)	0.59 (10.05)	-0.77% (-0.56)	0.487
ps_ttm	0.12 (4.33)	0.10 (1.72)	0.47 (17.13)	-0.11 (-2.34)	0.10 (2.37)	0.28 (4.95)	2.65% (1.66)	0.719
sales_to_price	0.12 (4.37)	0.09 (1.44)	0.48 (15.10)	-0.10 (-1.95)	0.11 (2.25)	0.30 (4.88)	2.38% (1.50)	0.715
log_market_cap	0.03 (0.77)	0.94 (12.98)	0.21 (4.44)	-0.03 (-0.61)	0.03 (0.37)	-0.12 (-1.52)	7.07% (4.49)	0.751
market_cap	0.03 (0.77)	0.94 (12.98)	0.21 (4.44)	-0.03 (-0.61)	0.03 (0.37)	-0.12 (-1.52)	7.07% (4.49)	0.751
revenue_growth_yoy	0.03 (1.15)	-0.15 (-2.60)	-0.14 (-2.58)	0.13 (2.96)	0.05 (0.83)	-0.07 (-0.90)	1.42% (0.83)	0.322
altman_z	0.04 (0.96)	-0.07 (-1.08)	0.39 (6.27)	-0.05 (-0.95)	0.07 (1.07)	-0.32 (-4.48)	5.66% (3.62)	0.572
ma_dist_63d	0.13 (2.92)	-0.05 (-0.80)	-0.03 (-0.77)	-0.31 (-6.49)	-0.17 (-3.22)	-0.14 (-2.15)	6.12% (3.74)	0.462
gpoa	-0.01 (-0.19)	0.04 (0.81)	-0.16 (-4.65)	-0.05 (-1.04)	0.04 (0.71)	0.49 (9.67)	-0.06% (-0.04)	0.515
fcf_to_equity	-0.01 (-0.17)	-0.34 (-6.71)	-0.01 (-0.17)	0.03 (0.65)	0.07 (1.68)	0.43 (6.13)	0.29% (0.22)	0.632
equity_growth_yoy	-0.09 (-2.26)	0.08 (1.27)	0.19 (3.20)	-0.07 (-1.12)	0.07 (0.84)	-0.05 (-0.77)	4.25% (3.59)	0.308
assets_to_mc	0.08 (3.02)	-0.04 (-0.64)	0.54 (13.82)	-0.08 (-1.93)	0.07 (1.84)	0.01 (0.22)	3.82% (2.71)	0.755
accruals_wc_proxy	-0.08 (-3.78)	0.01 (0.26)	0.06 (1.50)	-0.04 (-0.90)	0.10 (1.66)	-0.12 (-2.61)	2.96% (3.28)	0.178
ev	0.03 (0.83)	0.94 (12.79)	0.14 (3.20)	0.01 (0.25)	-0.01 (-0.12)	-0.04 (-0.53)	5.57% (3.88)	0.745
dso	0.02 (0.36)	-0.04 (-0.79)	-0.01 (-0.14)	0.01 (0.16)	0.09 (1.66)	0.08 (1.82)	0.55% (0.42)	0.072
is_negative_invested_capital	-0.01 (-0.35)	0.10 (1.59)	-0.35 (-6.16)	-0.01 (-0.19)	0.09 (1.18)	0.18 (2.00)	0.84% (0.49)	0.413
interest_to_debt	-0.03 (-1.72)	0.14 (2.67)	0.08 (1.81)	-0.01 (-0.21)	0.11 (2.22)	-0.29 (-5.85)	3.63% (3.02)	0.442
ncav_to_mc	0.10 (3.34)	-0.36 (-5.51)	0.33 (6.49)	-0.01 (-0.24)	0.14 (2.00)	-0.14 (-2.58)	1.83% (1.17)	0.591
accruals	-0.11 (-3.20)	0.09 (1.56)	0.05 (1.03)	-0.01 (-0.23)	0.09 (1.18)	-0.30 (-4.42)	4.77% (3.20)	0.316
evs_ttm	0.10 (3.15)	0.17 (3.06)	0.39 (14.92)	-0.09 (-1.78)	0.03 (0.68)	0.43 (8.02)	0.94% (0.62)	0.679
surprisePercentage	0.01 (0.47)	0.10 (2.49)	0.02 (0.81)	0.09 (2.66)	0.10 (2.69)	0.15 (2.87)	-0.64% (-0.61)	0.243
bab_252v_1260c_FRENCH_CRSP	0.08 (1.25)	-0.08 (-1.36)	0.11 (2.45)	0.01 (0.18)	-0.16 (-3.22)	-0.07 (-0.97)	3.64% (1.85)	0.149
bab_shrunk_252v_1260c_FRENCH_CRSP	0.08 (1.25)	-0.08 (-1.36)	0.11 (2.45)	0.01 (0.18)	-0.16 (-3.22)	-0.07 (-0.97)	3.64% (1.85)	0.149
beta_252v_1260c_FRENCH_CRSP	0.08 (1.25)	-0.08 (-1.36)	0.11 (2.45)	0.01 (0.18)	-0.16 (-3.22)	-0.07 (-0.97)	3.64% (1.85)	0.149
beta_shrunk_252v_1260c_FRENCH_CRSP	0.08 (1.25)	-0.08 (-1.36)	0.11 (2.45)	0.01 (0.18)	-0.16 (-3.22)	-0.07 (-0.97)	3.64% (1.85)	0.149
bab_252v_1260c_VTI	0.08 (1.14)	-0.05 (-0.90)	0.08 (1.72)	-0.00 (-0.06)	-0.16 (-3.11)	-0.09 (-1.20)	3.59% (1.82)	0.137
bab_shrunk_252v_1260c_VTI	0.08 (1.14)	-0.05 (-0.90)	0.08 (1.72)	-0.00 (-0.06)	-0.16 (-3.11)	-0.09 (-1.20)	3.59% (1.82)	0.137
beta_252v_1260c_VTI	0.08 (1.14)	-0.05 (-0.90)	0.08 (1.72)	-0.00 (-0.06)	-0.16 (-3.11)	-0.09 (-1.20)	3.59% (1.82)	0.137
beta_shrunk_252v_1260c_VTI	0.08 (1.14)	-0.05 (-0.90)	0.08 (1.72)	-0.00 (-0.06)	-0.16 (-3.11)	-0.09 (-1.20)	3.59% (1.82)	0.137
ebit_growth_yoy_ttm	0.05 (1.58)	-0.11 (-2.40)	0.04 (0.95)	0.14 (3.22)	0.13 (2.52)	0.14 (2.08)	-1.25% (-0.93)	0.307
debt_issuance_yoy	-0.03 (-1.01)	0.07 (1.86)	0.09 (2.83)	0.06 (2.16)	0.04 (0.58)	0.06 (1.06)	1.11% (1.07)	0.122
asset_growth_yoy	-0.09 (-2.25)	0.11 (1.60)	0.28 (4.14)	-0.03 (-0.53)	0.05 (0.45)	0.03 (0.38)	3.87% (2.61)	0.376
bab_252v_1260c_SPY	0.08 (1.18)	0.04 (0.61)	0.07 (1.49)	-0.01 (-0.12)	-0.16 (-2.81)	-0.11 (-1.56)	3.67% (1.81)	0.161

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bab_shrunk_252v_1260c_SPY	0.08 (1.18)	0.04 (0.61)	0.07 (1.49)	-0.01 (-0.12)	-0.16 (-2.81)	-0.11 (-1.56)	3.67% (1.81)	0.161
beta_252v_1260c_SPY	0.08 (1.18)	0.04 (0.61)	0.07 (1.49)	-0.01 (-0.12)	-0.16 (-2.81)	-0.11 (-1.56)	3.67% (1.81)	0.161
beta_shrunk_252v_1260c_SPY	0.08 (1.18)	0.04 (0.61)	0.07 (1.49)	-0.01 (-0.12)	-0.16 (-2.81)	-0.11 (-1.56)	3.67% (1.81)	0.161
fcf_margin	-0.02 (-0.43)	-0.32 (-4.58)	-0.04 (-0.72)	0.01 (0.20)	-0.07 (-0.77)	0.33 (4.48)	1.01% (0.69)	0.531
cash_ratio	0.08 (3.25)	-0.08 (-1.90)	0.27 (9.40)	0.02 (0.56)	0.16 (4.18)	0.16 (3.15)	-1.18% (-1.02)	0.517
dollar_volume_21d	0.03 (0.81)	0.67 (11.84)	0.17 (4.73)	-0.08 (-1.75)	-0.10 (-1.75)	-0.00 (-0.03)	4.54% (3.74)	0.665
ev_to_assets	0.05 (1.64)	0.05 (0.93)	0.50 (13.48)	-0.09 (-2.08)	-0.02 (-0.45)	0.10 (1.69)	3.12% (2.18)	0.731
gp_ev	0.05 (1.54)	0.08 (1.43)	0.31 (9.53)	-0.12 (-2.57)	0.06 (0.90)	0.44 (7.46)	0.06% (0.04)	0.637
gross_profit_to_ev	0.05 (1.54)	0.08 (1.43)	0.31 (9.53)	-0.12 (-2.57)	0.06 (0.90)	0.44 (7.46)	0.06% (0.04)	0.637
value_proxy_1260d	0.01 (0.26)	0.25 (2.68)	0.42 (6.60)	-0.17 (-2.67)	0.04 (0.34)	-0.30 (-2.51)	6.30% (2.93)	0.576
pe_ttm	0.04 (1.11)	-0.03 (-0.47)	0.34 (5.88)	-0.16 (-2.69)	0.06 (0.53)	0.10 (1.56)	2.32% (1.46)	0.556
fcf_to_market_cap	-0.00 (-0.09)	-0.21 (-3.91)	0.24 (5.78)	-0.03 (-0.61)	0.07 (1.50)	0.38 (6.68)	0.03% (0.02)	0.622
net_operating_assets_proxy	-0.02 (-0.49)	0.91 (12.16)	-0.12 (-2.72)	0.02 (0.38)	0.02 (0.26)	-0.13 (-1.61)	4.28% (2.80)	0.739
current_ratio	0.05 (1.73)	-0.34 (-5.61)	0.21 (4.85)	0.00 (0.07)	0.15 (2.19)	-0.06 (-1.01)	0.23% (0.15)	0.479
amihud_illiq_21d	0.00 (0.00)	0.73 (13.75)	0.16 (3.85)	-0.10 (-1.93)	-0.01 (-0.26)	-0.13 (-2.48)	4.98% (3.77)	0.711
pb	0.03 (1.21)	0.10 (2.25)	0.45 (17.95)	-0.08 (-2.44)	-0.08 (-2.05)	0.10 (2.00)	3.37% (3.09)	0.747
roe_volatility	0.02 (0.78)	0.20 (2.67)	-0.03 (-0.56)	-0.01 (-0.10)	0.17 (2.52)	-0.58 (-7.29)	3.74% (2.37)	0.665
dio	0.02 (0.41)	-0.07 (-1.79)	0.06 (1.28)	0.08 (2.02)	-0.04 (-0.49)	-0.00 (-0.07)	0.94% (0.68)	0.057
gross_profit_margin_ttm	0.08 (2.98)	0.16 (2.98)	0.21 (7.35)	0.04 (1.03)	0.03 (0.87)	0.00 (0.09)	0.26% (0.19)	0.375
net_income_margin_ttm	-0.03 (-0.69)	0.51 (6.88)	0.07 (1.28)	-0.05 (-0.84)	0.16 (1.93)	-0.54 (-5.21)	5.32% (3.11)	0.643
roic_nopat	-0.03 (-1.06)	-0.23 (-6.27)	-0.24 (-6.92)	-0.03 (-0.86)	0.00 (0.04)	0.10 (2.15)	0.34% (0.30)	0.559
growth_gmar_5y	0.03 (0.79)	0.18 (3.25)	0.34 (8.96)	0.08 (1.91)	-0.02 (-0.22)	0.01 (0.13)	1.60% (1.39)	0.491
capital_intensity	0.03 (1.12)	-0.05 (-1.09)	0.19 (5.49)	0.06 (1.43)	-0.03 (-0.43)	0.04 (0.86)	0.71% (0.54)	0.243
is_positive_revenue	-0.03 (-0.62)	0.01 (0.10)	0.03 (0.68)	0.13 (2.12)	-0.18 (-3.12)	-0.25 (-2.64)	4.26% (2.10)	0.188
skew_63d	0.06 (1.82)	-0.18 (-3.68)	-0.02 (-0.52)	-0.03 (-0.70)	-0.12 (-2.49)	0.11 (2.12)	0.68% (0.41)	0.244
accruals_percent	-0.03 (-1.27)	-0.05 (-1.26)	0.17 (4.30)	0.01 (0.30)	0.01 (0.29)	-0.00 (-0.09)	1.52% (1.21)	0.293
rd_to_opex	-0.03 (-0.76)	-0.02 (-0.31)	-0.31 (-8.56)	-0.06 (-1.21)	-0.14 (-2.32)	-0.13 (-1.74)	3.42% (2.23)	0.361
net_income_to_market_cap	-0.08 (-1.96)	0.39 (4.53)	-0.17 (-2.54)	0.01 (0.11)	0.06 (0.68)	-0.51 (-4.96)	5.34% (2.47)	0.586
corr_1260d_FRENCH_CRSP	-0.06 (-0.78)	0.66 (5.58)	-0.03 (-0.46)	-0.12 (-1.64)	0.14 (0.94)	-0.71 (-5.65)	7.40% (3.25)	0.626
corr_1260d_VTI	-0.06 (-0.79)	0.66 (5.62)	-0.04 (-0.68)	-0.12 (-1.76)	0.15 (0.95)	-0.71 (-5.60)	7.35% (3.24)	0.627
corr_1260d_SPY	-0.06 (-0.74)	0.69 (5.90)	-0.05 (-0.77)	-0.13 (-1.79)	0.15 (0.95)	-0.69 (-5.56)	7.25% (3.25)	0.637
growth_gpoa_5y	0.00 (0.03)	0.15 (2.64)	0.33 (7.85)	0.08 (1.81)	0.01 (0.07)	-0.11 (-1.49)	2.26% (1.85)	0.464
ebit_margin_ttm	-0.01 (-0.11)	0.56 (7.75)	0.04 (0.72)	-0.03 (-0.48)	0.10 (1.23)	-0.45 (-4.55)	4.41% (2.61)	0.639
working_capital_proxy	-0.02 (-0.83)	-0.03 (-0.65)	0.04 (1.17)	0.02 (0.59)	0.03 (0.55)	-0.27 (-5.28)	2.34% (1.93)	0.247
momentum_21_756	-0.07 (-1.83)	0.21 (2.70)	0.37 (5.95)	-0.19 (-2.98)	-0.03 (-0.36)	-0.16 (-1.56)	5.83% (3.07)	0.533
lottery_max_5_21d	0.09 (1.67)	-0.47 (-4.54)	0.01 (0.23)	0.05 (0.69)	-0.22 (-1.56)	0.58 (4.60)	-1.33% (-0.69)	0.585
growth_acc_5y	0.04 (1.28)	0.06 (1.14)	0.10 (2.15)	0.05 (0.99)	0.03 (0.73)	0.17 (3.54)	-0.94% (-0.79)	0.145
cash_earnings_ttm	0.04 (1.38)	0.67 (10.68)	-0.10 (-2.39)	0.00 (0.06)	-0.01 (-0.14)	0.04 (0.74)	0.88% (0.75)	0.704
equity_issuance_yoy	0.01 (0.41)	-0.24 (-7.32)	0.17 (5.50)	-0.03 (-0.90)	-0.06 (-1.22)	0.43 (9.79)	-0.70% (-0.65)	0.736
surprise	-0.03 (-0.89)	0.06 (1.03)	-0.03 (-1.23)	-0.12 (-3.57)	-0.03 (-0.37)	-0.24 (-3.09)	3.31% (2.60)	0.315
sga_to_revenue	0.03 (0.96)	-0.31 (-5.32)	0.07 (1.70)	0.00 (0.02)	-0.06 (-1.09)	0.14 (1.69)	0.13% (0.07)	0.390
earnings_yield	-0.08 (-1.83)	0.36 (4.25)	-0.21 (-3.41)	0.03 (0.56)	0.05 (0.53)	-0.55 (-5.84)	4.73% (2.27)	0.607
ebit_to_ev	-0.08 (-1.83)	0.36 (4.25)	-0.21 (-3.41)	0.03 (0.56)	0.05 (0.53)	-0.55 (-5.84)	4.73% (2.27)	0.607
net_income_growth_yoy_sign_flip	-0.03 (-0.44)	-0.03 (-0.42)	0.02 (0.39)	0.00 (0.00)	-0.04 (-0.69)	-0.03 (-0.44)	1.61% (0.99)	0.018
ret_volume_corr_63d	0.06 (2.03)	-0.27 (-4.22)	-0.04 (-0.93)	-0.09 (-2.33)	-0.15 (-3.78)	0.11 (1.60)	-0.02% (-0.01)	0.355

Return Contributions

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ret_zscore_63d	0.44%	-0.18%	-0.17%	-0.24%	1.10%	0.39%	6.91%	8.24%
resid_ret_252v_1260c_SPY	0.47%	-0.17%	-0.17%	-0.37%	1.24%	-0.30%	7.70%	8.40%
stoch_k_14d	1.58%	-0.22%	-0.10%	-0.53%	0.26%	-0.53%	8.49%	8.96%
bb_pctb_20d	1.71%	-0.13%	-0.06%	-0.57%	0.07%	-0.11%	7.23%	8.13%
eps_growth_yoy	0.55%	0.04%	-0.26%	1.15%	0.75%	0.38%	2.88%	5.49%
capex_to_revenue	0.16%	-0.23%	0.06%	0.01%	0.51%	1.45%	2.47%	4.42%
net_income_growth_yoy	0.33%	0.05%	-0.24%	0.95%	0.66%	0.84%	1.96%	4.55%
rsi_14d	1.42%	-0.11%	-0.03%	-0.56%	-0.85%	-0.30%	6.15%	5.72%
dpo	0.34%	-0.09%	0.15%	0.17%	0.35%	1.04%	2.08%	4.05%
price_zscore_63d	1.18%	-0.02%	0.08%	-1.31%	-0.90%	-0.60%	7.15%	5.57%
debt_to_equity	1.10%	0.29%	-0.33%	-0.07%	1.22%	-0.31%	2.47%	4.36%
debt_to_assets	0.70%	0.13%	-0.36%	0.08%	1.32%	-0.55%	2.73%	4.05%
gp_to_equity	0.69%	0.16%	0.18%	-0.23%	1.28%	1.81%	0.51%	4.40%
momentum_0_21	1.87%	-0.03%	0.01%	-0.98%	-0.48%	-0.51%	5.38%	5.26%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_total	-1.25%	-0.17%	-0.04%	0.05%	1.07%	-1.55%	5.81%	3.92%
fcf_growth_yoy_ttm	-0.25%	0.04%	-0.06%	0.24%	0.76%	0.48%	1.31%	2.52%
momentum_21_252_vol_63d	0.40%	0.11%	0.12%	2.49%	1.55%	-0.26%	1.12%	5.52%
accruals_noa_proxy	-1.44%	-0.06%	-0.45%	0.32%	0.40%	-0.33%	4.94%	3.38%
momentum_21_126	0.56%	-0.09%	-0.10%	2.33%	0.97%	0.35%	1.03%	5.05%
momentum_21_252	0.30%	0.05%	0.09%	2.68%	1.69%	-0.53%	1.31%	5.60%
resid_momentum_21_252	0.20%	0.04%	0.09%	2.59%	1.71%	-0.77%	1.61%	5.46%
share_turnover_21d	-0.16%	0.27%	0.01%	-0.17%	-2.39%	2.52%	5.47%	5.54%
asset_turnover	1.03%	-0.15%	-0.07%	-0.04%	0.68%	3.37%	-0.77%	4.05%
ps_ttm	1.55%	-0.09%	-1.08%	-0.56%	0.87%	1.62%	2.65%	4.97%
sales_to_price	1.58%	-0.07%	-1.10%	-0.48%	0.97%	1.74%	2.38%	5.01%
log_market_cap	0.38%	-0.83%	-0.47%	-0.15%	0.26%	-0.66%	7.07%	5.61%
market_cap	0.38%	-0.83%	-0.47%	-0.15%	0.26%	-0.66%	7.07%	5.61%
revenue_growth_yoy	0.40%	0.13%	0.33%	0.67%	0.45%	-0.41%	1.42%	2.98%
altman_z	0.50%	0.06%	-0.89%	-0.24%	0.62%	-1.80%	5.66%	3.92%
ma_dist_63d	1.66%	0.05%	0.07%	-1.57%	-1.53%	-0.80%	6.12%	4.00%
gpoa	-0.08%	-0.04%	0.37%	-0.23%	0.32%	2.79%	-0.06%	3.07%
fcf_to_equity	-0.08%	0.30%	0.02%	0.15%	0.60%	2.47%	0.29%	3.74%
equity_growth_yoy	-1.09%	-0.07%	-0.44%	-0.34%	0.62%	-0.30%	4.25%	2.63%
assets_to_mc	1.05%	0.03%	-1.23%	-0.42%	0.63%	0.07%	3.82%	3.97%
accruals_wc_proxy	-1.06%	-0.01%	-0.14%	-0.20%	0.92%	-0.67%	2.96%	1.79%
ev	0.37%	-0.83%	-0.31%	0.05%	-0.09%	-0.23%	5.57%	4.54%
dso	0.23%	0.03%	0.01%	0.04%	0.83%	0.47%	0.55%	2.17%
is_negative_invested_capital	-0.14%	-0.09%	0.80%	-0.07%	0.81%	1.05%	0.84%	3.20%
interest_to_debt	-0.43%	-0.12%	-0.19%	-0.04%	1.00%	-1.67%	3.63%	2.17%
ncav_to_mc	1.28%	0.32%	-0.76%	-0.05%	1.21%	-0.79%	1.83%	3.05%
accruals	-1.35%	-0.08%	-0.11%	-0.05%	0.80%	-1.71%	4.77%	2.27%
evs_ttm	1.22%	-0.15%	-0.90%	-0.44%	0.27%	2.43%	0.94%	3.38%
surprisePercentage	0.19%	-0.09%	-0.05%	0.46%	0.93%	0.87%	-0.64%	1.67%
bab_252v_1260c_FRENCH_CRSP	1.05%	0.07%	-0.25%	0.06%	-1.44%	-0.41%	3.64%	2.71%
bab_shrunk_252v_1260c_FRENCH_CRSP	0.07%	0.07%	-0.25%	0.06%	-1.44%	-0.41%	3.64%	2.71%
beta_252v_1260c_FRENCH_CRSP	1.05%	0.07%	-0.25%	0.06%	-1.44%	-0.41%	3.64%	2.71%
beta_shrunk_252v_1260c_FRENCH_CRSP	0.07%	0.07%	-0.25%	0.06%	-1.44%	-0.41%	3.64%	2.71%
bab_252v_1260c_VTI	0.98%	0.05%	-0.18%	-0.02%	-1.42%	-0.49%	3.59%	2.50%
bab_shrunk_252v_1260c_VTI	0.98%	0.05%	-0.18%	-0.02%	-1.42%	-0.49%	3.59%	2.50%
beta_252v_1260c_VTI	0.98%	0.05%	-0.18%	-0.02%	-1.42%	-0.49%	3.59%	2.50%
beta_shrunk_252v_1260c_VTI	0.98%	0.05%	-0.18%	-0.02%	-1.42%	-0.49%	3.59%	2.50%
ebit_growth_yoy_ttm	0.58%	0.09%	-0.09%	0.70%	1.17%	0.78%	-1.25%	1.97%
debt_issuance_yoy	-0.40%	-0.06%	-0.20%	0.28%	0.36%	0.35%	1.11%	1.44%
asset_growth_yoy	-1.20%	-0.10%	-0.64%	-0.15%	0.42%	0.18%	3.87%	2.39%
bab_252v_1260c_SPY	1.06%	-0.03%	-0.16%	-0.04%	-1.38%	-0.63%	3.67%	2.49%
bab_shrunk_252v_1260c_SPY	1.06%	-0.03%	-0.16%	-0.04%	-1.38%	-0.63%	3.67%	2.49%
beta_252v_1260c_SPY	1.06%	-0.03%	-0.16%	-0.04%	-1.38%	-0.63%	3.67%	2.49%
beta_shrunk_252v_1260c_SPY	1.06%	-0.03%	-0.16%	-0.04%	-1.38%	-0.63%	3.67%	2.49%
fcf_margin	-0.21%	0.28%	0.09%	0.06%	-0.65%	1.90%	1.01%	2.48%
cash_ratio	1.02%	0.07%	-0.61%	0.11%	1.46%	0.94%	-1.18%	1.80%
dollar_volume_21d	0.34%	-0.59%	-0.39%	-0.41%	-0.92%	-0.01%	4.54%	2.56%
ev_to_assets	0.65%	-0.05%	-1.15%	-0.44%	-0.17%	0.59%	3.12%	2.56%
gp_ev	0.58%	-0.07%	-0.70%	-0.62%	0.56%	2.50%	0.06%	2.30%
gross_profit_to_ev	0.58%	-0.07%	-0.70%	-0.62%	0.56%	2.50%	0.06%	2.30%
value_proxy_1260d	0.15%	-0.22%	-0.95%	-0.85%	0.33%	-1.69%	6.30%	3.08%
pe_ttm	0.52%	0.03%	-0.78%	-0.83%	0.53%	0.54%	2.32%	2.33%
fcf_to_market_cap	-0.04%	0.19%	-0.55%	-0.16%	0.60%	2.15%	0.03%	2.22%
net_operating_assets_proxy	-0.21%	-0.80%	0.27%	0.09%	0.18%	-0.72%	4.28%	3.10%
current_ratio	0.70%	-0.49%	0.30%	0.02%	1.35%	-0.33%	0.23%	1.77%
amihud_illiq_21d	0.00%	-0.65%	-0.37%	-0.49%	-0.10%	-0.77%	4.98%	2.60%
pb	0.42%	-0.09%	-1.04%	-0.40%	-0.70%	0.56%	3.37%	2.11%
roe_volatility	0.29%	-0.17%	0.06%	-0.03%	1.52%	-3.34%	3.74%	2.07%
dio	0.21%	0.07%	-0.13%	0.41%	-0.37%	-0.03%	0.94%	1.11%
gross_profit_margin_ttm	1.01%	-0.14%	-0.48%	0.23%	0.29%	0.02%	0.26%	1.19%
net_income_margin_ttm	-0.41%	-0.44%	-0.16%	-0.27%	1.42%	-3.08%	5.32%	2.37%
roic_nopat	-0.34%	0.20%	0.56%	-0.15%	0.02%	0.57%	0.34%	1.20%
growth_gmar_5y	0.36%	-0.16%	-0.78%	0.40%	-0.14%	0.05%	1.60%	1.33%
capital_intensity	0.40%	0.05%	-0.43%	0.31%	-0.24%	0.25%	0.71%	1.05%
is_positive_revenue	-0.34%	-0.01%	-0.08%	0.66%	-1.57%	-1.44%	4.26%	1.48%
skew_63d	0.72%	0.16%	0.05%	-0.14%	-1.02%	0.62%	0.68%	1.07%
accruals_percent	-0.44%	0.04%	-0.38%	0.04%	0.12%	-0.02%	1.52%	0.89%
rd_to_opex	-0.38%	0.02%	0.70%	-0.29%	-1.28%	-0.76%	3.42%	1.43%
net_income_to_market_cap	-1.04%	-0.34%	0.38%	0.03%	0.53%	-2.94%	5.34%	1.97%
corr_1260d_FRENCH_CRSP	-0.78%	-0.58%	0.07%	-0.59%	1.28%	-4.05%	7.40%	2.75%
corr_1260d_VTI	-0.80%	-0.58%	0.10%	-0.63%	1.30%	-4.04%	7.35%	2.71%
corr_1260d_SPY	-0.74%	-0.61%	0.12%	-0.63%	1.29%	-3.97%	7.25%	2.71%
growth_gpoa_5y	0.02%	-0.13%	-0.75%	0.39%	0.05%	-0.65%	2.26%	1.18%
ebit_margin_ttm	-0.07%	-0.49%	-0.09%	-0.14%	0.88%	-2.56%	4.41%	1.93%
working_capital_proxy	-0.31%	0.03%	-0.10%	0.12%	0.30%	-1.52%	2.34%	0.85%
momentum_21_756	-0.94%	-0.18%	-0.85%	-0.94%	-0.27%	-0.91%	5.83%	1.74%
lottery_max_5_21d	1.18%	0.41%	-0.03%	0.23%	-1.99%	3.29%	-1.33%	1.75%
growth_acc_5y	0.48%	-0.05%	-0.22%	0.24%	0.26%	0.99%	-0.94%	0.76%
cash_earnings_ttm	0.49%	-0.58%	0.22%	0.01%	-0.06%	0.22%	0.88%	1.18%
equity_issuance_yoy	0.15%	0.21%	-0.40%	-0.14%	-0.50%	2.46%	-0.70%	1.09%
surprise	-0.39%	-0.05%	0.08%	-0.60%	-0.23%	-1.38%	3.31%	0.74%
sga_to_revenue	0.36%	0.28%	-0.16%	0.01%	-0.54%	0.79%	0.13%	0.86%
earnings_yield	-0.97%	-0.32%	0.47%	0.16%	0.42%	-3.12%	4.73%	1.37%
ebit_to_ev	-0.97%	-0.32%	0.47%	0.16%	0.42%	-3.12%	4.73%	1.37%
net_income_growth_yoy_sign_flip	-0.34%	0.03%	-0.05%	0.00%	-0.39%	-0.19%	1.61%	0.66%
ret_volume_corr_63d	0.83%	0.24%	0.09%	-0.45%	-1.37%	0.62%	-0.02%	-0.07%

1. ret_zscore_63d

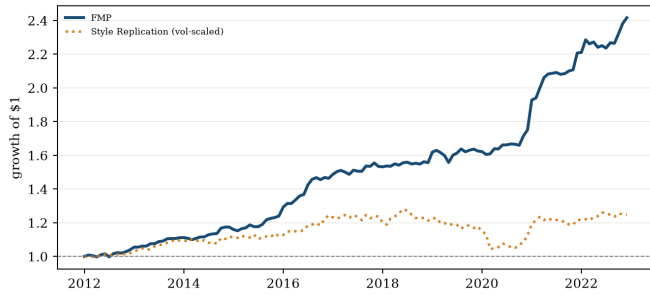


Figure: Common-window cumulative growth of \$1.

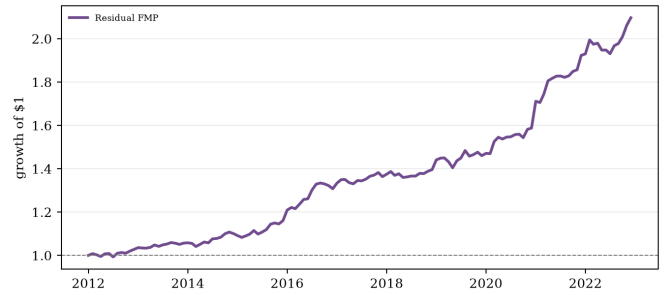


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ret_zscore_63d	8.24%	5.21%	1.58	4.51	-4.40%	0.179	7.32%	5.21%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ret_zscore_63d	0.03 (1.07)	0.21 (3.15)	0.07 (1.60)	-0.05 (-1.04)	0.12 (1.06)	0.07 (0.81)	6.91% (4.89)	0.188

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ret_zscore_63d	0.44%	-0.18%	-0.17%	-0.24%	1.10%	0.39%	6.91%	8.24%

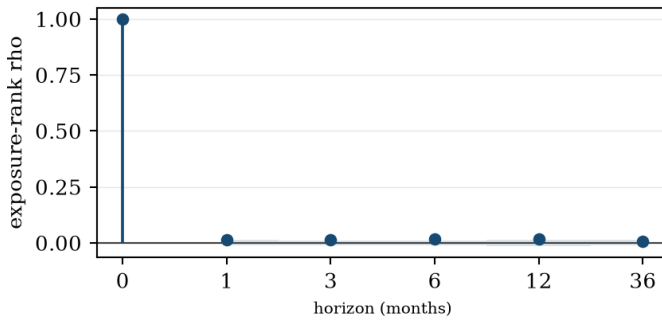


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

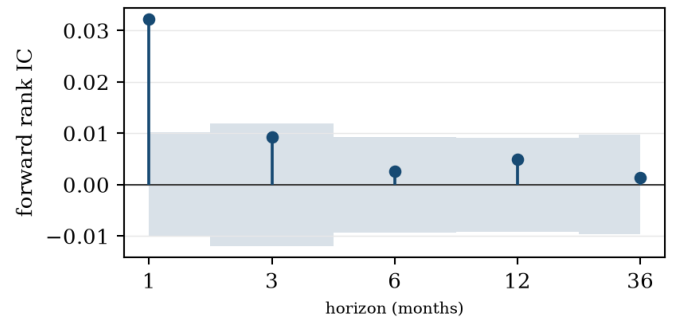


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	ret_zscore_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1267 / 1582 / 2103
Available-history Sharpe	1.58
Bootstrap Sharpe	1.58 [1.14, 2.06]; Pr(S<=0)=0.0%
Bootstrap tails	skew=2.29 [0.50, 3.05]; max DD=-4.4% [-5.5%, -1.9%]

2. resid_ret_252v_1260c_SPY

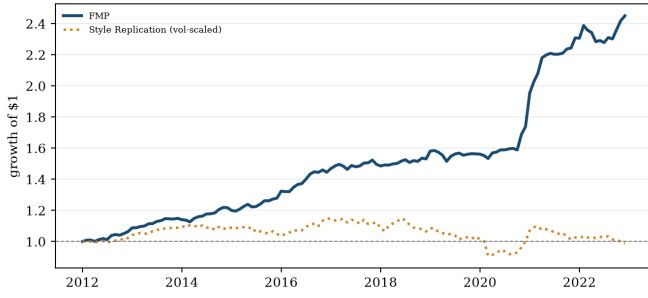


Figure: Common-window cumulative growth of \$1.

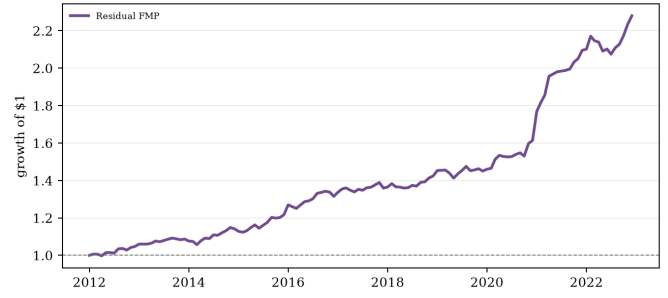


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
resid_ret_252v_1260c_SPY	8.40%	5.88%	1.43	3.53	-4.59%	0.197	8.37%	5.88%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
resid_ret_252v_1260c_SPY	0.04	0.19	0.07	-0.07	0.14	-0.05	7.70%	0.213
	(1.03)	(2.26)	(1.39)	(-1.44)	(1.08)	(-0.52)	(4.21)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
resid_ret_252v_1260c_SPY	0.47%	-0.17%	-0.17%	-0.37%	1.24%	-0.30%	7.70%	8.40%

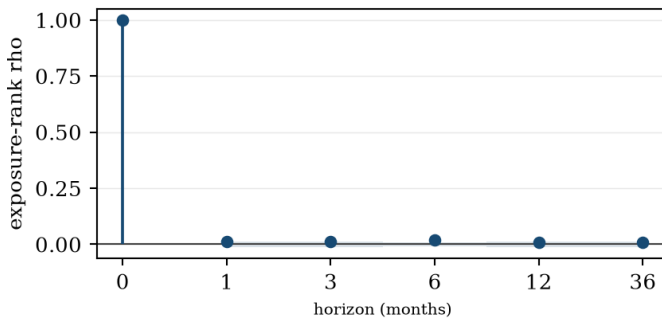


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

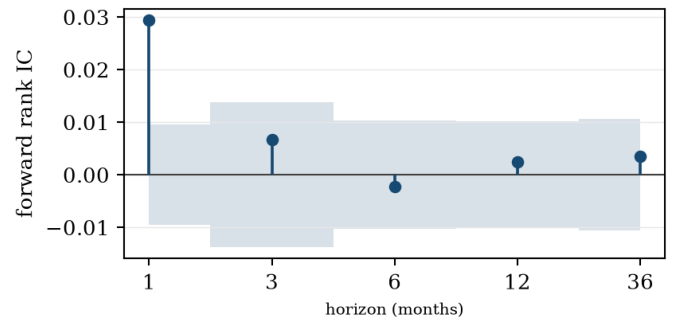


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	resid_ret_252v_1260c_SPY(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	1.43
Bootstrap Sharpe	1.43 [1.04, 1.97]; Pr(S<=0)=0.0%
Bootstrap tails	skew=2.96 [-0.03, 3.51]; max DD=-4.6% [-7.1%, -2.3%]

3. stoch_k_14d

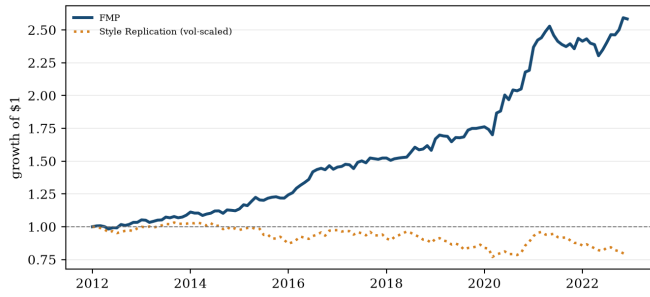


Figure: Common-window cumulative growth of \$1.

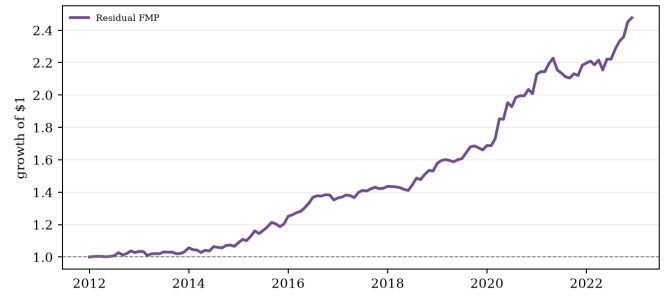


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
stoch_k_14d	8.96%	6.90%	1.30	4.22	-8.91%	0.294	10.04%	6.90%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
stoch_k_14d	0.12 (2.44)	0.25 (3.97)	0.04 (0.64)	-0.11 (-1.53)	0.03 (0.33)	-0.09 (-1.30)	8.49% (4.89)	0.335

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
stoch_k_14d	1.58%	-0.22%	-0.10%	-0.53%	0.26%	-0.53%	8.49%	8.96%

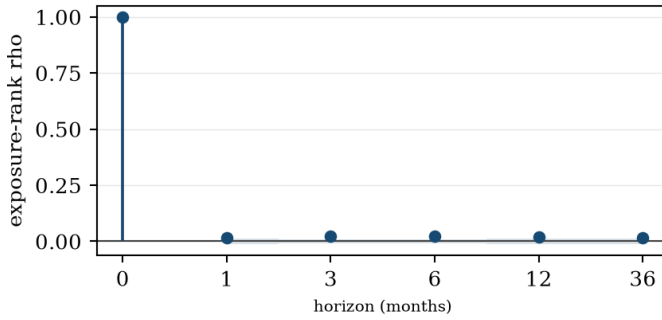


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

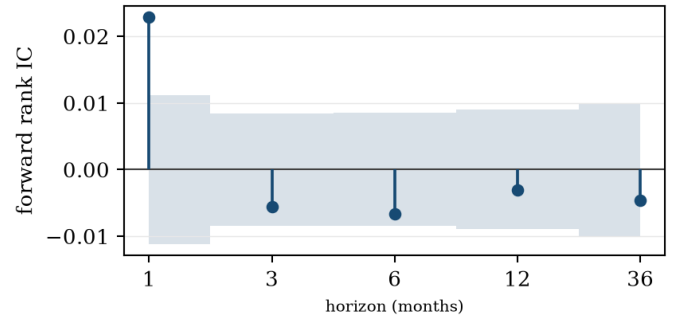


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	stoch_k_14d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1266 / 1578 / 2103
Available-history Sharpe	1.30
Bootstrap Sharpe	1.30 [0.93, 1.74]; Pr(S<=0)=0.0%
Bootstrap tails	skew=1.35 [0.03, 1.77]; max DD=-8.9% [-10.8%, -3.0%]

4. bb_pctb_20d

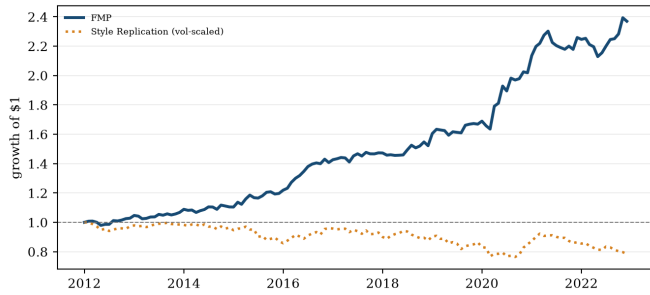


Figure: Common-window cumulative growth of \$1.

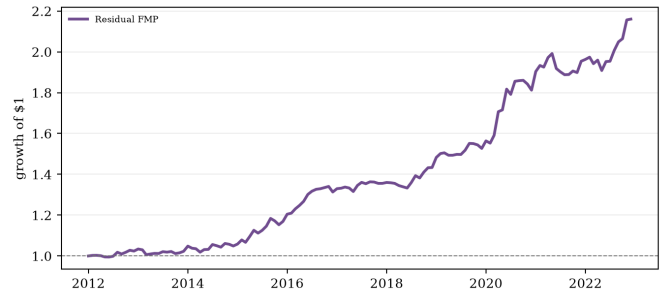


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bb_pctb_20d	8.13%	6.49%	1.25	4.20	-7.53%	0.189	9.00%	6.49%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bb_pctb_20d	0.13 (2.67)	0.15 (2.65)	0.03 (0.37)	-0.11 (-1.56)	0.01 (0.09)	-0.02 (-0.36)	7.23% (4.11)	0.250

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bb_pctb_20d	1.71%	-0.13%	-0.06%	-0.57%	0.07%	-0.11%	7.23%	8.13%

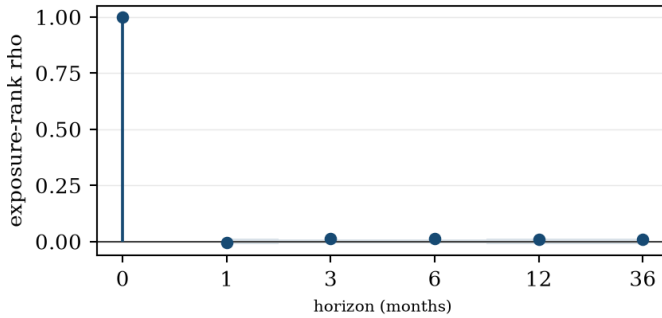


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

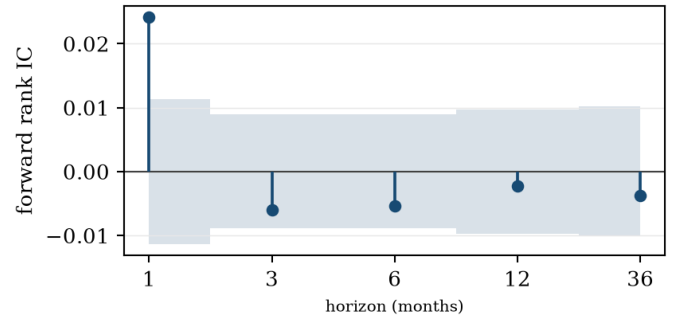


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	bb_pctb_20d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1266 / 1579 / 2103
Available-history Sharpe	1.25
Bootstrap Sharpe	1.25 [0.87, 1.72]; Pr(S<=0)=0.0%
Bootstrap tails	skew=1.26 [0.14, 1.81]; max DD=-7.5% [-9.8%, -3.1%]

5. eps_growth_yoy

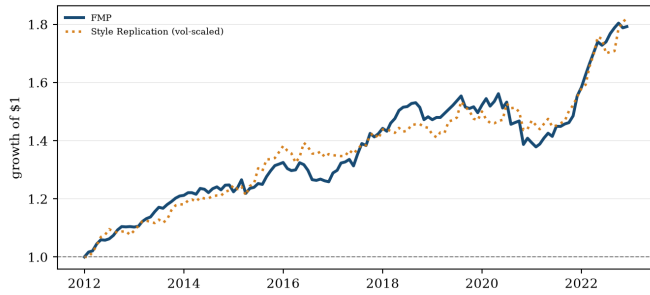


Figure: Common-window cumulative growth of \$1.

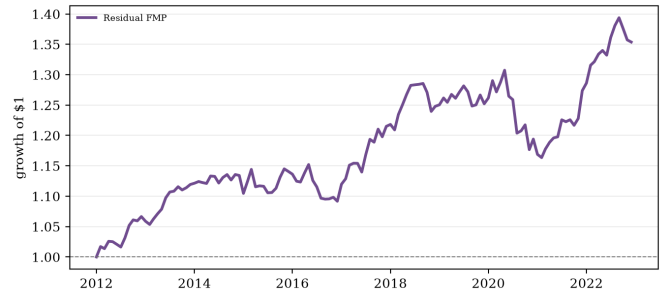


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
eps_growth_yoy	5.49%	5.14%	1.07	3.04	-11.71%	0.241	2.72%	5.14%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
eps_growth_yoy	0.04 (1.27)	-0.05 (-0.89)	0.11 (2.36)	0.23 (4.68)	0.08 (1.57)	0.07 (0.87)	2.88% (1.71)	0.243

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
eps_growth_yoy	0.55%	0.04%	-0.26%	1.15%	0.75%	0.38%	2.88%	5.49%

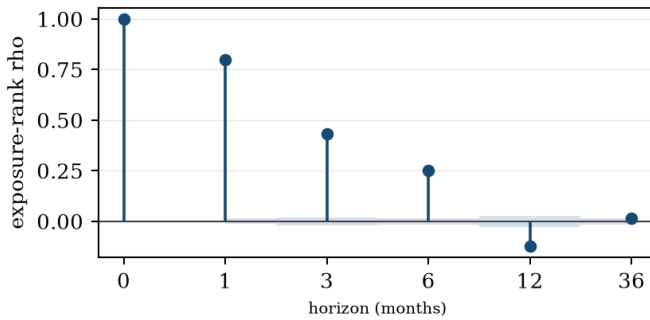


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

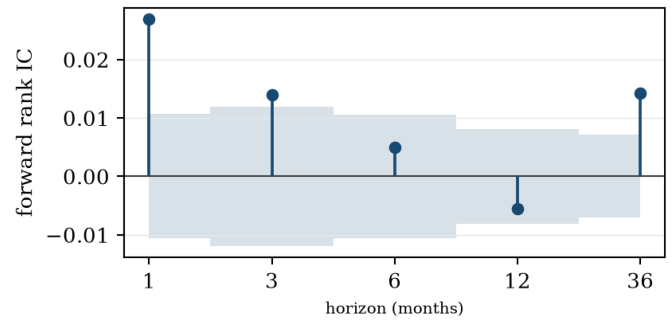


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	eps_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1235 / 1540 / 2060
Available-history Sharpe	1.07
Bootstrap Sharpe	1.07 [0.30, 2.07]; Pr(S<=0)=0.5%
Bootstrap tails	skew=-1.05 [-1.65, -0.06]; max DD=-11.7% [-18.6%, -3.8%]

6. capex_to_revenue

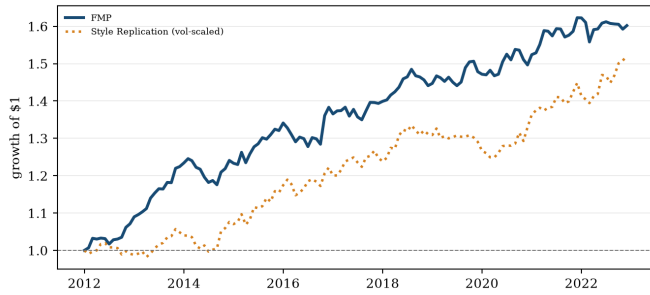


Figure: Common-window cumulative growth of \$1.

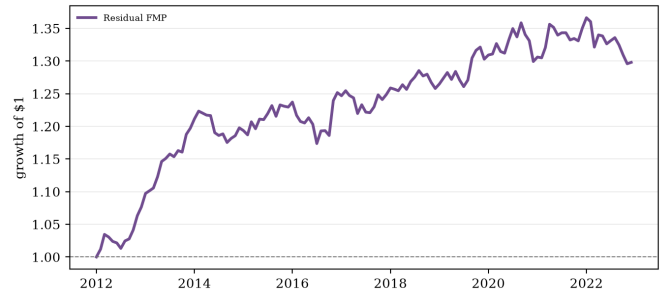


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
capex_to_revenue	4.42%	4.53%	0.98	3.38	-5.61%	0.263	2.40%	4.53%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
capex_to_revenue	0.01 (0.50)	0.27 (4.01)	-0.03 (-0.90)	0.00 (0.04)	0.06 (1.33)	0.25 (4.95)	2.47% (1.96)	0.264

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
capex_to_revenue	0.16%	-0.23%	0.06%	0.01%	0.51%	1.45%	2.47%	4.42%

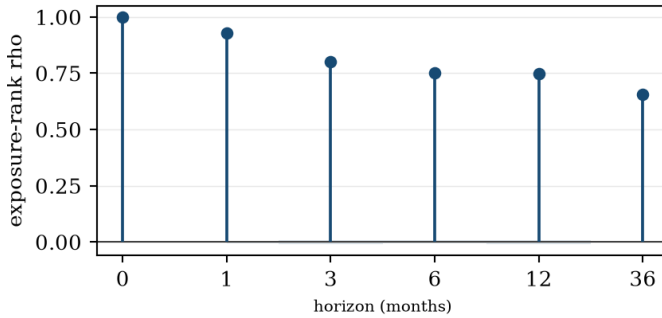


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

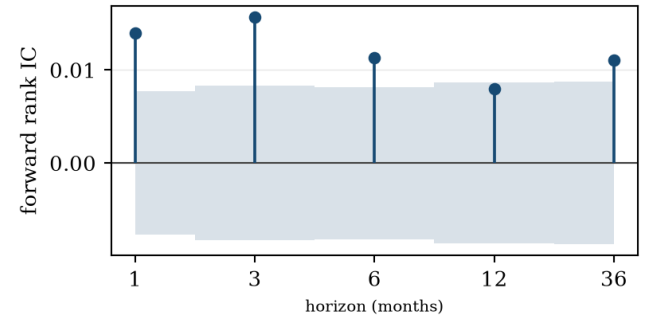


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	capex_to_revenue(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1258 / 1556 / 2035
Available-history Sharpe	0.98
Bootstrap Sharpe	0.98 [0.55, 1.47]; Pr(S<=0)=0.0%
Bootstrap tails	skew=0.52 [-0.23, 1.18]; max DD=-5.6% [-9.2%, -3.9%]

7. net_income_growth_yoy

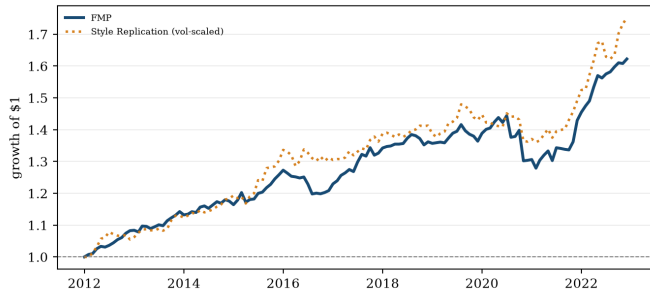


Figure: Common-window cumulative growth of \$1.

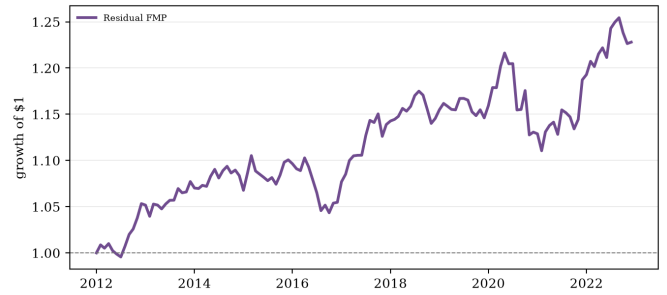


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_income_growth_yoy	4.55%	4.67%	0.97	2.88	-11.44%	0.309	1.63%	4.67%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_income_growth_yoy	0.03	-0.06	0.10	0.19	0.07	0.15	1.96%	0.313
	(0.85)	(-1.55)	(3.01)	(4.82)	(1.51)	(2.05)	(1.36)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_income_growth_yoy	0.33%	0.05%	-0.24%	0.95%	0.66%	0.84%	1.96%	4.55%

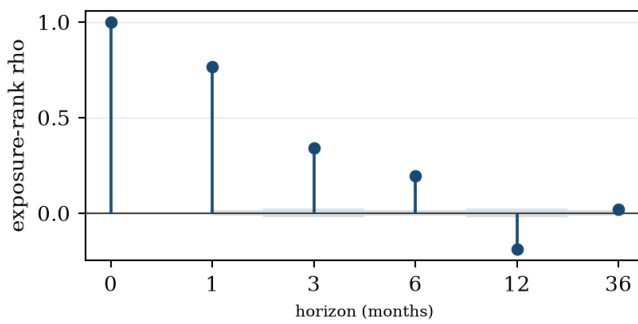


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

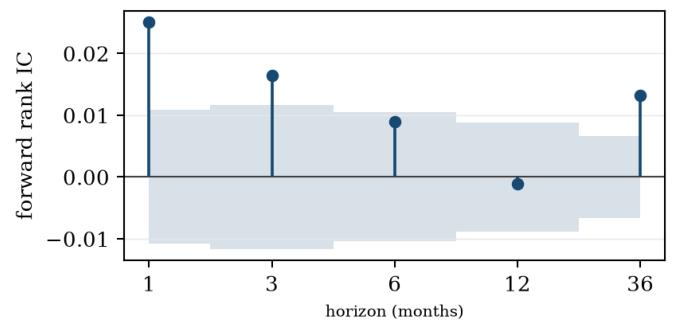


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	net_income_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1259 / 1565 / 2085
Available-history Sharpe	0.97
Bootstrap Sharpe	0.97 [0.28, 1.92]; Pr(S<=0)=0.4%
Bootstrap tails	skew=-1.44 [-2.45, 0.50]; max DD=-11.4% [-17.7%, -3.5%]

8. rsi_14d

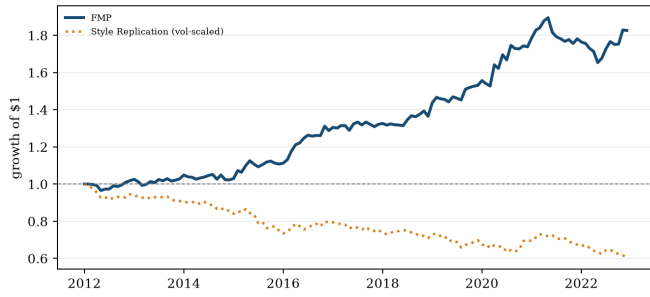


Figure: Common-window cumulative growth of \$1.

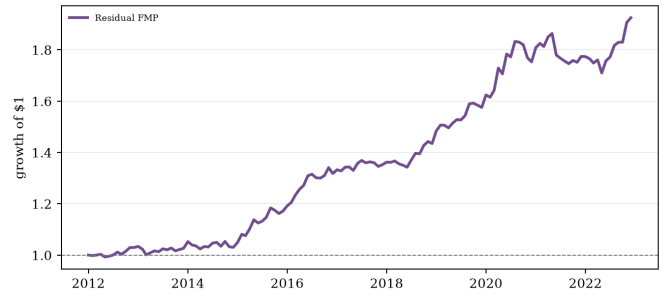


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
rsi_14d	5.72%	6.20%	0.92	3.26	-12.74%	0.220	7.76%	6.20%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
rsi_14d	0.11 (2.64)	0.13 (2.41)	0.01 (0.20)	-0.11 (-1.48)	-0.10 (-1.42)	-0.05 (-1.05)	6.15% (3.38)	0.264

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
rsi_14d	1.42%	-0.11%	-0.03%	-0.56%	-0.85%	-0.30%	6.15%	5.72%

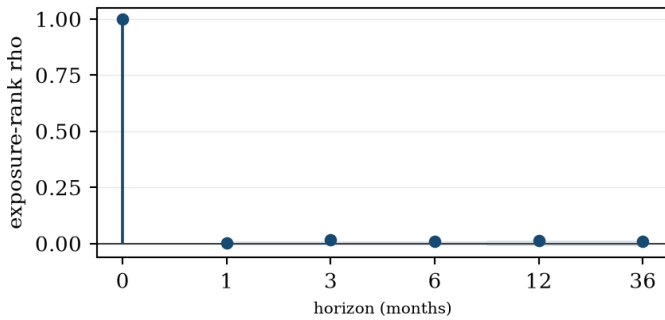


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

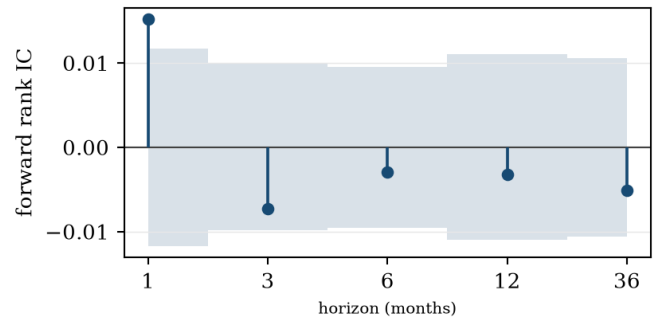


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	rsi_14d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1266 / 1578 / 2103
Available-history Sharpe	0.92
Bootstrap Sharpe	0.92 [0.43, 1.40]; Pr(S<=0)=0.0%
Bootstrap tails	skew=0.77 [0.10, 1.25]; max DD=-12.7% [-15.9%, -3.2%]

9. dpo

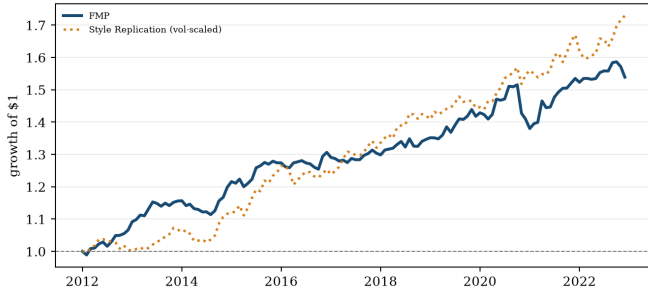


Figure: Common-window cumulative growth of \$1.

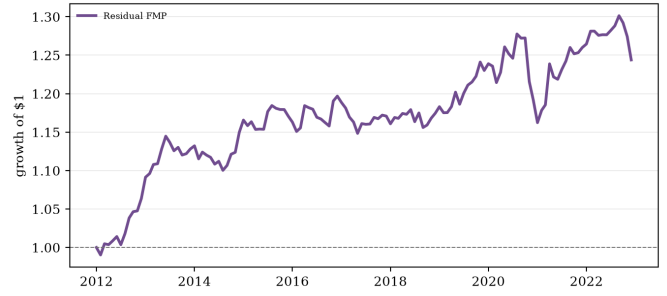


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dpo	4.05%	4.43%	0.92	3.03	-8.94%	0.158	2.02%	4.43%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dpo	0.03 (0.87)	0.10 (1.54)	-0.07 (-2.40)	0.03 (1.09)	0.04 (0.62)	0.18 (2.27)	2.08% (1.38)	0.158

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dpo	0.34%	-0.09%	0.15%	0.17%	0.35%	1.04%	2.08%	4.05%

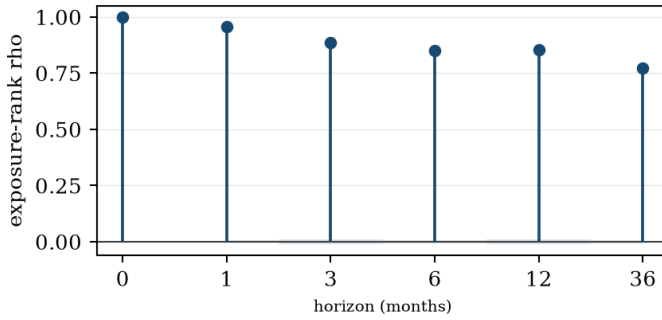


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

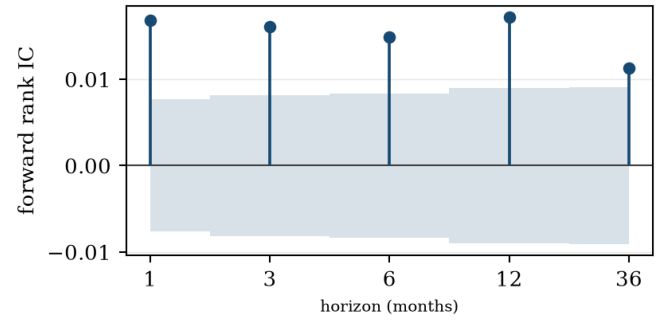


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	dpo(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1208 / 1548 / 2016
Available-history Sharpe	0.92
Bootstrap Sharpe	0.92 [0.48, 1.48]; Pr(S<=0)=0.0%
Bootstrap tails	skew=-0.37 [-1.08, 0.80]; max DD=-8.9% [-12.7%, -3.2%]

10. price_zscore_63d

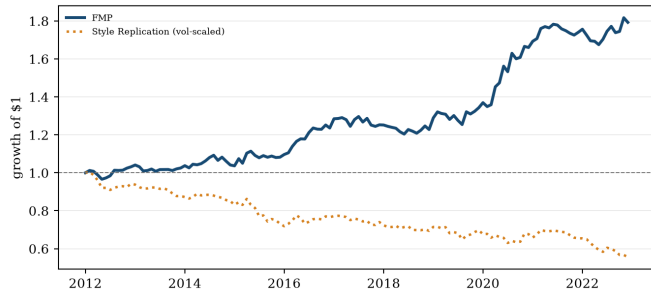


Figure: Common-window cumulative growth of \$1.

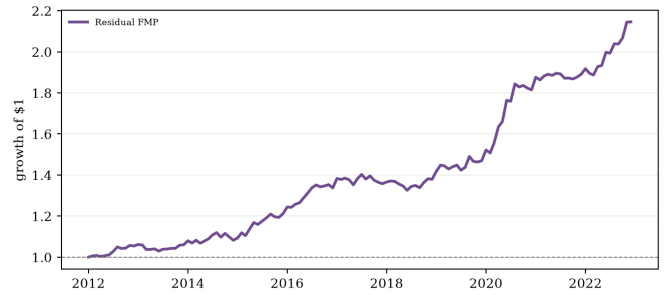


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
price_zscore_63d	5.57%	6.65%	0.84	2.83	-7.14%	0.340	8.57%	6.65%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
price_zscore_63d	0.09 (2.27)	0.02 (0.27)	-0.03 (-0.71)	-0.26 (-5.60)	-0.10 (-1.54)	-0.11 (-1.87)	7.15% (4.23)	0.368

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
price_zscore_63d	1.18%	-0.02%	0.08%	-1.31%	-0.90%	-0.60%	7.15%	5.57%

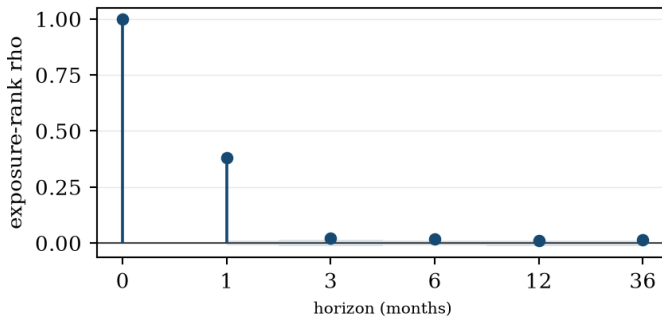


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

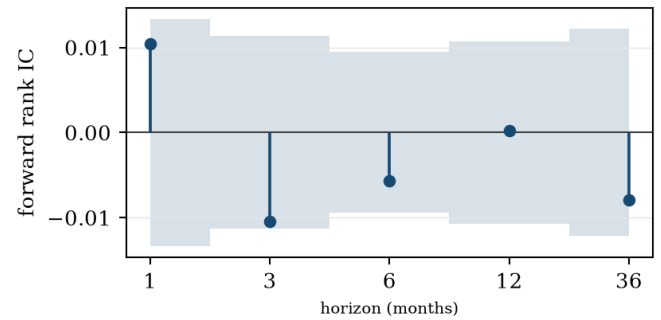


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	price_zscore_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1267 / 1582 / 2103
Available-history Sharpe	0.84
Bootstrap Sharpe	0.84 [0.36, 1.31]; Pr(S<=0)=0.0%
Bootstrap tails	skew=0.92 [0.32, 1.19]; max DD=-7.1% [-12.6%, -5.1%]

11. debt_to_equity

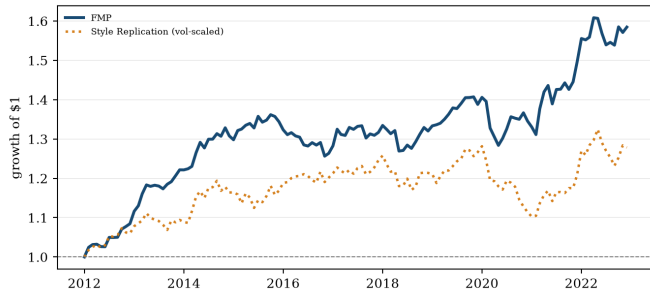


Figure: Common-window cumulative growth of \$1.

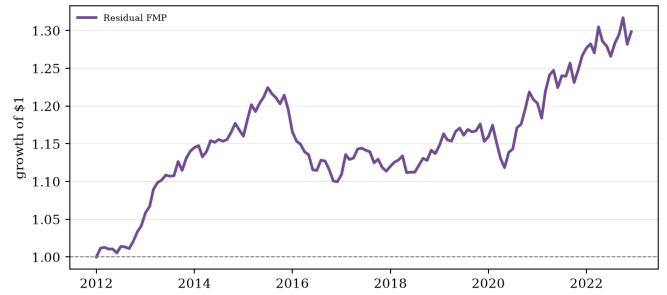


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_to_equity	4.36%	5.28%	0.83	2.62	-8.76%	0.445	2.77%	5.28%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_to_equity	0.09 (2.86)	-0.33 (-6.93)	0.14 (3.49)	-0.01 (-0.33)	0.14 (2.33)	-0.06 (-1.11)	2.47% (1.95)	0.450

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_to_equity	1.10%	0.29%	-0.33%	-0.07%	1.22%	-0.31%	2.47%	4.36%

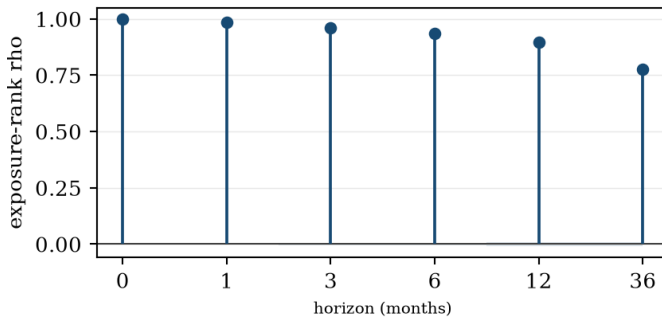


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

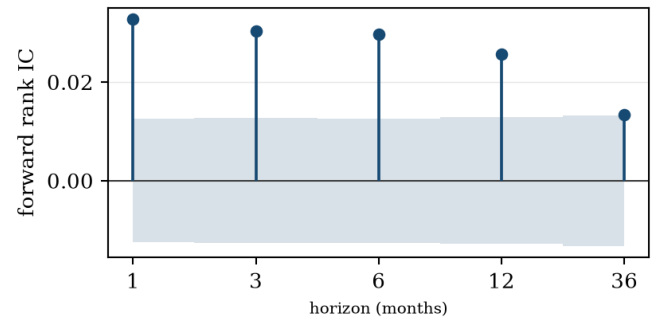


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	debt_to_equity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1229 / 1513 / 1982
Available-history Sharpe	0.83
Bootstrap Sharpe	0.83 [0.17, 1.56]; Pr(S<=0)=0.7%
Bootstrap tails	skew=-0.12 [-0.84, 0.40]; max DD=-8.8% [-16.1%, -4.6%]

12. debt_to_assets

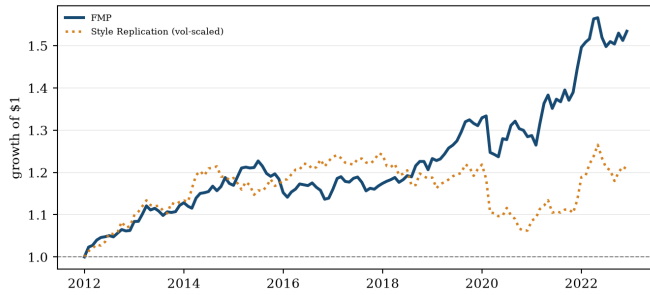


Figure: Common-window cumulative growth of \$1.

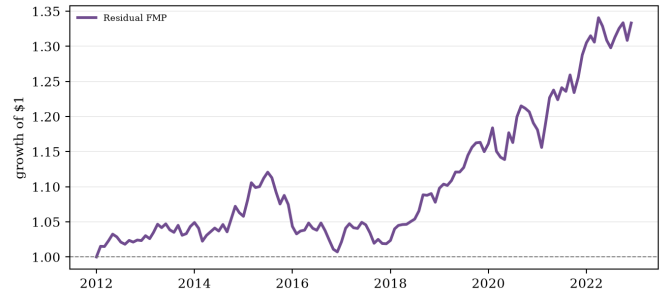


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_to_assets	4.05%	4.97%	0.81	2.68	-7.38%	0.252	3.12%	4.97%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_to_assets	0.06 (1.76)	-0.15 (-2.62)	0.16 (3.62)	0.02 (0.43)	0.15 (2.20)	-0.10 (-1.54)	2.73% (2.18)	0.261

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_to_assets	0.70%	0.13%	-0.36%	0.08%	1.32%	-0.55%	2.73%	4.05%

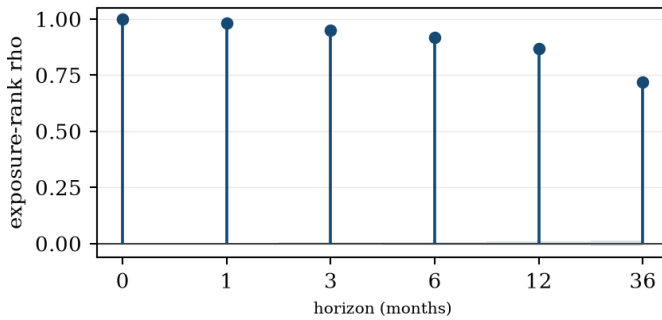


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

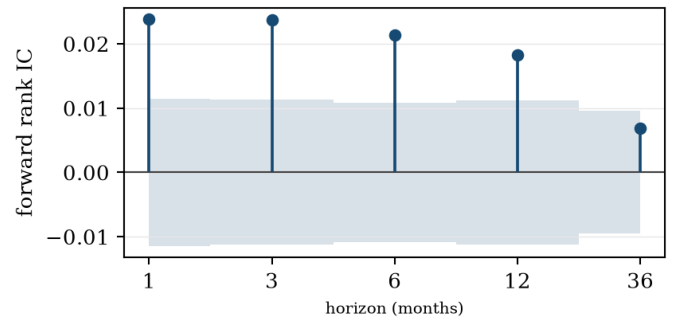


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	debt_to_assets(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1071 / 1381 / 2061
Available-history Sharpe	0.81
Bootstrap Sharpe	0.81 [0.27, 1.35]; Pr(S<=0)=0.2%
Bootstrap tails	skew=-0.50 [-1.61, 0.46]; max DD=-7.4% [-13.6%, -4.4%]

13. gp_to_equity

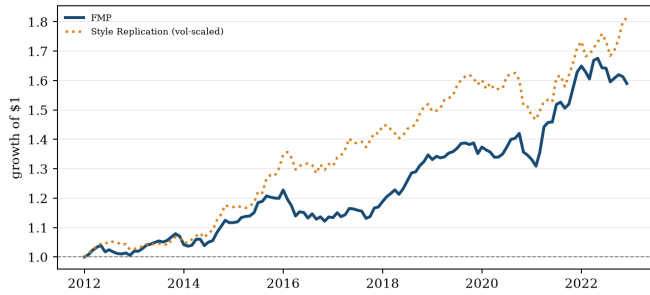


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gp_to_equity	4.40%	5.41%	0.81	2.51	-8.67%	0.497	0.44%	5.41%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gp_to_equity	0.05 (1.82)	-0.18 (-4.54)	-0.08 (-2.75)	-0.04 (-1.52)	0.14 (3.48)	0.32 (6.17)	0.51% (0.37)	0.498

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gp_to_equity	0.69%	0.16%	0.18%	-0.23%	1.28%	1.81%	0.51%	4.40%

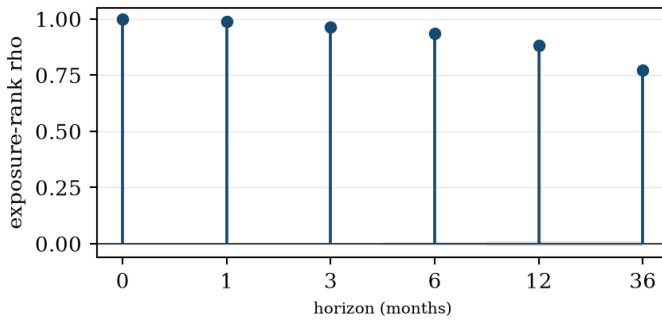


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

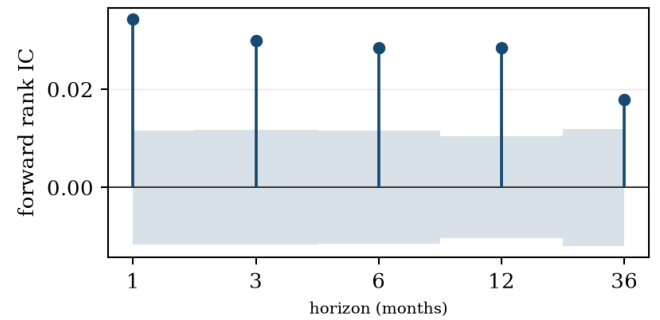


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	gp_to_equity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1210 / 1505 / 1965
Available-history Sharpe	0.81
Bootstrap Sharpe	0.81 [0.26, 1.37]; Pr(S<=0)=0.2%
Bootstrap tails	skew=0.22 [-0.68, 0.70]; max DD=-8.7% [-15.3%, -5.0%]

14. momentum_0_21

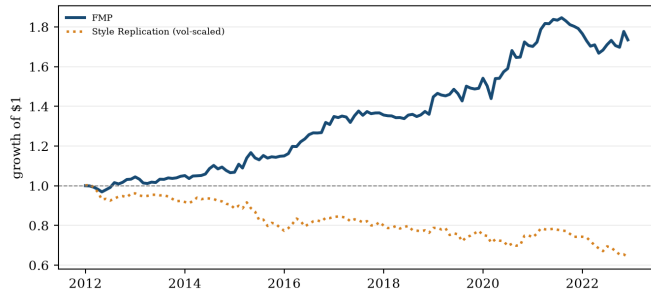


Figure: Common-window cumulative growth of \$1.

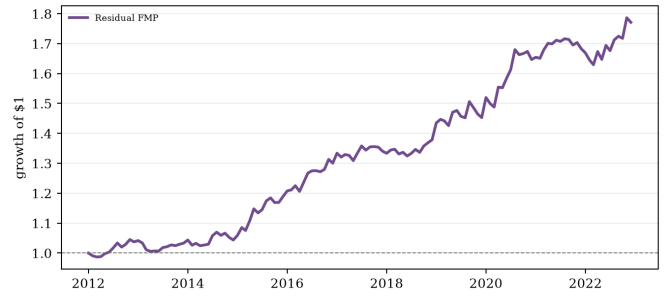


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_0_21	5.26%	6.47%	0.81	3.05	-9.68%	0.309	7.40%	6.47%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_0_21	0.15 (2.74)	0.03 (0.53)	-0.00 (-0.05)	-0.19 (-3.52)	-0.05 (-0.90)	-0.09 (-1.47)	5.38% (3.61)	0.365

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_0_21	1.87%	-0.03%	0.01%	-0.98%	-0.48%	-0.51%	5.38%	5.26%

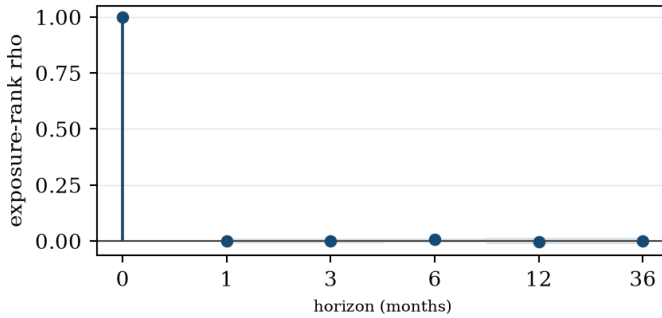


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

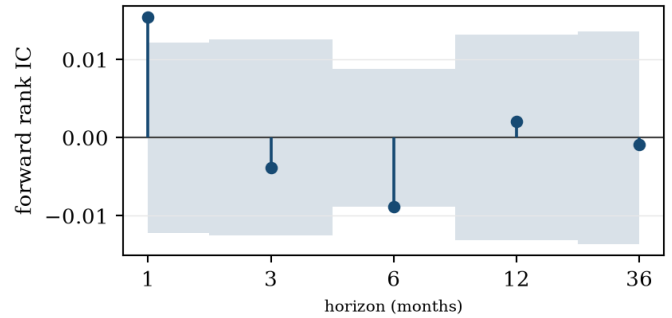


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	momentum_0_21(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.81
Bootstrap Sharpe	0.81 [0.38, 1.27]; Pr(S<=0)=0.0%
Bootstrap tails	skew=0.99 [0.58, 1.33]; max DD=-9.7% [-15.3%, -3.9%]

15. accruals_total

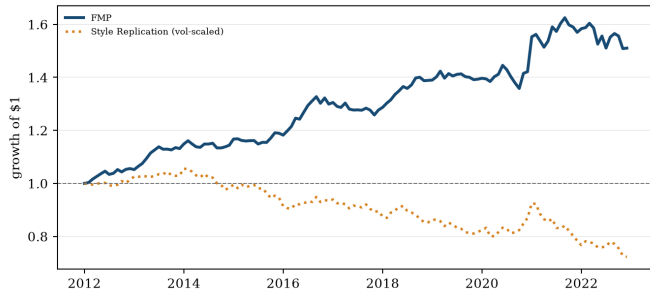


Figure: Common-window cumulative growth of \$1.

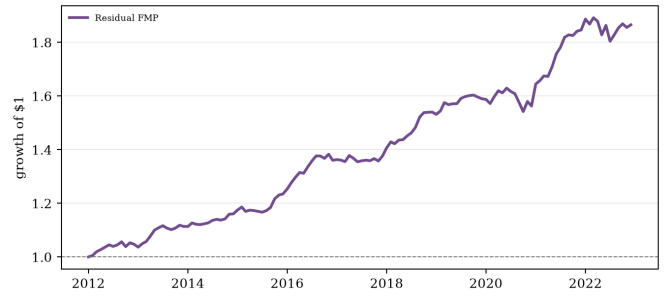


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_total	3.92%	5.23%	0.75	2.62	-7.17%	0.394	5.70%	5.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_total	-0.10 (-3.26)	0.19 (3.70)	0.02 (0.40)	0.01 (0.25)	0.12 (1.72)	-0.27 (-4.25)	5.81% (4.03)	0.395

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_total	-1.25%	-0.17%	-0.04%	0.05%	1.07%	-1.55%	5.81%	3.92%

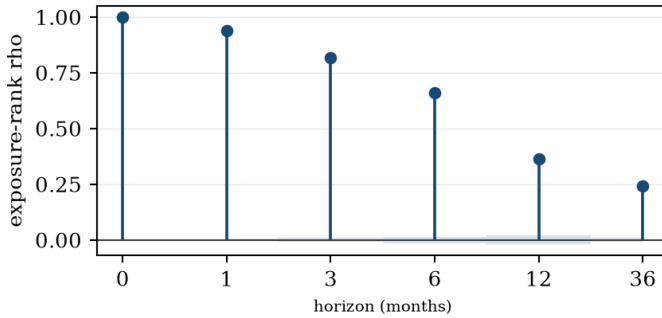


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

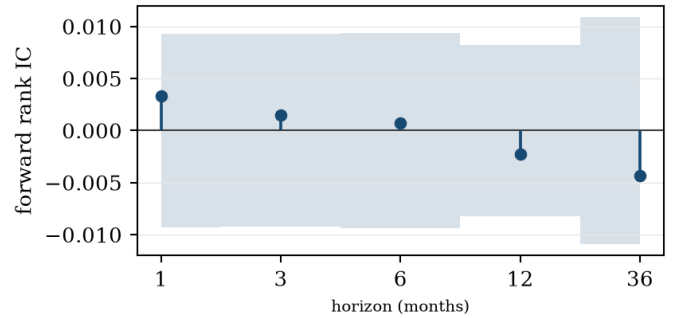


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	accruals_total(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1231 / 1556 / 2060
Available-history Sharpe	0.75
Bootstrap Sharpe	0.75 [0.33, 1.31]; Pr(S<=0)=0.1%
Bootstrap tails	skew=1.41 [-0.79, 2.42]; max DD=-7.2% [-12.2%, -4.0%]

16. fcf_growth_yoy_ttm

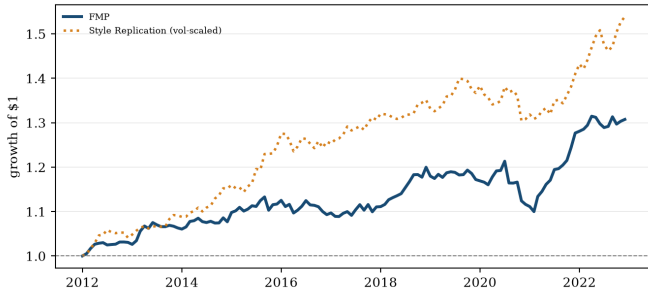


Figure: Common-window cumulative growth of \$1.

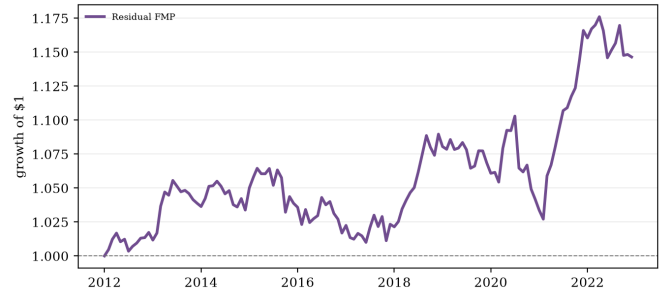


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
fcf_growth_yoy_ttm	2.52%	3.62%	0.70	2.14	-9.35%	0.178	0.83%	3.62%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
fcf_growth_yoy_ttm	-0.02 (-0.59)	-0.05 (-1.23)	0.02 (0.93)	0.05 (1.36)	0.09 (1.92)	0.08 (1.50)	1.31% (1.26)	0.192

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
fcf_growth_yoy_ttm	-0.25%	0.04%	-0.06%	0.24%	0.76%	0.48%	1.31%	2.52%

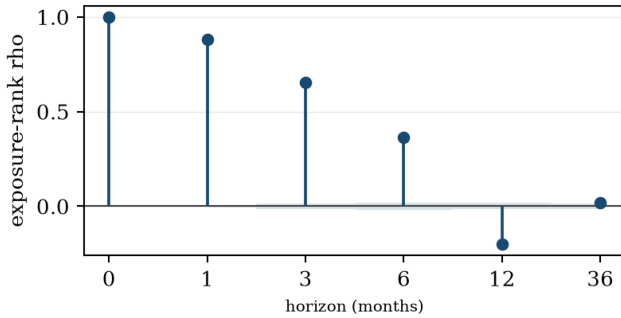


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

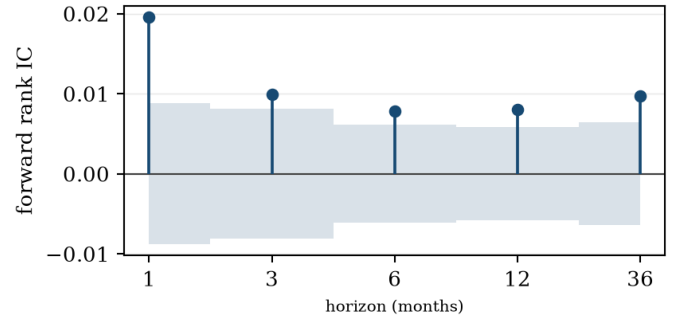


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	fcf_growth_yoy_ttm(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1222 / 1542 / 2050
Available-history Sharpe	0.70
Bootstrap Sharpe	0.70 [0.09, 1.34]; Pr(S<=0)=0.9%
Bootstrap tails	skew=-0.70 [-1.23, 0.29]; max DD=-9.3% [-14.2%, -2.8%]

17. momentum_21_252_vol_63d

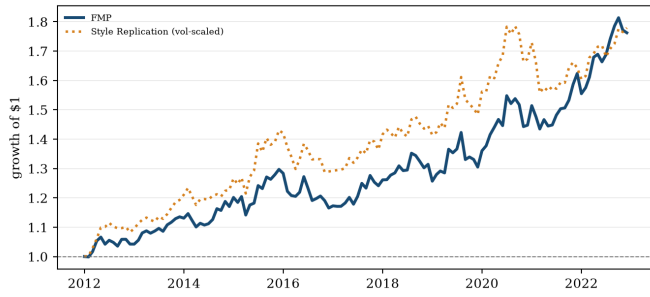


Figure: Common-window cumulative growth of \$1.

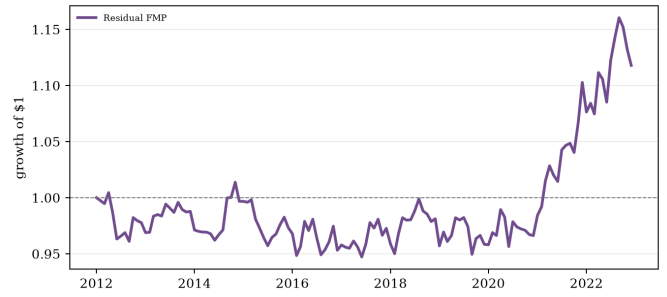


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_252_vol_63d	5.52%	7.95%	0.69	2.81	-10.16%	0.679	0.90%	7.95%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_252_vol_63d	0.03	-0.12	-0.05	0.49	0.17	-0.05	1.12%	0.680
	(0.98)	(-2.33)	(-1.20)	(9.65)	(3.37)	(-0.60)	(0.84)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_252_vol_63d	0.40%	0.11%	0.12%	2.49%	1.55%	-0.26%	1.12%	5.52%

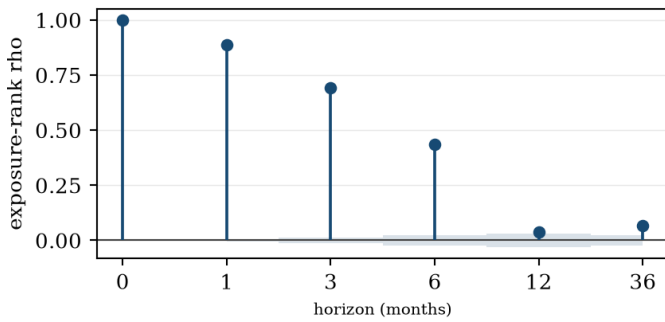


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

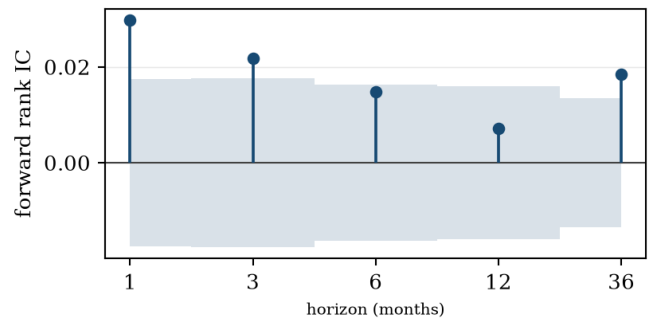


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	momentum_21_252_vol_63d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1267 / 1582 / 2103
Available-history Sharpe	0.69
Bootstrap Sharpe	0.69 [0.30, 1.13]; Pr(S<=0)=0.0%
Bootstrap tails	skew=-0.13 [-0.46, 0.21]; max DD=-10.2% [-17.4%, -7.0%]

18. accruals_noa_proxy

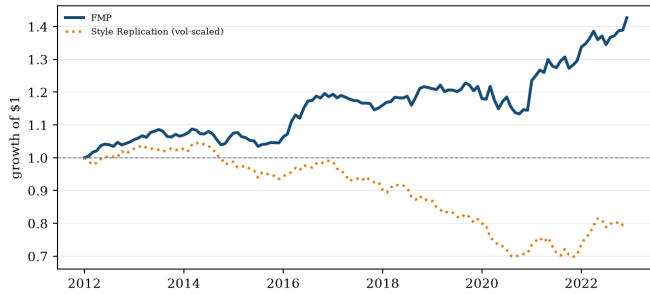


Figure: Common-window cumulative growth of \$1.

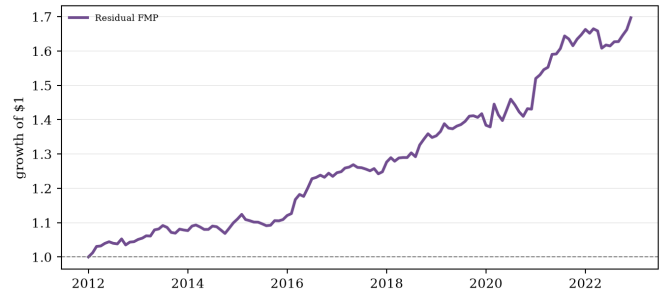


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_noa_proxy	3.38%	4.99%	0.68	2.19	-7.68%	0.241	4.32%	4.99%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_noa_proxy	-0.11 (-3.11)	0.07 (1.16)	0.20 (3.56)	0.06 (1.39)	0.05 (0.41)	-0.06 (-0.80)	4.94% (3.57)	0.265

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_noa_proxy	-1.44%	-0.06%	-0.45%	0.32%	0.40%	-0.33%	4.94%	3.38%

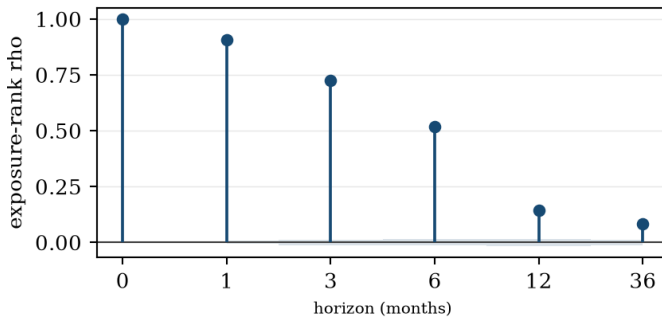


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

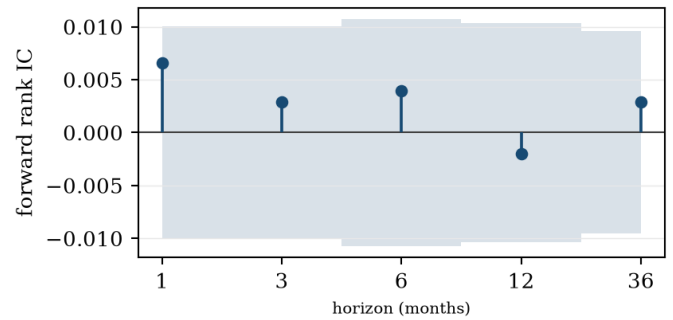


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	accruals_noa_proxy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1033 / 1338 / 2031
Available-history Sharpe	0.68
Bootstrap Sharpe	0.68 [0.15, 1.24]; Pr(S<=0)=0.5%
Bootstrap tails	skew=0.99 [-0.33, 1.88]; max DD=-7.7% [-13.5%, -4.2%]

19. momentum_21_126

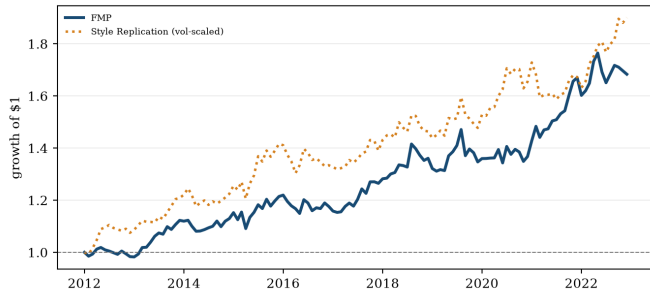


Figure: Common-window cumulative growth of \$1.

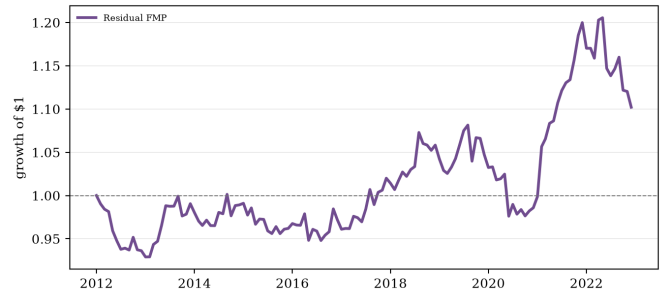


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_126	5.05%	7.47%	0.68	2.90	-8.70%	0.485	0.76%	7.47%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_126	0.04 (1.16)	0.10 (1.22)	0.04 (0.61)	0.46 (6.82)	0.11 (2.43)	0.06 (0.81)	1.03% (0.61)	0.486

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_126	0.56%	-0.09%	-0.10%	2.33%	0.97%	0.35%	1.03%	5.05%

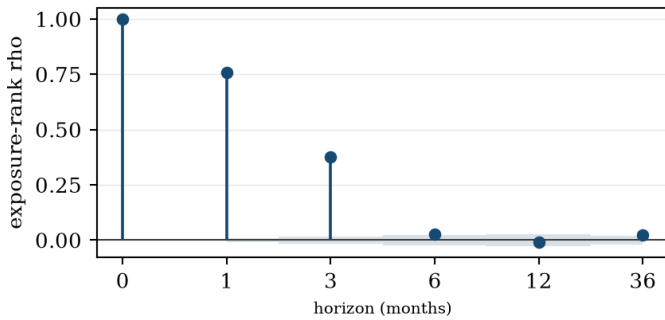


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

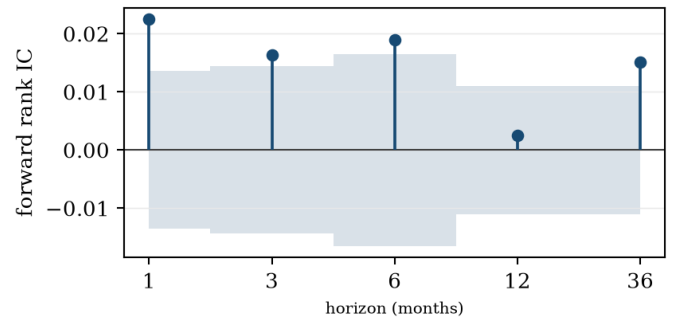


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	momentum_21_126(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.68
Bootstrap Sharpe	0.68 [0.30, 1.06]; Pr(S<=0)=0.0%
Bootstrap tails	skew=-0.15 [-0.61, 0.48]; max DD=-8.7% [-15.2%, -6.5%]

20. momentum_21_252

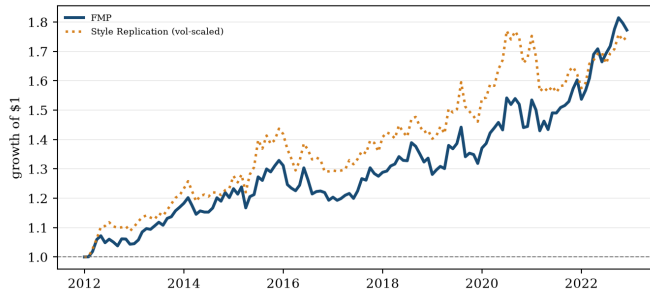


Figure: Common-window cumulative growth of \$1.

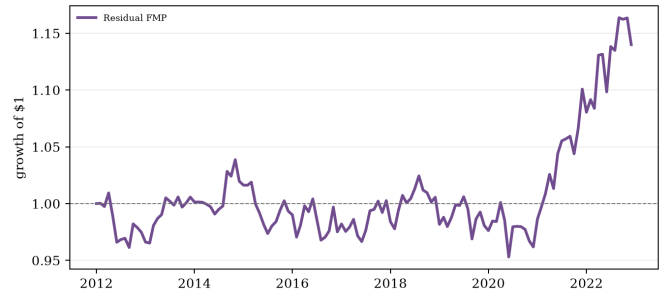


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_252	5.60%	8.38%	0.67	2.87	-10.23%	0.678	1.09%	8.38%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_252	0.02 (0.77)	-0.06 (-1.11)	-0.04 (-0.70)	0.53 (9.02)	0.19 (3.21)	-0.09 (-1.65)	1.31% (1.03)	0.679

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_252	0.30%	0.05%	0.09%	2.68%	1.69%	-0.53%	1.31%	5.60%

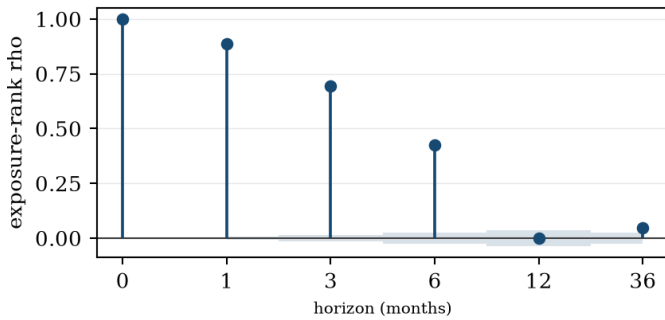


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

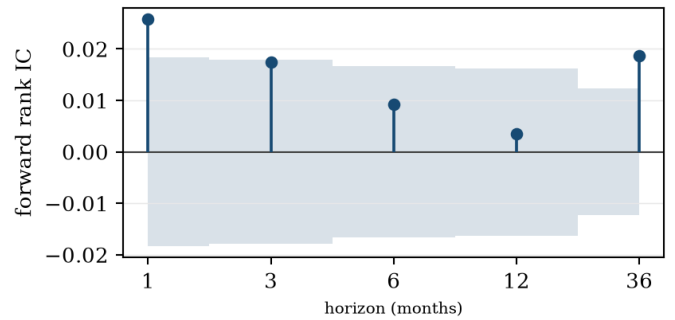


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	momentum_21_252(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.67
Bootstrap Sharpe	0.67 [0.26, 1.13]; Pr(S<=0)=0.0%
Bootstrap tails	skew=-0.15 [-0.54, 0.21]; max DD=-10.2% [-18.1%, -7.3%]

21. resid_momentum_21_252

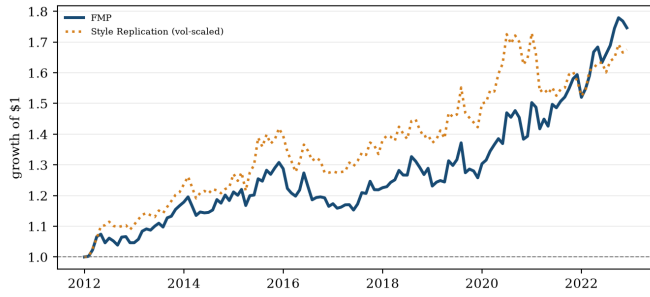


Figure: Common-window cumulative growth of \$1.

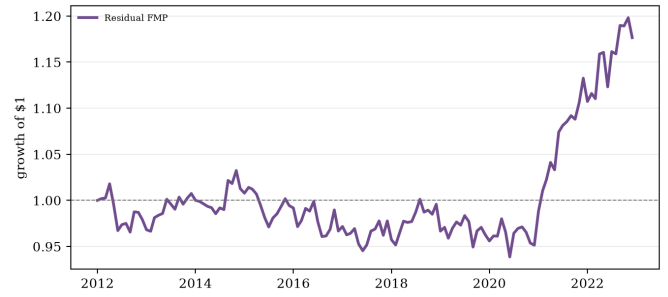


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
resid_momentum_21_252	5.46%	8.35%	0.65	2.90	-11.86%	0.643	1.39%	8.35%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
resid_momentum_21_252	0.02	-0.04	-0.04	0.51	0.19	-0.13	1.61%	0.644
	(0.45)	(-0.66)	(-0.64)	(8.68)	(2.72)	(-2.46)	(1.23)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
resid_momentum_21_252	0.20%	0.04%	0.09%	2.59%	1.71%	-0.77%	1.61%	5.46%

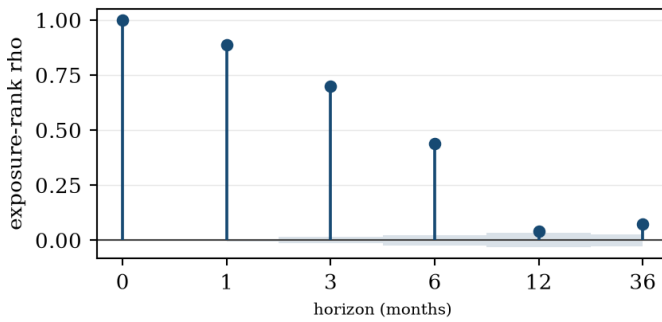


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

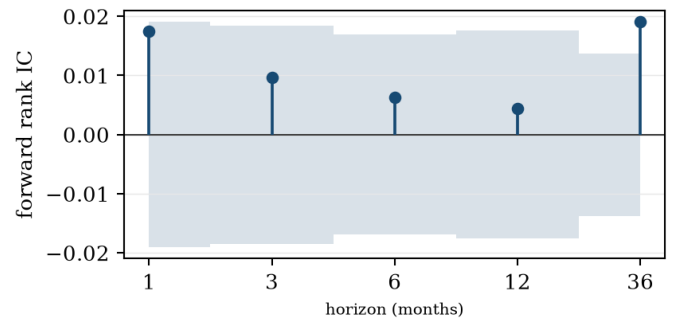


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	resid_momentum_21_252(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1248 / 1496 / 2000
Available-history Sharpe	0.65
Bootstrap Sharpe	0.65 [0.23, 1.10]; Pr(S<=0)=0.1%
Bootstrap tails	skew=-0.04 [-0.52, 0.36]; max DD=-11.9% [-19.4%, -6.5%]

22. share_turnover_21d

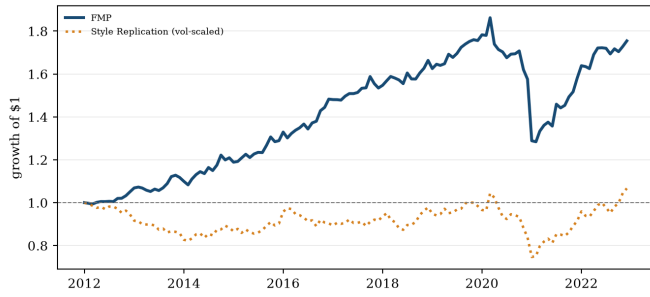


Figure: Common-window cumulative growth of \$1.

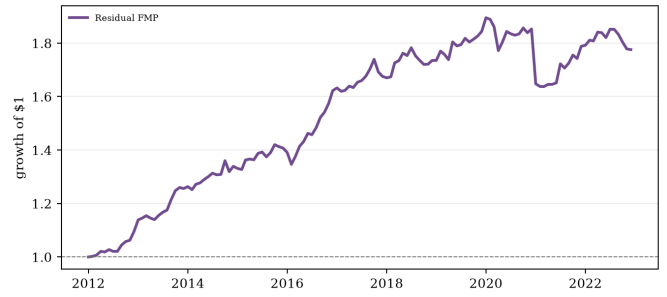


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
share_turnover_21d	5.54%	8.54%	0.65	2.05	-31.05%	0.463	4.89%	8.54%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
share_turnover_21d	-0.01 (-0.23)	-0.31 (-2.63)	-0.00 (-0.05)	-0.03 (-0.54)	-0.27 (-1.48)	0.44 (3.61)	5.47% (2.97)	0.469

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
share_turnover_21d	-0.16%	0.27%	0.01%	-0.17%	-2.39%	2.52%	5.47%	5.54%

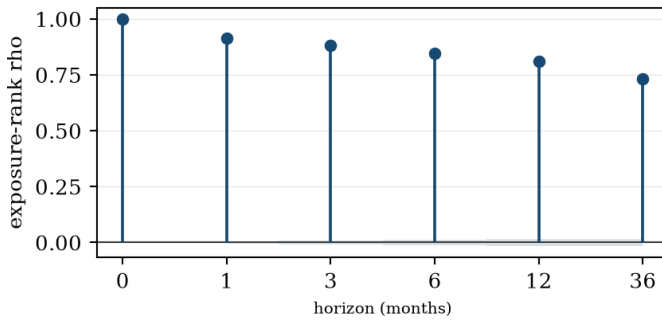


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

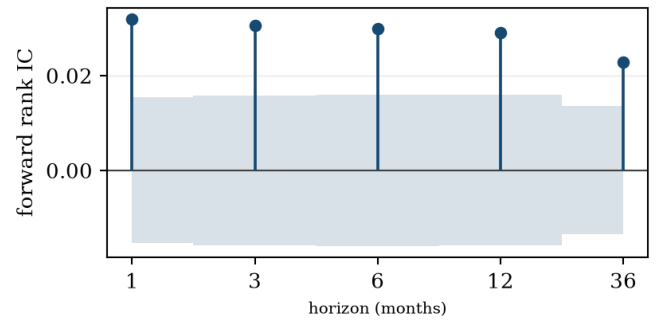


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	share_turnover_21d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.65
Bootstrap Sharpe	0.65 [-0.03, 1.70]; Pr(S<=0)=2.9%
Bootstrap tails	skew=-3.33 [-4.26, 0.68]; max DD=-31.1% [-45.8%, -3.7%]

23. asset_turnover

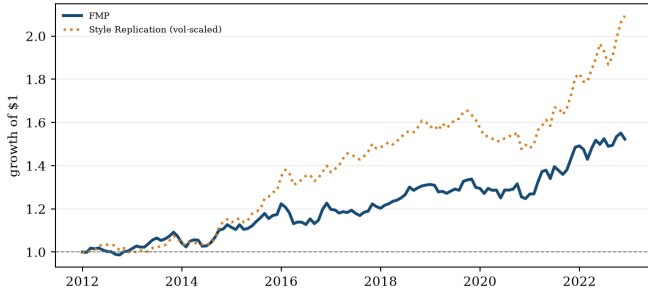


Figure: Common-window cumulative growth of \$1.

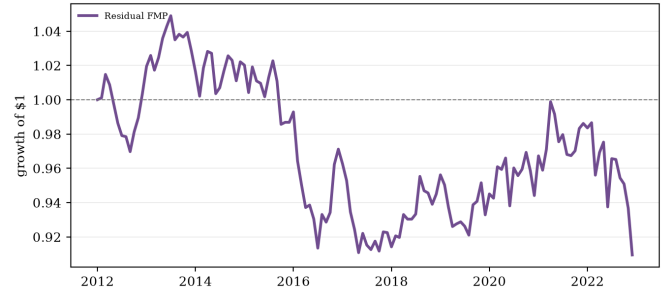


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
asset_turnover	4.05%	6.26%	0.65	2.58	-7.85%	0.487	-0.82%	6.26%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
asset_turnover	0.08 (2.45)	0.17 (3.03)	0.03 (0.75)	-0.01 (-0.20)	0.08 (1.35)	0.59 (10.05)	-0.77% (-0.56)	0.487

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
asset_turnover	1.03%	-0.15%	-0.07%	-0.04%	0.68%	3.37%	-0.77%	4.05%

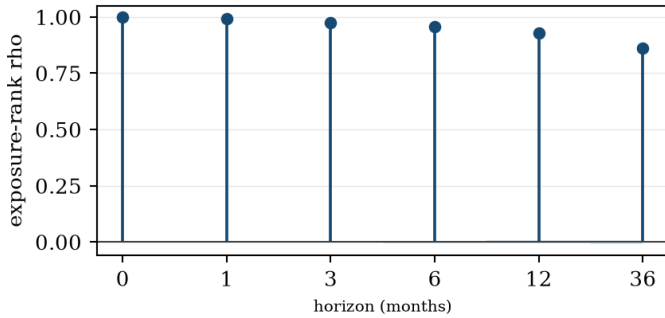


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

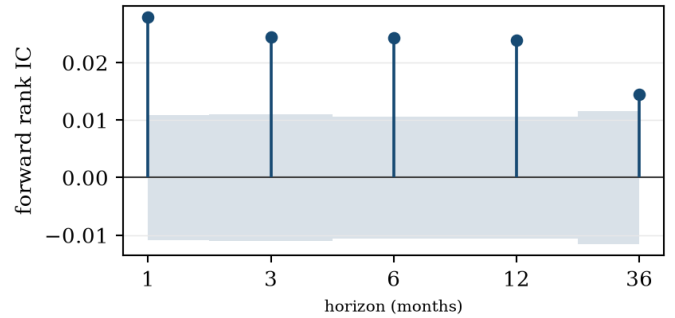


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	asset_turnover(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1246 / 1571 / 2083
Available-history Sharpe	0.65
Bootstrap Sharpe	0.65 [0.26, 1.03]; Pr(S<=0)=0.1%
Bootstrap tails	skew=-0.03 [-0.31, 0.21]; max DD=-7.9% [-13.9%, -6.2%]

24. ps_ttm

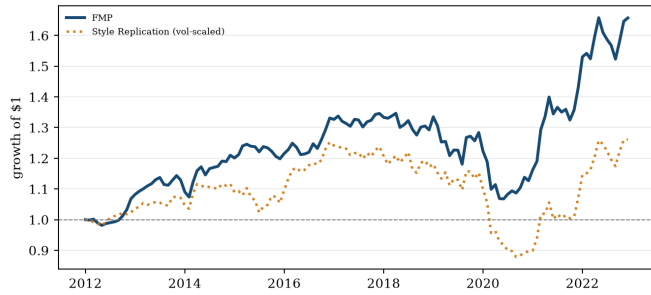


Figure: Common-window cumulative growth of \$1.

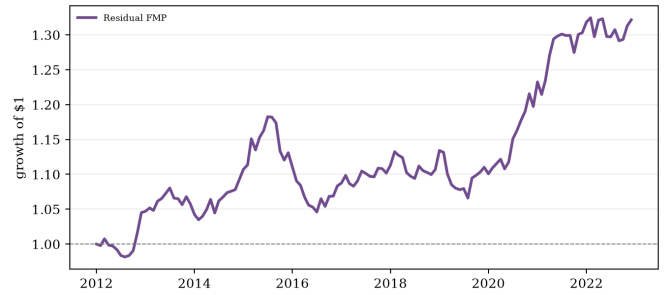


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ps_ttm	4.97%	8.25%	0.60	1.74	-20.72%	0.718	2.88%	8.25%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ps_ttm	0.12 (4.33)	0.10 (1.72)	0.47 (17.13)	-0.11 (-2.34)	0.10 (2.37)	0.28 (4.95)	2.65% (1.66)	0.719

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ps_ttm	1.55%	-0.09%	-1.08%	-0.56%	0.87%	1.62%	2.65%	4.97%

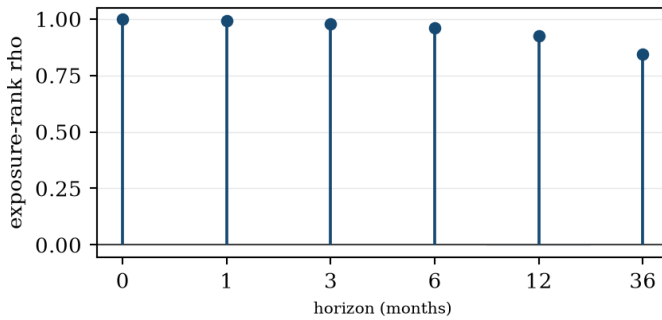


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

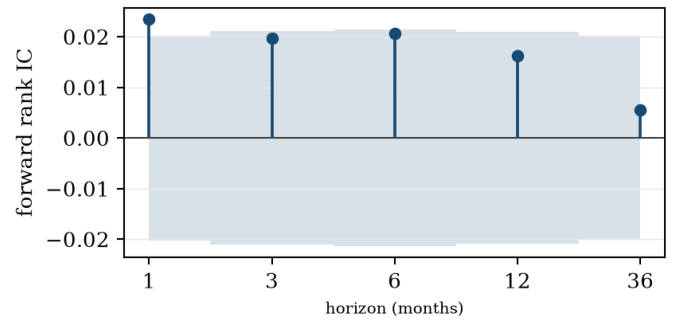


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	ps_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1236 / 1560 / 2036
Available-history Sharpe	0.60
Bootstrap Sharpe	0.60 [-0.07, 1.28]; Pr(S<=0)=3.9%
Bootstrap tails	skew=0.26 [-0.42, 0.83]; max DD=-20.7% [-34.0%, -6.2%]

25. sales_to_price

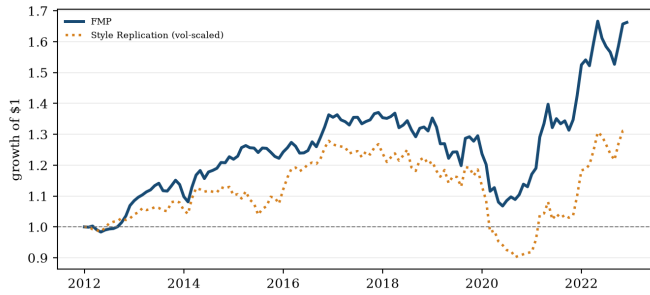


Figure: Common-window cumulative growth of \$1.

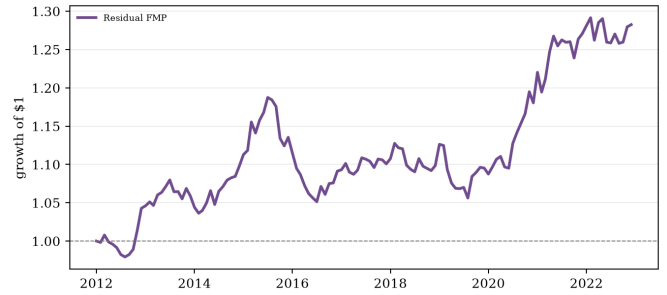


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
sales_to_price	5.01%	8.35%	0.60	1.74	-22.11%	0.714	2.60%	8.35%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
sales_to_price	0.12 (4.37)	0.09 (1.44)	0.48 (15.10)	-0.10 (-1.95)	0.11 (2.25)	0.30 (4.88)	2.38% (1.50)	0.715

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
sales_to_price	1.58%	-0.07%	-1.10%	-0.48%	0.97%	1.74%	2.38%	5.01%

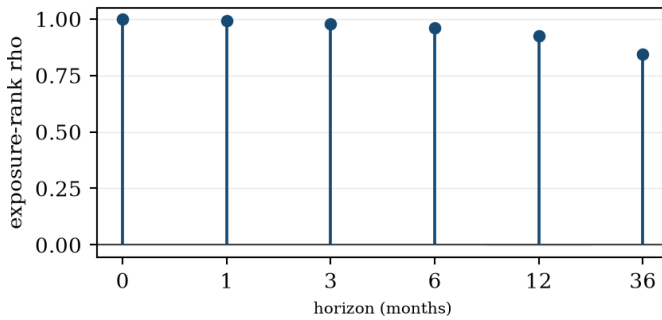


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

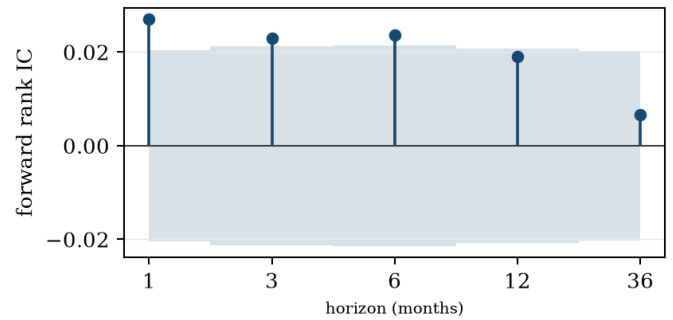


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	sales_to_price(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1247 / 1571 / 2083
Available-history Sharpe	0.60
Bootstrap Sharpe	0.60 [-0.09, 1.28]; Pr(S<=0)=3.9%
Bootstrap tails	skew=0.25 [-0.35, 0.73]; max DD=-22.1% [-35.5%, -6.6%]

26. log_market_cap

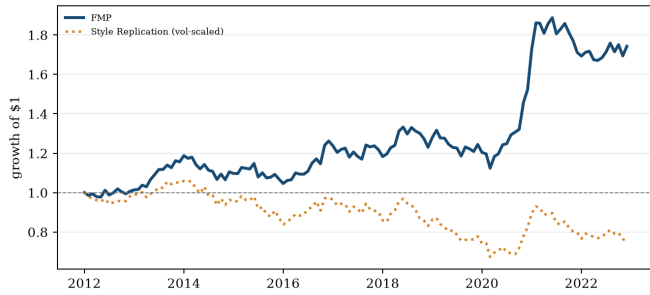


Figure: Common-window cumulative growth of \$1.

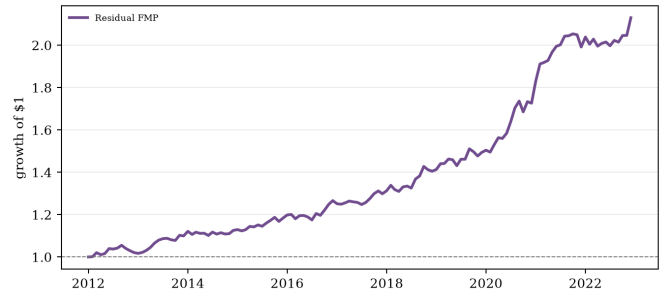


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
log_market_cap	5.61%	10.24%	0.55	1.56	-15.67%	0.750	7.39%	10.24%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
log_market_cap	0.03 (0.77)	0.94 (12.98)	0.21 (4.44)	-0.03 (-0.61)	0.03 (0.37)	-0.12 (-1.52)	7.07% (4.49)	0.751

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
log_market_cap	0.38%	-0.83%	-0.47%	-0.15%	0.26%	-0.66%	7.07%	5.61%

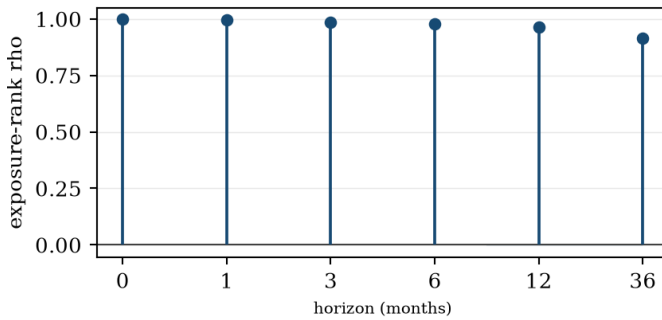


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

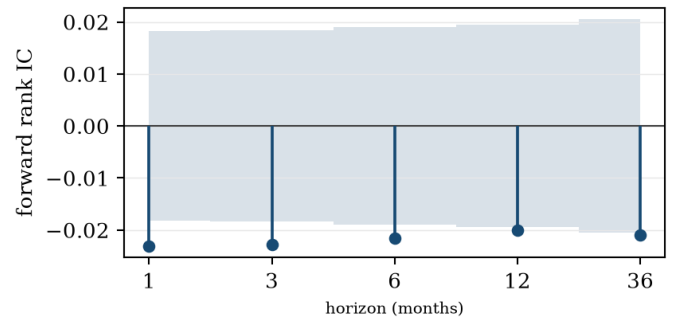


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	log_market_cap(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.55
Bootstrap Sharpe	0.55 [0.00, 1.04]; Pr(S<=0)=2.5%
Bootstrap tails	skew=1.02 [-0.07, 1.42]; max DD=-15.7% [-28.8%, -10.3%]

27. market_cap

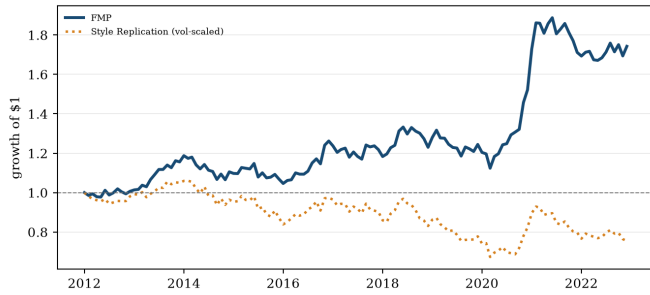


Figure: Common-window cumulative growth of \$1.

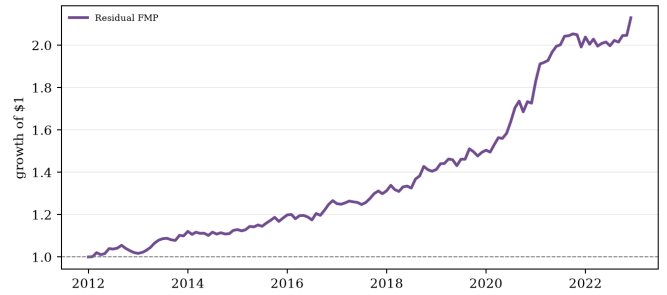


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
market_cap	5.61%	10.24%	0.55	1.56	-15.67%	0.750	7.39%	10.24%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
market_cap	0.03 (0.77)	0.94 (12.98)	0.21 (4.44)	-0.03 (-0.61)	0.03 (0.37)	-0.12 (-1.52)	7.07% (4.49)	0.751

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
market_cap	0.38%	-0.83%	-0.47%	-0.15%	0.26%	-0.66%	7.07%	5.61%

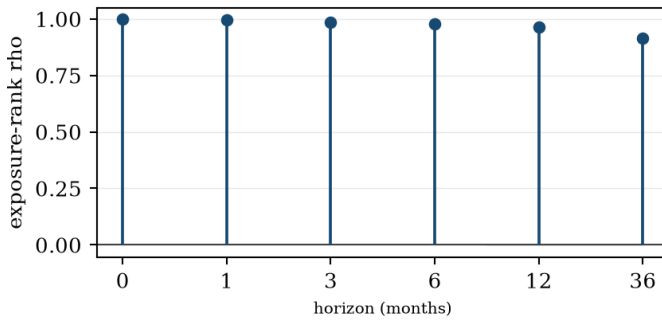


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

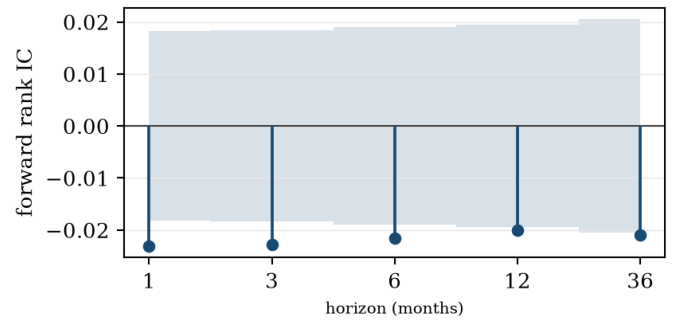


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	market_cap(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.55
Bootstrap Sharpe	0.55 [0.00, 1.04]; Pr(S<=0)=2.5%
Bootstrap tails	skew=1.02 [-0.07, 1.42]; max DD=-15.7% [-28.8%, -10.3%]

28. revenue_growth_yoy

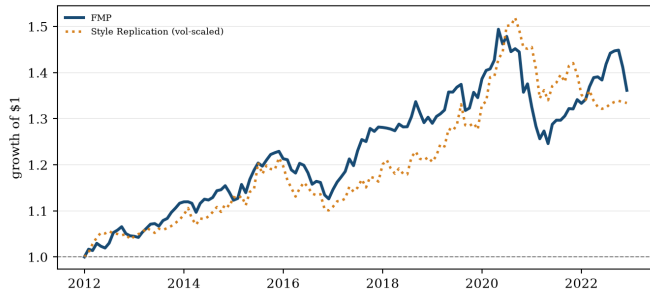


Figure: Common-window cumulative growth of \$1.

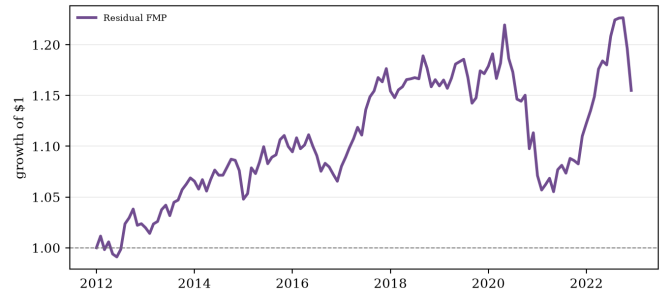


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
revenue_growth_yoy	2.98%	5.51%	0.54	1.61	-16.61%	0.322	1.40%	5.51%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
revenue_growth_yoy	0.03 (1.15)	-0.15 (-2.60)	-0.14 (-2.58)	0.13 (2.96)	0.05 (0.83)	-0.07 (-0.90)	1.42% (0.83)	0.322

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
revenue_growth_yoy	0.40%	0.13%	0.33%	0.67%	0.45%	-0.41%	1.42%	2.98%

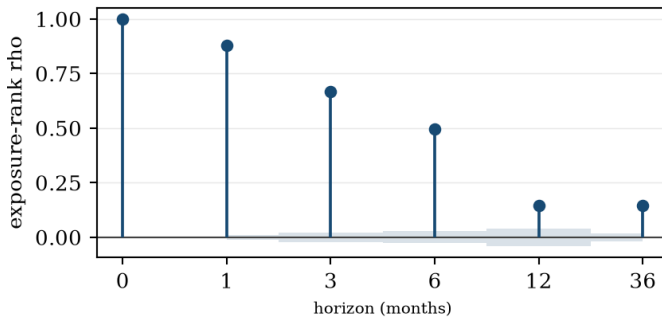


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

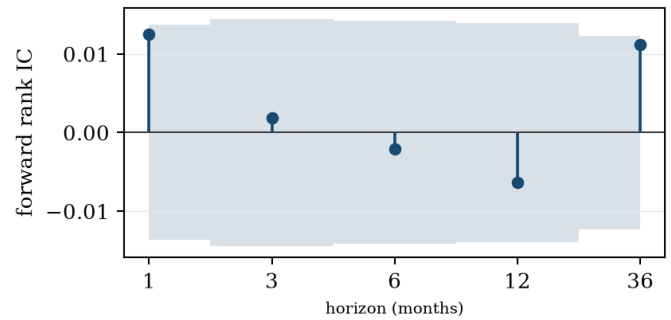


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	revenue_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1251 / 1542 / 2024
Available-history Sharpe	0.54
Bootstrap Sharpe	0.54 [0.00, 1.19]; Pr(S<=0)=2.5%
Bootstrap tails	skew=-0.68 [-1.09, -0.02]; max DD=-16.6% [-26.0%, -6.0%]

29. altman_z

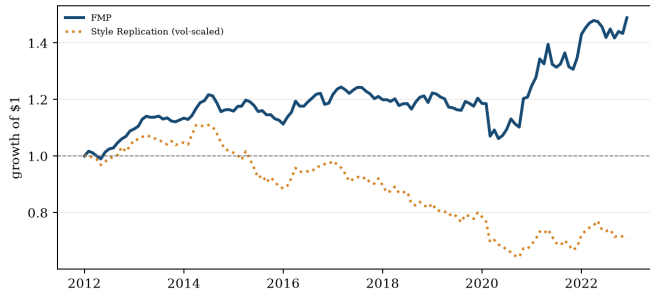


Figure: Common-window cumulative growth of \$1.

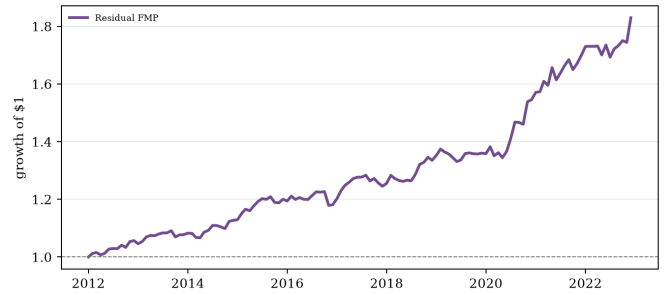


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
altman_z	3.92%	7.31%	0.54	1.70	-14.66%	0.568	6.11%	7.31%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
altman_z	0.04 (0.96)	-0.07 (-1.08)	0.39 (6.27)	-0.05 (-0.95)	0.07 (1.07)	-0.32 (-4.48)	5.66% (3.62)	0.572

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
altman_z	0.50%	0.06%	-0.89%	-0.24%	0.62%	-1.80%	5.66%	3.92%

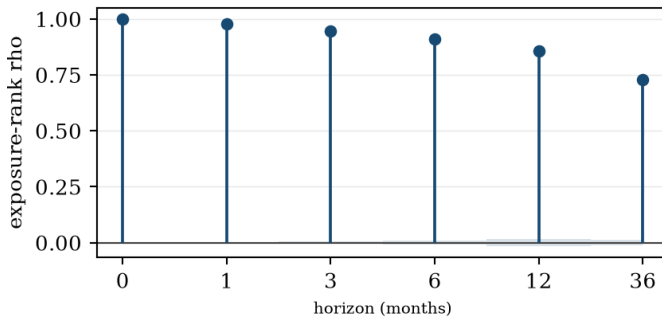


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

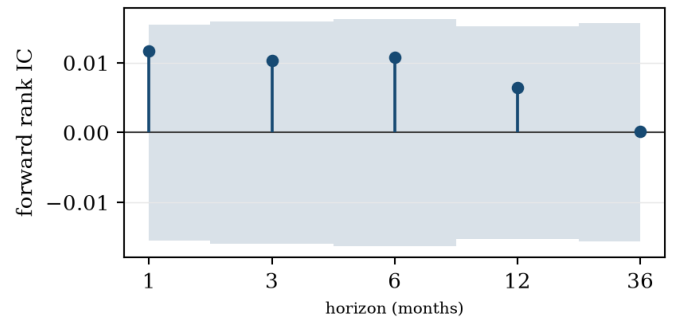


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	altman_z(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1213 / 1542 / 2049
Available-history Sharpe	0.54
Bootstrap Sharpe	0.54 [0.02, 1.10]; Pr(S<=0)=2.1%
Bootstrap tails	skew=-0.08 [-1.72, 1.20]; max DD=-14.7% [-24.2%, -6.3%]

30. ma_dist_63d

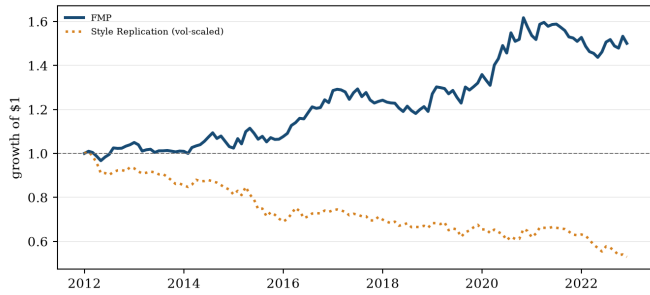


Figure: Common-window cumulative growth of \$1.

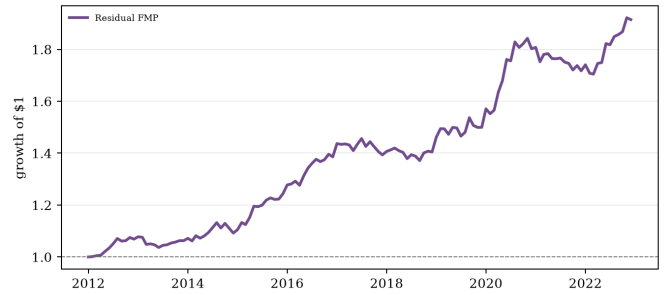


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ma_dist_63d	4.00%	7.57%	0.53	1.94	-11.14%	0.438	7.66%	7.57%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ma_dist_63d	0.13 (2.92)	-0.05 (-0.80)	-0.03 (-0.77)	-0.31 (-6.49)	-0.17 (-3.22)	-0.14 (-2.15)	6.12% (3.74)	0.462

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ma_dist_63d	1.66%	0.05%	0.07%	-1.57%	-1.53%	-0.80%	6.12%	4.00%

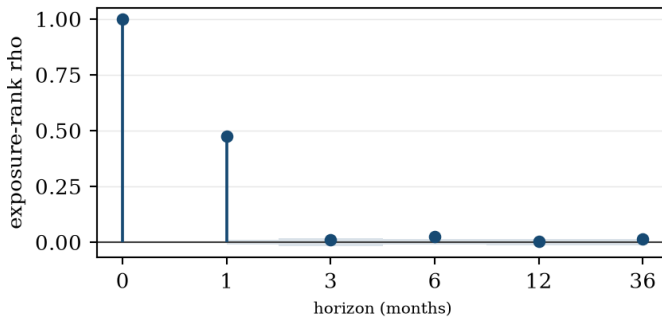


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

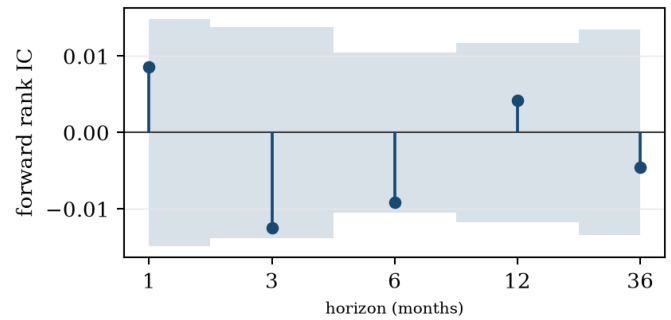


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	ma_dist_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.53
Bootstrap Sharpe	0.53 [0.08, 0.95]; Pr(S<=0)=1.1%
Bootstrap tails	skew=0.92 [0.36, 1.17]; max DD=-11.1% [-20.1%, -6.4%]

31. gpoa

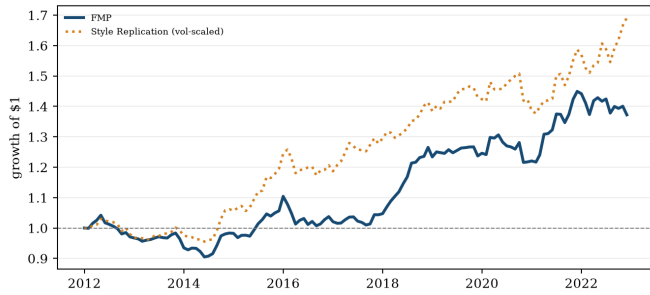


Figure: Common-window cumulative growth of \$1.

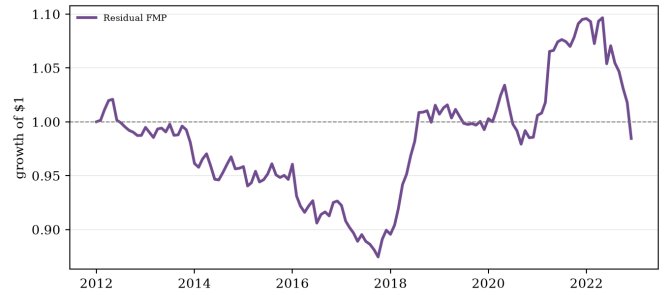


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gpoa	3.07%	5.90%	0.52	1.58	-13.17%	0.511	-0.50%	5.90%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gpoa	-0.01 (-0.19)	0.04 (0.81)	-0.16 (-4.65)	-0.05 (-1.04)	0.04 (0.71)	0.49 (9.67)	-0.06% (-0.04)	0.515

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gpoa	-0.08%	-0.04%	0.37%	-0.23%	0.32%	2.79%	-0.06%	3.07%

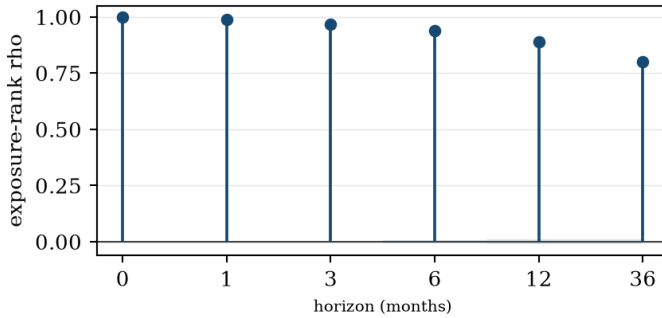


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

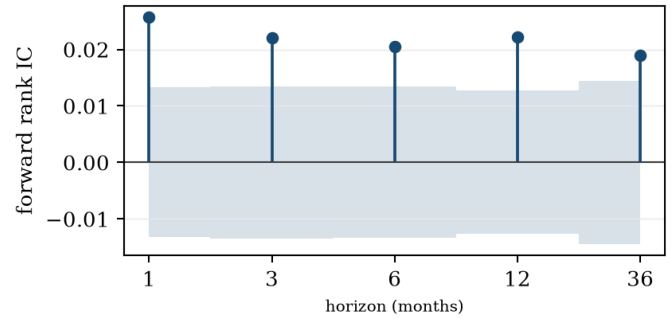


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	gpoa(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1246 / 1572 / 2078
Available-history Sharpe	0.52
Bootstrap Sharpe	0.52 [-0.14, 1.13]; Pr(S<=0)=5.7%
Bootstrap tails	skew=0.13 [-0.25, 0.49]; max DD=-13.2% [-23.1%, -7.2%]

32. fcf_to_equity

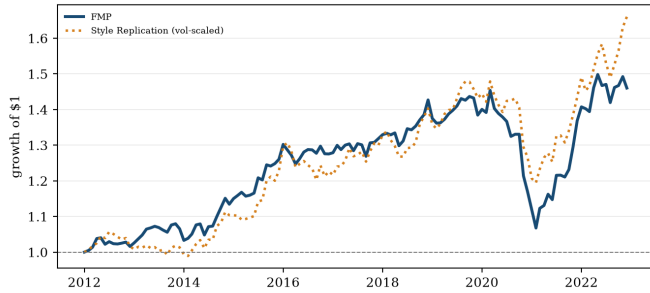


Figure: Common-window cumulative growth of \$1.

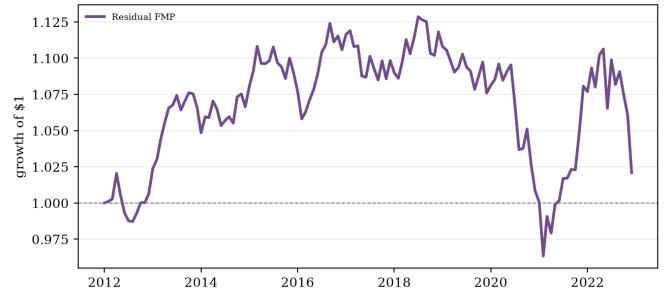


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
fcf_to_equity	3.74%	7.35%	0.51	1.49	-26.56%	0.629	-0.16%	7.35%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
fcf_to_equity	-0.01 (-0.17)	-0.34 (-6.71)	-0.01 (-0.17)	0.03 (0.65)	0.07 (1.68)	0.43 (6.13)	0.29% (0.22)	0.632

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
fcf_to_equity	-0.08%	0.30%	0.02%	0.15%	0.60%	2.47%	0.29%	3.74%

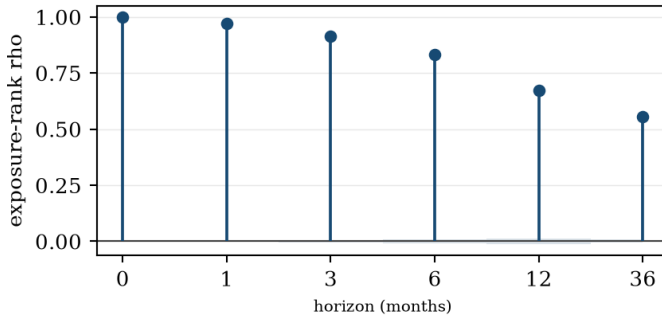


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

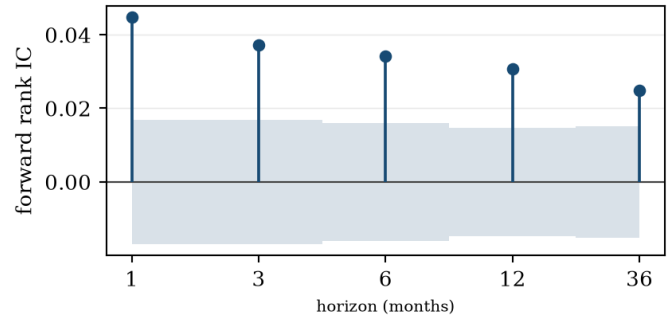


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	fcf_to_equity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1194 / 1489 / 1954
Available-history Sharpe	0.51
Bootstrap Sharpe	0.51 [-0.16, 1.21]; Pr(S<=0)=6.5%
Bootstrap tails	skew=-0.40 [-1.09, 0.47]; max DD=-26.6% [-40.6%, -4.6%]

33. equity_growth_yoy

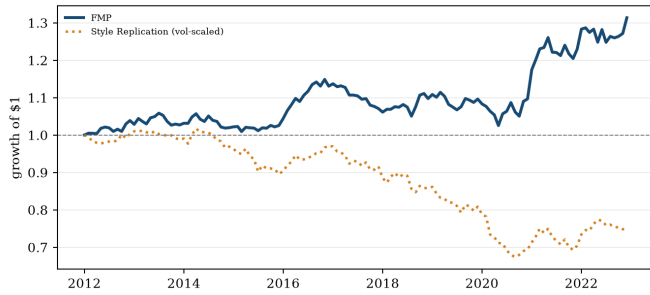


Figure: Common-window cumulative growth of \$1.

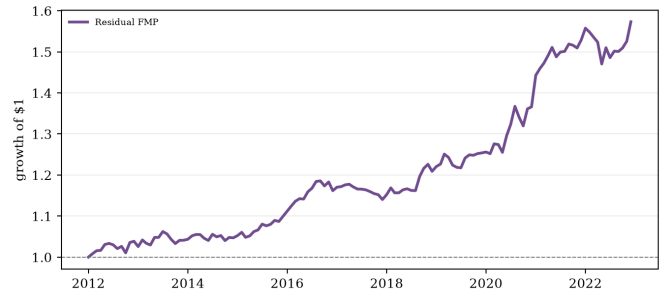


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
equity_growth_yoy	2.63%	5.20%	0.51	1.58	-10.74%	0.305	4.02%	5.20%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
equity_growth_yoy	-0.09 (-2.26)	0.08 (1.27)	0.19 (3.20)	-0.07 (-1.12)	0.07 (0.84)	-0.05 (-0.77)	4.25% (3.59)	0.308

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
equity_growth_yoy	-1.09%	-0.07%	-0.44%	-0.34%	0.62%	-0.30%	4.25%	2.63%

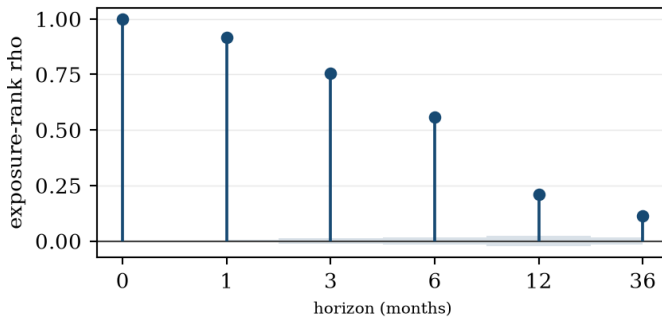


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

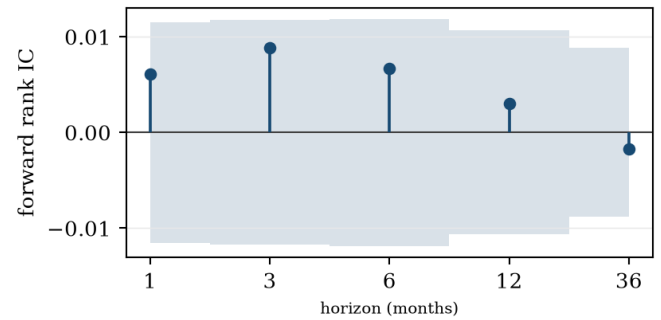


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	equity_growth_yoy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1257 / 1569 / 2088
Available-history Sharpe	0.51
Bootstrap Sharpe	0.51 [-0.05, 0.97]; Pr(S<=0)=3.6%
Bootstrap tails	skew=0.92 [0.08, 1.39]; max DD=-10.7% [-17.0%, -4.8%]

34. assets_to_mc

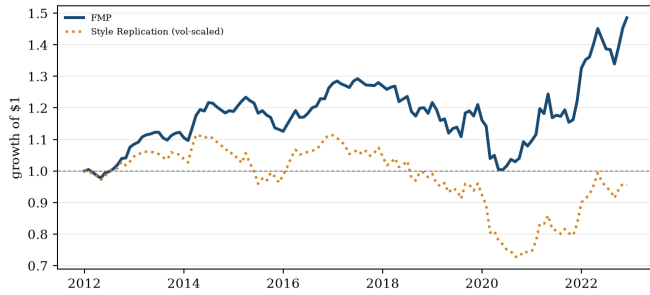


Figure: Common-window cumulative growth of \$1.

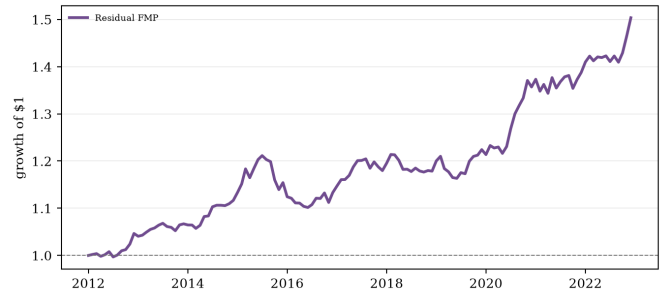


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
assets_to_mc	3.97%	8.26%	0.48	1.41	-22.36%	0.754	4.03%	8.26%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
assets_to_mc	0.08 (3.02)	-0.04 (-0.64)	0.54 (13.82)	-0.08 (-1.93)	0.07 (1.84)	0.01 (0.22)	3.82% (2.71)	0.755

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
assets_to_mc	1.05%	0.03%	-1.23%	-0.42%	0.63%	0.07%	3.82%	3.97%

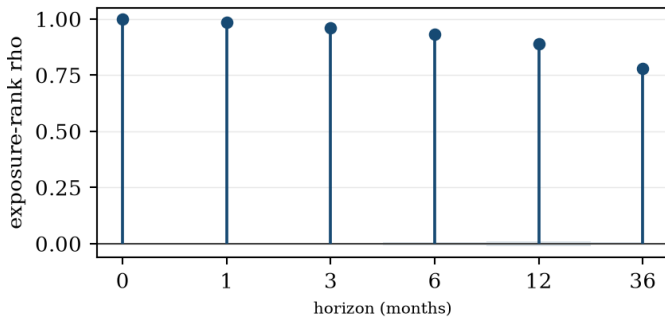


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

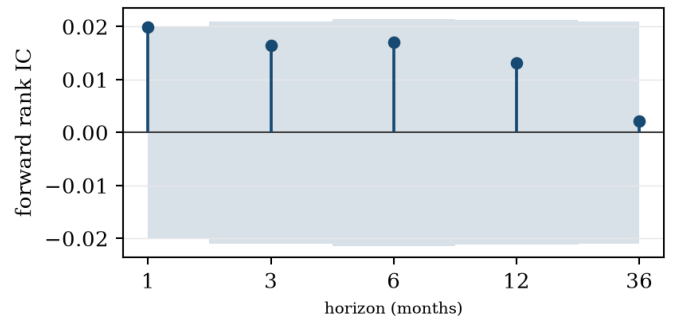


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	assets_to_mc(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1267 / 1582 / 2102
Available-history Sharpe	0.48
Bootstrap Sharpe	0.48 [-0.17, 1.19]; Pr(S<=0)=7.0%
Bootstrap tails	skew=0.03 [-0.83, 0.84]; max DD=-22.4% [-36.1%, -8.1%]

35. accruals_wc_proxy

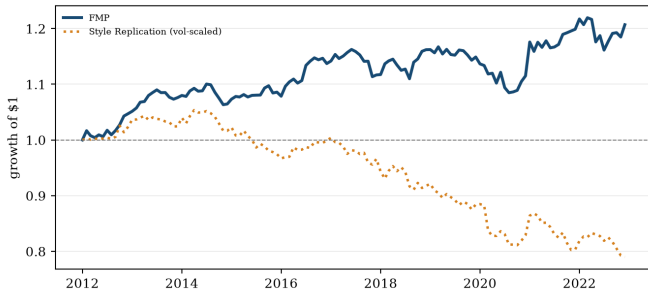


Figure: Common-window cumulative growth of \$1.

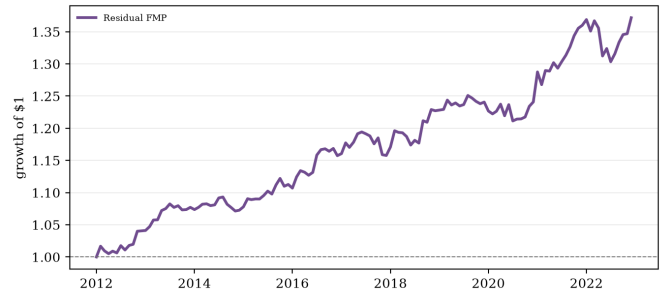


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_wc_proxy	1.79%	3.73%	0.48	1.62	-7.06%	0.168	2.65%	3.73%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_wc_proxy	-0.08 (-3.78)	0.01 (0.26)	0.06 (1.50)	-0.04 (-0.90)	0.10 (1.66)	-0.12 (-2.61)	2.96% (3.28)	0.178

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_wc_proxy	-1.06%	-0.01%	-0.14%	-0.20%	0.92%	-0.67%	2.96%	1.79%

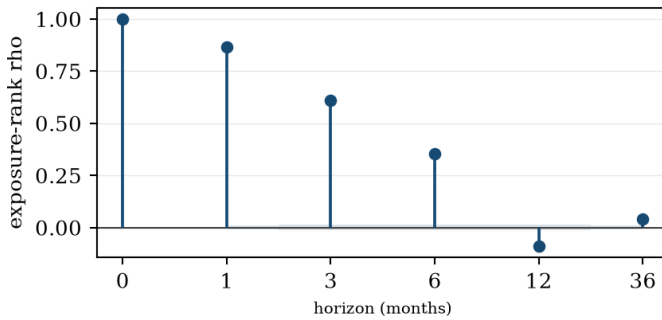


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

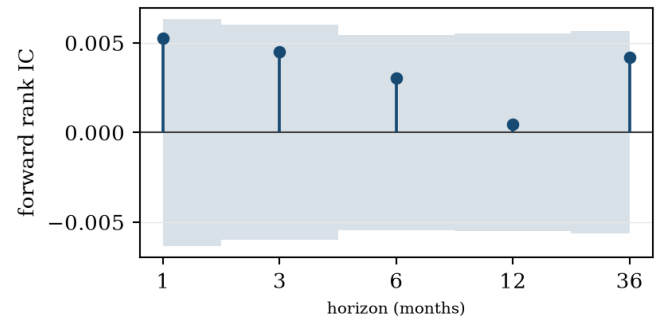


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	accruals_wc_proxy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1255 / 1570 / 2081
Available-history Sharpe	0.48
Bootstrap Sharpe	0.48 [0.03, 0.99]; Pr(S<=0)=2.2%
Bootstrap tails	skew=0.54 [-0.78, 1.48]; max DD=-7.1% [-12.7%, -4.1%]

36. ev

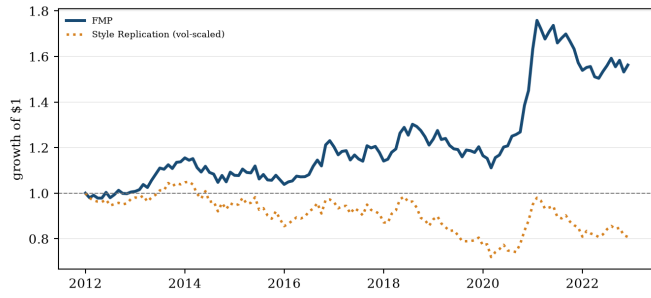


Figure: Common-window cumulative growth of \$1.

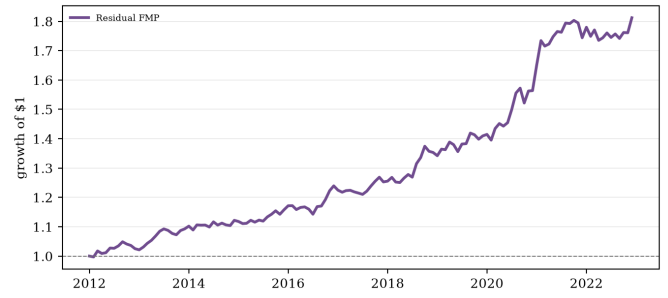


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ev	4.54%	9.61%	0.47	1.35	-14.69%	0.744	5.83%	9.61%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ev	0.03 (0.83)	0.94 (12.79)	0.14 (3.20)	0.01 (0.25)	-0.01 (-0.12)	-0.04 (-0.53)	5.57% (3.88)	0.745

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ev	0.37%	-0.83%	-0.31%	0.05%	-0.09%	-0.23%	5.57%	4.54%

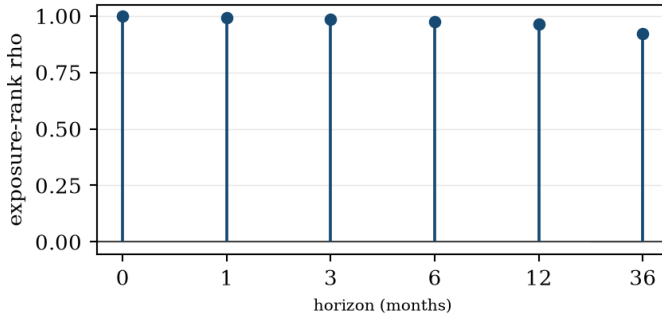


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

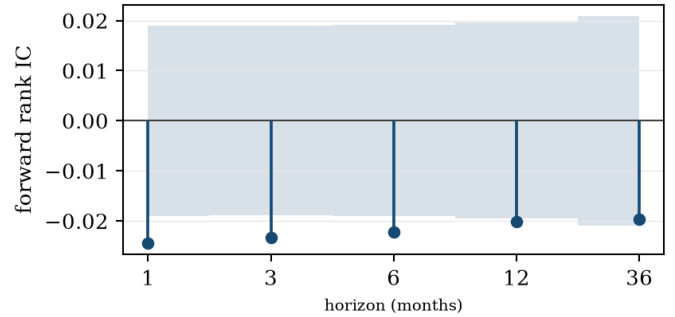


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	ev(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1069 / 1376 / 2057
Available-history Sharpe	0.47
Bootstrap Sharpe	0.47 [-0.07, 0.96]; Pr(S<=0)=4.2%
Bootstrap tails	skew=1.07 [0.01, 1.40]; max DD=-14.7% [-30.3%, -9.8%]

37. dso

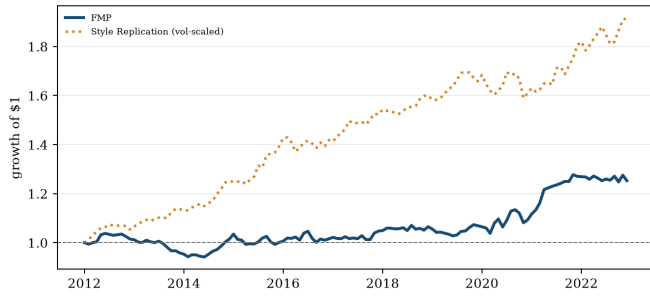


Figure: Common-window cumulative growth of \$1.

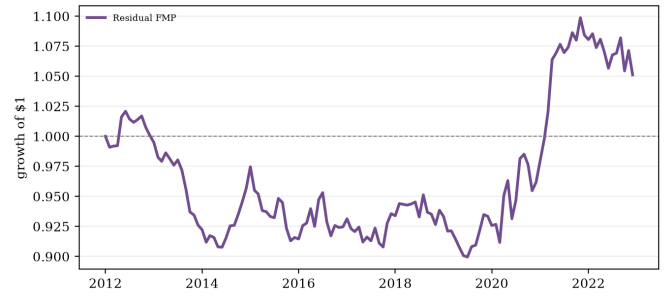


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dso	2.17%	4.63%	0.47	1.54	-9.24%	0.072	0.54%	4.63%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dso	0.02 (0.36)	-0.04 (-0.79)	-0.01 (-0.14)	0.01 (0.16)	0.09 (1.66)	0.08 (1.82)	0.55% (0.42)	0.072

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dso	0.23%	0.03%	0.01%	0.04%	0.83%	0.47%	0.55%	2.17%

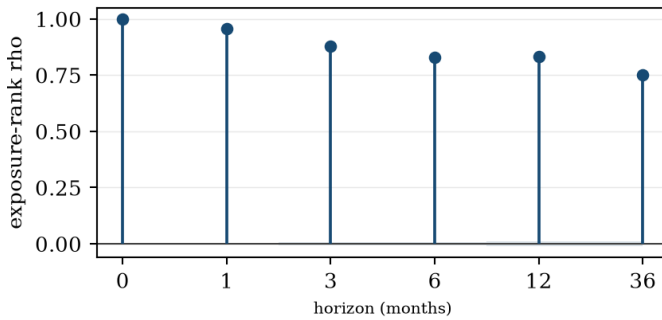


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

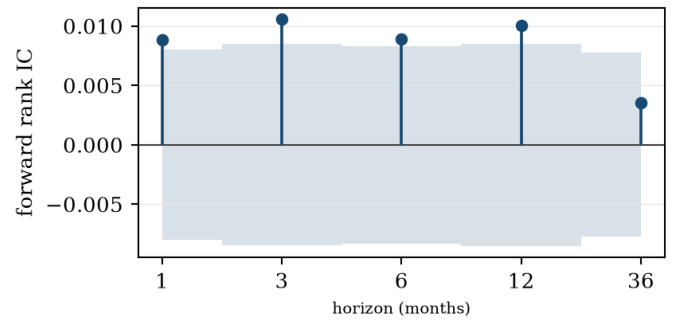


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	dso(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1214 / 1534 / 1997
Available-history Sharpe	0.47
Bootstrap Sharpe	0.47 [-0.07, 0.94]; Pr(S<=0)=4.5%
Bootstrap tails	skew=0.33 [-0.12, 0.75]; max DD=-9.2% [-16.7%, -4.6%]

38. is_negative_invested_capital

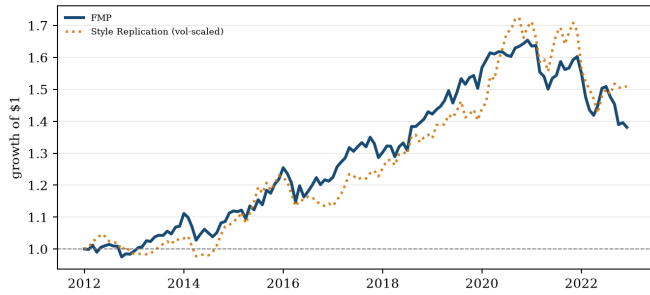


Figure: Common-window cumulative growth of \$1.

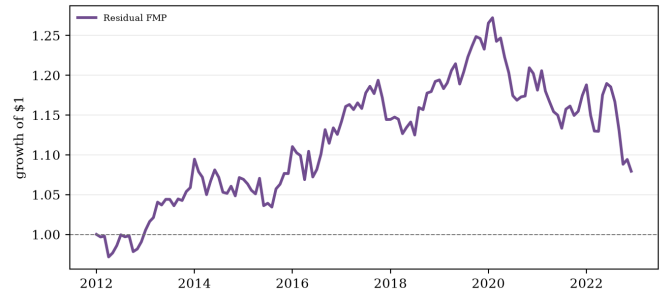


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
is_negative_invested_capital	3.20%	7.00%	0.46	1.60	-16.52%	0.412	0.62%	7.00%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
is_negative_invested_capital	-0.01	0.10	-0.35	-0.01	0.09	0.18	0.84%	0.413
	(-0.35)	(1.59)	(-6.16)	(-0.19)	(1.18)	(2.00)	(0.49)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
is_negative_invested_capital	-0.14%	-0.09%	0.80%	-0.07%	0.81%	1.05%	0.84%	3.20%

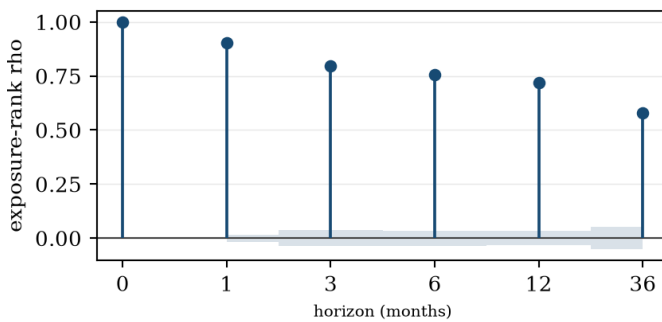


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

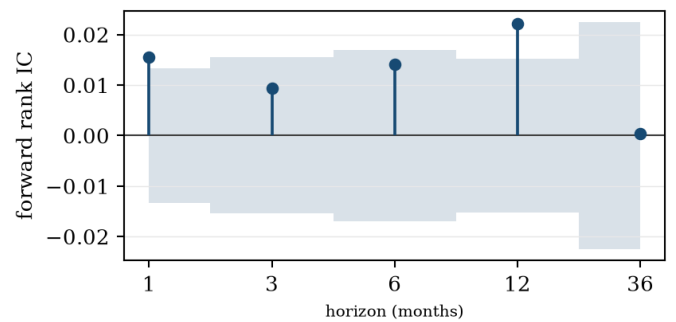


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	is_negative_invested_capital(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1145 / 1439 / 2013
Available-history Sharpe	0.46
Bootstrap Sharpe	0.46 [-0.15, 1.02]; Pr(S<=0)=6.5%
Bootstrap tails	skew=-0.33 [-0.60, -0.05]; max DD=-16.5% [-30.1%, -8.5%]

39. interest_to_debt

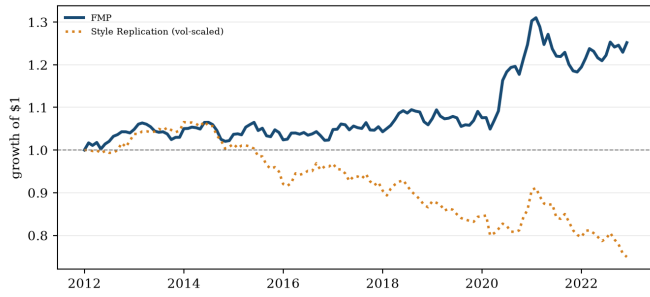


Figure: Common-window cumulative growth of \$1.

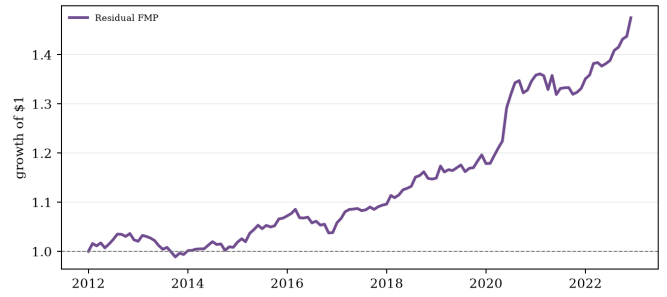


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
interest_to_debt	2.17%	4.79%	0.45	1.46	-9.71%	0.440	3.85%	4.79%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
interest_to_debt	-0.03 (-1.72)	0.14 (2.67)	0.08 (1.81)	-0.01 (-0.21)	0.11 (2.22)	-0.29 (-5.85)	3.63% (3.02)	0.442

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
interest_to_debt	-0.43%	-0.12%	-0.19%	-0.04%	1.00%	-1.67%	3.63%	2.17%

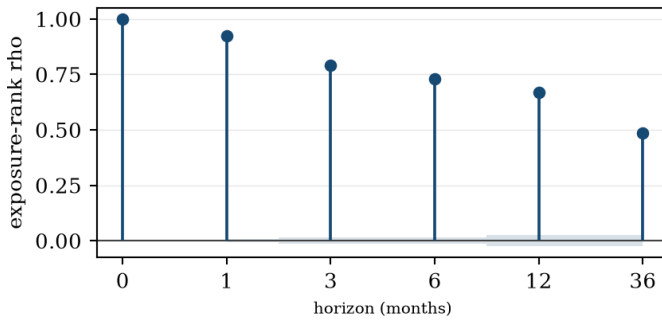


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

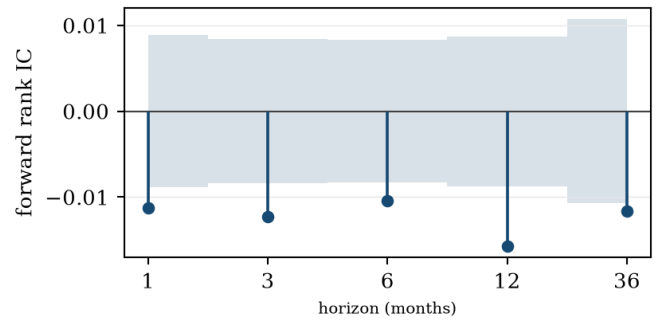


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	interest_to_debt(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1064 / 1369 / 2039
Available-history Sharpe	0.45
Bootstrap Sharpe	0.45 [-0.03, 0.90]; Pr(S<=0)=3.1%
Bootstrap tails	skew=0.80 [-0.21, 1.32]; max DD=-9.7% [-15.7%, -4.2%]

40. ncav_to_mc

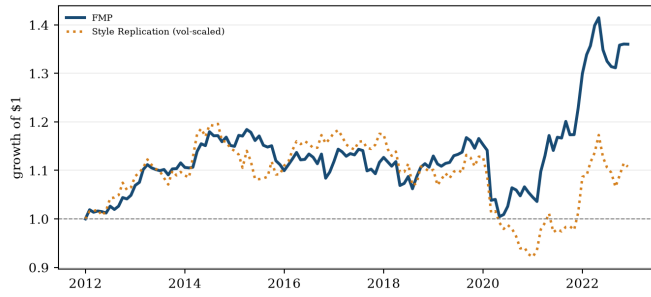


Figure: Common-window cumulative growth of \$1.

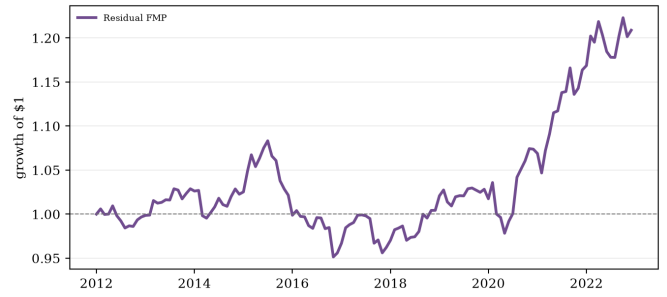


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ncav_to_mc	3.05%	6.80%	0.45	1.37	-15.17%	0.588	2.16%	6.80%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ncav_to_mc	0.10 (3.34)	-0.36 (-5.51)	0.33 (6.49)	-0.01 (-0.24)	0.14 (2.00)	-0.14 (-2.58)	1.83% (1.17)	0.591

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ncav_to_mc	1.28%	0.32%	-0.76%	-0.05%	1.21%	-0.79%	1.83%	3.05%

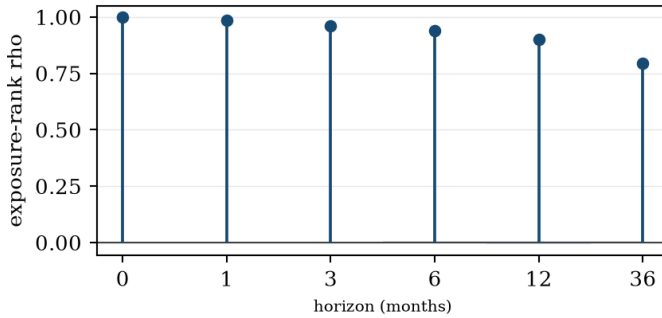


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

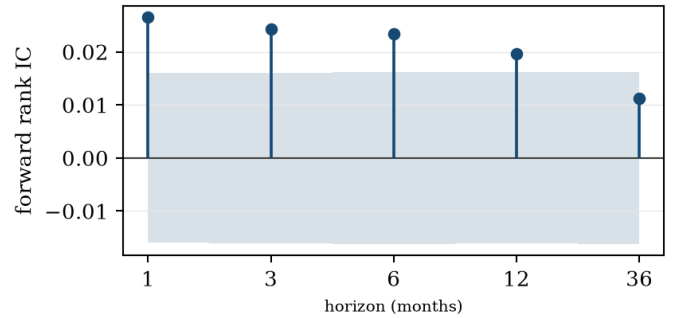


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	ncav_to_mc(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1261 / 1580 / 2097
Available-history Sharpe	0.45
Bootstrap Sharpe	0.45 [-0.14, 1.09]; Pr(S<=0)=7.3%
Bootstrap tails	skew=-0.62 [-1.81, 0.49]; max DD=-15.2% [-26.9%, -7.2%]

41. accruals

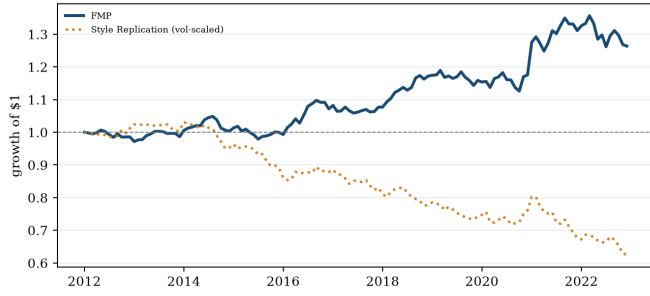


Figure: Common-window cumulative growth of \$1.

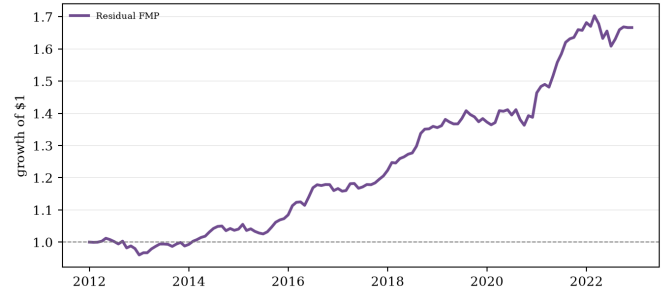


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals	2.27%	5.08%	0.45	1.53	-7.00%	0.315	4.61%	5.08%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals	-0.11 (-3.20)	0.09 (1.56)	0.05 (1.03)	-0.01 (-0.23)	0.09 (1.18)	-0.30 (-4.42)	4.77% (3.20)	0.316

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals	-1.35%	-0.08%	-0.11%	-0.05%	0.80%	-1.71%	4.77%	2.27%

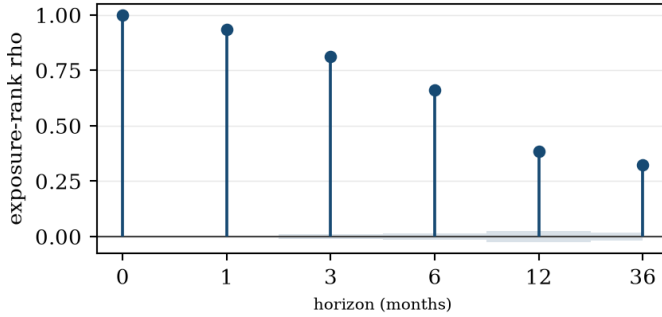


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

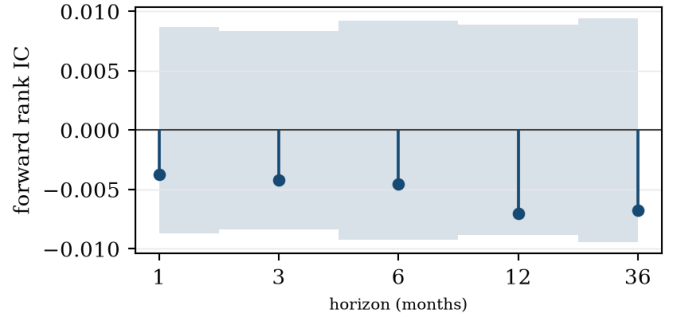


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	accruals(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1231 / 1556 / 2060
Available-history Sharpe	0.45
Bootstrap Sharpe	0.45 [-0.08, 0.92]; Pr(S<=0)=4.5%
Bootstrap tails	skew=1.36 [-0.40, 2.18]; max DD=-7.0% [-17.4%, -5.3%]

42. evs_ttm

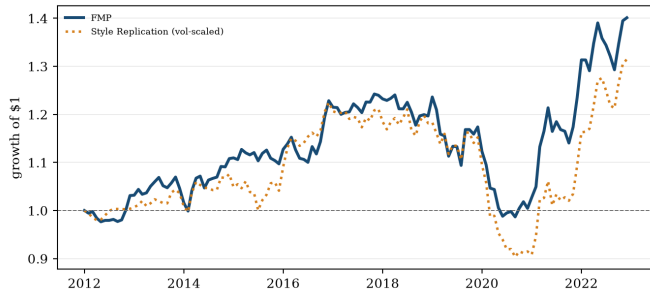


Figure: Common-window cumulative growth of \$1.

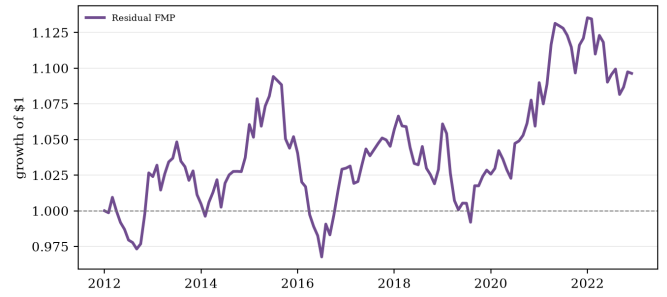


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
evs_ttm	3.38%	7.64%	0.44	1.33	-20.57%	0.678	1.08%	7.64%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
evs_ttm	0.10 (3.15)	0.17 (3.06)	0.39 (14.92)	-0.09 (-1.78)	0.03 (0.68)	0.43 (8.02)	0.94% (0.62)	0.679

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
evs_ttm	1.22%	-0.15%	-0.90%	-0.44%	0.27%	2.43%	0.94%	3.38%

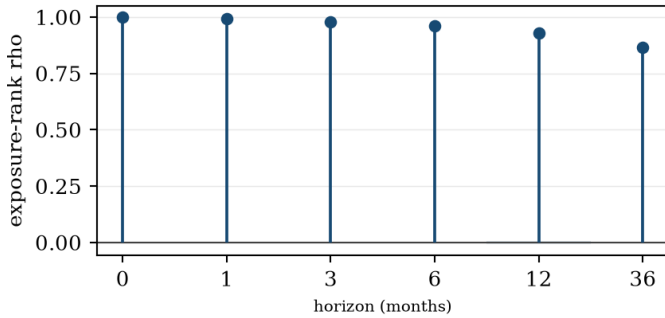


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

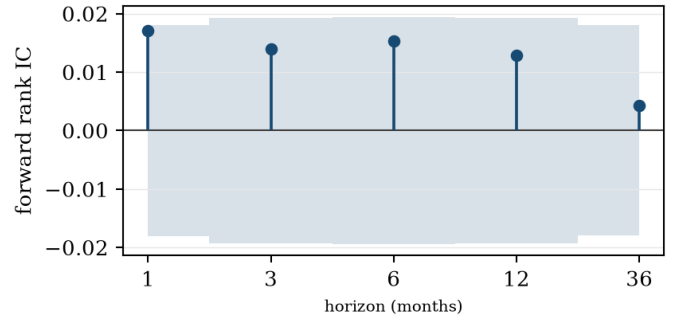


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	evs_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1044 / 1362 / 1985
Available-history Sharpe	0.44
Bootstrap Sharpe	0.44 [-0.20, 1.01]; Pr(S<=0)=8.0%
Bootstrap tails	skew=0.60 [0.17, 0.90]; max DD=-20.6% [-34.9%, -6.7%]

43. surprisePercentage

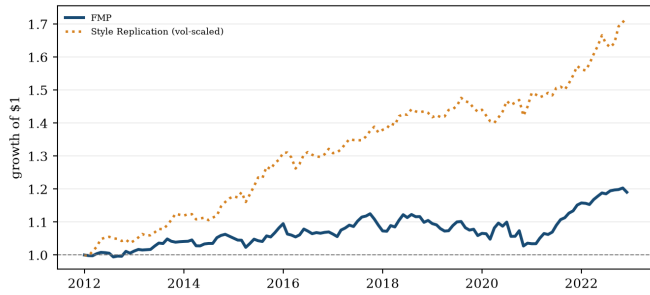


Figure: Common-window cumulative growth of \$1.

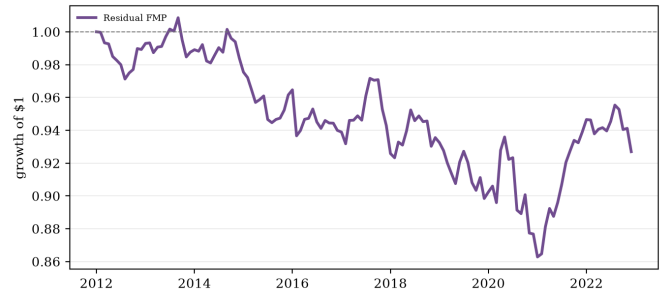


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
surprisePercentage	1.67%	3.84%	0.43	1.47	-8.69%	0.241	-0.81%	3.84%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
surprisePercentage	0.01 (0.47)	0.10 (2.49)	0.02 (0.81)	0.09 (2.66)	0.10 (2.69)	0.15 (2.87)	-0.64% (-0.61)	0.243

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
surprisePercentage	0.19%	-0.09%	-0.05%	0.46%	0.93%	0.87%	-0.64%	1.67%

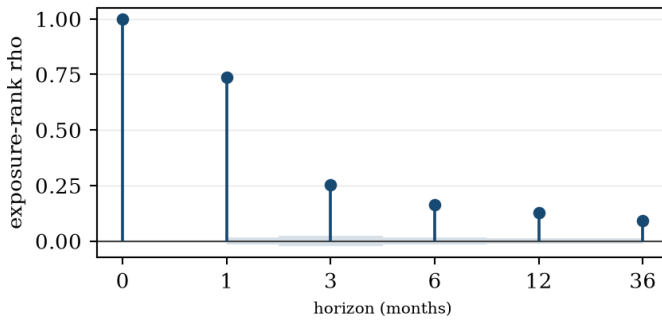


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

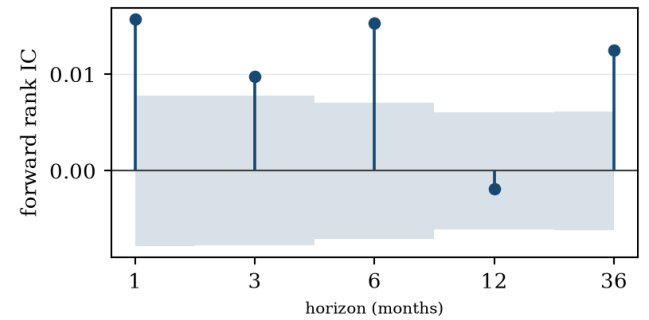


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	surprisePercentage(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1154 / 1435 / 1975
Available-history Sharpe	0.43
Bootstrap Sharpe	0.43 [-0.06, 1.07]; Pr(S<=0)=4.3%
Bootstrap tails	skew=-0.86 [-1.50, 0.02]; max DD=-8.7% [-14.3%, -4.2%]

44. bab_252v_1260c_FRENCH_CRSP

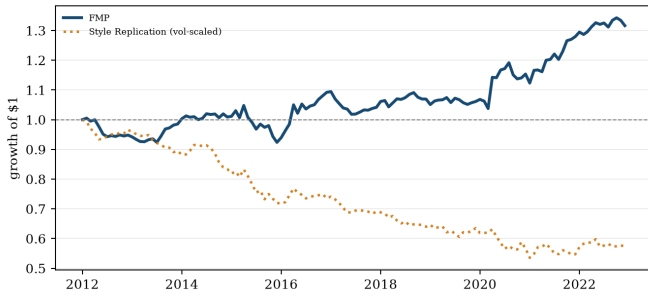


Figure: Common-window cumulative growth of \$1.

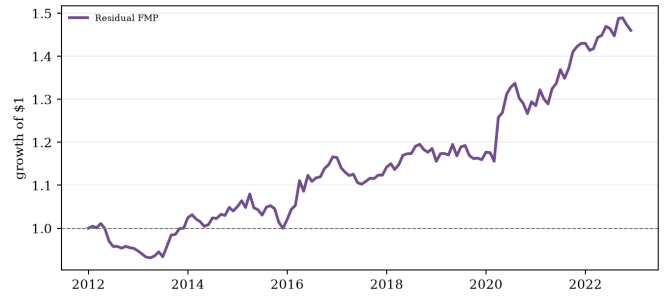


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bab_252v_1260c_FRENCH_CRSP	2.71%	6.28%	0.43	1.46	-11.84%	0.132	4.49%	6.28%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bab_252v_1260c_FRENCH_CRSP	0.08	-0.08	0.11	0.01	-0.16	-0.07	3.64%	0.149
	(1.25)	(-1.36)	(2.45)	(0.18)	(-3.22)	(-0.97)	(1.85)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bab_252v_1260c_FRENCH_CRSP	1.05%	0.07%	-0.25%	0.06%	-1.44%	-0.41%	3.64%	2.71%

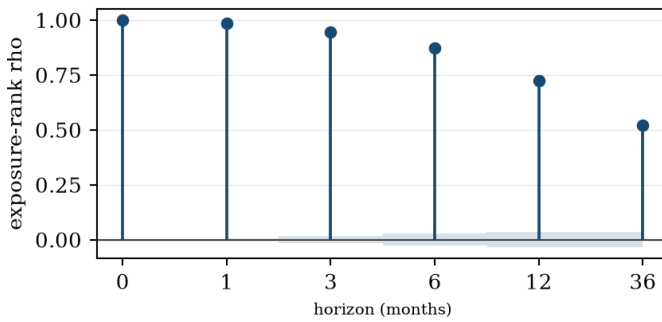


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

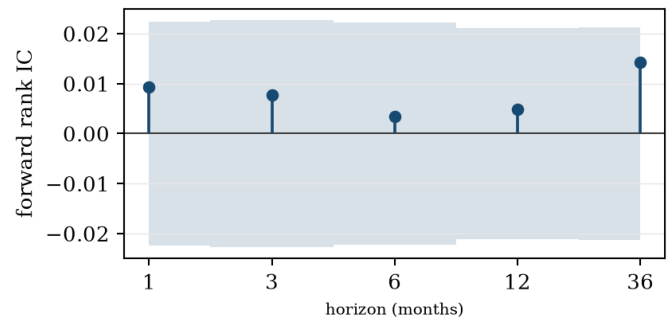


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	bab_252v_1260c_FRENCH_CRSP(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.43
Bootstrap Sharpe	0.43 [-0.03, 0.86]; Pr(S<=0)=3.5%
Bootstrap tails	skew=1.41 [-0.08, 2.54]; max DD=-11.8% [-21.6%, -7.0%]

45. bab_shrunk_252v_1260c_FRENCH_CRSP

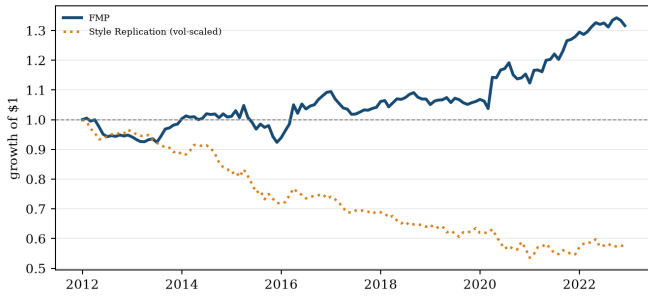


Figure: Common-window cumulative growth of \$1.

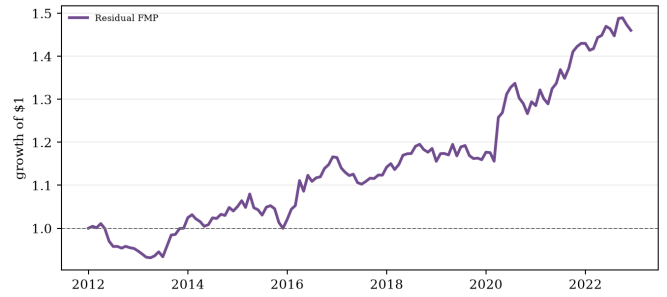


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bab_shrunk_252v_1260c_FRENCH_CRSP	2.71%	6.28%	0.43	1.46	-11.84%	0.132	4.49%	6.28%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bab_shrunk_252v_1260c_FRENCH_CRSP	0.08 (1.25)	-0.08 (-1.36)	0.11 (2.45)	0.01 (0.18)	-0.16 (-3.22)	-0.07 (-0.97)	3.64% (1.85)	0.149

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bab_shrunk_252v_1260c_FRENCH_CRSP	1.05%	0.07%	-0.25%	0.06%	-1.44%	-0.41%	3.64%	2.71%

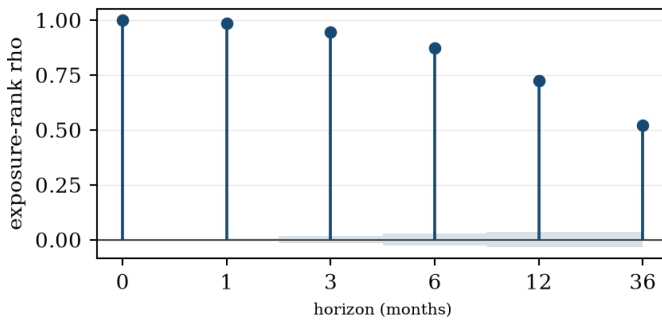


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

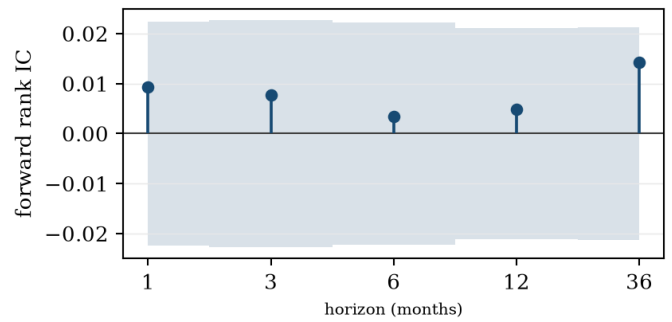


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	bab_shrunk_252v_1260c_FRENCH_CRSP(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.43
Bootstrap Sharpe	0.43 [-0.03, 0.86]; Pr(S<=0)=3.5%
Bootstrap tails	skew=1.41 [-0.08, 2.54]; max DD=-11.8% [-21.6%, -7.0%]

46. beta_252v_1260c_FRENCH_CRSP

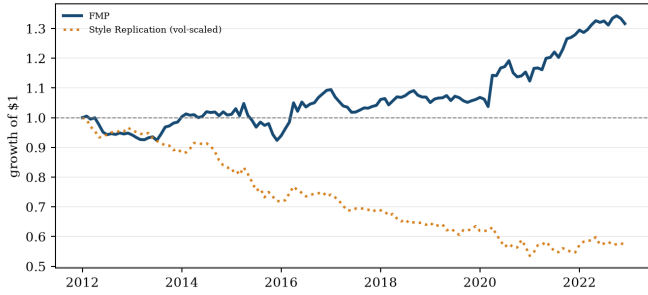


Figure: Common-window cumulative growth of \$1.

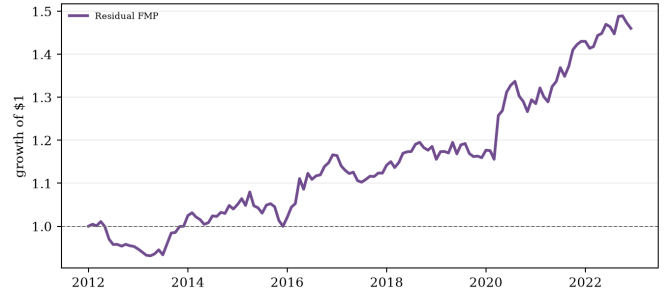


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
beta_252v_1260c_FRENCH_CRSP	2.71%	6.28%	0.43	1.46	-11.84%	0.132	4.49%	6.28%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
beta_252v_1260c_FRENCH_CRSP	0.08	-0.08	0.11	0.01	-0.16	-0.07	3.64%	0.149
	(1.25)	(-1.36)	(2.45)	(0.18)	(-3.22)	(-0.97)	(1.85)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
beta_252v_1260c_FRENCH_CRSP	1.05%	0.07%	-0.25%	0.06%	-1.44%	-0.41%	3.64%	2.71%

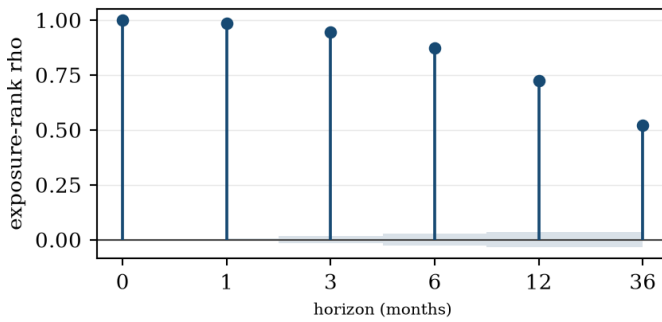


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

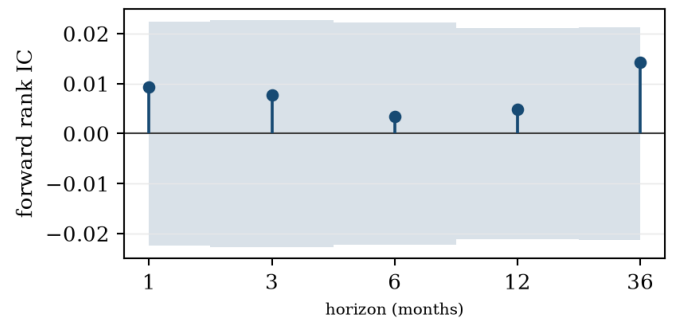


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	beta_252v_1260c_FRENCH_CRSP(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.43
Bootstrap Sharpe	0.43 [-0.03, 0.86]; Pr(S<=0)=3.5%
Bootstrap tails	skew=1.41 [-0.08, 2.54]; max DD=-11.8% [-21.6%, -7.0%]

47. beta_shrunk_252v_1260c_FRENCH_CRSP

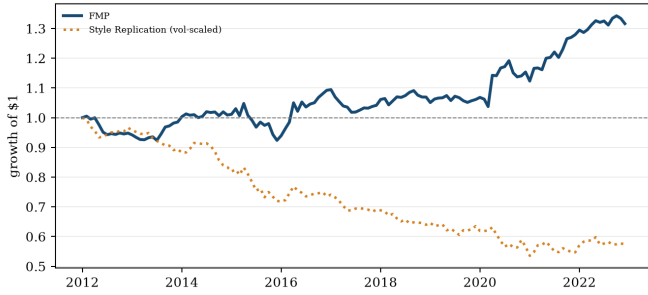


Figure: Common-window cumulative growth of \$1.

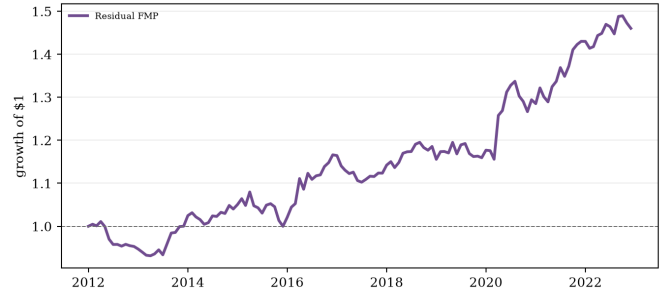


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
beta_shrunk_252v_1260c_FRENCH_CRSP	2.71%	6.28%	0.43	1.46	-11.84%	0.132	4.49%	6.28%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
beta_shrunk_252v_1260c_FRENCH_CRSP	0.08 (1.25)	-0.08 (-1.36)	0.11 (2.45)	0.01 (0.18)	-0.16 (-3.22)	-0.07 (-0.97)	3.64% (1.85)	0.149

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
beta_shrunk_252v_1260c_FRENCH_CRSP	1.05%	0.07%	-0.25%	0.06%	-1.44%	-0.41%	3.64%	2.71%

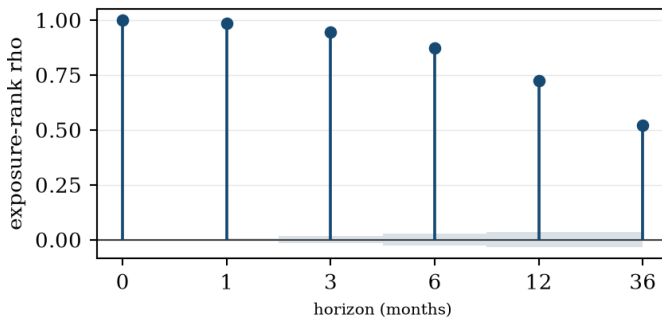


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

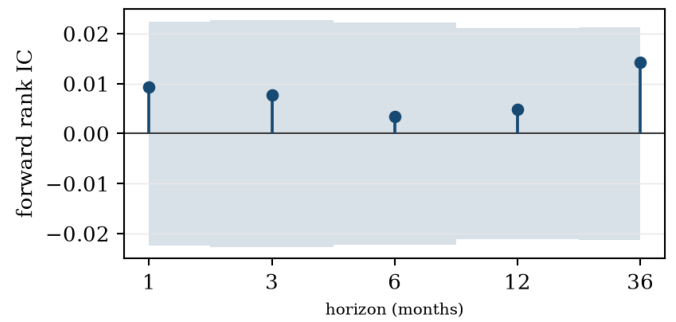


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	beta_shrunk_252v_1260c_FRENCH_CRSP(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.43
Bootstrap Sharpe	0.43 [-0.03, 0.86]; Pr(S<=0)=3.5%
Bootstrap tails	skew=1.41 [-0.08, 2.54]; max DD=-11.8% [-21.6%, -7.0%]

48. bab_252v_1260c_VTI

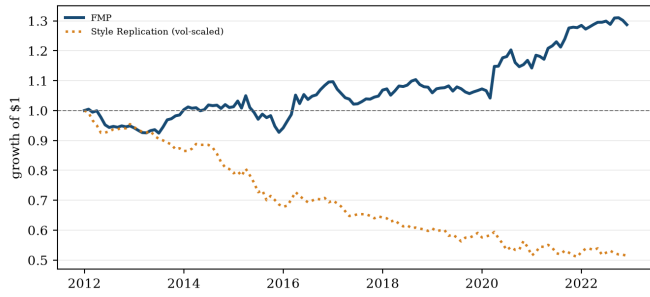


Figure: Common-window cumulative growth of \$1.

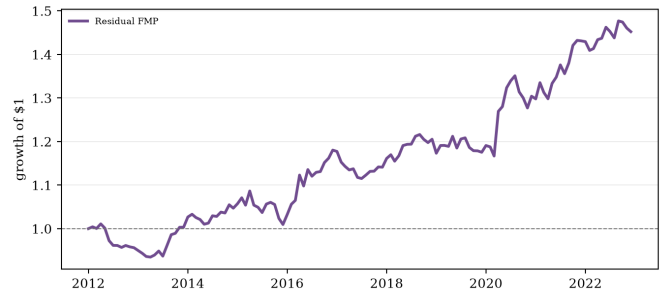


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bab_252v_1260c_VTI	2.50%	6.23%	0.40	1.37	-11.66%	0.118	4.52%	6.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bab_252v_1260c_VTI	0.08 (1.14)	-0.05 (-0.90)	0.08 (1.72)	-0.00 (-0.06)	-0.16 (-3.11)	-0.09 (-1.20)	3.59% (1.82)	0.137

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bab_252v_1260c_VTI	0.98%	0.05%	-0.18%	-0.02%	-1.42%	-0.49%	3.59%	2.50%

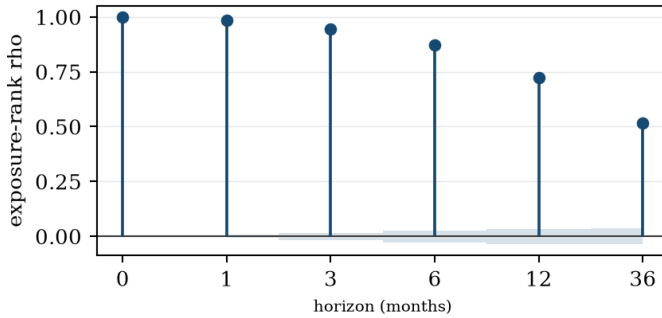


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

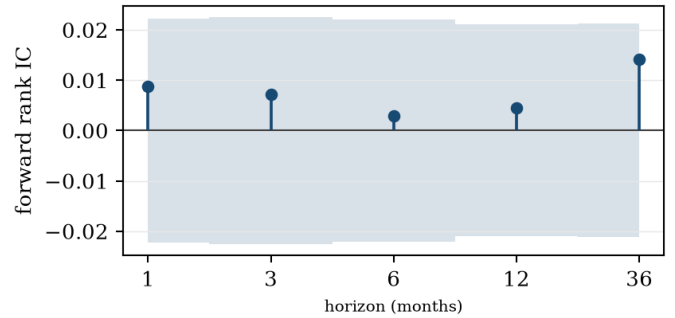


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	bab_252v_1260c_VTI(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.40
Bootstrap Sharpe	0.40 [-0.05, 0.81]; Pr(S<=0)=4.5%
Bootstrap tails	skew=1.50 [-0.02, 2.63]; max DD=-11.7% [-21.3%, -6.9%]

49. bab_shrunk_252v_1260c_VTI

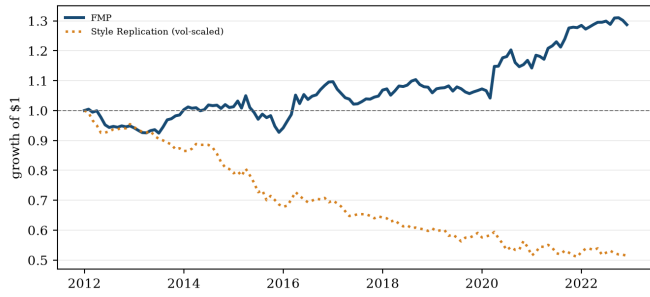


Figure: Common-window cumulative growth of \$1.

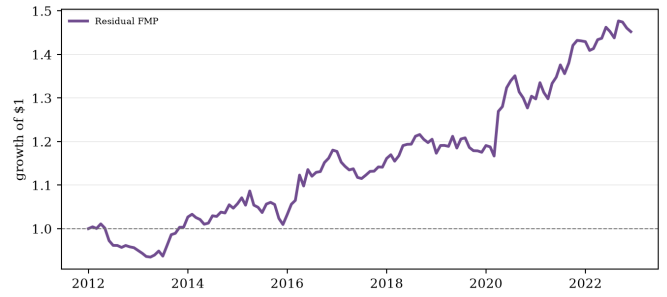


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bab_shrunk_252v_1260c_VTI	2.50%	6.23%	0.40	1.37	-11.66%	0.118	4.52%	6.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bab_shrunk_252v_1260c_VTI	0.08	-0.05	0.08	-0.00	-0.16	-0.09	3.59%	0.137
	(1.14)	(-0.90)	(1.72)	(-0.06)	(-3.11)	(-1.20)	(1.82)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bab_shrunk_252v_1260c_VTI	0.98%	0.05%	-0.18%	-0.02%	-1.42%	-0.49%	3.59%	2.50%

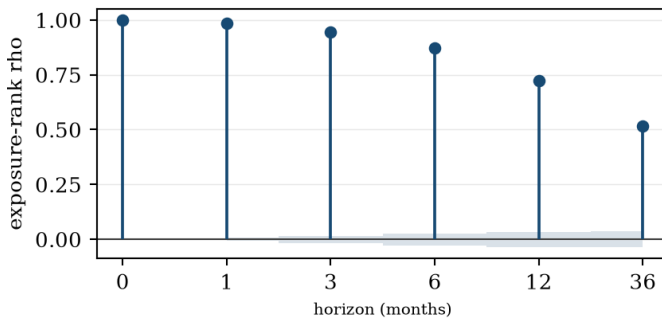


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

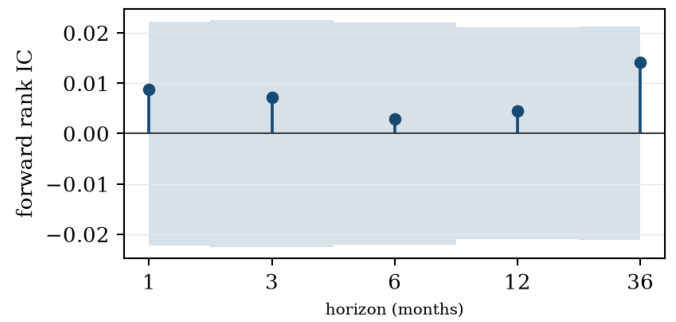


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	bab_shrunk_252v_1260c_VTI(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.40
Bootstrap Sharpe	0.40 [-0.05, 0.81]; Pr(S<=0)=4.5%
Bootstrap tails	skew=1.50 [-0.02, 2.63]; max DD=-11.7% [-21.3%, -6.9%]

50. beta_252v_1260c_VTI

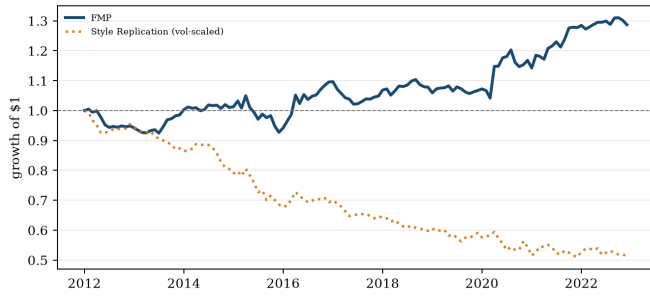


Figure: Common-window cumulative growth of \$1.

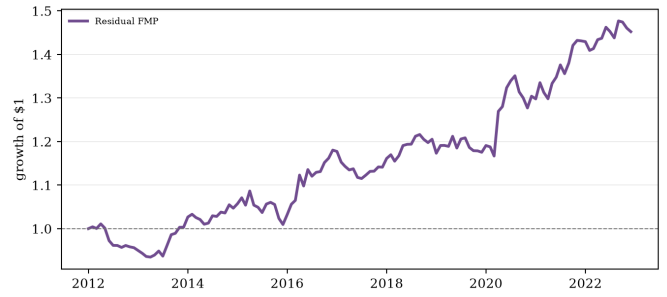


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
beta_252v_1260c_VTI	2.50%	6.23%	0.40	1.37	-11.66%	0.118	4.52%	6.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
beta_252v_1260c_VTI	0.08 (1.14)	-0.05 (-0.90)	0.08 (1.72)	-0.00 (-0.06)	-0.16 (-3.11)	-0.09 (-1.20)	3.59% (1.82)	0.137

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
beta_252v_1260c_VTI	0.98%	0.05%	-0.18%	-0.02%	-1.42%	-0.49%	3.59%	2.50%

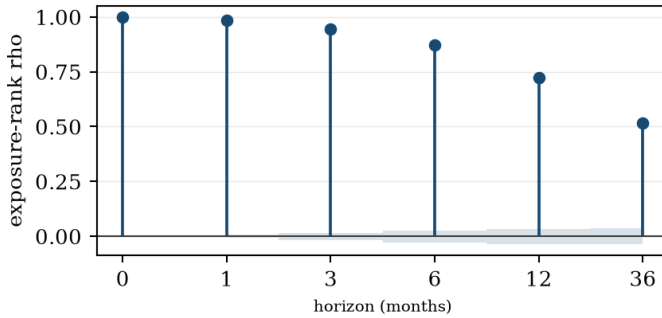


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

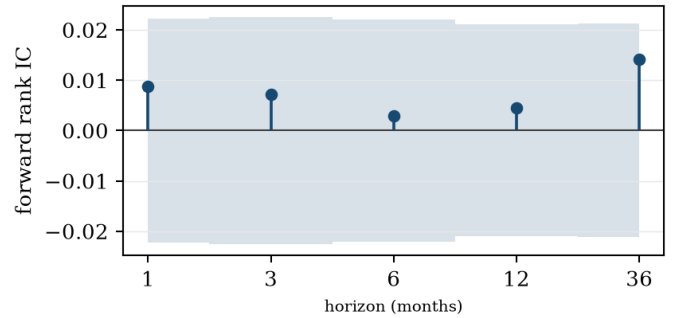


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	beta_252v_1260c_VTI(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.40
Bootstrap Sharpe	0.40 [-0.05, 0.81]; Pr(S<=0)=4.5%
Bootstrap tails	skew=1.50 [-0.02, 2.63]; max DD=-11.7% [-21.3%, -6.9%]

51. beta_shrunk_252v_1260c_VTI

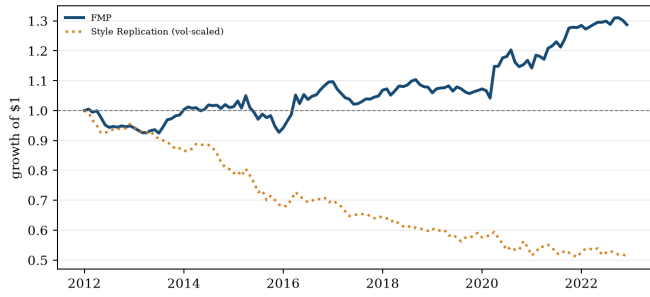


Figure: Common-window cumulative growth of \$1.

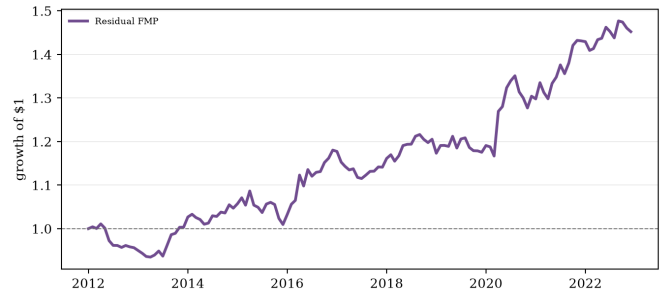


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
beta_shrunk_252v_1260c_VTI	2.50%	6.23%	0.40	1.37	-11.66%	0.118	4.52%	6.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
beta_shrunk_252v_1260c_VTI	0.08	-0.05	0.08	-0.00	-0.16	-0.09	3.59%	0.137
	(1.14)	(-0.90)	(1.72)	(-0.06)	(-3.11)	(-1.20)	(1.82)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
beta_shrunk_252v_1260c_VTI	0.98%	0.05%	-0.18%	-0.02%	-1.42%	-0.49%	3.59%	2.50%

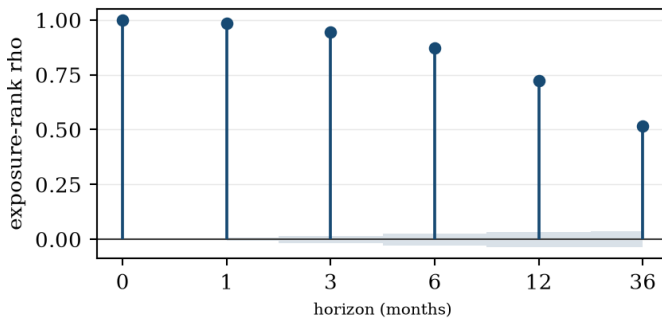


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

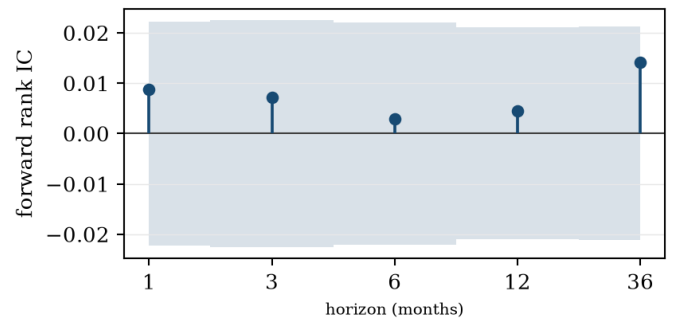


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	beta_shrunk_252v_1260c_VTI(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.40
Bootstrap Sharpe	0.40 [-0.05, 0.81]; Pr(S<=0)=4.5%
Bootstrap tails	skew=1.50 [-0.02, 2.63]; max DD=-11.7% [-21.3%, -6.9%]

52. ebit_growth_yoy_ttm

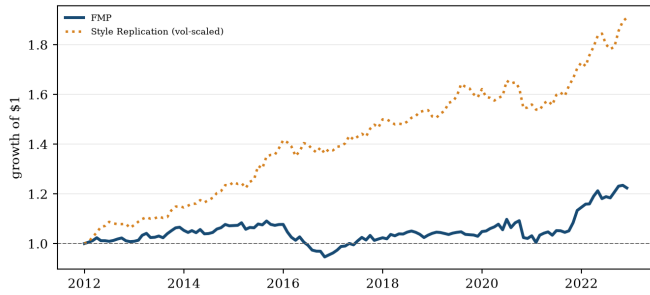


Figure: Common-window cumulative growth of \$1.

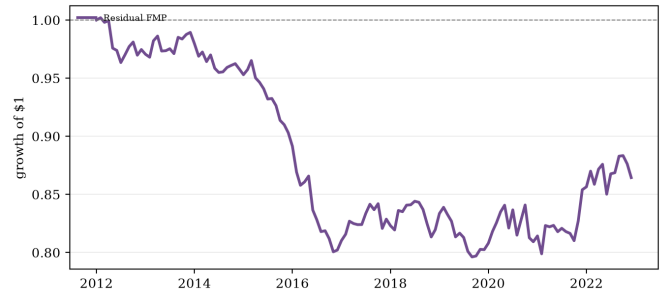


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ebit_growth_yoy_ttm	1.97%	4.91%	0.40	1.33	-13.25%	0.306	-1.38%	4.91%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ebit_growth_yoy_ttm	0.05 (1.58)	-0.11 (-2.40)	0.04 (0.95)	0.14 (3.22)	0.13 (2.52)	0.14 (2.08)	-1.25% (-0.93)	0.307

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ebit_growth_yoy_ttm	0.58%	0.09%	-0.09%	0.70%	1.17%	0.78%	-1.25%	1.97%

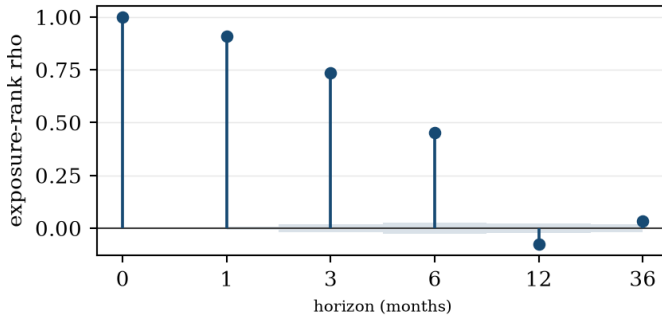


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

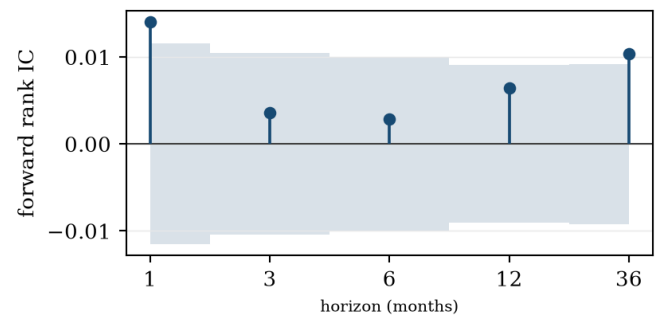


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	ebit_growth_yoy_ttm(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1234 / 1560 / 2083
Available-history Sharpe	0.40
Bootstrap Sharpe	0.40 [-0.21, 1.00]; Pr(S<=0)=11.1%
Bootstrap tails	skew=-0.59 [-1.31, 0.33]; max DD=-13.2% [-23.8%, -3.5%]

53. debt_issuance_yoy

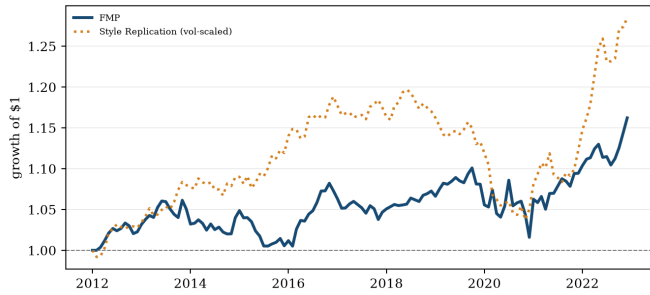


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_issuance_yoy	1.44%	3.61%	0.40	1.52	-7.71%	0.107	0.67%	3.61%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_issuance_yoy	-0.03 (-1.01)	0.07 (1.86)	0.09 (2.83)	0.06 (2.16)	0.04 (0.58)	0.06 (1.06)	1.11% (1.07)	0.122

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_issuance_yoy	-0.40%	-0.06%	-0.20%	0.28%	0.36%	0.35%	1.11%	1.44%

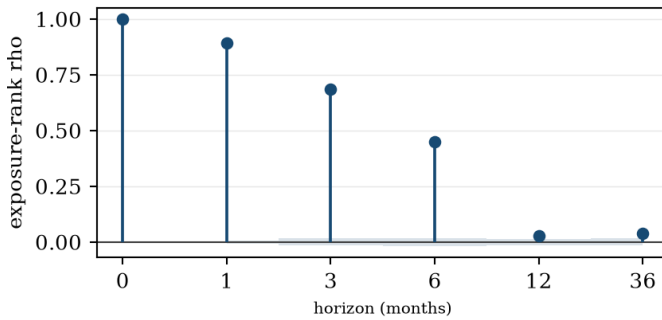


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

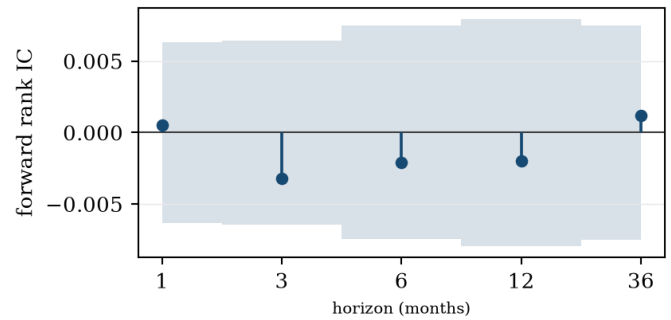


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	debt_issuance_yoy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1029 / 1330 / 2027
Available-history Sharpe	0.40
Bootstrap Sharpe	0.40 [-0.06, 0.98]; Pr(S<=0)=4.6%
Bootstrap tails	skew=0.24 [-0.52, 0.84]; max DD=-7.7% [-13.0%, -4.1%]

54. asset_growth_yoy

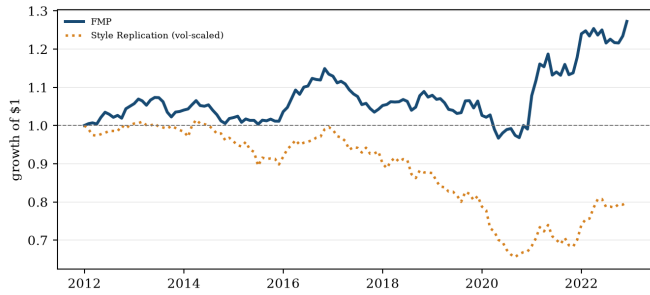


Figure: Common-window cumulative growth of \$1.

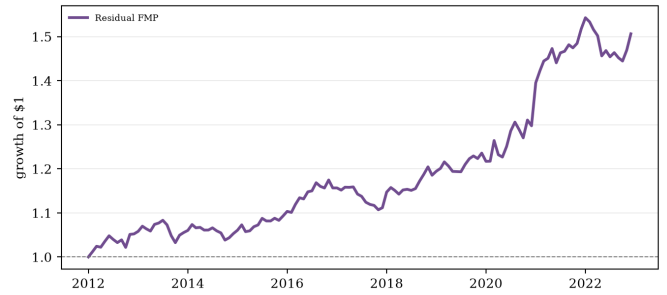


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
asset_growth_yoy	2.39%	6.03%	0.40	1.14	-15.85%	0.372	3.53%	6.03%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
asset_growth_yoy	-0.09 (-2.25)	0.11 (1.60)	0.28 (4.14)	-0.03 (-0.53)	0.05 (0.45)	0.03 (0.38)	3.87% (2.61)	0.376

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
asset_growth_yoy	-1.20%	-0.10%	-0.64%	-0.15%	0.42%	0.18%	3.87%	2.39%

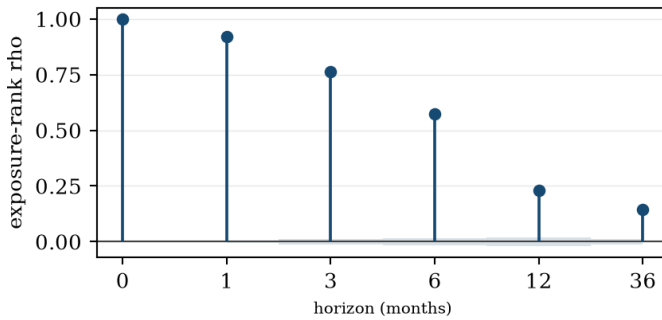


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

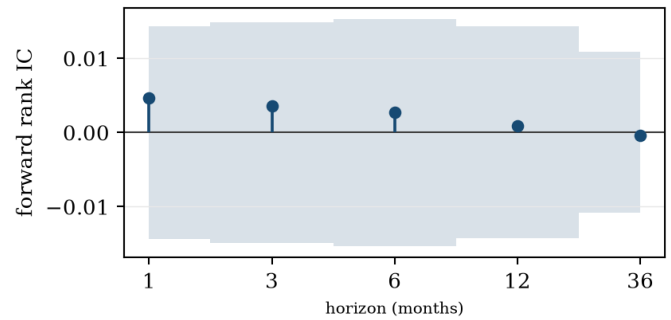


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	asset_growth_yoy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1261 / 1574 / 2090
Available-history Sharpe	0.40
Bootstrap Sharpe	0.40 [-0.23, 0.93]; Pr(S<=0)=9.6%
Bootstrap tails	skew=1.03 [-0.19, 1.65]; max DD=-15.8% [-24.2%, -6.3%]

55. bab_252v_1260c_SPY

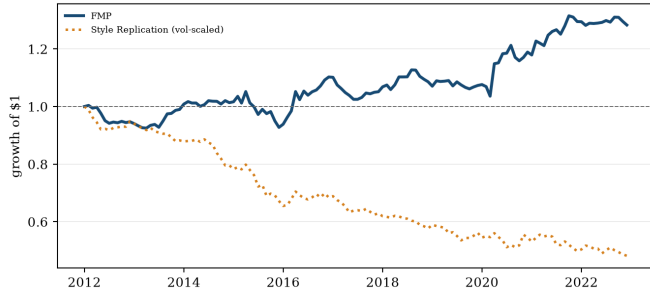


Figure: Common-window cumulative growth of \$1.

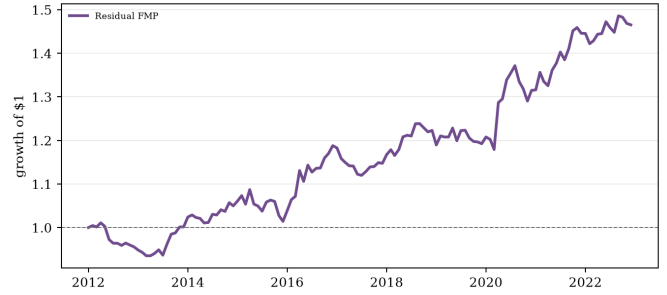


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bab_252v_1260c_SPY	2.49%	6.44%	0.39	1.30	-11.82%	0.136	4.87%	6.44%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bab_252v_1260c_SPY	0.08 (1.18)	0.04 (0.61)	0.07 (1.49)	-0.01 (-0.12)	-0.16 (-2.81)	-0.11 (-1.56)	3.67% (1.81)	0.161

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bab_252v_1260c_SPY	1.06%	-0.03%	-0.16%	-0.04%	-1.38%	-0.63%	3.67%	2.49%

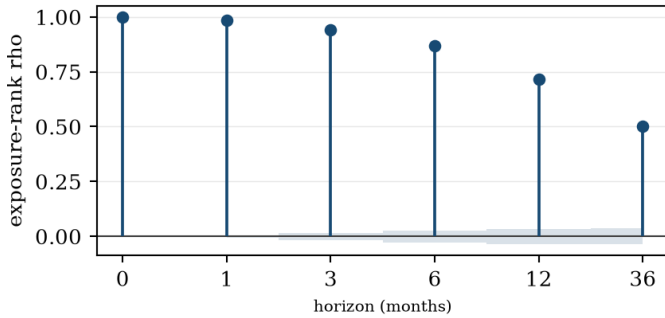


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

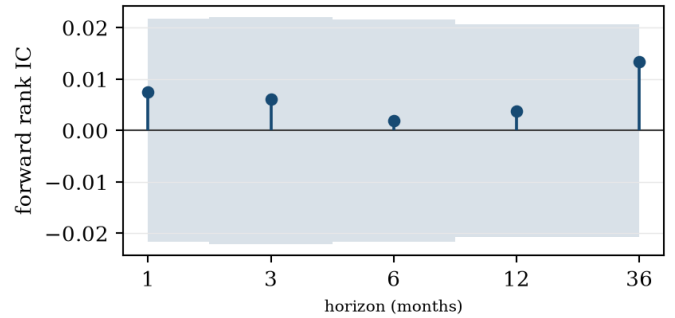


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	bab_252v_1260c_SPY(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.39
Bootstrap Sharpe	0.39 [-0.10, 0.79]; Pr(S<=0)=5.0%
Bootstrap tails	skew=1.71 [0.13, 2.76]; max DD=-11.8% [-22.6%, -7.5%]

56. bab_shrunk_252v_1260c_SPY

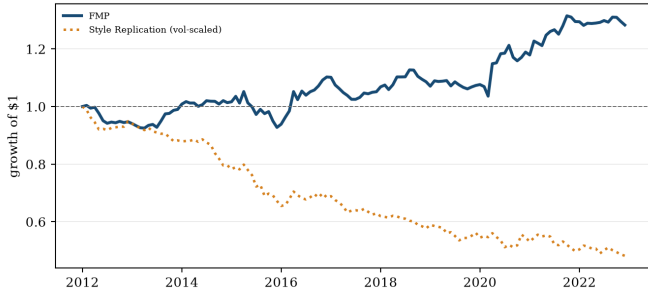


Figure: Common-window cumulative growth of \$1.

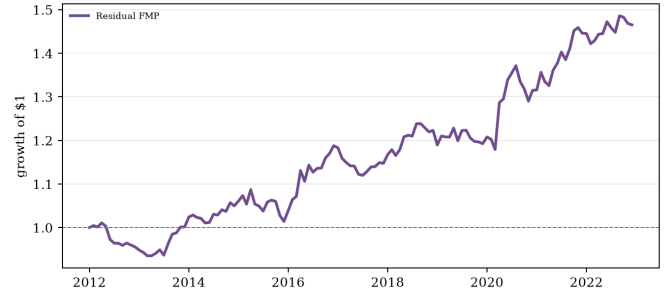


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bab_shrunk_252v_1260c_SPY	2.49%	6.44%	0.39	1.30	-11.82%	0.136	4.87%	6.44%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bab_shrunk_252v_1260c_SPY	0.08	0.04	0.07	-0.01	-0.16	-0.11	3.67%	0.161
	(1.18)	(0.61)	(1.49)	(-0.12)	(-2.81)	(-1.56)	(1.81)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bab_shrunk_252v_1260c_SPY	1.06%	-0.03%	-0.16%	-0.04%	-1.38%	-0.63%	3.67%	2.49%

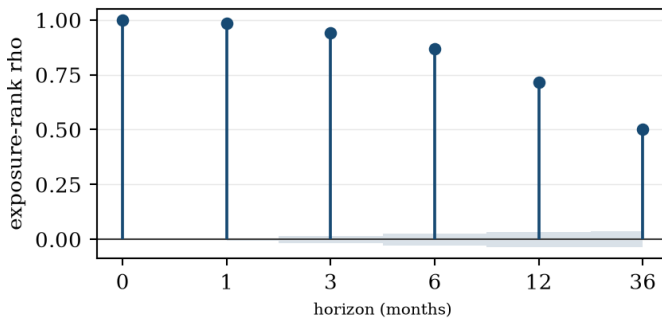


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

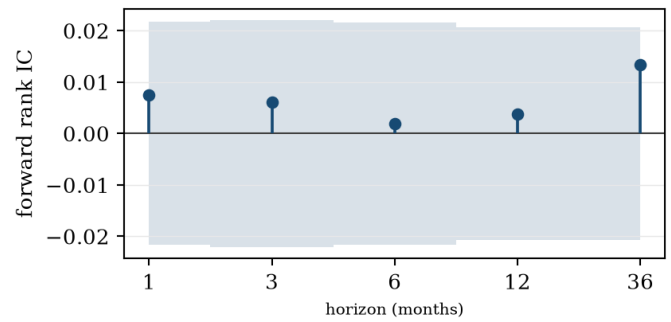


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	bab_shrunk_252v_1260c_SPY(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.39
Bootstrap Sharpe	0.39 [-0.10, 0.79]; Pr(S<=0)=5.0%
Bootstrap tails	skew=1.71 [0.13, 2.76]; max DD=-11.8% [-22.6%, -7.5%]

57. beta_252v_1260c_SPY

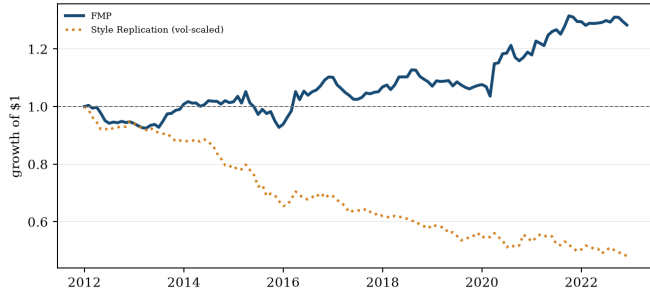


Figure: Common-window cumulative growth of \$1.

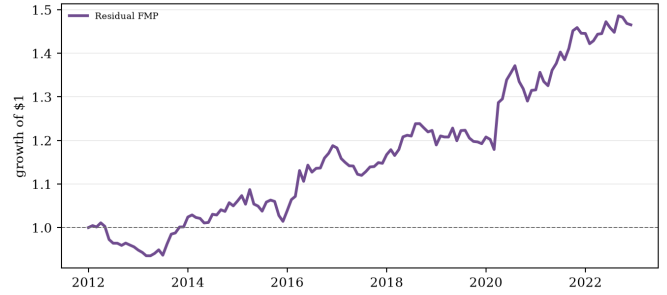


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
beta_252v_1260c_SPY	2.49%	6.44%	0.39	1.30	-11.82%	0.136	4.87%	6.44%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
beta_252v_1260c_SPY	0.08 (1.18)	0.04 (0.61)	0.07 (1.49)	-0.01 (-0.12)	-0.16 (-2.81)	-0.11 (-1.56)	3.67% (1.81)	0.161

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
beta_252v_1260c_SPY	1.06%	-0.03%	-0.16%	-0.04%	-1.38%	-0.63%	3.67%	2.49%

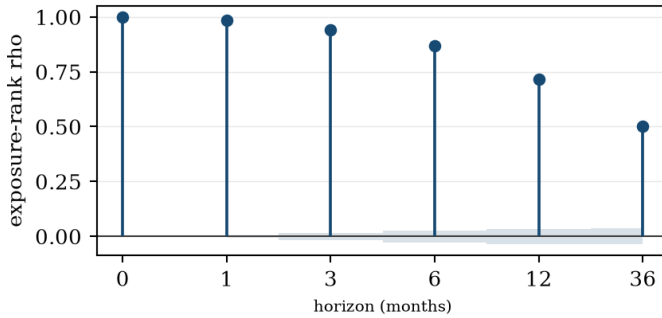


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

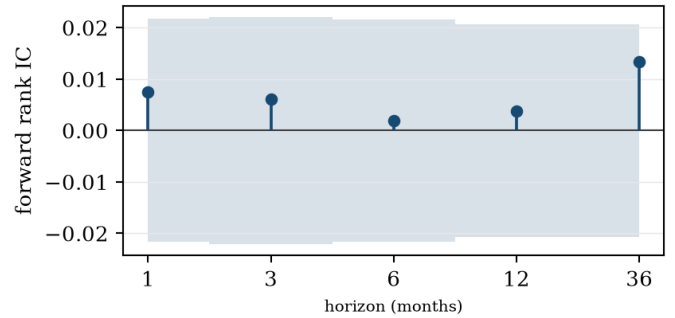


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	beta_252v_1260c_SPY(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.39
Bootstrap Sharpe	0.39 [-0.10, 0.79]; Pr(S<=0)=5.0%
Bootstrap tails	skew=1.71 [0.13, 2.76]; max DD=-11.8% [-22.6%, -7.5%]

58. beta_shrunk_252v_1260c_SPY

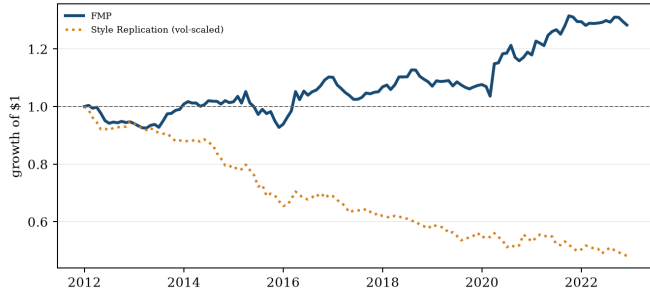


Figure: Common-window cumulative growth of \$1.

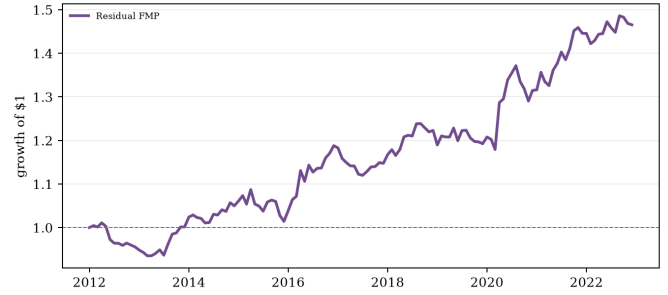


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
beta_shrunk_252v_1260c_SPY	2.49%	6.44%	0.39	1.30	-11.82%	0.136	4.87%	6.44%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
beta_shrunk_252v_1260c_SPY	0.08	0.04	0.07	-0.01	-0.16	-0.11	3.67%	0.161
	(1.18)	(0.61)	(1.49)	(-0.12)	(-2.81)	(-1.56)	(1.81)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
beta_shrunk_252v_1260c_SPY	1.06%	-0.03%	-0.16%	-0.04%	-1.38%	-0.63%	3.67%	2.49%

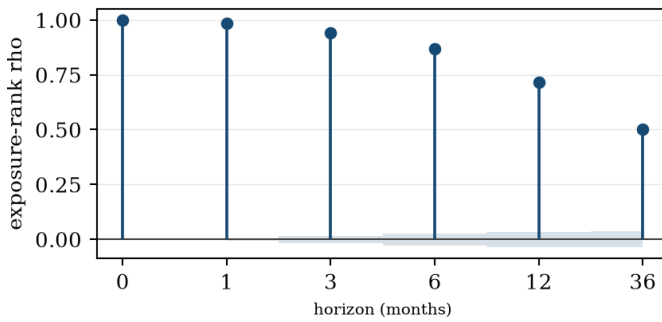


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

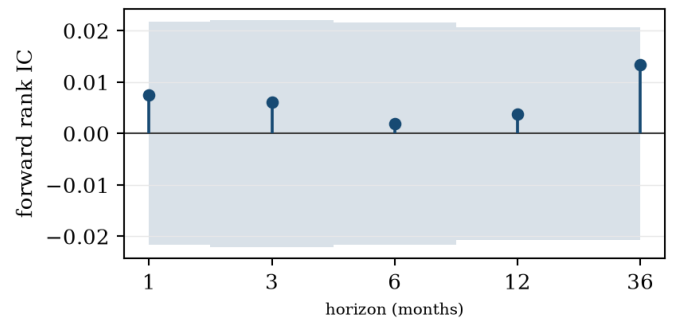


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	beta_shrunk_252v_1260c_SPY(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.39
Bootstrap Sharpe	0.39 [-0.10, 0.79]; Pr(S<=0)=5.0%
Bootstrap tails	skew=1.71 [0.13, 2.76]; max DD=-11.8% [-22.6%, -7.5%]

59. fcf_margin

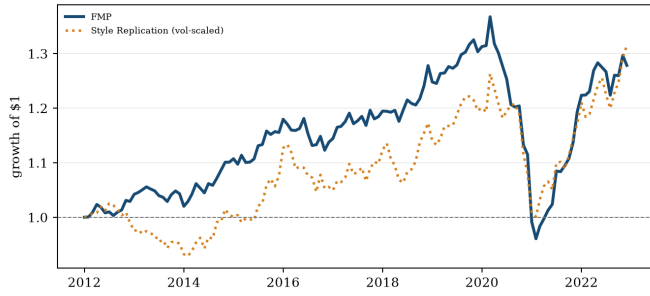


Figure: Common-window cumulative growth of \$1.

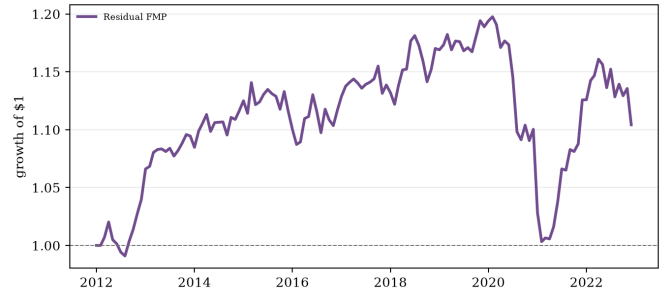


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
fcf_margin	2.48%	6.66%	0.37	0.99	-29.76%	0.525	0.50%	6.66%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
fcf_margin	-0.02 (-0.43)	-0.32 (-4.58)	-0.04 (-0.72)	0.01 (0.20)	-0.07 (-0.77)	0.33 (4.48)	1.01% (0.69)	0.531

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
fcf_margin	-0.21%	0.28%	0.09%	0.06%	-0.65%	1.90%	1.01%	2.48%

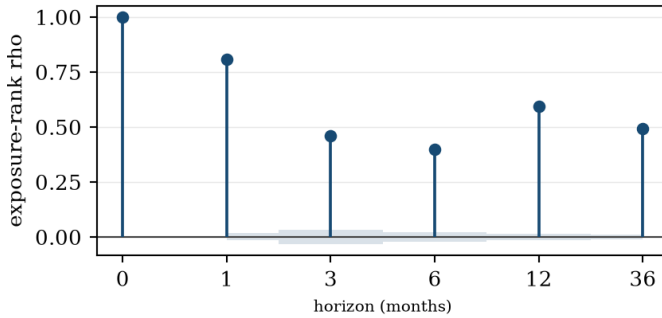


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

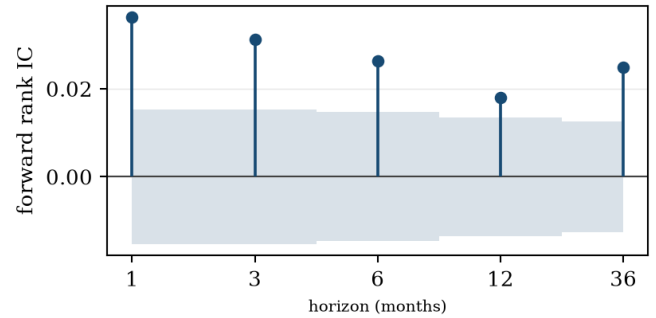


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	fcf_margin(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1243 / 1547 / 2018
Available-history Sharpe	0.37
Bootstrap Sharpe	0.37 [-0.32, 1.22]; Pr(S<=0)=14.4%
Bootstrap tails	skew=-1.60 [-2.50, 0.53]; max DD=-29.8% [-45.6%, -4.6%]

60. cash_ratio

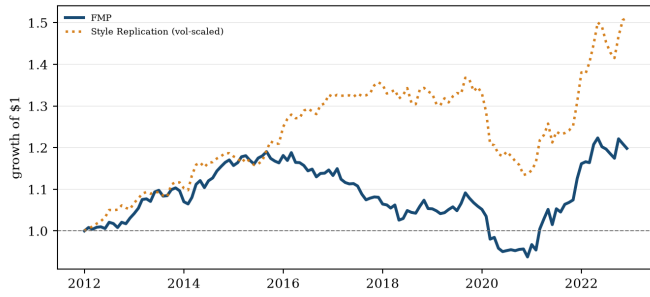


Figure: Common-window cumulative growth of \$1.

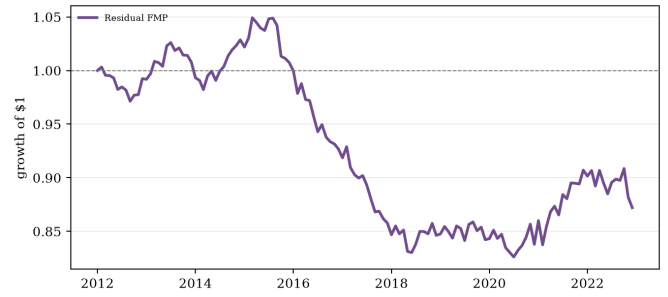


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
cash_ratio	1.80%	5.44%	0.33	0.99	-21.24%	0.516	-1.03%	5.44%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
cash_ratio	0.08 (3.25)	-0.08 (-1.90)	0.27 (9.40)	0.02 (0.56)	0.16 (4.18)	0.16 (3.15)	-1.18% (-1.02)	0.517

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
cash_ratio	1.02%	0.07%	-0.61%	0.11%	1.46%	0.94%	-1.18%	1.80%

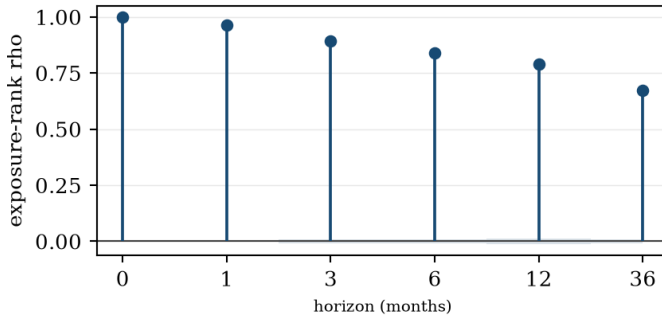


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

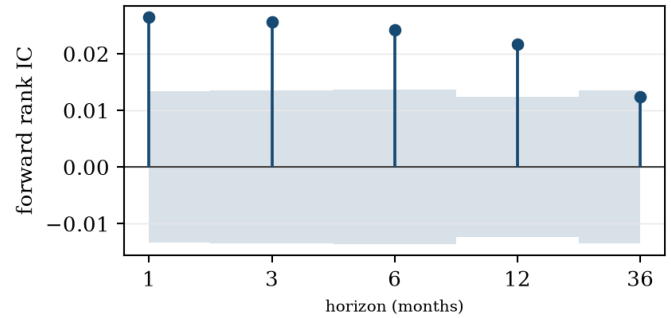


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	cash_ratio(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1262 / 1575 / 2088
Available-history Sharpe	0.33
Bootstrap Sharpe	0.33 [-0.47, 1.04]; Pr(S<=0)=19.9%
Bootstrap tails	skew=0.29 [-0.62, 0.81]; max DD=-21.2% [-31.0%, -5.1%]

61. dollar_volume_21d

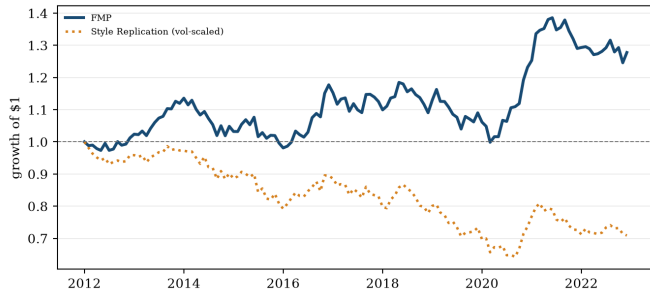


Figure: Common-window cumulative growth of \$1.

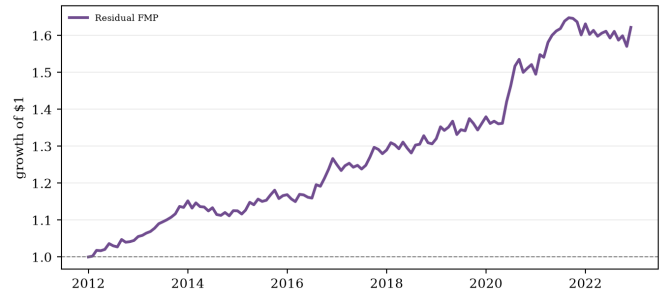


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dollar_volume_21d	2.56%	7.92%	0.32	1.00	-15.71%	0.663	4.86%	7.92%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dollar_volume_21d	0.03 (0.81)	0.67 (11.84)	0.17 (4.73)	-0.08 (-1.75)	-0.10 (-1.75)	-0.00 (-0.03)	4.54% (3.74)	0.665

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dollar_volume_21d	0.34%	-0.59%	-0.39%	-0.41%	-0.92%	-0.01%	4.54%	2.56%

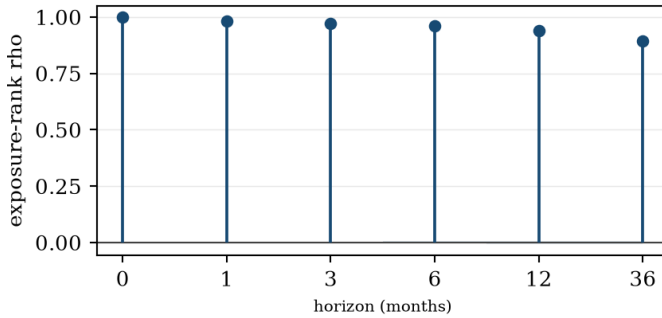


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

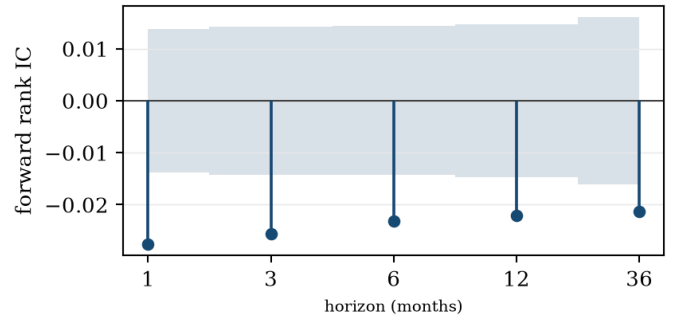


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	dollar_volume_21d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.32
Bootstrap Sharpe	0.32 [-0.23, 0.86]; Pr(S<=0)=10.9%
Bootstrap tails	skew=0.39 [-0.11, 0.67]; max DD=-15.7% [-32.3%, -10.3%]

62. ev_to_assets

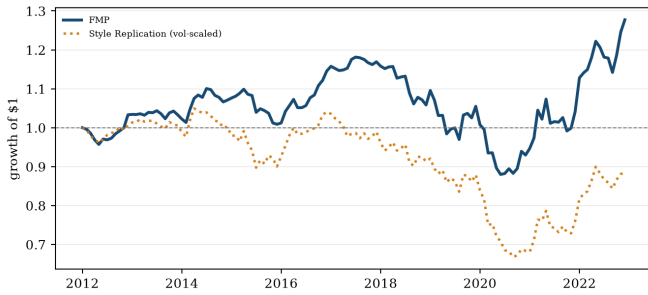


Figure: Common-window cumulative growth of \$1.

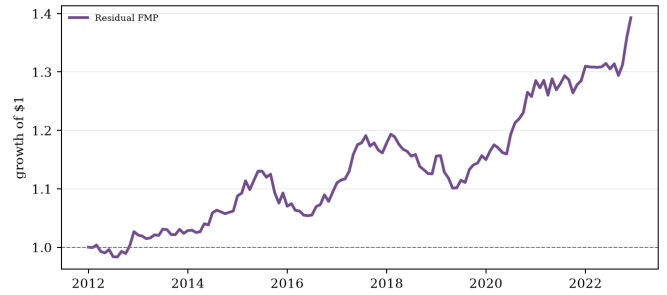


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ev_to_assets	2.56%	7.98%	0.32	0.93	-25.52%	0.731	3.27%	7.98%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ev_to_assets	0.05 (1.64)	0.05 (0.93)	0.50 (13.48)	-0.09 (-2.08)	-0.02 (-0.45)	0.10 (1.69)	3.12% (2.18)	0.731

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ev_to_assets	0.65%	-0.05%	-1.15%	-0.44%	-0.17%	0.59%	3.12%	2.56%

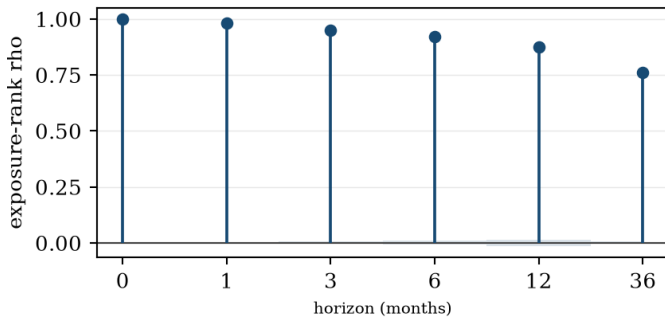


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

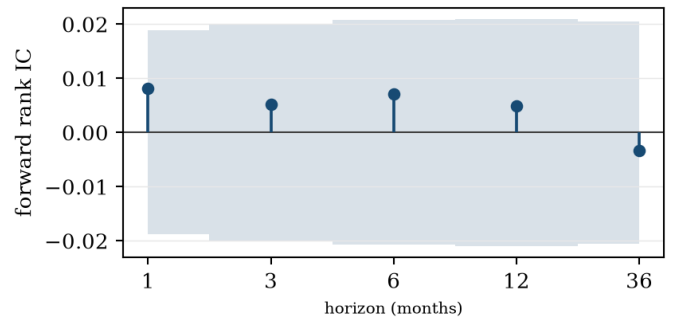


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	ev_to_assets(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1068 / 1376 / 2057
Available-history Sharpe	0.32
Bootstrap Sharpe	0.32 [-0.34, 0.98]; Pr(S<=0)=16.4%
Bootstrap tails	skew=0.39 [-0.29, 0.88]; max DD=-25.5% [-40.5%, -8.2%]

63. gp_ev

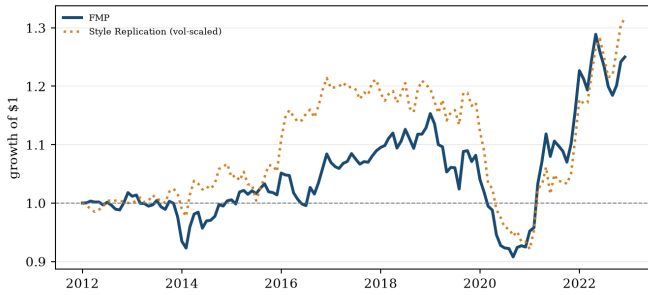


Figure: Common-window cumulative growth of \$1.

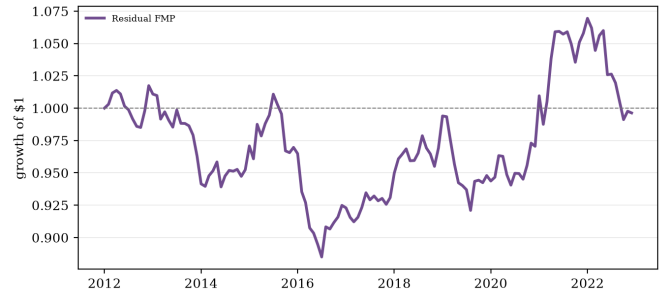


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gp_ev	2.30%	7.20%	0.32	0.95	-21.28%	0.637	0.08%	7.20%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gp_ev	0.05 (1.54)	0.08 (1.43)	0.31 (9.53)	-0.12 (-2.57)	0.06 (0.90)	0.44 (7.46)	0.06% (0.04)	0.637

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gp_ev	0.58%	-0.07%	-0.70%	-0.62%	0.56%	2.50%	0.06%	2.30%

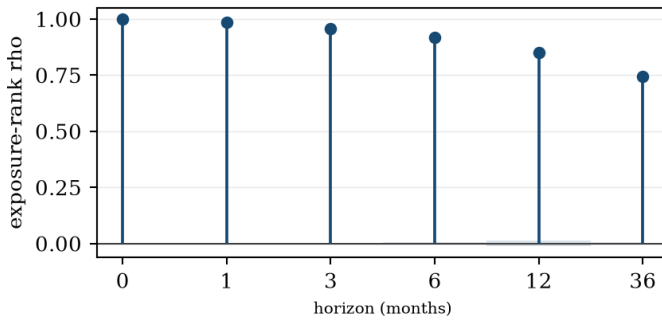


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

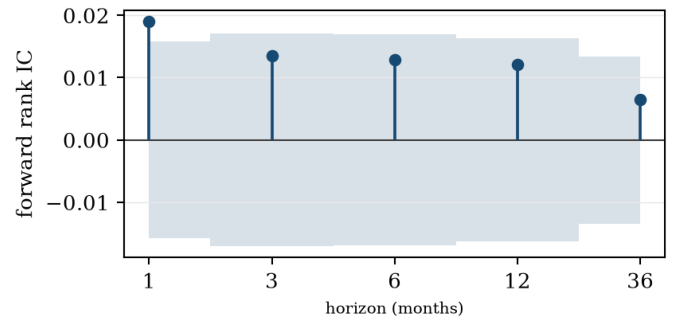


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	gp_ev(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1050 / 1368 / 2023
Available-history Sharpe	0.32
Bootstrap Sharpe	0.32 [-0.31, 0.90]; Pr(S<=0)=15.2%
Bootstrap tails	skew=0.70 [-0.02, 0.99]; max DD=-21.3% [-36.4%, -8.1%]

64. gross_profit_to_ev

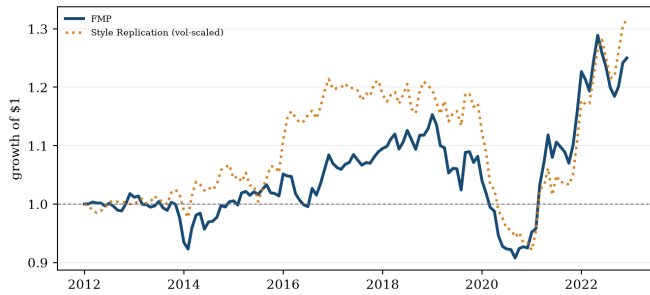


Figure: Common-window cumulative growth of \$1.

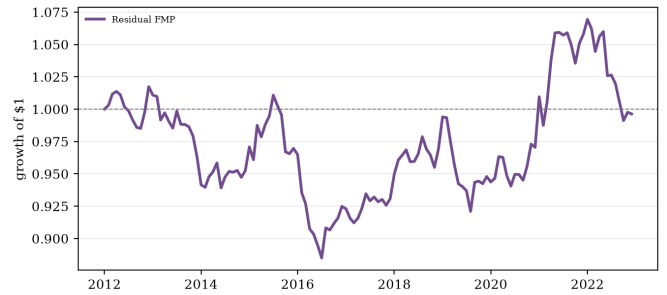


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gross_profit_to_ev	2.30%	7.20%	0.32	0.95	-21.28%	0.637	0.08%	7.20%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gross_profit_to_ev	0.05 (1.54)	0.08 (1.43)	0.31 (9.53)	-0.12 (-2.57)	0.06 (0.90)	0.44 (7.46)	0.06% (0.04)	0.637

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gross_profit_to_ev	0.58%	-0.07%	-0.70%	-0.62%	0.56%	2.50%	0.06%	2.30%

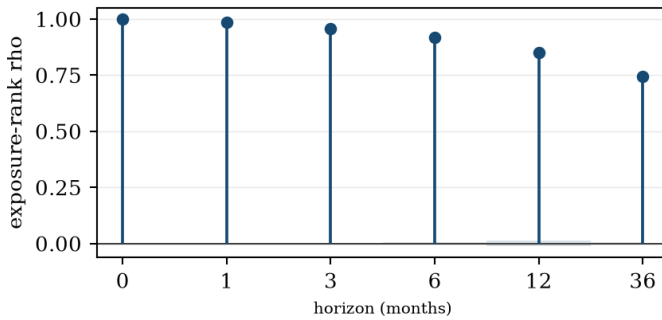


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

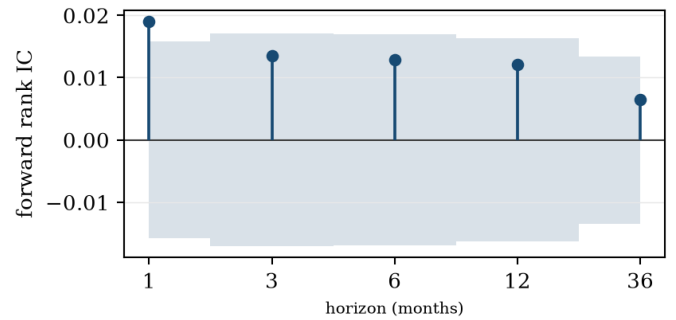


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	gross_profit_to_ev(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1050 / 1368 / 2023
Available-history Sharpe	0.32
Bootstrap Sharpe	0.32 [-0.31, 0.90]; Pr(S<=0)=15.2%
Bootstrap tails	skew=0.70 [-0.02, 0.99]; max DD=-21.3% [-36.4%, -8.1%]

65. value_proxy_1260d

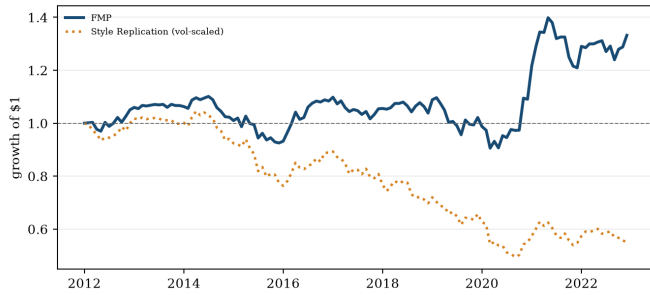


Figure: Common-window cumulative growth of \$1.

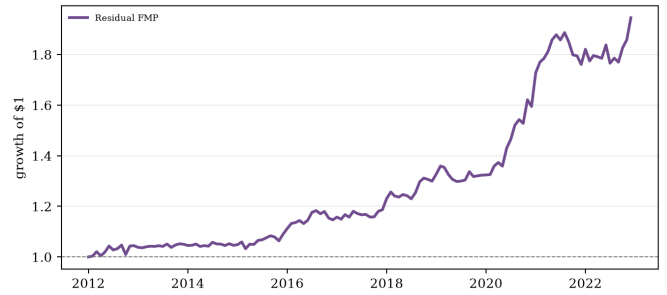


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
value_proxy_1260d	3.08%	9.66%	0.32	0.90	-17.75%	0.572	6.87%	9.66%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
value_proxy_1260d	0.01 (0.26)	0.25 (2.68)	0.42 (6.60)	-0.17 (-2.67)	0.04 (0.34)	-0.30 (-2.51)	6.30% (2.93)	0.576

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
value_proxy_1260d	0.15%	-0.22%	-0.95%	-0.85%	0.33%	-1.69%	6.30%	3.08%

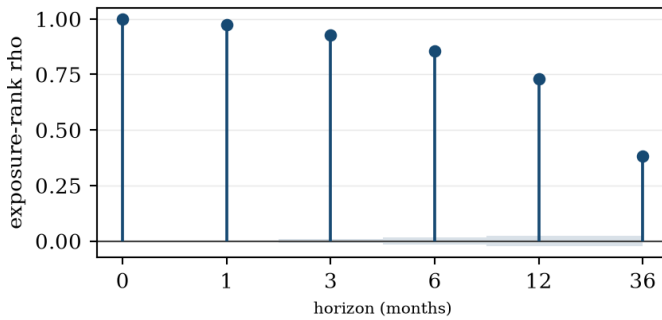


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

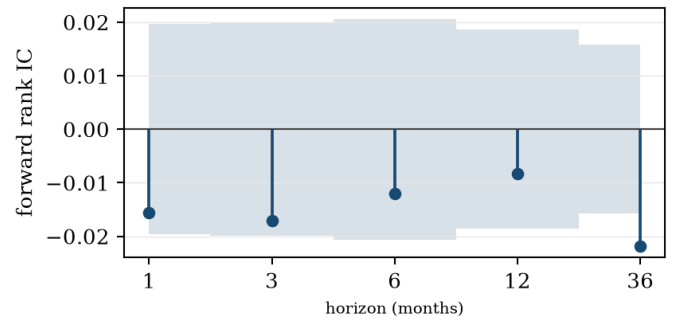


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	value_proxy_1260d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1204 / 1445 / 1886
Available-history Sharpe	0.32
Bootstrap Sharpe	0.32 [-0.31, 0.81]; Pr(S<=0)=14.4%
Bootstrap tails	skew=1.12 [-0.12, 1.67]; max DD=-17.7% [-38.2%, -11.6%]

66. pe_ttm

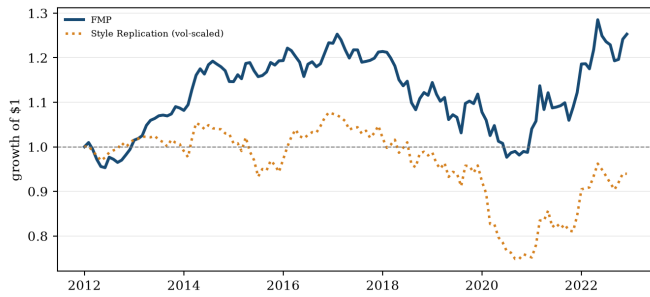


Figure: Common-window cumulative growth of \$1.

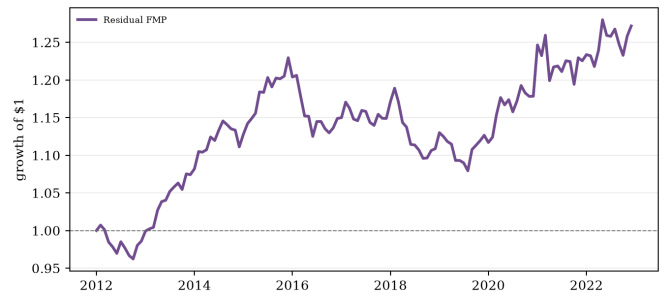


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
pe_ttm	2.33%	7.36%	0.32	1.03	-22.05%	0.555	2.56%	7.36%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
pe_ttm	0.04 (1.11)	-0.03 (-0.47)	0.34 (5.88)	-0.16 (-2.69)	0.06 (0.53)	0.10 (1.56)	2.32% (1.46)	0.556

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
pe_ttm	0.52%	0.03%	-0.78%	-0.83%	0.53%	0.54%	2.32%	2.33%

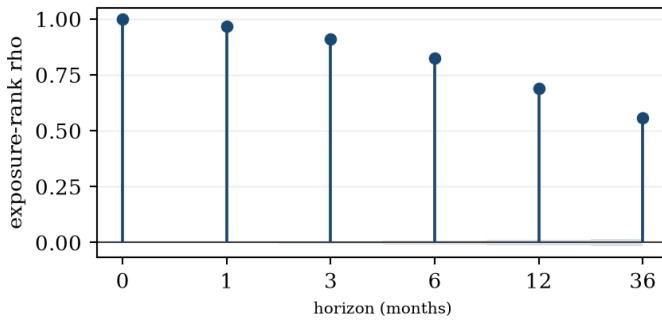


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

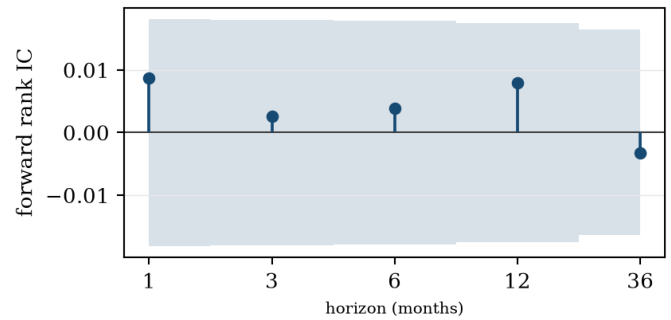


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	pe_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1098 / 1300 / 1593
Available-history Sharpe	0.32
Bootstrap Sharpe	0.32 [-0.28, 0.93]; Pr(S<=0)=14.9%
Bootstrap tails	skew=0.48 [-0.13, 0.73]; max DD=-22.1% [-33.4%, -7.2%]

67. fcf_to_market_cap

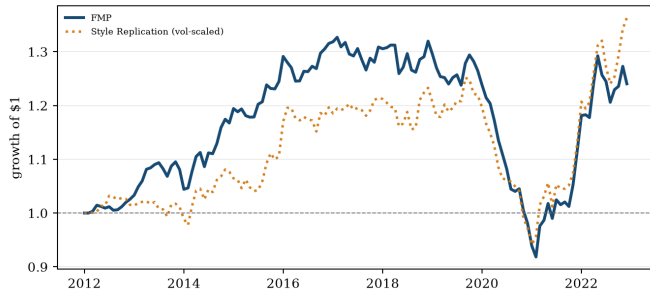


Figure: Common-window cumulative growth of \$1.

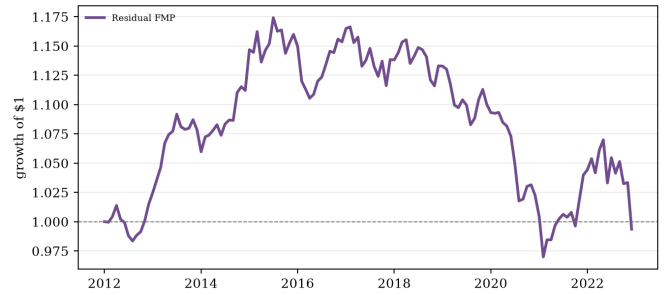


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
fcf_to_market_cap	2.22%	7.01%	0.32	0.87	-30.79%	0.620	-0.21%	7.01%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
fcf_to_market_cap	-0.00 (-0.09)	-0.21 (-3.91)	0.24 (5.78)	-0.03 (-0.61)	0.07 (1.50)	0.38 (6.68)	0.03% (0.02)	0.622

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
fcf_to_market_cap	-0.04%	0.19%	-0.55%	-0.16%	0.60%	2.15%	0.03%	2.22%

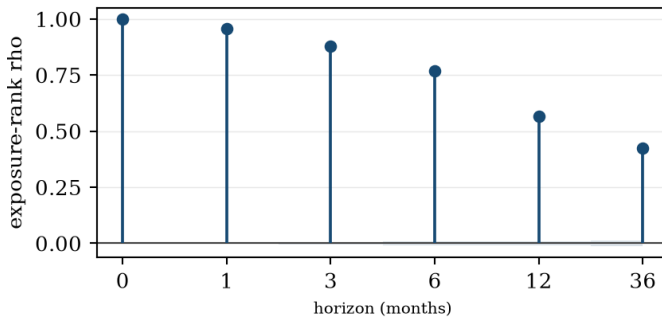


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

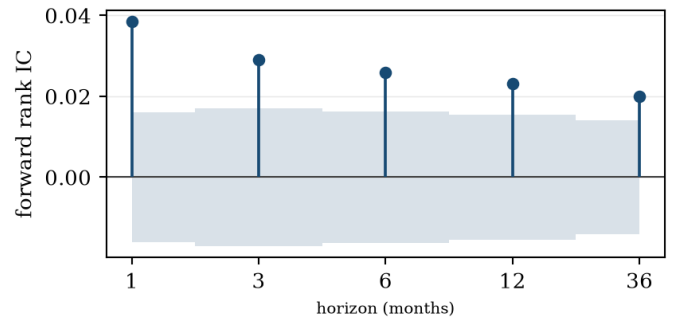


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	fcf_to_market_cap(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1232 / 1556 / 2069
Available-history Sharpe	0.32
Bootstrap Sharpe	0.32 [-0.53, 1.08]; Pr(S<=0)=21.9%
Bootstrap tails	skew=0.46 [-0.30, 0.76]; max DD=-30.8% [-47.2%, -6.4%]

68. net_operating_assets_proxy

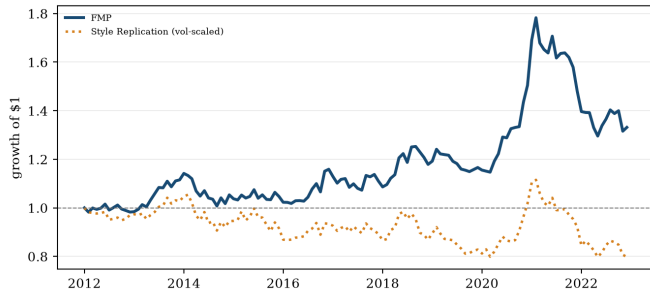


Figure: Common-window cumulative growth of \$1.

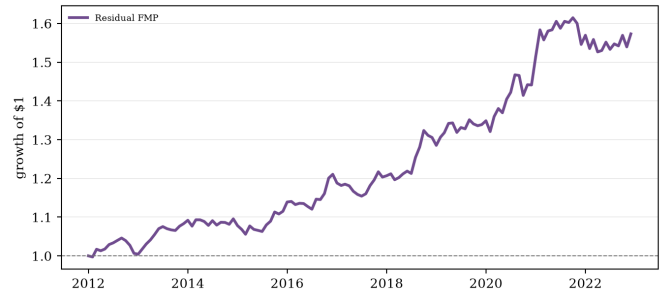


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_operating_assets_proxy	3.10%	9.86%	0.31	0.91	-27.32%	0.738	4.46%	9.86%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_operating_assets_proxy	-0.02	0.91	-0.12	0.02	0.02	-0.13	4.28%	0.739
	(-0.49)	(12.16)	(-2.72)	(0.38)	(0.26)	(-1.61)	(2.80)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_operating_assets_proxy	-0.21%	-0.80%	0.27%	0.09%	0.18%	-0.72%	4.28%	3.10%

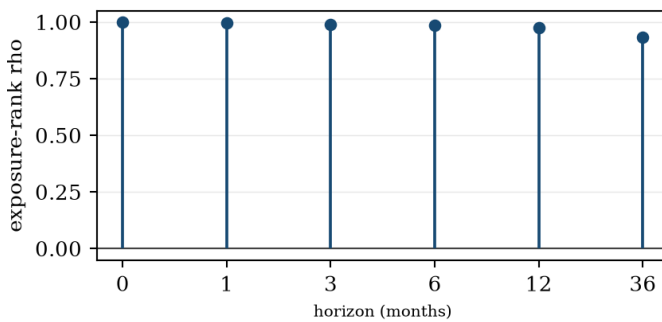


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

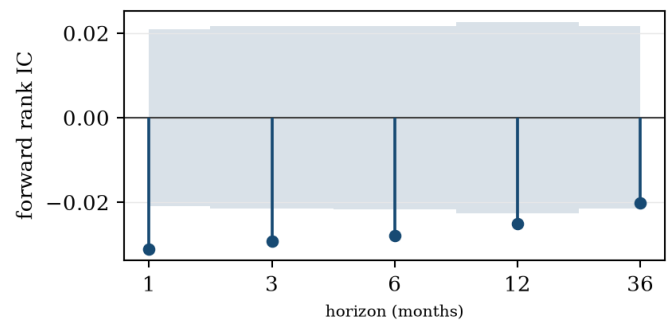


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	net_operating_assets_proxy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1069 / 1380 / 2059
Available-history Sharpe	0.31
Bootstrap Sharpe	0.31 [-0.28, 0.89]; Pr(S<=0)=15.7%
Bootstrap tails	skew=0.72 [-0.19, 1.22]; max DD=-27.3% [-44.4%, -10.0%]

69. current_ratio

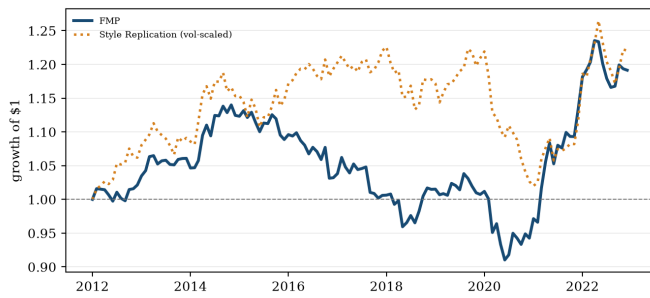


Figure: Common-window cumulative growth of \$1.

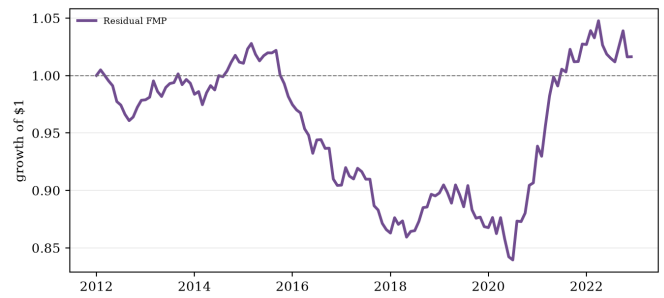


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
current_ratio	1.77%	5.72%	0.31	0.93	-20.15%	0.478	0.39%	5.72%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
current_ratio	0.05 (1.73)	-0.34 (-5.61)	0.21 (4.85)	0.00 (0.07)	0.15 (2.19)	-0.06 (-1.01)	0.23% (0.15)	0.479

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
current_ratio	0.70%	0.30%	-0.49%	0.02%	1.35%	-0.33%	0.23%	1.77%

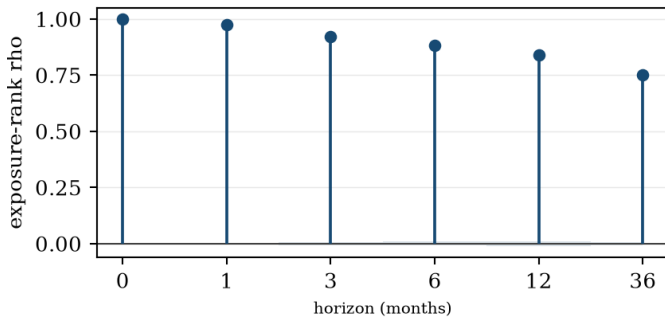


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

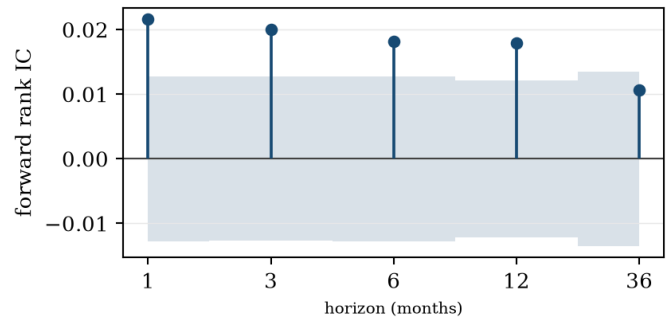


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	current_ratio(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1261 / 1580 / 2095
Available-history Sharpe	0.31
Bootstrap Sharpe	0.31 [-0.40, 0.97]; Pr(S<=0)=19.8%
Bootstrap tails	skew=0.07 [-0.66, 0.58]; max DD=-20.1% [-29.8%, -6.4%]

70. amihud_illiq_21d

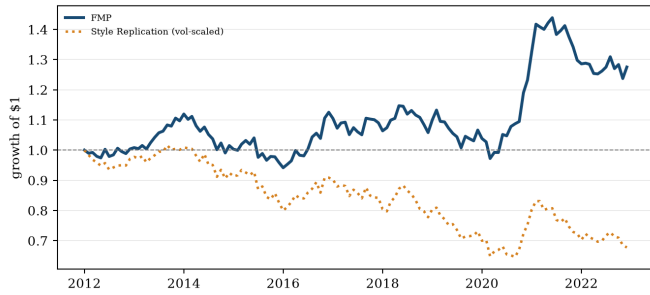


Figure: Common-window cumulative growth of \$1.

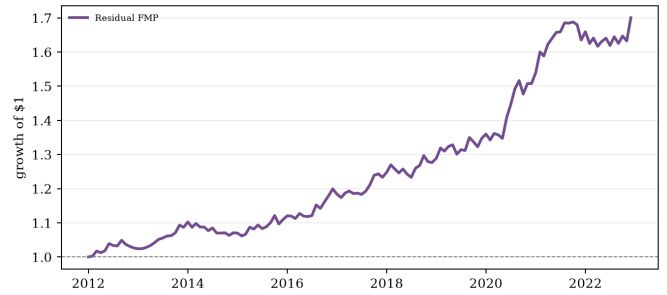


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
amihud_illiq_21d	2.60%	8.72%	0.30	0.87	-15.89%	0.710	5.28%	8.72%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
amihud_illiq_21d	0.00 (0.00)	0.73 (13.75)	0.16 (3.85)	-0.10 (-1.93)	-0.01 (-0.26)	-0.13 (-2.48)	4.98% (3.77)	0.711

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
amihud_illiq_21d	0.00%	-0.65%	-0.37%	-0.49%	-0.10%	-0.77%	4.98%	2.60%

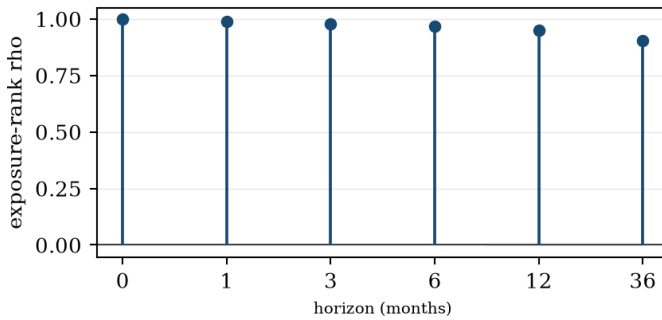


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

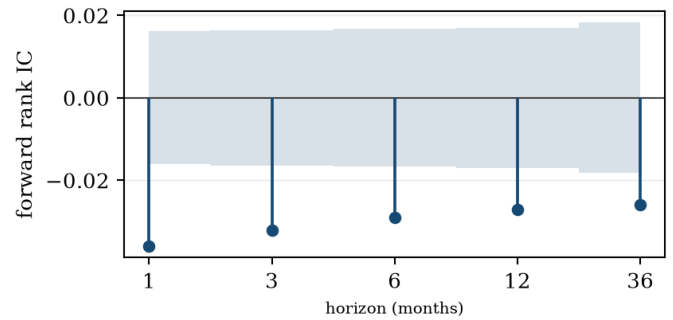


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	amihud_illiq_21d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1254 / 1561 / 2094
Available-history Sharpe	0.30
Bootstrap Sharpe	0.30 [-0.29, 0.84]; Pr(S<=0)=15.4%
Bootstrap tails	skew=0.57 [-0.15, 0.89]; max DD=-15.9% [-36.3%, -11.5%]

71. pb

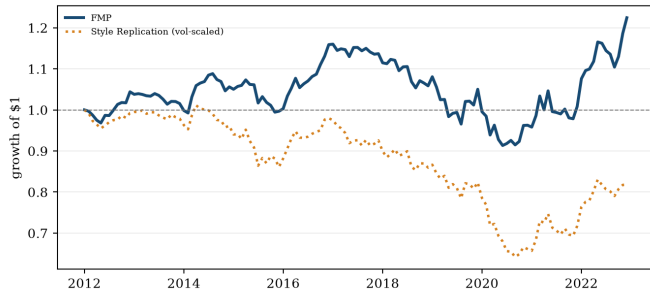


Figure: Common-window cumulative growth of \$1.

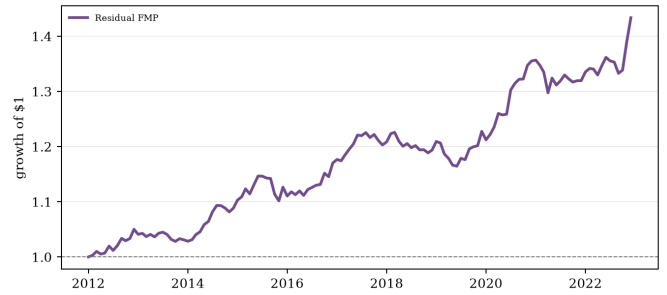


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
pb	2.11%	7.23%	0.29	0.91	-21.26%	0.747	3.47%	7.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
pb	0.03 (1.21)	0.10 (2.25)	0.45 (17.95)	-0.08 (-2.44)	-0.08 (-2.05)	0.10 (2.00)	3.37% (3.09)	0.747

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
pb	0.42%	-0.09%	-1.04%	-0.40%	-0.70%	0.56%	3.37%	2.11%

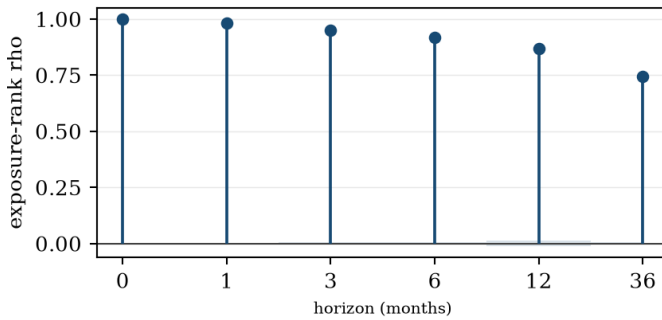


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

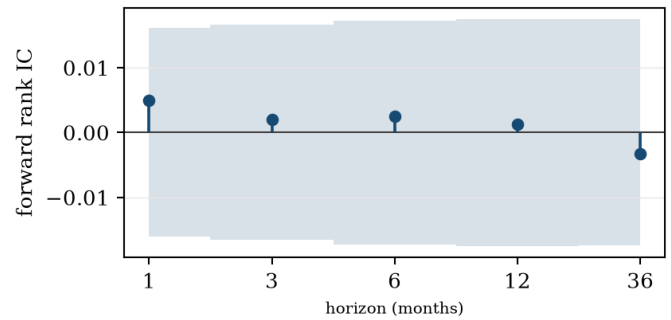


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	pb(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1229 / 1515 / 1982
Available-history Sharpe	0.29
Bootstrap Sharpe	0.29 [-0.32, 0.92]; Pr(S<=0)=16.4%
Bootstrap tails	skew=0.26 [-0.17, 0.72]; max DD=-21.3% [-34.9%, -7.8%]

72. roe_volatility

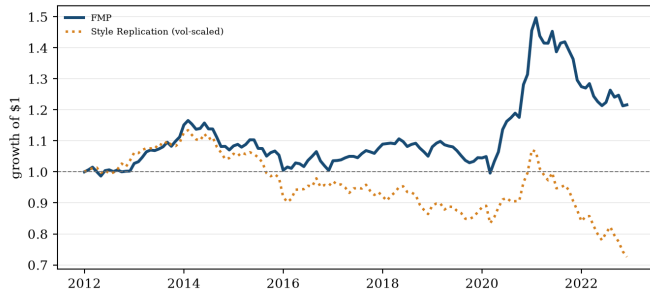


Figure: Common-window cumulative growth of \$1.

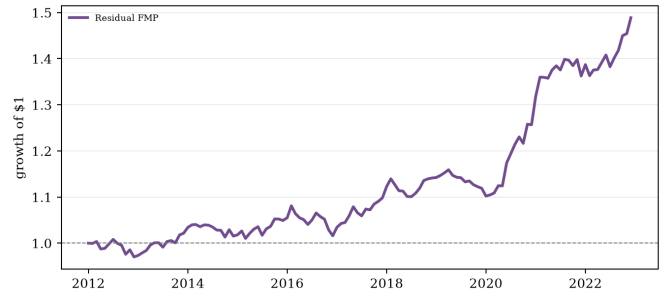


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
roe_volatility	2.07%	7.51%	0.28	0.77	-18.98%	0.662	4.23%	7.51%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
roe_volatility	0.02 (0.78)	0.20 (2.67)	-0.03 (-0.56)	-0.01 (-0.10)	0.17 (2.52)	-0.58 (-7.29)	3.74% (2.37)	0.665

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
roe_volatility	0.29%	-0.17%	0.06%	-0.03%	1.52%	-3.34%	3.74%	2.07%

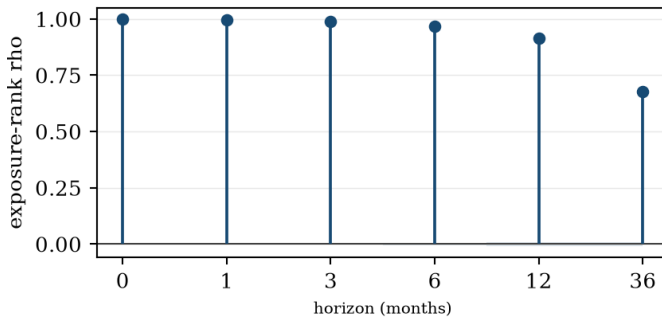


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

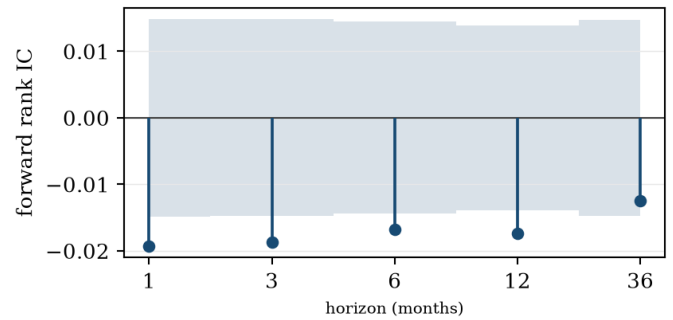


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	roe_volatility(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1198 / 1486 / 1949
Available-history Sharpe	0.28
Bootstrap Sharpe	0.28 [-0.44, 0.87]; Pr(S<=0)=21.3%
Bootstrap tails	skew=1.25 [-0.64, 1.85]; max DD=-19.0% [-36.8%, -9.3%]

73. dio

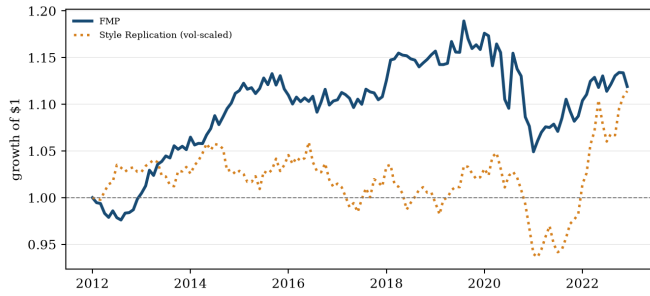


Figure: Common-window cumulative growth of \$1.

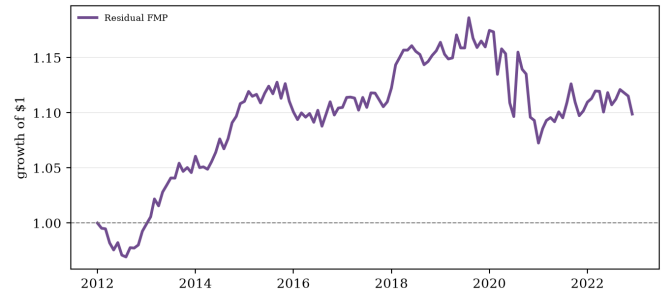


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dio	1.11%	4.06%	0.27	1.08	-11.78%	0.056	0.86%	4.06%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dio	0.02 (0.41)	-0.07 (-1.79)	0.06 (1.28)	0.08 (2.02)	-0.04 (-0.49)	-0.00 (-0.07)	0.94% (0.68)	0.057

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dio	0.21%	0.07%	-0.13%	0.41%	-0.37%	-0.03%	0.94%	1.11%

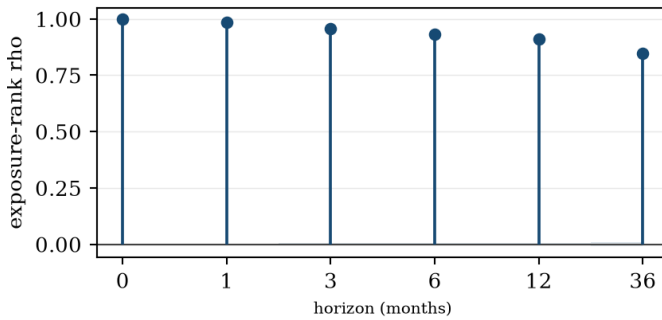


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

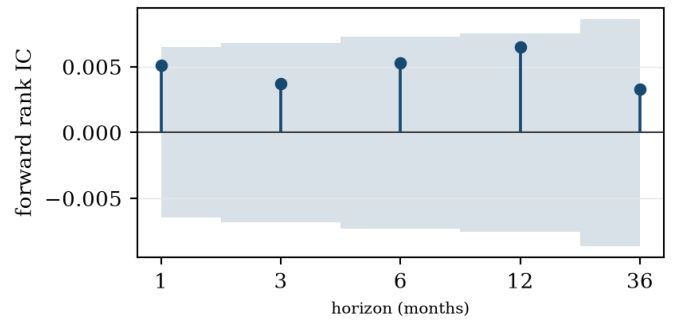


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	dio(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1176 / 1505 / 1798
Available-history Sharpe	0.27
Bootstrap Sharpe	0.27 [-0.19, 0.98]; Pr(S<=0)=14.3%
Bootstrap tails	skew=-0.02 [-0.93, 0.59]; max DD=-11.8% [-20.4%, -3.6%]

74. gross_profit_margin_ttm

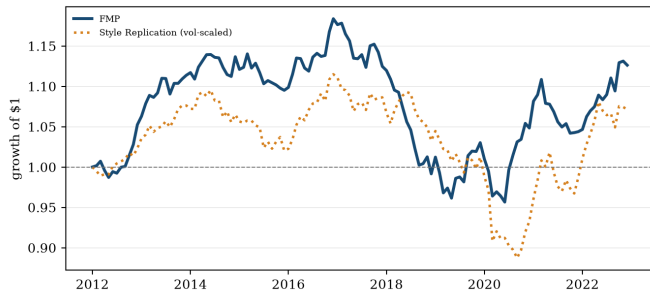


Figure: Common-window cumulative growth of \$1.

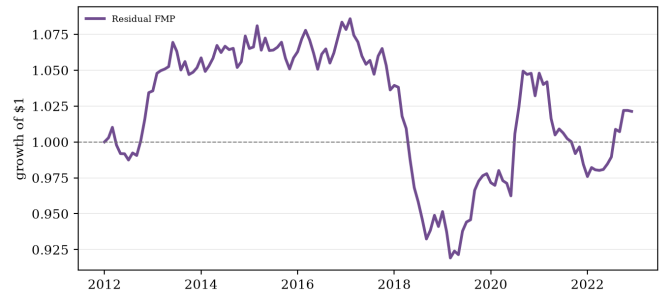


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gross_profit_margin_ttm	1.19%	4.58%	0.26	0.71	-19.19%	0.360	0.75%	4.58%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gross_profit_margin_ttm	0.08	0.16	0.21	0.04	0.03	0.00	0.26%	0.375
	(2.98)	(2.98)	(7.35)	(1.03)	(0.87)	(0.09)	(0.19)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gross_profit_margin_ttm	1.01%	-0.14%	-0.48%	0.23%	0.29%	0.02%	0.26%	1.19%

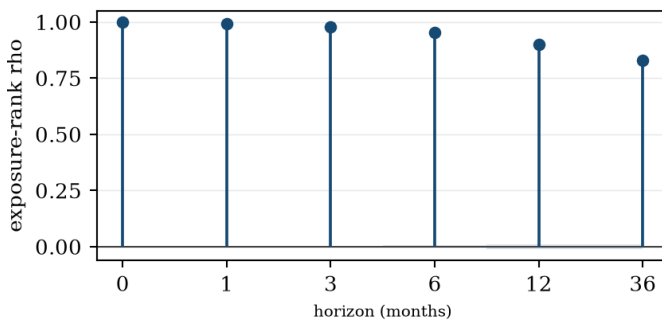


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

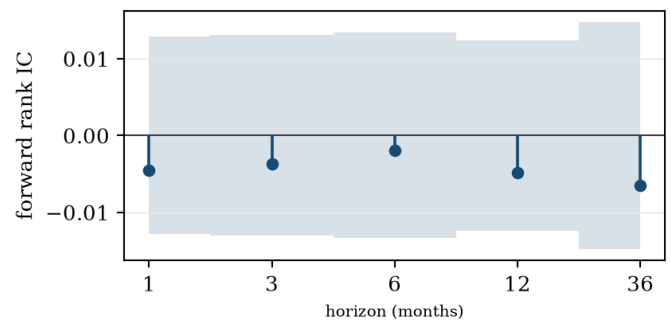


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	gross_profit_margin_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1240 / 1560 / 2041
Available-history Sharpe	0.26
Bootstrap Sharpe	0.26 [-0.47, 1.10]; Pr(S<=0)=22.9%
Bootstrap tails	skew=0.32 [-0.00, 0.63]; max DD=-19.2% [-31.4%, -5.5%]

75. net_income_margin_ttm

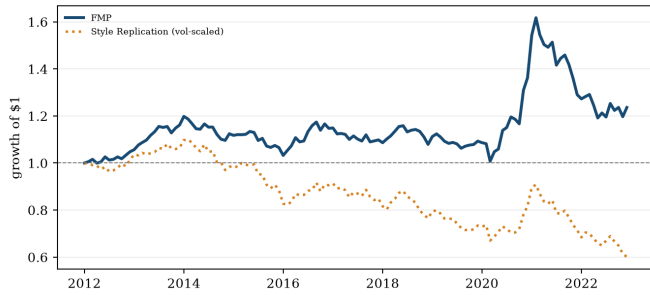


Figure: Common-window cumulative growth of \$1.

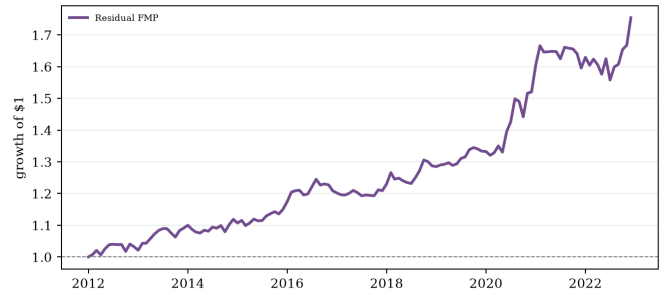


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_income_margin_ttm	2.37%	9.44%	0.25	0.72	-26.34%	0.641	5.75%	9.44%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_income_margin_ttm	-0.03	0.51	0.07	-0.05	0.16	-0.54	5.32%	0.643
	(-0.69)	(6.88)	(1.28)	(-0.84)	(1.93)	(-5.21)	(3.11)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_income_margin_ttm	-0.41%	-0.44%	-0.16%	-0.27%	1.42%	-3.08%	5.32%	2.37%

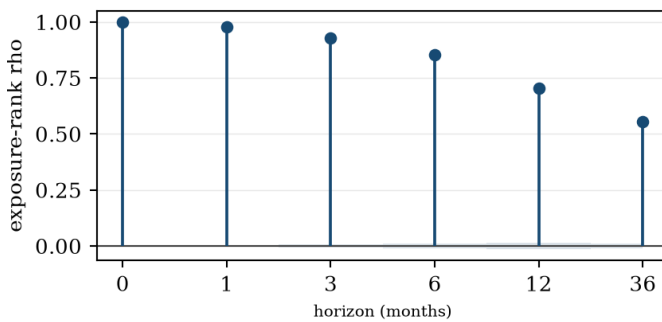


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

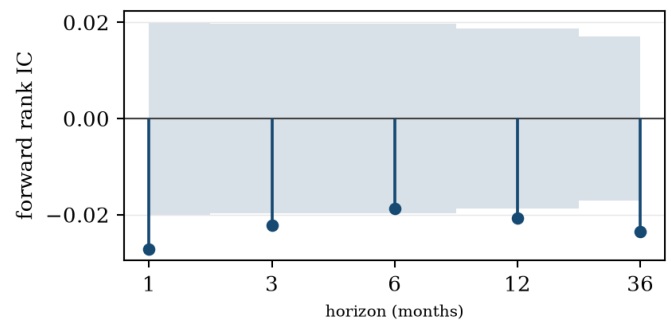


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	net_income_margin_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1240 / 1560 / 2042
Available-history Sharpe	0.25
Bootstrap Sharpe	0.25 [-0.37, 0.79]; Pr(S<=0)=19.3%
Bootstrap tails	skew=1.31 [-0.57, 1.91]; max DD=-26.3% [-43.2%, -11.6%]

76. roic_nopat

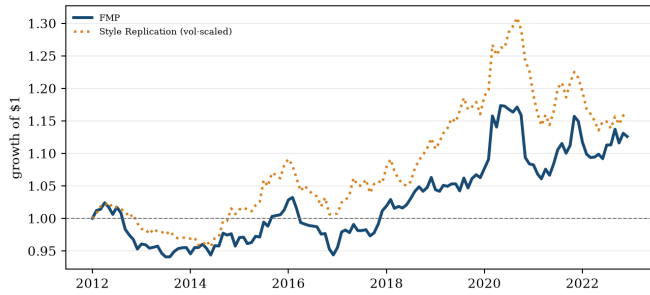


Figure: Common-window cumulative growth of \$1.

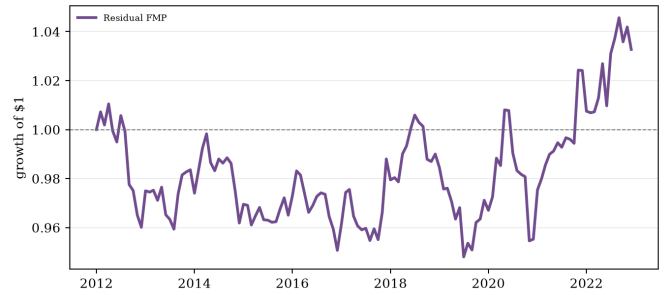


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
roic_nopat	1.20%	4.77%	0.25	0.78	-9.59%	0.556	0.08%	4.77%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
roic_nopat	-0.03 (-1.06)	-0.23 (-6.27)	-0.24 (-6.92)	-0.03 (-0.86)	0.00 (0.04)	0.10 (2.15)	0.34% (0.30)	0.559

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
roic_nopat	-0.34%	0.20%	0.56%	-0.15%	0.02%	0.57%	0.34%	1.20%

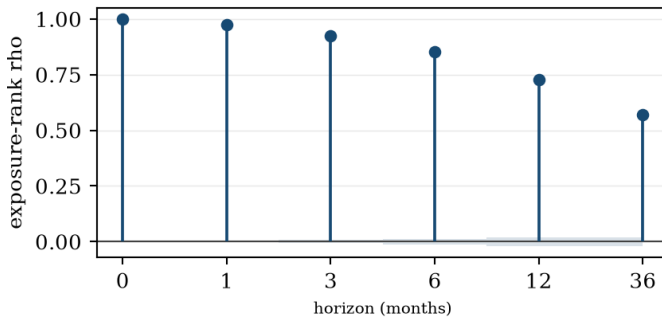


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

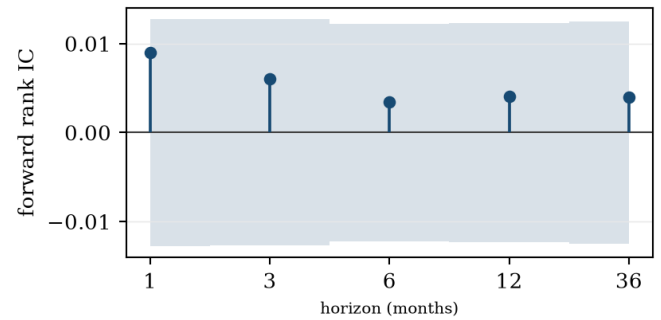


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	roic_nopat(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	814 / 955 / 1331
Available-history Sharpe	0.25
Bootstrap Sharpe	0.25 [-0.26, 0.73]; Pr(S<=0)=16.8%
Bootstrap tails	skew=0.23 [-0.80, 1.20]; max DD=-9.6% [-22.3%, -6.7%]

77. growth_gmar_5y

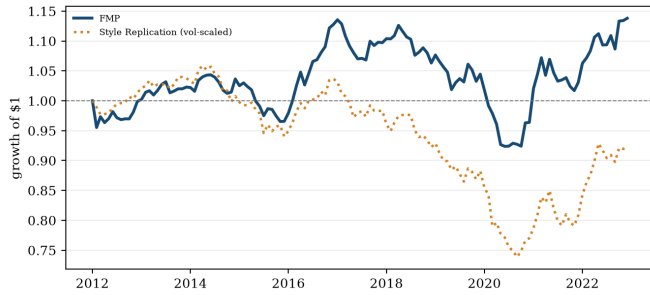


Figure: Common-window cumulative growth of \$1.

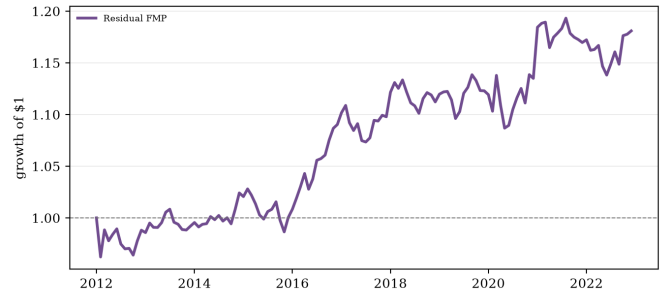


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
growth_gmar_5y	1.33%	5.48%	0.24	0.66	-18.65%	0.490	1.75%	5.48%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
growth_gmar_5y	0.03 (0.79)	0.18 (3.25)	0.34 (8.96)	0.08 (1.91)	-0.02 (-0.22)	0.01 (0.13)	1.60% (1.39)	0.491

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
growth_gmar_5y	0.36%	-0.16%	-0.78%	0.40%	-0.14%	0.05%	1.60%	1.33%

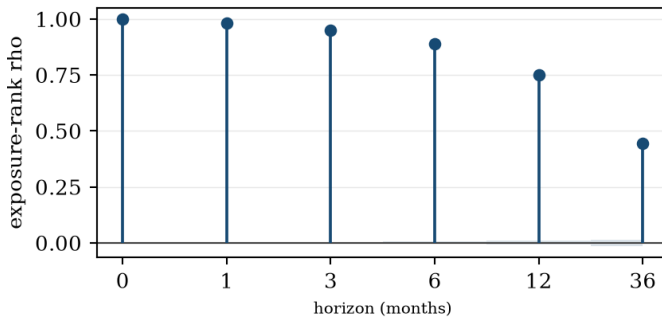


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

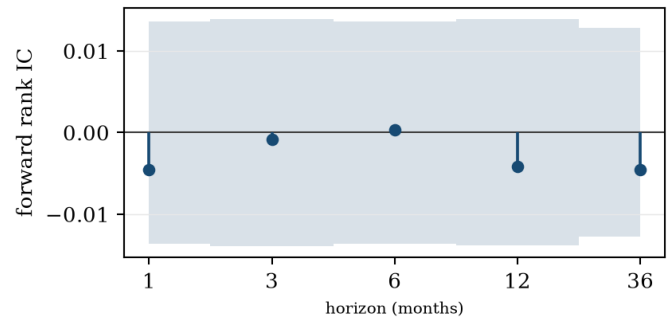


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	growth_gmar_5y(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	24 / 1418 / 1890
Available-history Sharpe	0.24
Bootstrap Sharpe	0.24 [-0.40, 0.83]; Pr(S<=0)=19.6%
Bootstrap tails	skew=0.39 [-0.26, 0.84]; max DD=-18.7% [-31.1%, -6.9%]

78. capital_intensity

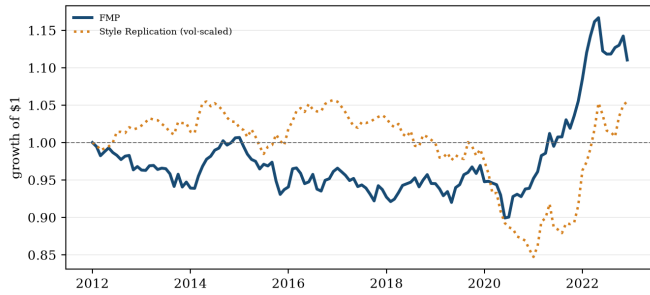


Figure: Common-window cumulative growth of \$1.

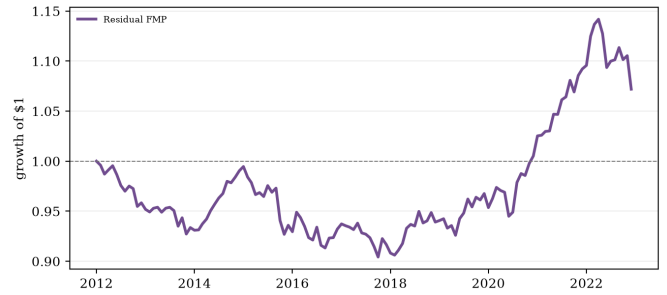


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
capital_intensity	1.05%	4.34%	0.24	0.71	-10.68%	0.243	0.76%	4.34%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
capital_intensity	0.03 (1.12)	-0.05 (-1.09)	0.19 (5.49)	0.06 (1.43)	-0.03 (-0.43)	0.04 (0.86)	0.71% (0.54)	0.243

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
capital_intensity	0.40%	0.05%	-0.43%	0.31%	-0.24%	0.25%	0.71%	1.05%

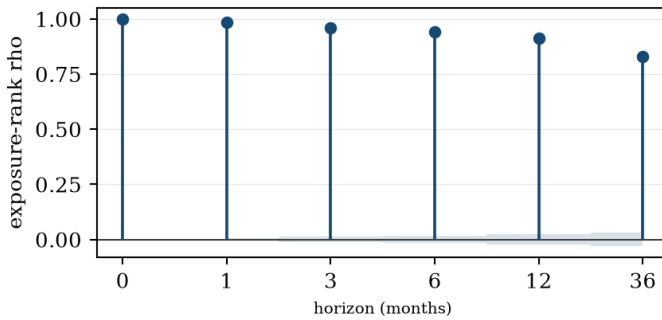


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

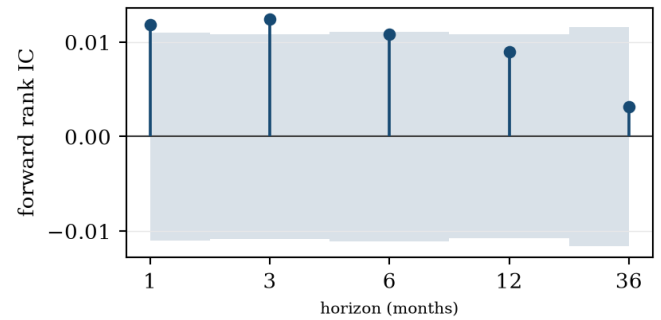


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	capital_intensity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1145 / 1439 / 2013
Available-history Sharpe	0.24
Bootstrap Sharpe	0.24 [-0.41, 0.80]; Pr(S<=0)=24.6%
Bootstrap tails	skew=-0.15 [-0.52, 0.22]; max DD=-10.7% [-22.6%, -7.2%]

79. is_positive_revenue

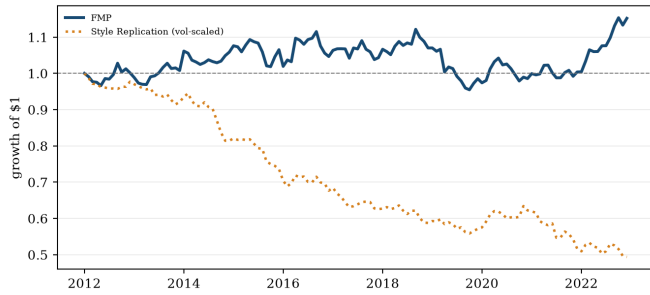


Figure: Common-window cumulative growth of \$1.

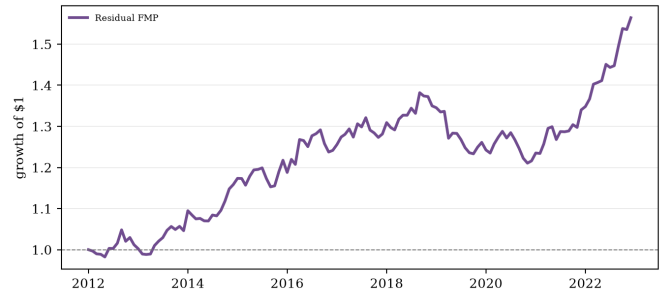


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
is_positive_revenue	1.48%	6.16%	0.24	0.87	-14.86%	0.188	4.19%	6.16%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
is_positive_revenue	-0.03 (-0.62)	0.01 (0.10)	0.03 (0.68)	0.13 (2.12)	-0.18 (-3.12)	-0.25 (-2.64)	4.26% (2.10)	0.188

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
is_positive_revenue	-0.34%	-0.01%	-0.08%	0.66%	-1.57%	-1.44%	4.26%	1.48%

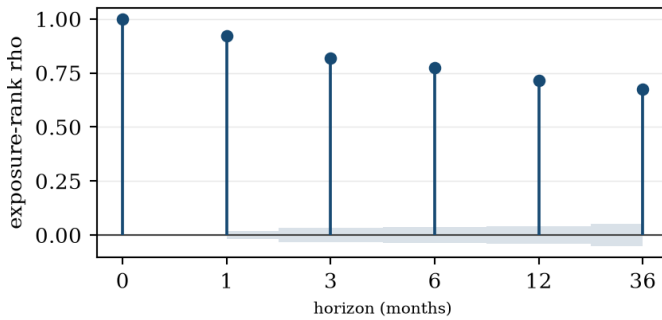


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

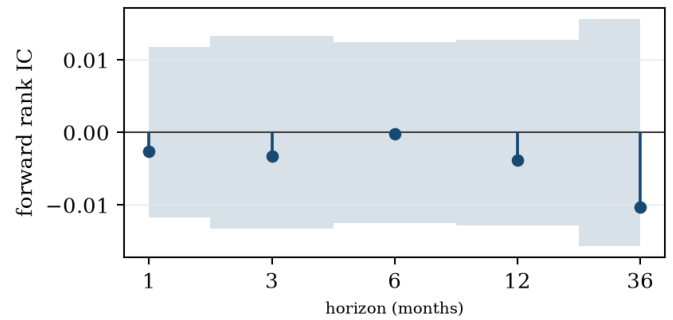


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	is_positive_revenue(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1267 / 1577 / 2099
Available-history Sharpe	0.24
Bootstrap Sharpe	0.24 [-0.23, 0.76]; Pr(S<=0)=15.0%
Bootstrap tails	skew=0.15 [-0.41, 0.75]; max DD=-14.9% [-26.5%, -6.6%]

80. skew_63d

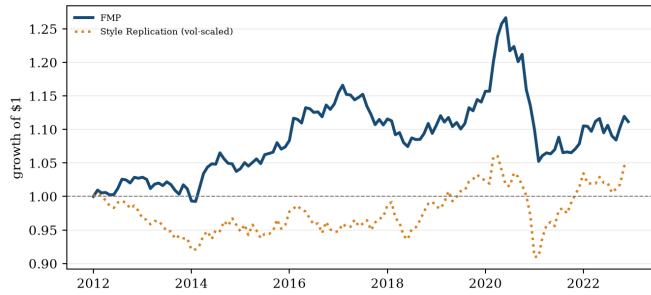


Figure: Common-window cumulative growth of \$1.

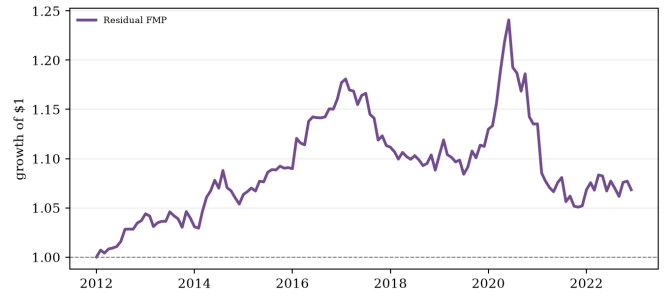


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
skew_63d	1.07%	4.53%	0.24	0.66	-16.90%	0.243	0.81%	4.53%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
skew_63d	0.06 (1.82)	-0.18 (-3.68)	-0.02 (-0.52)	-0.03 (-0.70)	-0.12 (-2.49)	0.11 (2.12)	0.68% (0.41)	0.244

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
skew_63d	0.72%	0.16%	0.05%	-0.14%	-1.02%	0.62%	0.68%	1.07%

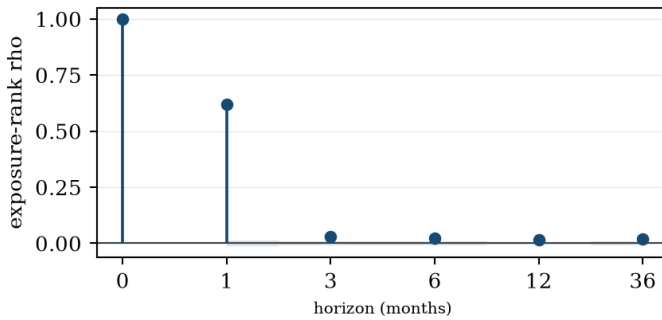


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

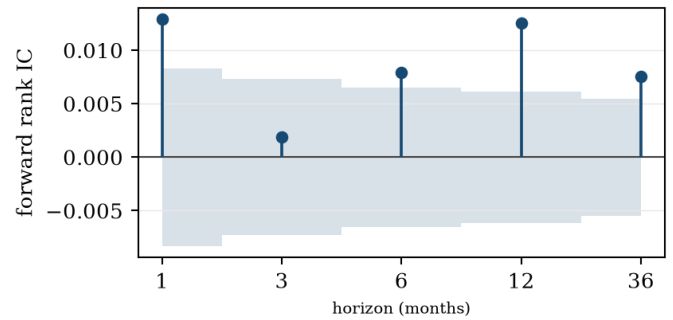


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	skew_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1267 / 1582 / 2103
Available-history Sharpe	0.24
Bootstrap Sharpe	0.24 [-0.34, 0.93]; Pr(S<=0)=21.1%
Bootstrap tails	skew=-0.53 [-0.99, 0.60]; max DD=-16.9% [-29.9%, -4.2%]

81. accruals_percent

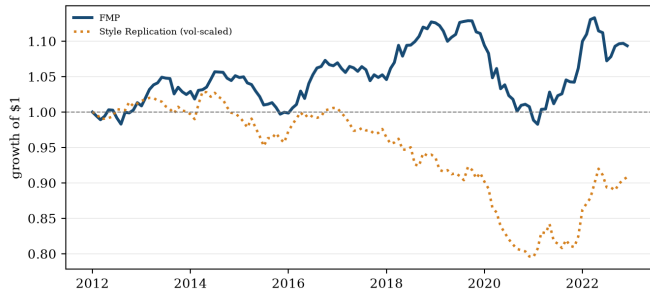


Figure: Common-window cumulative growth of \$1.

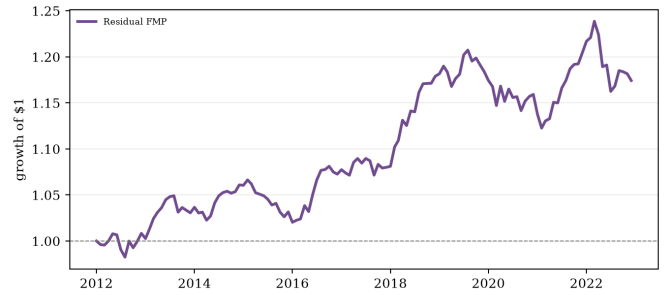


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_percent	0.89%	3.81%	0.23	0.66	-12.95%	0.289	1.33%	3.81%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_percent	-0.03 (-1.27)	-0.05 (-1.26)	0.17 (4.30)	0.01 (0.30)	0.01 (0.29)	-0.00 (-0.09)	1.52% (1.21)	0.293

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_percent	-0.44%	0.04%	-0.38%	0.04%	0.12%	-0.02%	1.52%	0.89%

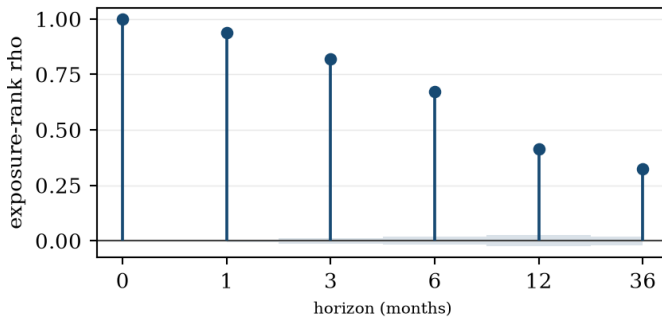


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

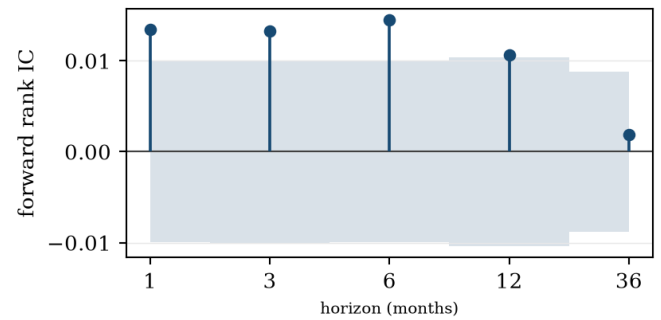


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	accruals_percent(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1215 / 1514 / 2023
Available-history Sharpe	0.23
Bootstrap Sharpe	0.23 [-0.46, 0.87]; Pr(S<=0)=22.2%
Bootstrap tails	skew=-0.20 [-0.72, 0.38]; max DD=-12.9% [-25.5%, -5.3%]

82. rd_to_opex

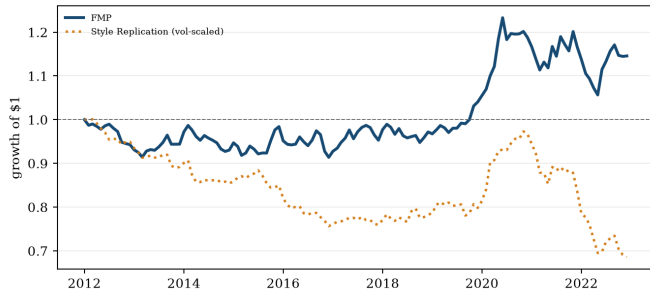


Figure: Common-window cumulative growth of \$1.

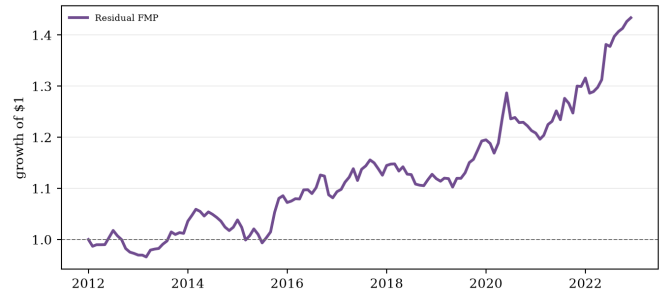


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
rd_to_opex	1.43%	6.16%	0.23	0.73	-14.31%	0.361	3.39%	6.16%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
rd_to_opex	-0.03 (-0.76)	-0.02 (-0.31)	-0.31 (-8.56)	-0.06 (-1.21)	-0.14 (-2.32)	-0.13 (-1.74)	3.42% (2.23)	0.361

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
rd_to_opex	-0.38%	0.02%	0.70%	-0.29%	-1.28%	-0.76%	3.42%	1.43%

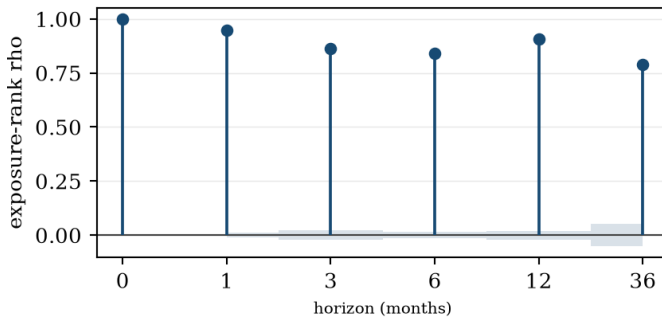


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

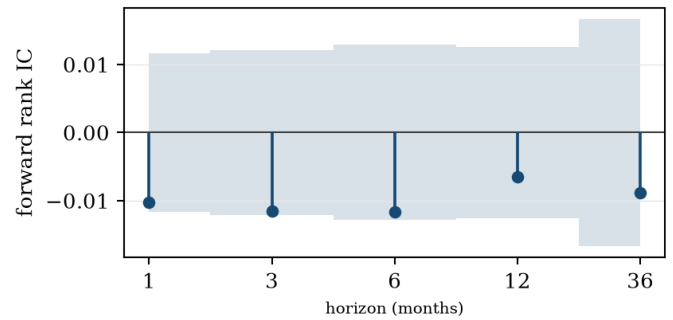


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	rd_to_opex(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	798 / 1266 / 1596
Available-history Sharpe	0.23
Bootstrap Sharpe	0.23 [-0.31, 0.80]; Pr(S<=0)=21.1%
Bootstrap tails	skew=0.46 [-0.14, 0.86]; max DD=-14.3% [-27.3%, -7.7%]

83. net_income_to_market_cap

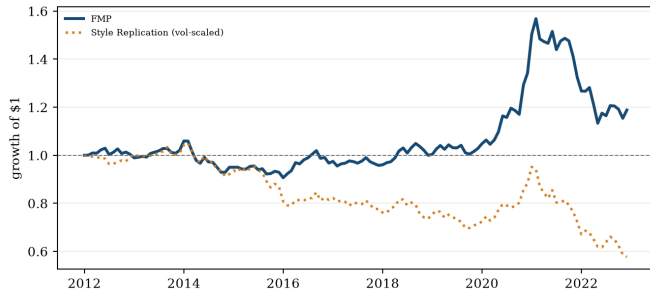


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_income_to_market_cap	1.97%	8.91%	0.22	0.63	-27.74%	0.585	5.51%	8.91%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_income_to_market_cap	-0.08	0.39	-0.17	0.01	0.06	-0.51	5.34%	0.586
	(-1.96)	(4.53)	(-2.54)	(0.11)	(0.68)	(-4.96)	(2.47)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_income_to_market_cap	-1.04%	-0.34%	0.38%	0.03%	0.53%	-2.94%	5.34%	1.97%

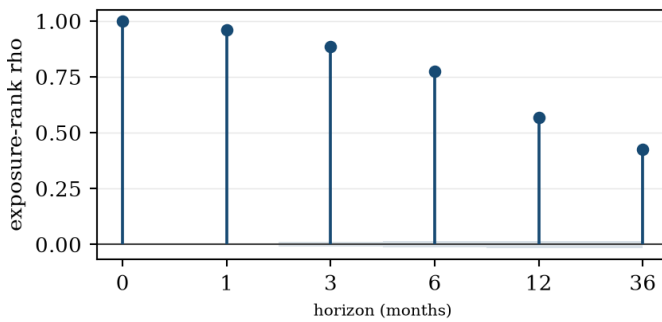


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

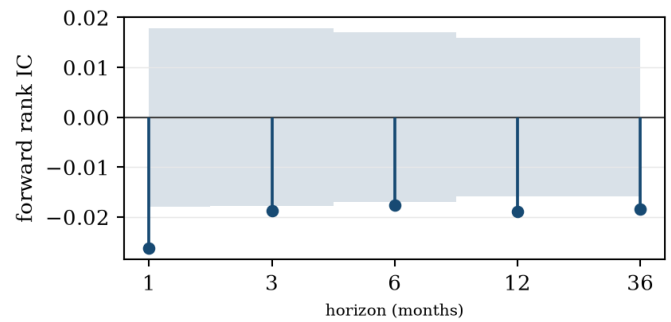


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	net_income_to_market_cap(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1249 / 1576 / 2088
Available-history Sharpe	0.22
Bootstrap Sharpe	0.22 [-0.45, 0.87]; Pr(S<=0)=26.1%
Bootstrap tails	skew=0.86 [-0.77, 1.77]; max DD=-27.7% [-47.4%, -8.8%]

84. corr_1260d_FRENCH_CRSP

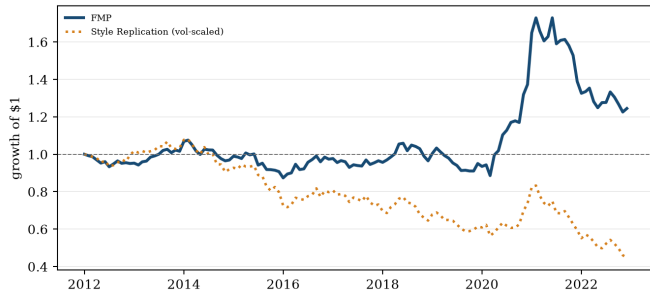


Figure: Common-window cumulative growth of \$1.

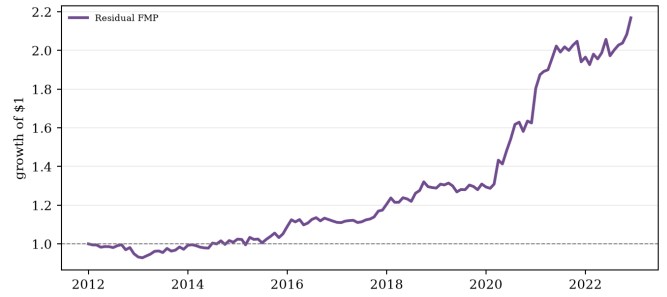


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
corr_1260d_FRENCH_CRSP	2.75%	12.49%	0.22	0.62	-29.11%	0.624	7.95%	12.49%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
corr_1260d_FRENCH_CRSP	-0.06	0.66	-0.03	-0.12	0.14	-0.71	7.40%	0.626
	(-0.78)	(5.58)	(-0.46)	(-1.64)	(0.94)	(-5.65)	(3.25)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
corr_1260d_FRENCH_CRSP	-0.78%	-0.58%	0.07%	-0.59%	1.28%	-4.05%	7.40%	2.75%

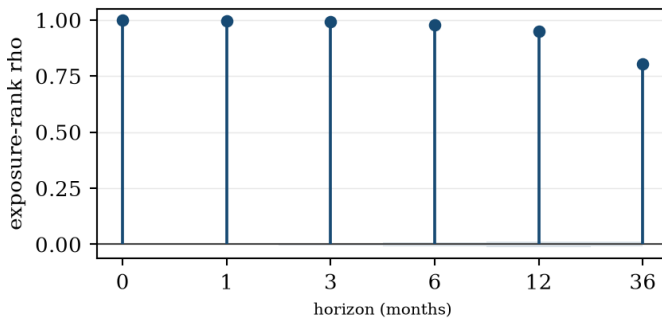


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

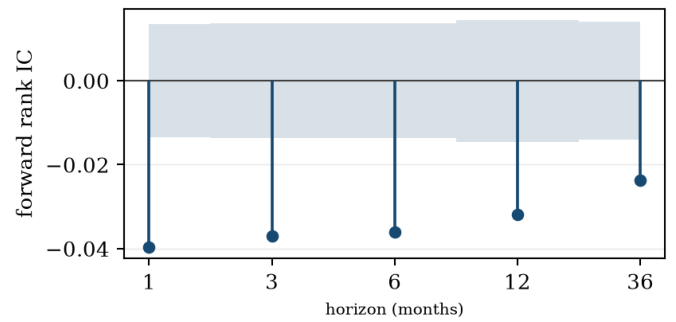


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	corr_1260d_FRENCH_CRSP(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.22
Bootstrap Sharpe	0.22 [-0.52, 0.81]; Pr(S<=0)=28.0%
Bootstrap tails	skew=1.72 [-0.42, 2.41]; max DD=-29.1% [-54.8%, -15.1%]

85. corr_1260d_VTI

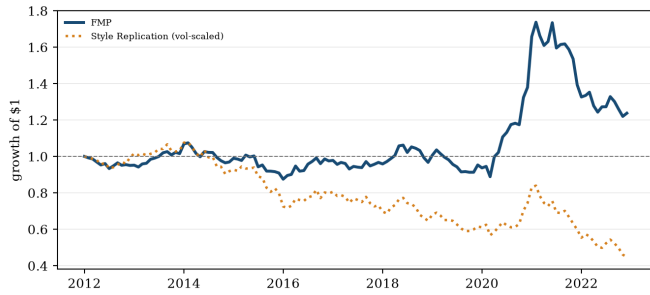


Figure: Common-window cumulative growth of \$1.

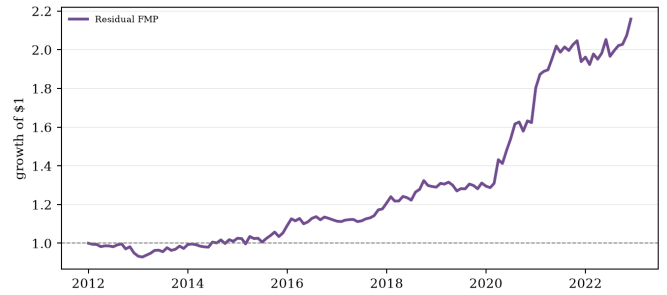


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
corr_1260d_VTI	2.71%	12.55%	0.22	0.60	-29.78%	0.626	7.90%	12.55%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
corr_1260d_VTI	-0.06 (-0.79)	0.66 (5.62)	-0.04 (-0.68)	-0.12 (-1.76)	0.15 (0.95)	-0.71 (-5.60)	7.35% (3.24)	0.627

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
corr_1260d_VTI	-0.80%	-0.58%	0.10%	-0.63%	1.30%	-4.04%	7.35%	2.71%

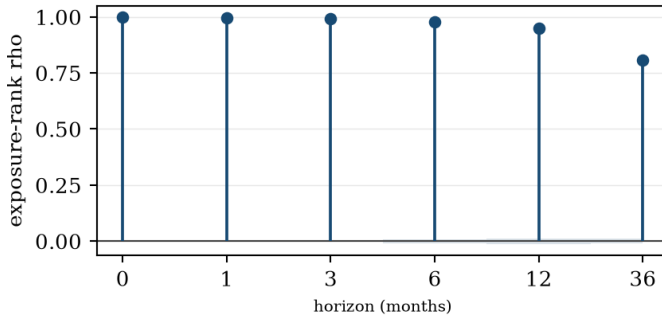


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

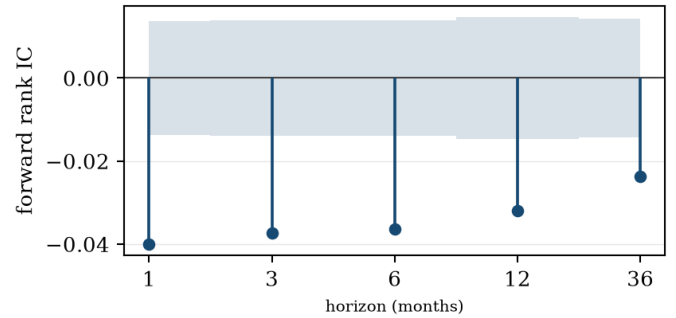


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	corr_1260d_VTI(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.22
Bootstrap Sharpe	0.22 [-0.53, 0.81]; Pr(S<=0)=28.3%
Bootstrap tails	skew=1.70 [-0.44, 2.41]; max DD=-29.8% [-55.6%, -15.1%]

86. corr_1260d_SPY

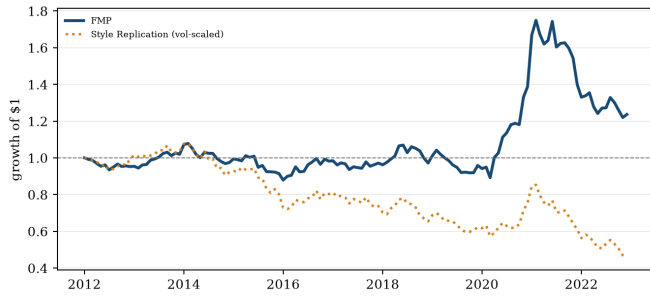


Figure: Common-window cumulative growth of \$1.

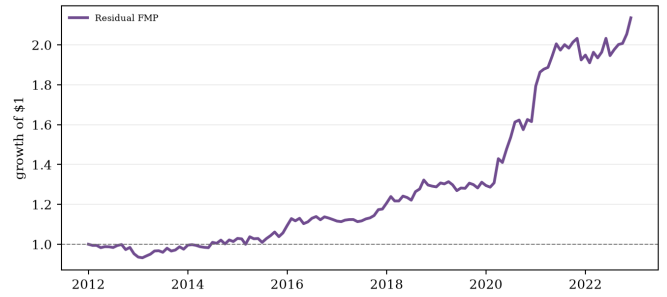


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
corr_1260d_SPY	2.71%	12.62%	0.21	0.60	-30.26%	0.635	7.80%	12.62%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
corr_1260d_SPY	-0.06 (-0.74)	0.69 (5.90)	-0.05 (-0.77)	-0.13 (-1.79)	0.15 (0.95)	-0.69 (-5.56)	7.25% (3.25)	0.637

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
corr_1260d_SPY	-0.74%	-0.61%	0.12%	-0.63%	1.29%	-3.97%	7.25%	2.71%

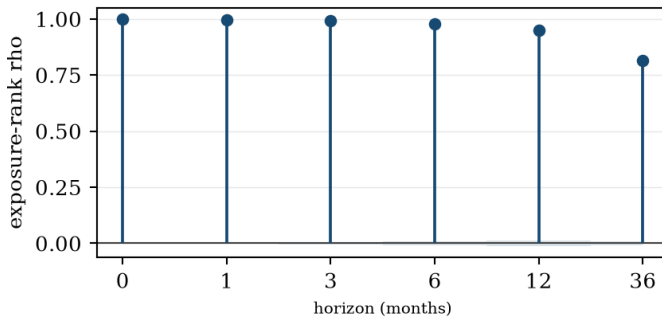


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

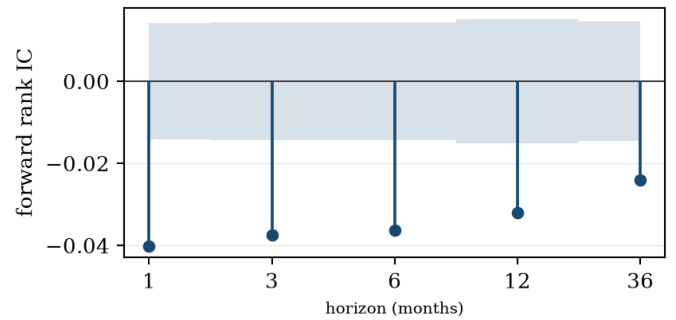


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	corr_1260d_SPY(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.21
Bootstrap Sharpe	0.21 [-0.52, 0.81]; Pr(S<=0)=28.1%
Bootstrap tails	skew=1.68 [-0.42, 2.37]; max DD=-30.3% [-56.2%, -15.1%]

87. growth_gpoa_5y

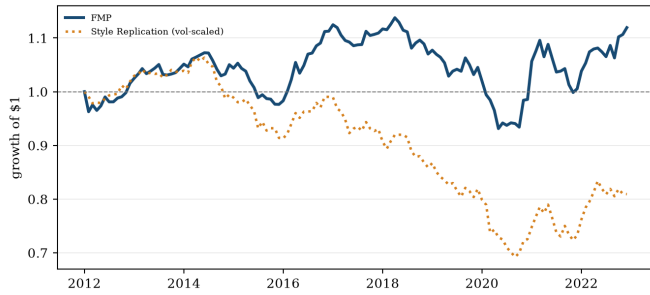


Figure: Common-window cumulative growth of \$1.

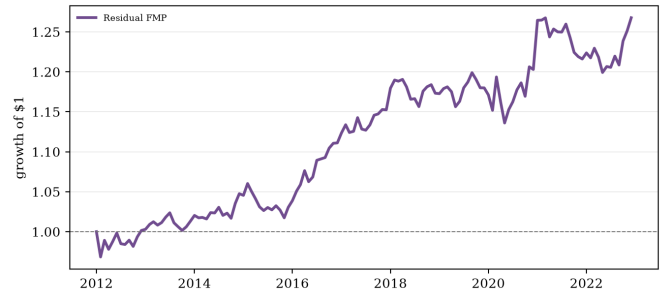


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
growth_gpoa_5y	1.18%	5.54%	0.21	0.58	-18.14%	0.463	2.40%	5.54%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
growth_gpoa_5y	0.00 (0.03)	0.15 (2.64)	0.33 (7.85)	0.08 (1.81)	0.01 (0.07)	-0.11 (-1.49)	2.26% (1.85)	0.464

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
growth_gpoa_5y	0.02%	-0.13%	-0.75%	0.39%	0.05%	-0.65%	2.26%	1.18%

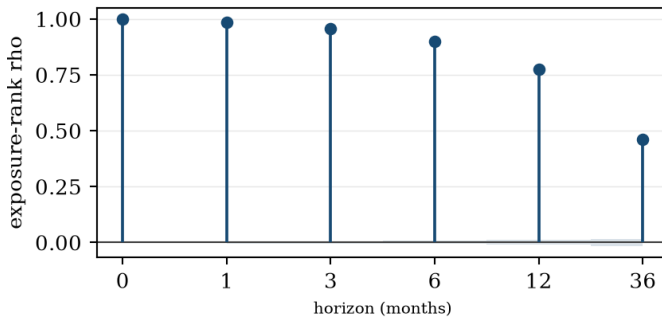


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

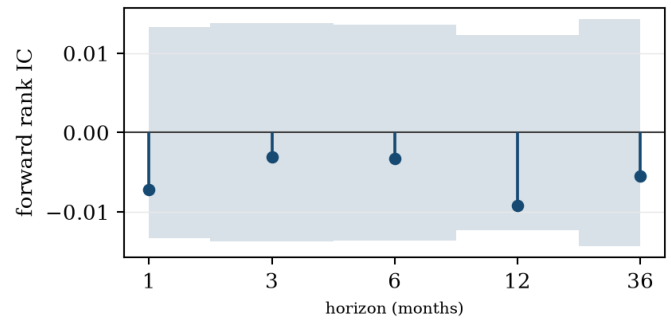


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	growth_gpoa_5y(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	23 / 1426 / 1901
Available-history Sharpe	0.21
Bootstrap Sharpe	0.21 [-0.41, 0.79]; Pr(S<=0)=21.5%
Bootstrap tails	skew=0.71 [-0.39, 1.29]; max DD=-18.1% [-30.6%, -7.8%]

88. ebit_margin_ttm

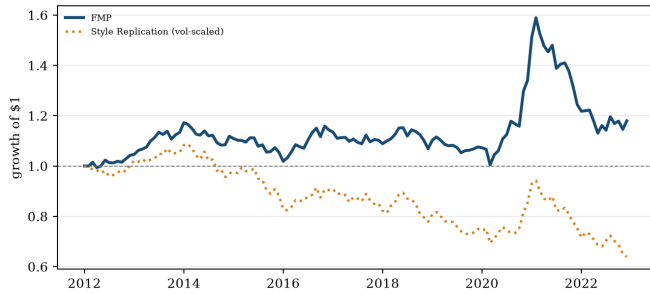


Figure: Common-window cumulative growth of \$1.

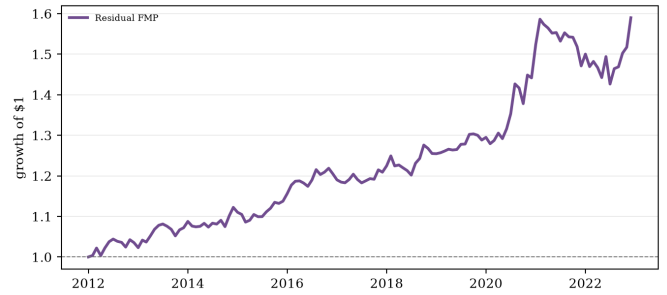


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ebit_margin_ttm	1.93%	9.23%	0.21	0.59	-28.86%	0.637	4.86%	9.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ebit_margin_ttm	-0.01 (-0.11)	0.56 (7.75)	0.04 (0.72)	-0.03 (-0.48)	0.10 (1.23)	-0.45 (-4.55)	4.41% (2.61)	0.639

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ebit_margin_ttm	-0.07%	-0.49%	-0.09%	-0.14%	0.88%	-2.56%	4.41%	1.93%

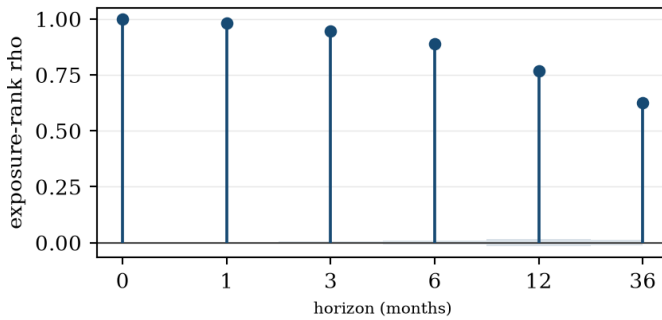


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

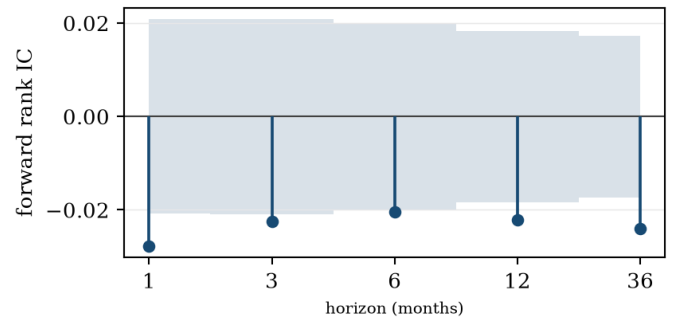


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	ebit_margin_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1239 / 1559 / 2041
Available-history Sharpe	0.21
Bootstrap Sharpe	0.21 [-0.44, 0.77]; Pr(S<=0)=23.4%
Bootstrap tails	skew=1.28 [-0.41, 1.86]; max DD=-28.9% [-46.6%, -11.1%]

89. working_capital_proxy

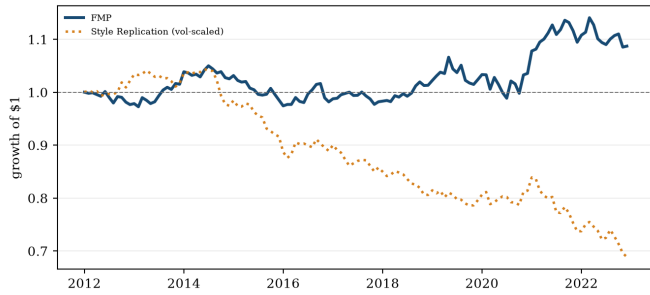


Figure: Common-window cumulative growth of \$1.

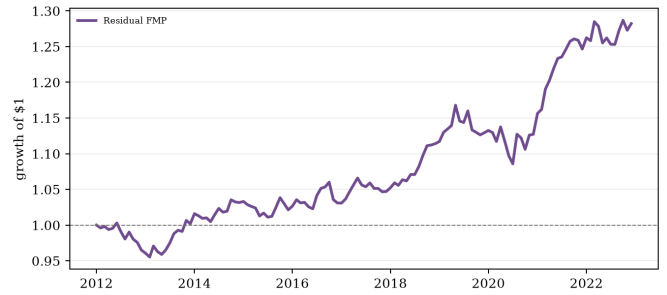


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
working_capital_proxy	0.85%	4.09%	0.21	0.77	-7.27%	0.245	2.50%	4.09%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
working_capital_proxy	-0.02 (-0.83)	-0.03 (-0.65)	0.04 (1.17)	0.02 (0.59)	0.03 (0.55)	-0.27 (-5.28)	2.34% (1.93)	0.247

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
working_capital_proxy	-0.31%	0.03%	-0.10%	0.12%	0.30%	-1.52%	2.34%	0.85%

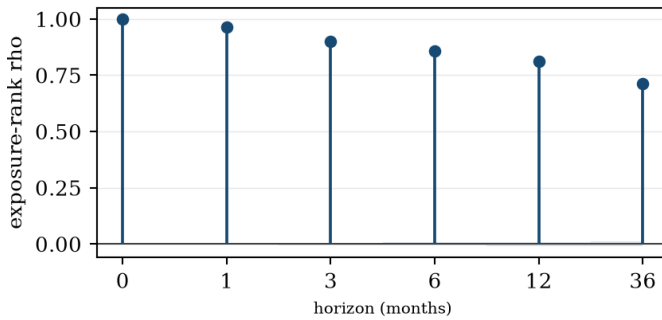


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

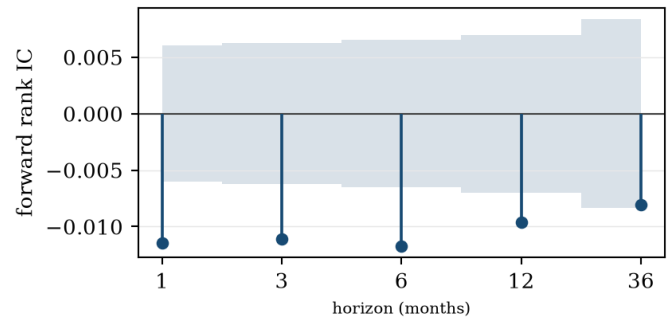


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	working_capital_proxy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1259 / 1579 / 2092
Available-history Sharpe	0.21
Bootstrap Sharpe	0.21 [-0.30, 0.64]; Pr(S<=0)=21.4%
Bootstrap tails	skew=0.41 [-0.30, 0.84]; max DD=-7.3% [-18.1%, -5.3%]

90. momentum_21_756

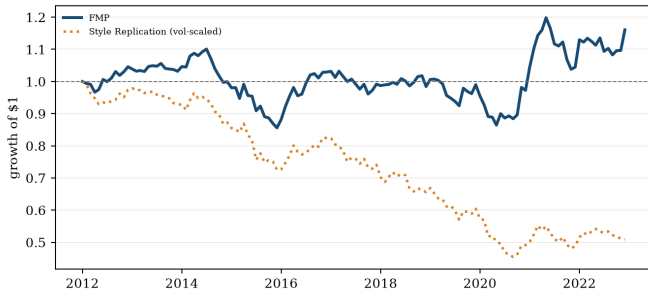


Figure: Common-window cumulative growth of \$1.

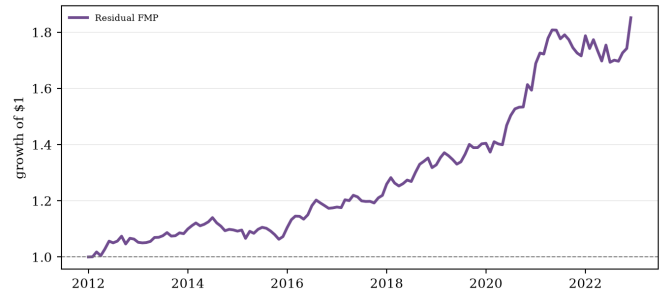


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_756	1.74%	8.74%	0.20	0.57	-22.25%	0.533	5.96%	8.74%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_756	-0.07 (-1.83)	0.21 (2.70)	0.37 (5.95)	-0.19 (-2.98)	-0.03 (-0.36)	-0.16 (-1.56)	5.83% (3.07)	0.533

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_756	-0.94%	-0.18%	-0.85%	-0.94%	-0.27%	-0.91%	5.83%	1.74%

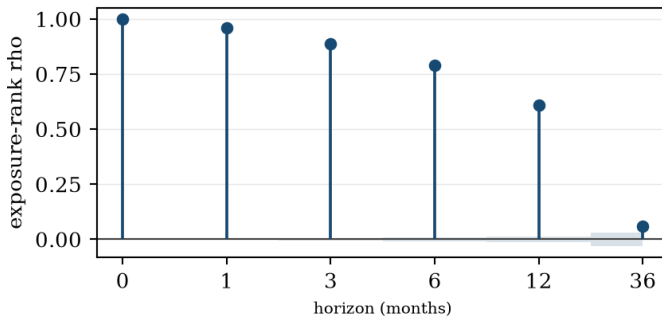


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

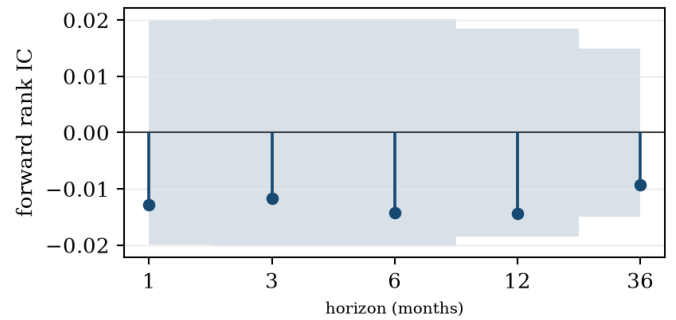


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	momentum_21_756(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.20
Bootstrap Sharpe	0.20 [-0.40, 0.68]; Pr(S<=0)=22.9%
Bootstrap tails	skew=0.87 [0.15, 1.22]; max DD=-22.2% [-41.8%, -13.3%]

91. lottery_max_5_21d

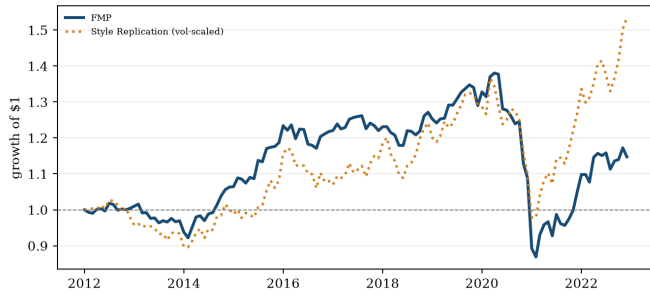


Figure: Common-window cumulative growth of \$1.

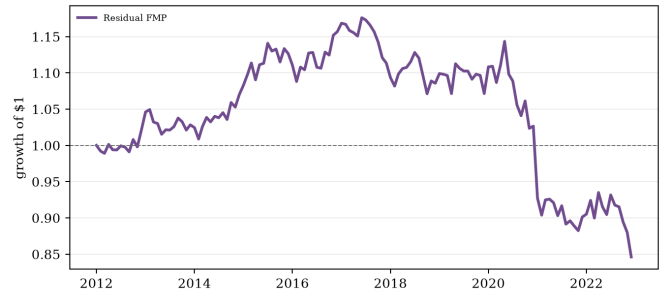


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
lottery_max_5_21d	1.75%	9.74%	0.18	0.54	-37.05%	0.584	-1.61%	9.74%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
lottery_max_5_21d	0.09 (1.67)	-0.47 (-4.54)	0.01 (0.23)	0.05 (0.69)	-0.22 (-1.56)	0.58 (4.60)	-1.33% (-0.69)	0.585

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
lottery_max_5_21d	1.18%	0.41%	-0.03%	0.23%	-1.99%	3.29%	-1.33%	1.75%

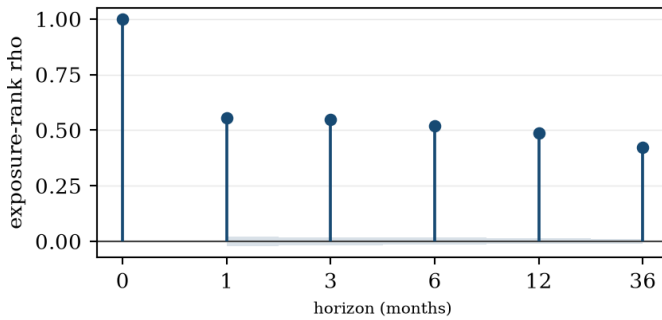


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

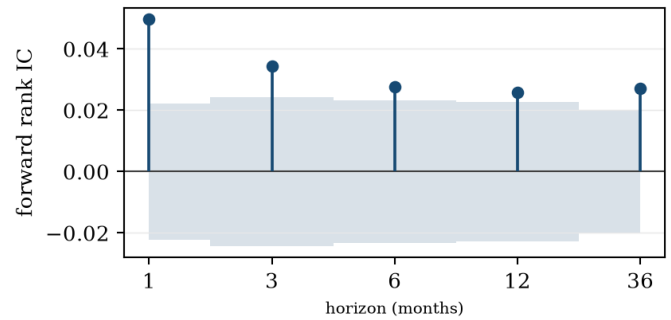


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	lottery_max_5_21d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.18
Bootstrap Sharpe	0.18 [-0.40, 1.04]; Pr(S<=0)=28.5%
Bootstrap tails	skew=-2.15 [-2.97, 0.58]; max DD=-37.0% [-60.1%, -6.8%]

92. growth_acc_5y

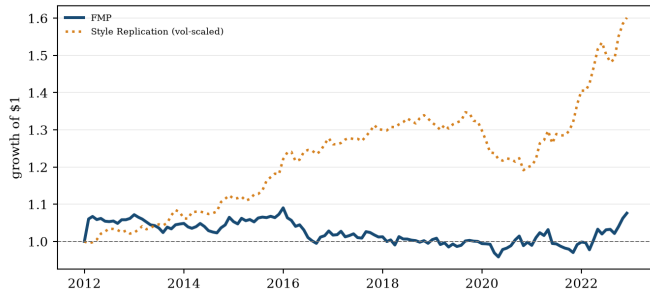


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
growth_acc_5y	0.76%	4.28%	0.18	0.65	-12.10%	0.145	-0.92%	4.28%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
growth_acc_5y	0.04 (1.28)	0.06 (1.14)	0.10 (2.15)	0.05 (0.99)	0.03 (0.73)	0.17 (3.54)	-0.94% (-0.79)	0.145

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
growth_acc_5y	0.48%	-0.05%	-0.22%	0.24%	0.26%	0.99%	-0.94%	0.76%

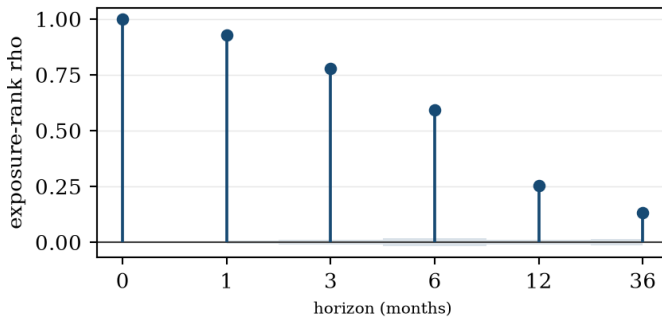


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

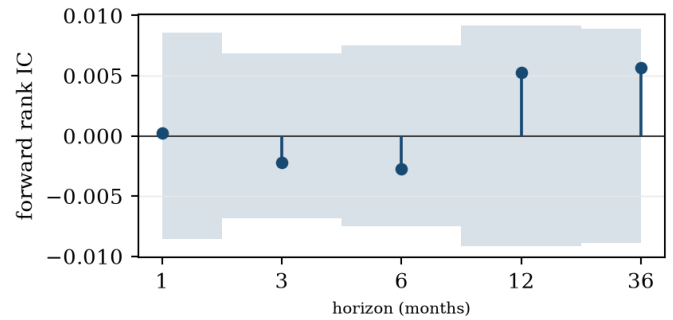


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	growth_acc_5y(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	22 / 1411 / 1893
Available-history Sharpe	0.18
Bootstrap Sharpe	0.18 [-0.39, 0.68]; Pr(S<=0)=29.0%
Bootstrap tails	skew=0.80 [-0.33, 1.73]; max DD=-12.1% [-19.7%, -5.6%]

93. cash_earnings_ttm

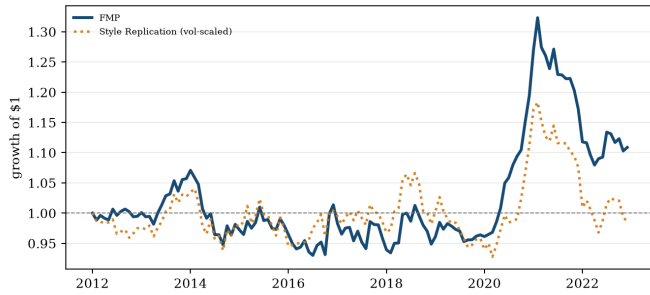


Figure: Common-window cumulative growth of \$1.

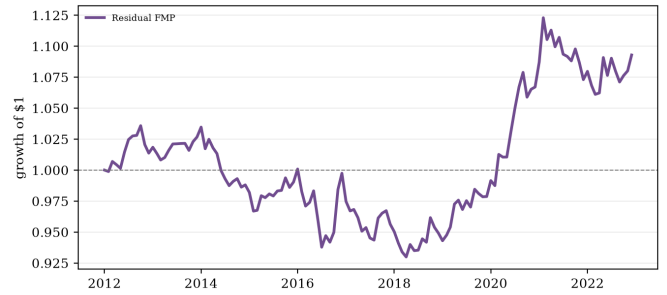


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
cash_earnings_ttm	1.18%	6.92%	0.17	0.49	-18.41%	0.703	1.09%	6.92%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
cash_earnings_ttm	0.04 (1.38)	0.67 (10.68)	-0.10 (-2.39)	0.00 (0.06)	-0.01 (-0.14)	0.04 (0.74)	0.88% (0.75)	0.704

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
cash_earnings_ttm	0.49%	-0.58%	0.22%	0.01%	-0.06%	0.22%	0.88%	1.18%

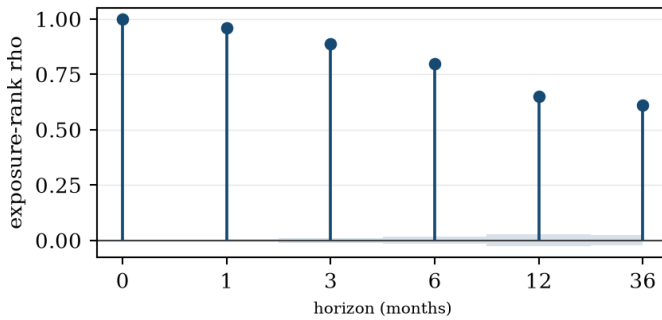


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

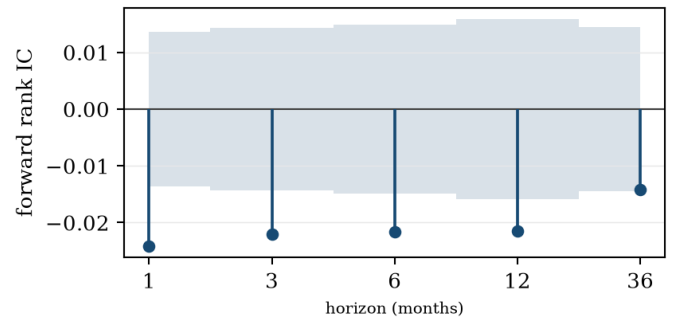


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	cash_earnings_ttm(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1232 / 1556 / 2060
Available-history Sharpe	0.17
Bootstrap Sharpe	0.17 [-0.42, 0.80]; Pr(S<=0)=30.4%
Bootstrap tails	skew=0.74 [0.17, 1.20]; max DD=-18.4% [-35.9%, -9.1%]

94. equity_issuance_yoy

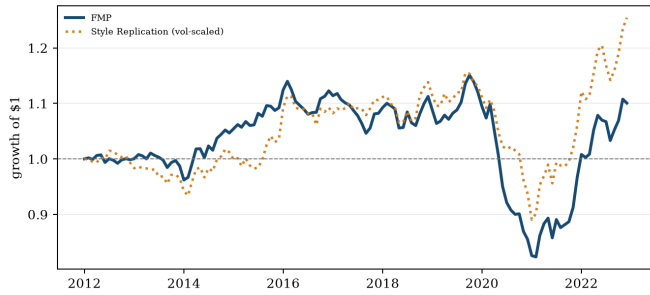


Figure: Common-window cumulative growth of \$1.

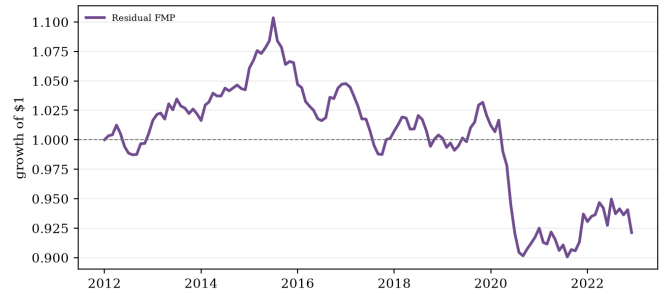


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
equity_issuance_yoy	1.09%	6.47%	0.17	0.44	-28.44%	0.735	-0.87%	6.47%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
equity_issuance_yoy	0.01 (0.41)	-0.24 (-7.32)	0.17 (5.50)	-0.03 (-0.90)	-0.06 (-1.22)	0.43 (9.79)	-0.70% (-0.65)	0.736

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
equity_issuance_yoy	0.15%	0.21%	-0.40%	-0.14%	-0.50%	2.46%	-0.70%	1.09%

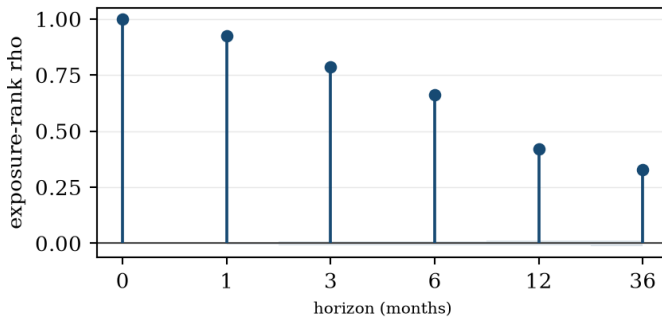


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

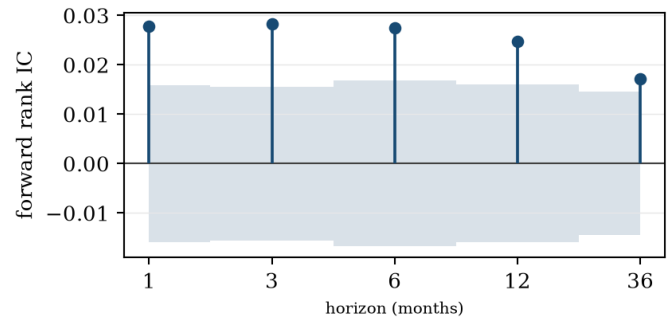


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	equity_issuance_yoy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1263 / 1575 / 2091
Available-history Sharpe	0.17
Bootstrap Sharpe	0.17 [-0.61, 0.93]; Pr(S<=0)=32.2%
Bootstrap tails	skew=0.05 [-0.52, 0.75]; max DD=-28.4% [-47.8%, -6.8%]

95. surprise

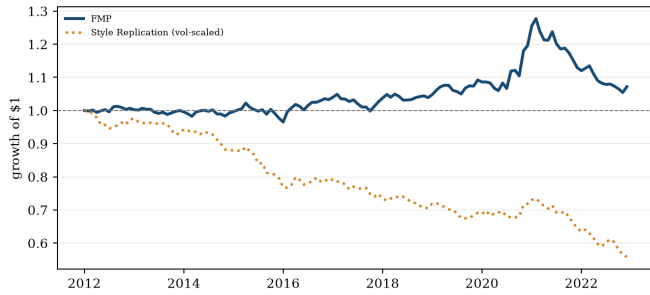


Figure: Common-window cumulative growth of \$1.

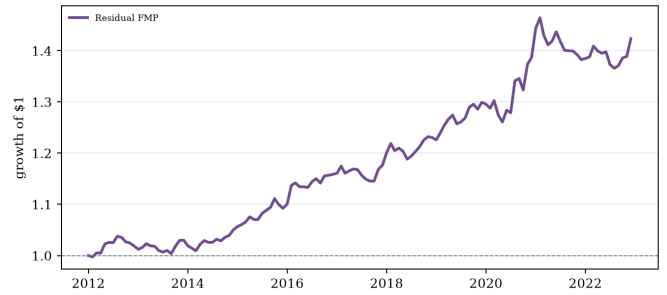


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
surprise	0.74%	4.70%	0.16	0.49	-17.47%	0.311	3.65%	4.70%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
surprise	-0.03 (-0.89)	0.06 (1.03)	-0.03 (-1.23)	-0.12 (-3.57)	-0.03 (-0.37)	-0.24 (-3.09)	3.31% (2.60)	0.315

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
surprise	-0.39%	-0.05%	0.08%	-0.60%	-0.23%	-1.38%	3.31%	0.74%

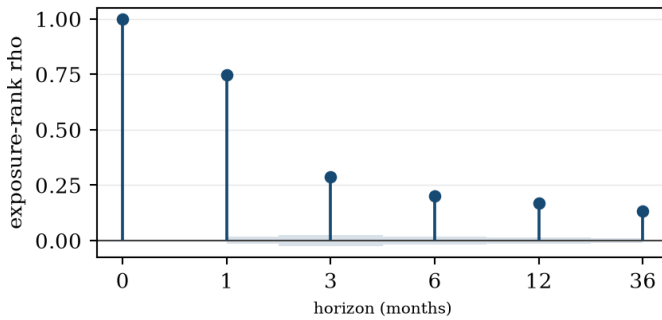


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

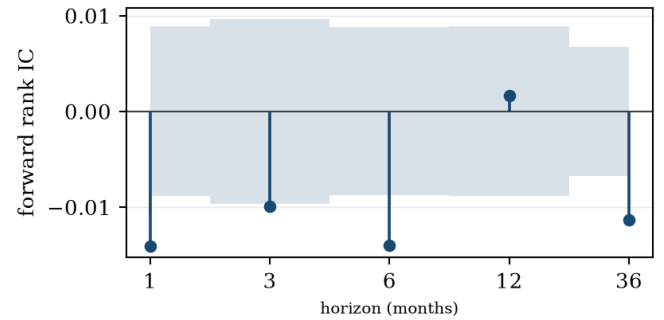


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	surprise(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1166 / 1460 / 2016
Available-history Sharpe	0.16
Bootstrap Sharpe	0.16 [-0.49, 0.68]; Pr(S<=0)=28.5%
Bootstrap tails	skew=1.52 [-0.07, 2.14]; max DD=-17.5% [-28.1%, -5.3%]

96. sga_to_revenue

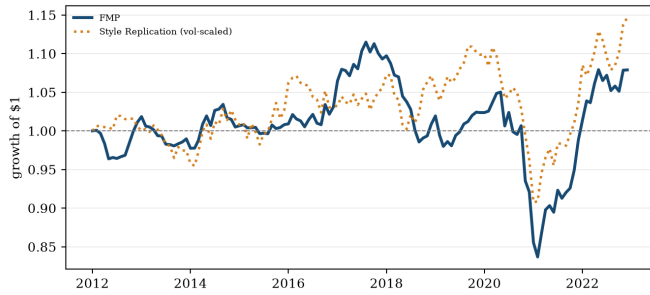


Figure: Common-window cumulative growth of \$1.

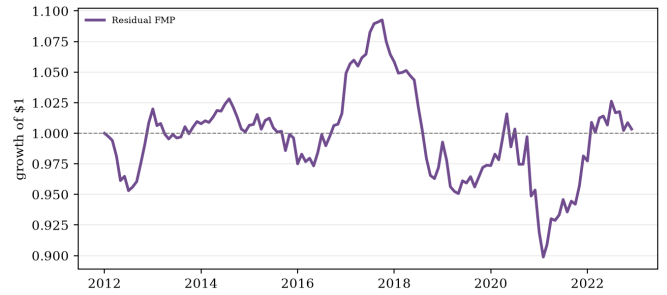


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
sga_to_revenue	0.86%	5.65%	0.15	0.41	-24.94%	0.389	-0.02%	5.65%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
sga_to_revenue	0.03 (0.96)	-0.31 (-5.32)	0.07 (1.70)	0.00 (0.02)	-0.06 (-1.09)	0.14 (1.69)	0.13% (0.07)	0.390

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
sga_to_revenue	0.36%	0.28%	-0.16%	0.01%	-0.54%	0.79%	0.13%	0.86%

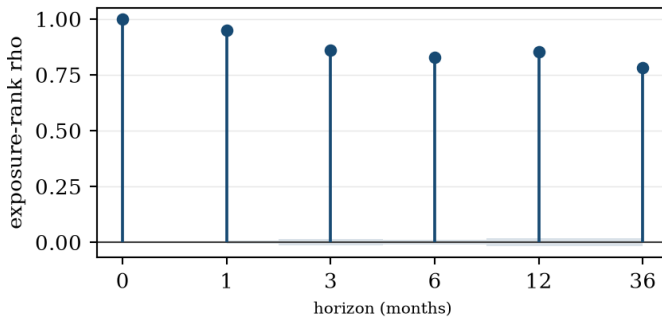


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

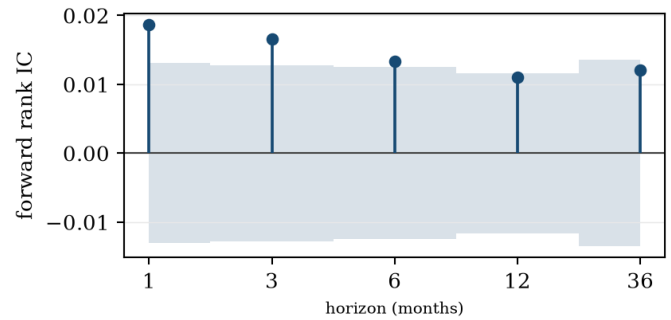


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	sga_to_revenue(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1220 / 1506 / 1964
Available-history Sharpe	0.15
Bootstrap Sharpe	0.15 [-0.47, 0.88]; Pr(S<=0)=32.0%
Bootstrap tails	skew=-1.06 [-1.76, 0.63]; max DD=-24.9% [-38.2%, -5.2%]

97. earnings_yield

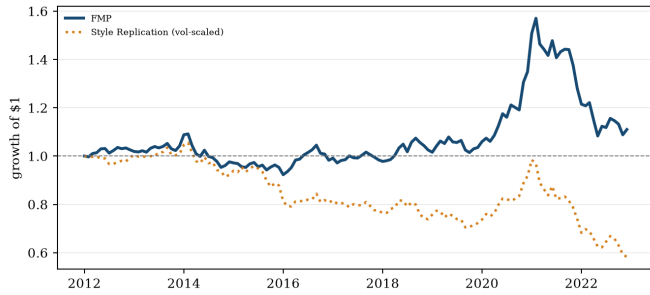


Figure: Common-window cumulative growth of \$1.

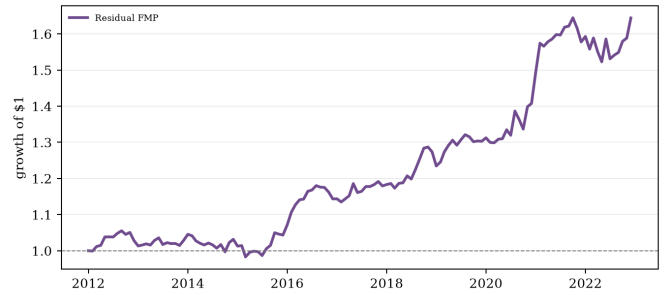


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
earnings_yield	1.37%	9.17%	0.15	0.44	-31.02%	0.607	4.89%	9.17%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
earnings_yield	-0.08 (-1.83)	0.36 (4.25)	-0.21 (-3.41)	0.03 (0.56)	0.05 (0.53)	-0.55 (-5.84)	4.73% (2.27)	0.607

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
earnings_yield	-0.97%	-0.32%	0.47%	0.16%	0.42%	-3.12%	4.73%	1.37%

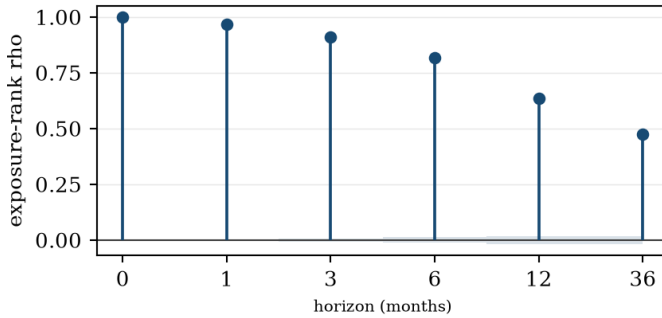


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

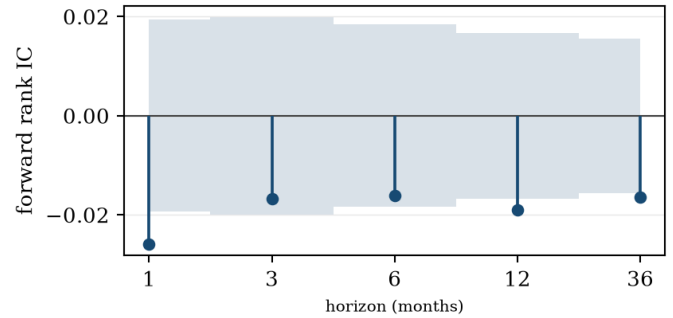


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	earnings_yield(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1050 / 1368 / 2035
Available-history Sharpe	0.15
Bootstrap Sharpe	0.15 [-0.50, 0.80]; Pr(S<=0)=30.9%
Bootstrap tails	skew=0.54 [-0.76, 1.34]; max DD=-31.0% [-51.4%, -10.4%]

98. ebit_to_ev

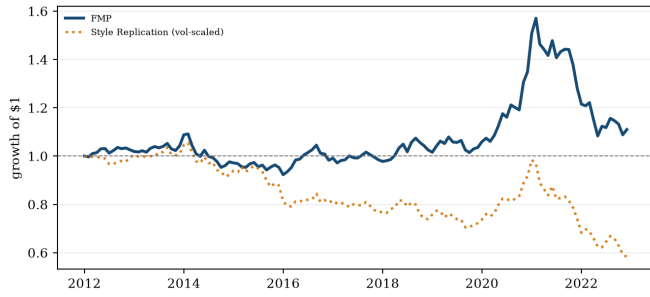


Figure: Common-window cumulative growth of \$1.

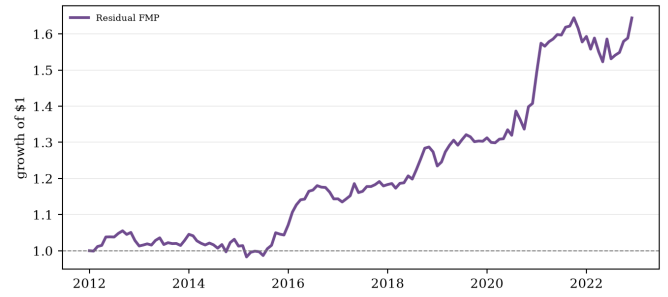


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ebit_to_ev	1.37%	9.17%	0.15	0.44	-31.02%	0.607	4.89%	9.17%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ebit_to_ev	-0.08 (-1.83)	0.36 (4.25)	-0.21 (-3.41)	0.03 (0.56)	0.05 (0.53)	-0.55 (-5.84)	4.73% (2.27)	0.607

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ebit_to_ev	-0.97%	-0.32%	0.47%	0.16%	0.42%	-3.12%	4.73%	1.37%

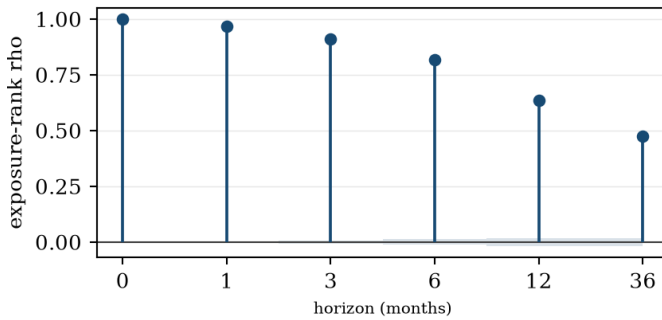


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

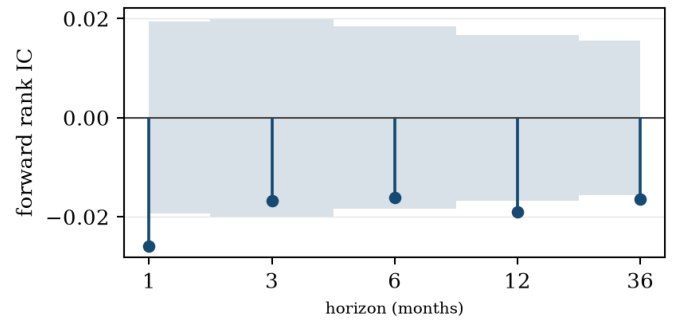


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	ebit_to_ev(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1050 / 1368 / 2035
Available-history Sharpe	0.15
Bootstrap Sharpe	0.15 [-0.50, 0.80]; Pr(S<=0)=30.9%
Bootstrap tails	skew=0.54 [-0.76, 1.34]; max DD=-31.0% [-51.4%, -10.4%]

99. net_income_growth_yoy_sign_flip

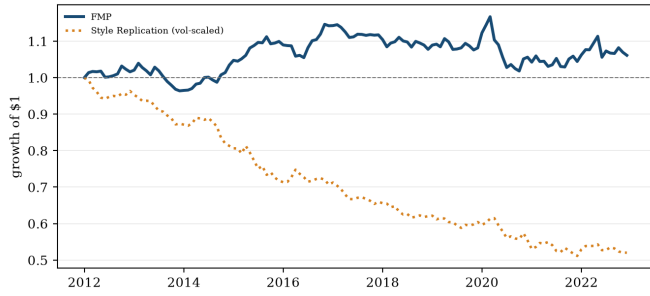


Figure: Common-window cumulative growth of \$1.

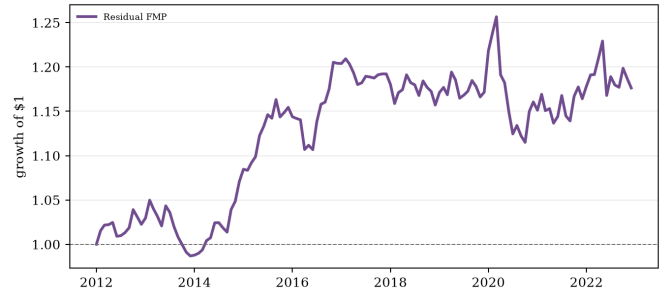


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_income_growth_yoy_sign_flip	0.66%	4.92%	0.13	0.45	-12.74%	0.015	1.37%	4.92%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_income_growth_yoy_sign_flip	-0.03	-0.03	0.02	0.00	-0.04	-0.03	1.61%	0.018
	(-0.44)	(-0.42)	(0.39)	(0.00)	(-0.69)	(-0.44)	(0.99)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_income_growth_yoy_sign_flip	-0.34%	0.03%	-0.05%	0.00%	-0.39%	-0.19%	1.61%	0.66%

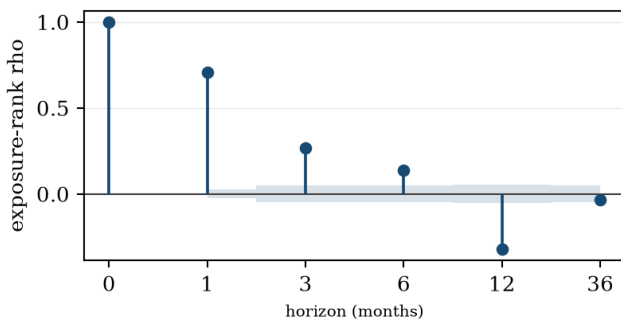


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

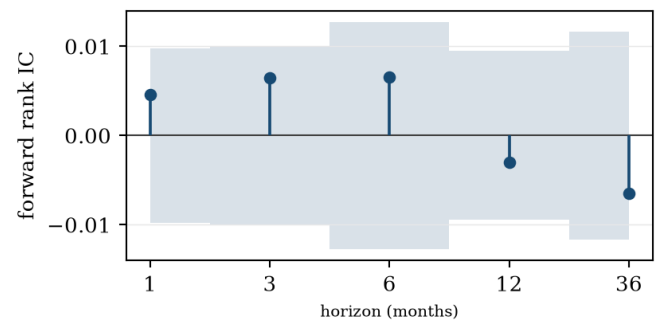


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	net_income_growth_yoy_sign_flip(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1263 / 1575 / 2090
Available-history Sharpe	0.13
Bootstrap Sharpe	0.13 [-0.31, 0.70]; Pr(S<=0)=29.6%
Bootstrap tails	skew=-0.58 [-1.07, 0.32]; max DD=-12.7% [-24.6%, -6.1%]

100. ret_volume_corr_63d

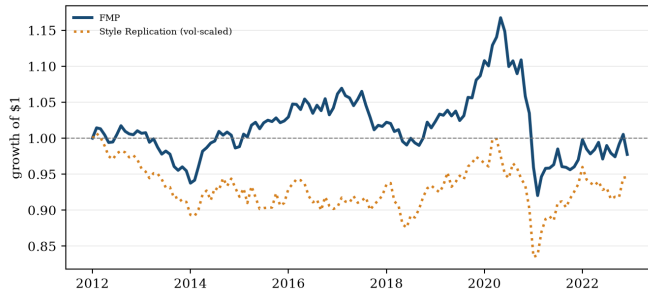


Figure: Common-window cumulative growth of \$1.

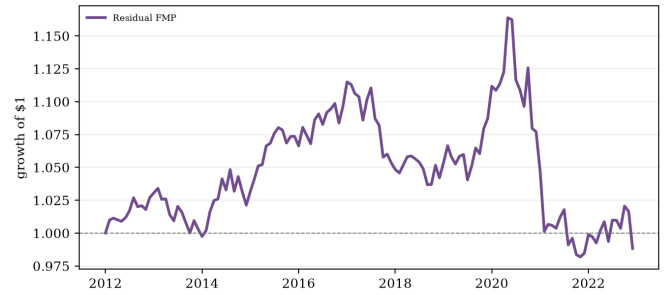


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ret_volume_corr_63d	-0.07%	5.31%	-0.01	-0.04	-21.17%	0.353	0.15%	5.31%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ret_volume_corr_63d	0.06 (2.03)	-0.27 (-4.22)	-0.04 (-0.93)	-0.09 (-2.33)	-0.15 (-3.78)	0.11 (1.60)	-0.02% (-0.01)	0.355

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ret_volume_corr_63d	0.83%	0.24%	0.09%	-0.45%	-1.37%	0.62%	-0.02%	-0.07%

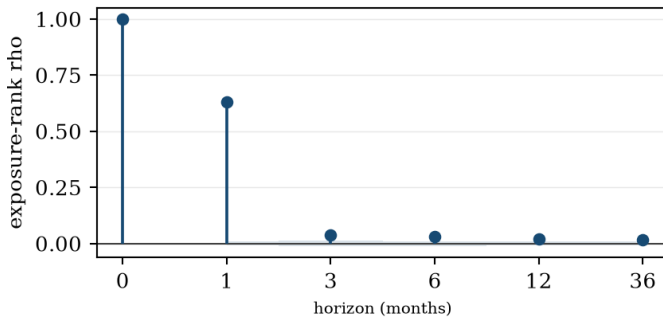


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

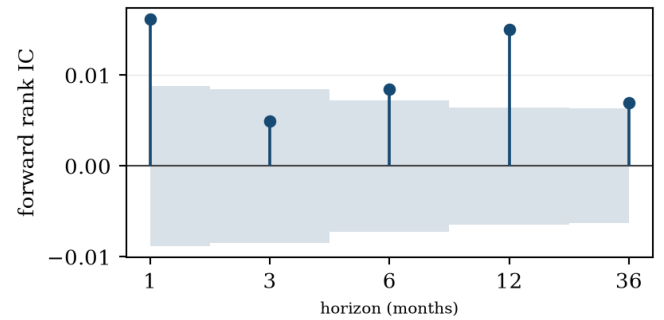


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	ret_volume_corr_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1267 / 1581 / 2103
Available-history Sharpe	-0.01
Bootstrap Sharpe	-0.01 [-0.48, 0.64]; Pr(S<=0)=51.7%
Bootstrap tails	skew=-1.14 [-1.50, 0.12]; max DD=-21.2% [-38.5%, -6.6%]

Reproducibility

Run ID: 20260906T120508Z_no-clustering-bootstrap-screen-sharpe-0-is-2012-2022_b858e9c574.

Gold snapshot: version=20260829T051952Z.

Configuration hash: b858e9c574.

Symbols requested / loaded: 0 / 3009.

Seed: 0. Comparison months: 131, 2012-02-01 to 2022-12-01.

Command:

```
C:/Users/User/Desktop/alpha/.venv/Scripts/python.exeC:/Users/User/Desktop/alpha/src/experiments/automatic_factor_grouping/screen_handoff.py--screen-sourcedata/outputs/automatic_factor_grouping/signal_screening/20260906T064949Z_e2fdcf7479--panel-sourcedata/outputs/automatic_factor_grouping/runs/20260906T065013Z_25145edccf--cutoff0.
```

Direction is frozen upstream; this report performs no return-based sign selection. The reported direction is applied consistently before every downstream regression, style calculation, plot, table, PCA coordinate, and index assignment. The run directory contains the source configuration, normalized configuration, manifest, canonical tables, monthly returns, PCA coordinates, plots, report source, and PDF.